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1 Introduction

The Communications Management Function (CMF) provides Aircraft Communications Addressing and Reporting System (ACARS) and Aeronautical Telecommunications Network (ATN) datalink capabilities to the Boeing 787 airplane.

CMF also provides Air Traffic Control (ATC) datalink applications, including Controller-Pilot Data Link Communications (CPDLC) and Automatic Dependent Surveillance – Contract (ADS-C) applications for Future Air Navigation System (FANS) environments and the Baseline 1 (B1) CPDLC application for ATN environments, together with their respective datalink initiation applications. CMF also provides Flight Information Services (FIS) applications for Digital Automated Terminal Information Service (D-ATIS) reports, oceanic clearances, and other similar reports and clearances. For the 787, Boeing defines these collective Air Traffic Services (ATS) capabilities in the certified CMF operational software and the capabilities are standard for all aircraft operators.

CMF does, however, offer the capability for aircraft operators to customize uplink message decoding, downlink message encoding, display format, print format, and related definitions for an Aeronautical Operational Communications (AOC) / Aeronautical Administrative Communications (AAC) datalink application (also known as the “Company” datalink application), as well as datalink routing definitions for the ATS (ATC and FIS) and AOC/AAC applications. This capability is implemented in a CMF Airline Modifiable Information (AMI) file that is created using the Ground-Based Software Tool (GBST). In order to support 787 airplane production, Boeing has developed a series of CMF “baseline” AMIs with generic functionality. There are two CMF baseline AMI configurations:

- Configuration 1 (Config 1) is a 787-8 CMF baseline AMI for CMF operational software versions before Block Point version (BPv) 2.
- Configuration 2 (Config 2) is a 787-8, 787-9, and 787-10 CMF baseline AMI for CMF operational software versions including and after BPv2.

A 787 aircraft operator may choose to:

- use the Boeing CMF baseline AMI;
- use a Boeing CMF “service ready” AMI, which is designed to prefer a specific datalink service provider (DSP) with specific datalink options;
- request Boeing (or a third party) to construct a partially or completely customized CMF AMI using GBST; or
- construct a partially or completely customized CMF AMI itself using GBST.

This document describes the Boeing CMF baseline AMI, specifically the CMF baseline AMI version F001 that Section 1.2 references.

1.1 CMF AMI Components and Functions

A CMF AMI is composed the following types of components:

- Dataset, which includes global settings for:
 - Function enables
 - System-generated reject messages
- Uplink message decoding definitions
- Downlink message encoding definitions
- Display definitions
- Print definitions
- Logic units
- Review logs definition
- Downlink buffers definition
- Customer parameter database
- ACARS end system uplink routing and downlink label tables
- Datalink options, which includes:
 - Geographic region tables
 - Timers, counters, and enables
 - Subnetwork preference table

In general, one or more components provide specific AMI functions. For example, for the Out, Off, On, In (OOOI) Reports function a logic unit component detects OOOI events and encoding definition components define a downlink message for each event. As another example, for the Free-Text Uplink function a decoding definition component defines the uplink message and a display definition component displays the content of the message.

This document is organized based on the specific functions that the AMI provides and on the supporting components that the AMI contains.

1.1.1 Logic Units

CMF executes logic units once per second in the order defined in the AMI, specifically in this order for the baseline AMI:

- Parameters processing (Section 12.3.1)
- Automatic emergency detection (Section 9.4.1)

-
- Automatic emergency reporting (Section 9.4.2)
 - OOOI state (Section 9.6.1)
 - OOOI events (Section 9.6.2)
 - Airports processing (Section 12.1.1)
 - Reminders (Section 10.2)
 - Automatic position reporting (Section 9.5.1)
 - Automatic diversion reporting (Section 9.3.1)
 - Airplane Heath Management (AHM) flight phase reporting (Section 9.1.1)
 - Notices to Airmen (NOTAMs) Request display SEND button (Section 5.2.1.2)
 - ACARS-over-IP (AoIP) link advisories (Section 9.2.1)
 - Plain Old ACARS (POA) autotune (Section 12.4.2)
 - End of flight processing (Section 12.2.1)

1.2 Config 2 CMF Baseline AMI Part Numbers

Table 1-1 lists the Config 2 CMF baseline AMI part numbers along with the target CMF operational software and other relevant information.

Table 1-1 Config 2 CMF Baseline AMI Part Numbers

Black Label LSAP	Description	Target CCS Hardware	Version	Type	Target CMF Operational Software	GBST / IPDB	Description Document	
BCG28-U000-0083	23 CMF AMI	CCR L8, R8	B000	CFG21307	BPv2 (HNP5E-AL21-5003) BPv2A (HNP55-AL21-5008) BPv2B (HNP54-AL21-5009)	PS4092978-905 / PS4092983-902	D613Z019-01 Rev B	
BCG2F-U000-0084	23 CMF AMI UADF DB	GGM L1, L2, R1, R2						
BCG23-U000-0123	23 CMF AMI	CCR L8, R8	B001	CFG21408	BPv2.5 (HNP5B-AL21-5006)	PS4092978-A06 / PS4092983-902	D613Z019-01 Rev C	
BCG24-U000-0124	23 CMF AMI UADF DB	GGM L1, L2, R1, R2						
BCG29-U000-0291	23 CMF AMI	CCR L8, R8	B002	CFG21505	BPv3 (HNP5A-AL21-5007)	PS4092978-906a / PS4092983-902	D613Z019-01 Rev D	
BCG2A-U000-0292	23 CMF AMI UADF DB	GGM L1, L2, R1, R2						
BCG4D-U000-05R9	23 CMF AMI	CCR L8, R8	D001	2017NOV	BPv5 (HNP5D-AL21-5011)	PS4092978-907 / PS4092983-903	D613Z019-01 Rev E	
BCG45-U000-05S0	23 CMF AMI UADF DB	GGM L1, L2, R1, R2					D002	2018OCT
BCG47-8CMF-A000	23 CMF AMI	CCR L8, R8	D003	2018DEC				
BCG53-8CMF-U000	23 CMF AMI UADF DB	GGM L1, L2, R1, R2					E001	2019APR
BCG42-8CMF-A005	23 CMF AMI	CCR L8, R8	E002	2019JUN				
BCG56-8CMF-U005	23 CMF AMI UADF DB	GGM L1, L2, R1, R2						
BCG47-8CMF-A011	23 CMF AMI	CCR L8, R8	E001	2019APR				
BCG53-8CMF-U011	23 CMF AMI UADF DB	GGM L1, L2, R1, R2						
BCG44-8CMF-A012	23 CMF AMI	CCR L8, R8	E002	2019JUN				
BCG50-8CMF-U012	23 CMF AMI UADF DB	GGM L1, L2, R1, R2						

Black Label LSAP	Description	Target CCS Hardware	Version	Type	Target CMF Operational Software	GBST / IPDB	Description Document
BCG45-8CMF-A013	23 CMF AMI	CCR L8, R8	F001	2020SEP	BPv6 (HNP5E-AL21-5012)	PS4092978-908 / PS4092983-903	D613Z019-06 NEW
BCG51-8CMF-U013	23 CMF AMI UADF DB	GGM L1, L2, R1, R2					

Note: CMF BPv4 software (HNP5C-AL21-5010) did not cause development of a new CMF baseline AMI.

1.3 Significant Changes

Significant changes made from CMF baseline AMI version E002 to version F001 include the following:

- Enhanced diversion report display to include fuel at diversion station and “ENROUTE WEATHER”, “DESTINATION WEATHER”, and “SECURITY” diversion reasons
- Enhanced diversion report downlink (version 2) to include standard initialized destination airport in message header and also diversion station and fuel at diversion station in message body
- Added automatic diversion report logic and downlink
- Added logic to send Airplane Health Management (AHM) flight phase report downlink at OOOI events, diversion event, and flight initialization event
- Added fuel on board, fuel at destination, and ETA at destination to AHM flight phase report downlink
- Added logic for CMF origin and destination airports to more closely follow FMF origin and destination airports as appropriate
- Corrected logic that calculates block fuel
- Changed “DEPARTURE STATION” to “ORIGIN STATION” on flight initialization displays and print formats (for consistency with Flight Management Function [FMF] and CMF ATC displays)
- Disabled AoIP link advisory downlink by default
- Updated VHF region definitions, including addition of Andaman Islands region

1.4 Message Viewing

Messages that a CMF AMI defines to be viewable (i.e., to have an associated display) are viewable in the NEW MESSAGES and REVIEW menus that are respectively accessible via the NEW MESSAGES and REVIEW main menu items.

CMF places a display referenced by an uplink decoding in the NEW MESSAGES menu when it receives the uplink message. CMF also places a display referenced by a logic unit in the NEW MESSAGES menu when it executes the associated logic (such as the Reminders function logic that Section 9.6 describes).

CMF moves a display that is defined to be accessible in a REVIEW menu category from the NEW MESSAGES menu to a category in the REVIEW menu when it has displayed an associated uplink message that does not require an ACCEPT or REJECT response or when the

flight crew has selected ACCEPT or REJECT for an associated uplink message that does require an ACCEPT or REJECT response. Additionally, CMF places a display that references a downlink encoding and is defined to be accessible in a REVIEW menu category when the flight crew has selected SEND for the associated downlink message.

Note: The ACCEPT, CANCEL, and REJECT glareshield buttons are reserved for use by the ATS applications; the flight crew cannot use them to respond to AMI-defined messages.

Section 13.5 describes the REVIEW menu categories that the CMF baseline AMI defines in the Review Logs component and also lists which displays are assigned to which REVIEW menu categories (if any).

1.5 CMF Baseline AMI Functions

The following sections in this document describe the functions that the CMF baseline AMI provides and also the supporting components that the AMI contains.

Note: As defined in the CMF baseline AMI with default values, the Automatic Position Report function that Section 9.5 describes meets the requirements for International Civil Aviation Organization (ICAO) “4D/15” normal aircraft tracking.

2 Global Settings

This section describes the global settings for the CMF baseline AMI, which include function enables that allow an aircraft operator to enable and disable certain certified CMF operational software features.

2.1 Start Display

The start display is the display that CMF shows when the COMPANY main menu item is selected. The start display in the CMF baseline AMI is the Company Menu that Section 3.1 describes.

2.2 ATC Enable

ATC applications are the FANS applications and the B1 applications.

This option allows an aircraft operator to collectively enable or disable these applications (if, for example, it does not wish to train pilots to use them).

In the CMF baseline AMI, this option is set to enable use of the ATC applications.

2.3 FIS Enable

FIS applications are those defined in ARINC Specification 623, *Character-Oriented Air Traffic Service (ATS) Applications* (and equivalent EUROCAE documents) to provide:

- D-ATIS reports
- Oceanic clearances
- Departure clearances
- Terminal Weather Information for Pilots (TWIP) reports
- Expected taxi clearances
- Pushback clearances

This option allows an aircraft operator to enable or disable these applications (if, for example, it does not wish to subscribe to the associated services).

In the CMF baseline AMI, this option is set to enable use of the FIS applications.

2.4 Automatic Master Reset Enable

This option determines whether CMF automatically resets 10 minutes after the end of a flight (i.e., after the airplane has transitioned from the in-air state to the on-ground state and at least one door has been opened) in preparation for the next flight. An automatic master reset clears all displays and message logs and also resets designated parameters to their default values.

In the CMF baseline AMI, this option is set to enable automatic master resets.

2.5 EICAS Memo Message Enable

787 provides Engine Indicating and Crew Alerting System (EICAS) memo messages to annunciate situations when the flight crew has disabled any of the very high frequency (VHF), satellite communications (SATCOM), and high frequency (HF) datalink subnetworks. This option determines whether CMF instructs EICAS to display those messages when appropriate.

In the CMF baseline AMI, this option is set to enable EICAS memo messages.

2.6 Maximum Uplink Delay Timer Enable and Value

The maximum uplink delay timer function provides the ability for CMF to indicate to the flight crew that a FANS CPDLC uplink message has been delayed by more than a certain number of seconds. (This function is inherent to B1 CPDLC uplink messages.)

This option allows an aircraft operator to enable or disable this function, and, if it is enabled, what the default value is (which may be overwritten by the flight crew). An entry of “0” in the AMI results in no default value.

In the CMF baseline AMI, this option is set to enable the maximum uplink delay timer with no default value.

2.7 Password Definition

The CMF baseline AMI sets the password for the DIAGNOSTICS display (which is accessed via DEBUG button on the COMM AUDIT display) to “GORDON”.

Note: The CMF baseline AMI also sets the password for the AIRCRAFT PARAMETERS display (which is accessed via the A/C PARAMS button on the SENSOR STATUS display) to “GORDON”.

Note: The password must be entered into the scratchpad every time the display is accessed.

2.8 Special Encodings

The CMF baseline AMI defines the three special encodings (reject downlinks) that Section 12.5 describes. These encodings respectively indicate that CMF received an uplink which:

- does not comply with the associated decoding definition;
- is intended for a particular datalink end system but CMF cannot deliver the uplink to the end system; or
- CMF is unable to process (for example, when it receives a printer uplink but the printer is unavailable).

3 Menus

The CMF baseline AMI implements two menus, which are displays that provide access to other displays.

3.1 Company Menu

The Company Menu serves as the main menu (the starting display) for the Company application.

3.1.1 Display Definition

Figure 3-1 shows the Company Menu display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	COMPANY	XXXXXXXXX
FLIGHT INIT	DEPARTURE REPORT	WEATHER REQUEST
AUTO FLIGHT INIT REQUEST	ARRIVAL REPORT	CREW REQUESTS...
DELAY REPORT	MESSAGE TO GROUND	MAINTENANCE REPORT
DIVERSION REPORT	ATC DATALINK PROBLEM REPORT	MISCELLANEOUS CODE REPORT
ETA REPORT		SITUATION REPORT
TIMES & FUELS		SENSOR STATUS
REMINDERS		E-MAIL
		EXIT MENU
SCRATCHPAD		

Figure 3-1 Company Menu Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “COMPANY”

Main Area

Table 3-1 describes the Company Menu display items.

Table 3-1 Company Menu Display Items

Menu Item	Display Selected
FLIGHT INIT	FLIGHT INITIALIZATION
AUTO FLIGHT INIT REQUEST	AUTOMATIC FLIGHT INITIALIZATION REQUEST
DELAY REPORT	DELAY REPORT
DIVERSION REPORT	DIVERSION REPORT
ETA REPORT	ETA REPORT
TIMES & FUELS	TIMES & FUELS
REMINDERS	REMINDERS
DEPARTURE REPORT	DEPARTURE REPORT
ARRIVAL REPORT	ARRIVAL REPORT
MESSAGE TO GROUND	MESSAGE TO GROUND
ATC DATALINK PROBLEM REPORT	ATC DATALINK PROBLEM REPORT
WEATHER REQUEST	WEATHER REQUEST
CREW REQUESTS...	CREW REQUESTS
MAINTENANCE REPORT	MAINTENANCE REPORT
MISCELLANEOUS CODE REPORT	MISCELLANEOUS CODE REPORT
SITUATION REPORT	SITUATION REPORT
SENSOR STATUS	SENSOR STATUS
E-MAIL	E-MAIL DOWNLINK MESSAGE

Command Buttons

Table 3-2 describes the Company Menu command buttons. If neither condition is satisfied, then the button is not displayed.

Table 3-2 Company Menu Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-

Position	Legend	Function	Displayed	Displayed as Inhibited
4	-	-	-	-
5	-	-	-	-
6	EXIT MENU	Exit	Always	-

3.2 Crew Requests Menu

The Crew Requests Menu is accessed from the Company Menu and provides access to displays that the flight crew uses to compose downlink requests.

3.2.1 Display Definition

Figure 3-2 shows the Crew Requests Menu display definition.



Figure 3-2 Crew Requests Menu Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “CREW REQUESTS”

Main Area

Table 3-3 describes the Crew Requests Menu display items.

Table 3-3 Crew Requests Menu Display Items

Menu Item	Display Selected
AUTO FLIGHT INIT	AUTOMATIC FLIGHT INITIALIZATION REQUEST
WEIGHT & BALANCE	WEIGHT & BALANCE REQUEST
FLIGHT RELEASE	FLIGHT RELEASE REQUEST
NOTAMS	NOTAMS REQUEST
GATE ASSIGNMENT	GATE ASSIGNMENT REQUEST
WEATHER	WEATHER REQUEST

Command Buttons

Table 3-4 describes the Crew Requests Menu command buttons. If neither condition is satisfied, then the button is not displayed.

Table 3-4 Crew Requests Menu Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-
4	-	-	-	-
5	RETURN	Return	Always	-
6	EXIT MENU	Exit	Always	-

4 Flight Initialization

This section describes the flight initialization functions that the AML implements.

4.1 Manual Flight Initialization

The Manual Flight Initialization function enables the flight crew to manually initialize flight information for the Company application.

4.1.1 Display Definition

Figure 4-1 shows the Manual Flight Initialization display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES

hmmz	FLIGHT INITIALIZATION	XXXXXXXX
------	-----------------------	----------

FLIGHT NUMBER:	-----
SCHEDULED DATE:	--
ORIGIN STATION:	----
DESTINATION STATION:	----
ALTERNATE STATION 1:	----
ALTERNATE STATION 2:	----
FLIGHT PLAN TIME:	--+--
FUEL ON BOARD:	-----
FUEL BOARDED:	-----

CREW PAYROLL NUMBERS			
CAPT:	-----	F/O 1:	-----
F/O 2:	-----	F/O 3:	-----

PRINT	RESET	RETURN	EXIT
-------	-------	--------	------

SCRATCHPAD

Figure 4-1 Manual Flight Initialization Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “FLIGHT INITIALIZATION”

Main Area

Table 4-1 describes the Manual Flight Initialization display entry boxes.

Table 4-1 Manual Flight Initialization Display Entry Boxes

Name	Mandatory	Length	Format	Range	Default	Parameter
FLIGHT NUMBER	No	1-10	XXX...		FMF_CMF_FLIGHT_NUMBER	ENTERED_FLTNO
SCHEDULED DATE	No	1-2	DD	1-31		INIT_SKED_DATE
ORIGIN STATION	No	3-4	XXXX	A-Z, 0-9	FMF_ACTRTEORIGIN	INIT_ORIG_STA
DESTINATION STATION	No	3-4	XXXX	A-Z, 0-9	FMF_ACTRTEDEST	INIT_DEST_STA
ALTERNATE STATION 1	No	3-4	XXXX	A-Z, 0-9		INIT_ALTN_STA_1
ALTERNATE STATION 2	No	3-4	XXXX	A-Z, 0-9		INIT_ALTN_STA_2
FLIGHT PLAN TIME	No	4	HHMM			INIT_FLIGHT_PLAN_TIME
FUEL ON BOARD	No	1-5	NNN.N	0.0-280.0	FUEL_ON_BOARD	INIT_FOB
FUEL BOARDED	No	1-6	NNNNNN	-99999 – 280000		INIT_BOARDED_FUEL
CAPTAIN	No	1-6	XXXXXXX			INIT_PAY_CAPT
F/O 1	No	1-6	XXXXXXX			INIT_PAY_FO_1
F/O 2	No	1-6	XXXXXXX			INIT_PAY_FO_2
F/O 3	No	1-6	XXXXXXX			INIT_PAY_FO_3

Command Buttons

Table 4-2 describes the Manual Flight Initialization display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 4-2 Manual Flight Initialization Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-

Position	Legend	Function	Displayed	Displayed as Inhibited
2	-	-	-	-
3	PRINT	Print	Always	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

4.1.2 Print Definition

Figure 4-2 shows the Manual Flight Initialization print definition.

```

TIME: hhmmz          FLIGHT INITIALIZATION      A/C REG: xxxxxxxx
DATE: dd-MMM-yyyy    FLT NO: xxxxxxxxxxxx

    FLIGHT NUMBER: xxxxxxxxxxxx
    SCHEDULED DATE: dd
    ORIGIN STATION: xxxx
    DESTINATION STATION: xxxx
    ALTERNATE STATION 1: xxxx
    ALTERNATE STATION 2: xxxx
    FLIGHT PLAN TIME: hh+mm
    FUEL ON BOARD: nnn.n
    FUEL BOARDED: nnnnnn

    CREW PAYROLL NUMBERS

    CAPTAIN: xxxxxx
    FIRST OFFICER 1: xxxxxx
    FIRST OFFICER 2: xxxxxx
    FIRST OFFICER 3: xxxxxx

```

Figure 4-2 Manual Flight Initialization Print Definition

4.2 Automatic Flight Initialization

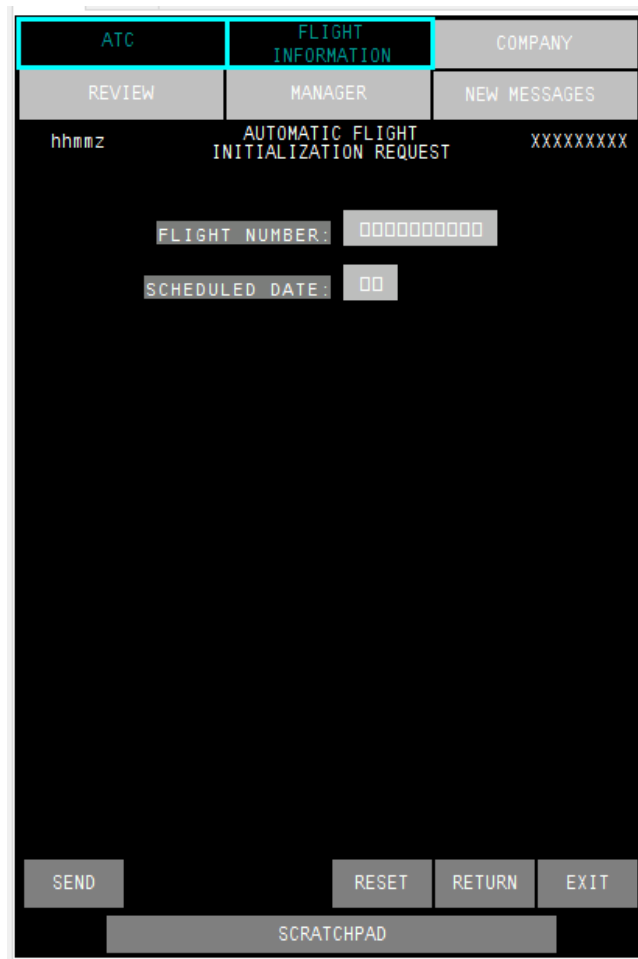
The Automatic Flight Initialization function enables the flight crew to request the aircraft operator to automatically initialize flight information for the Company application via a flight initialization uplink message. The aircraft operator may also automatically initialize flight information for the Company application via a flight initialization uplink message without a request from the flight crew.

4.2.1 Request

This section describes the Automatic Flight Initialization downlink request function.

4.2.1.1 Display Definition

Figure 4-3 shows the Automatic Flight Initialization request display definition.



The image shows a graphical user interface for an Automatic Flight Initialization Request. At the top, there is a header bar with three sections: 'ATC' (highlighted in cyan), 'FLIGHT INFORMATION' (highlighted in cyan), and 'COMPANY'. Below this is a row of three buttons: 'REVIEW', 'MANAGER', and 'NEW MESSAGES'. The main display area has a black background with white text. It shows the current time 'hhmmz' on the left, the title 'AUTOMATIC FLIGHT INITIALIZATION REQUEST' in the center, and 'XXXXXXXX' on the right. Below the title, there are two input fields: 'FLIGHT NUMBER' with a value of '0000000000' and 'SCHEDULED DATE' with a value of '00'. At the bottom, there is a row of four buttons: 'SEND', 'RESET', 'RETURN', and 'EXIT'. Below these buttons is a 'SCRATCHPAD' area.

Figure 4-3 Automatic Flight Initialization Request Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “AUTOMATIC FLIGHT INITIALIZATION REQUEST”

Main Area

Table 4-3 describes the request Automatic Flight Initialization request display entry boxes.

Table 4-3 Automatic Flight Initialization Request Display Entry Boxes

Name	Mandatory	Length	Format	Range	Default	Parameter
FLIGHT NUMBER	Yes	1-10	XXX...		FMF_CMF_FLIGHT_NUMBER	ENTERED_FLTNO
SCHEDULED DATE	Yes	2	DD	1-31		INIT_SKED_DATE

Command Buttons

Table 4-4 describes the Automatic Flight Initialization request display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 4-4 Automatic Flight Initialization Request Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	SEND	Send downlink	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	-	-	-	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

Review Log

When SEND is selected from this display, it becomes available for review in the COMPANY DOWNLINKS category on the REVIEW menu.

4.2.1.2 Downlink Encoding Definition

When SEND is selected from this display, the following downlink encoding is used:

- ACARS SMI: M20
- ACARS Label: 20
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REQUEST

- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 4-5 describes the Automatic Flight Initialization request downlink encoding definition.

Table 4-5 Automatic Flight Initialization Request Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	INI	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"

Sample Message

```
QU RTNBOCR
.DDLXCXA 221846
M20
FI BO24/AN N7873BO
DT DDL BFI 221846 M43A
- INI01 BOE24/22221846
```

4.2.2 Uplink

This section describes the Automatic Flight Initialization uplink function.

4.2.2.1 Uplink Decoding Definition

The decoding for the Automatic Flight Initialization uplink is defined as follows:

- ACARS SMI: CMD
- ACARS Label: RA
- Message Identifier: ~1INI
- Alert Level: Medium (visual “●COMM” and aural high-low chime)
- Optional Element Delimiter: “/” (forward slash)

Table 4-6 describes the Automatic Flight Initialization uplink decoding definition.

Table 4-6 Automatic Flight Initialization Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	2	Yes	<none>	"Priority code (XX)"
2	7	Yes	<none>	"Ground address (XXXXXXX)"
3	5	Yes	<none>	"Message identifier (~1INI)"
4	2	Yes	<none>	"Message version (XX)"
5	2	Yes	<none>	"Message date (DD)"
6	4	Yes	<none>	"Message time (HHMM)"
7	2	Yes	<none>	"Delimiter (<CR><LF>)"
8	3-4	No	INIT_ORIG_STA	"Origin station (3-4 characters [A-Z, 0-9])"
9	3-4	No	INIT_DEST_STA	"Destination station (3-4 characters [A-Z, 0-9])"
10	3-4	No	INIT_ALTN_STA_1	"Alternate station 1 (3-4 characters [A-Z, 0-9])"
11	3-4	No	INIT_ALTN_STA_2	"Alternate station 2 (3-4 characters [A-Z, 0-9])"
12	4	No	INIT_FLIGHT_PLAN_TIME	"Flight plan time (HHMM)"
13	1-6	No	INIT_BOARDED_FUEL	"Boarded fuel (-99999 – 280000)"
14	1-6	No	INIT_PAY_CAPT	"Captain ID (1-6 characters)"
15	1-6	No	INIT_PAY_FO_1	"First officer 1 ID (1-6 characters)"
16	1-6	No	INIT_PAY_FO_2	"First officer 2 ID (1-6 characters)"
17	1-6	No	INIT_PAY_FO_3	"First officer 3 ID (1-6 characters)"

Note: Elements 13 through 17 are variable-length elements and accordingly should be terminated with a delimiter character. Pre-defined element delimiters are <CR>, <CR><LF>, and <ETX> and the optional element delimiter defined for this uplink decoding is "/" (forward slash).

Sample Message

```
QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 001A
- QURTNBOCR~1INI01190457
KBFI/PHOG/PHNL//0456/28000/1/22/333//
```

Note: In this sample message, Alternate Station 2 and First Officer 3 ID are omitted.

4.2.2.2 Display Definition

Figure 4-4 shows the Automatic Flight Initialization uplink display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	FLIGHT INITIALIZATION	XXXXXXXXXX
FLIGHT NUMBER: XXXXXXXXXX SCHEDULED DATE: DD ORIGIN STATION: XXXX DESTINATION STATION: XXXX ALTERNATE STATION 1: XXXX ALTERNATE STATION 2: XXXX FLIGHT PLAN TIME: HH+MM FUEL ON BOARD: NNN.N FUEL BOARDED: NNNNNN CREW PAYROLL NUMBERS CAPT: XXXXXX F/O 1: XXXXXX F/O 2: XXXXXX F/O 3: XXXXXX		
PRINT FLIGHT INIT CANCEL SCRATCHPAD		

Figure 4-4 Automatic Flight Initialization Uplink Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “FLIGHT INITIALIZATION”

Main Area

Table 4-7 describes the Automatic Flight Initialization uplink display parameters.

Table 4-7 Automatic Flight Initialization Uplink Display Parameters

Name	Length	Format	Range	Parameter
FLIGHT NUMBER	3-10	XXXXXXXXXX		ENTERED_FLTNO
SCHEDULED DATE	2	DD	1-31	INIT_SKED_DATE

Name	Length	Format	Range	Parameter
ORIGIN STATION	3-4	XXXX		INIT_ORIG_STA
DESTINATION STATION	3-4	XXXX		INIT_DEST_STA
ALTERNATE STATION 1	3-4	XXXX		INIT_ALTN_STA_1
ALTERNATE STATION 2	3-4	XXXX		INIT_ALTN_STA_2
FLIGHT PLAN TIME	5	HH+MM		INIT_FLIGHT_PLAN_TIME
FUEL ON BOARD	3-5	NNN.N	0.0-280.0	INIT_FOB
FUEL BOARDED	1-6	NNNNNN	-99999 – 280000	INIT_BOARDED_FUEL
CAPTAIN	1-6	XXXXXX		INIT_PAY_CAPT
F/O 1	1-6	XXXXXX		INIT_PAY_FO_1
F/O 2	1-6	XXXXXX		INIT_PAY_FO_2
F/O 3	1-6	XXXXXX		INIT_PAY_FO_3

Command Buttons

Table 4-8 describes the Automatic Flight Initialization display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 4-8 Automatic Flight Initialization Uplink Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	PRINT	Print	Always	-
4	FLIGHT INIT	Show FLIGHT INIT display	-	-
5	-	-	-	-
6	CANCEL	Cancel	Always	-

Review Log

When the associated uplink decoding is received, this display becomes available for review in the COMPANY UPLINKS category on the REVIEW menu.

4.2.2.3 Print Definition

Figure 4-5 shows the Automatic Flight Initialization uplink print definition.

```
TIME: hhmmz          FLIGHT INITIALIZATION      A/C REG: xxxxxx
DATE: dd-MMM-yyyy    FLT NO: xxxxxxxxxxxx

    FLIGHT NUMBER: xxxxxxxxxxxx
    SCHEDULED DATE: dd
    ORIGIN STATION: xxxx
    DESTINATION STATION: xxxx
    ALTERNATE STATION 1: xxxx
    ALTERNATE STATION 2: xxxx
    FLIGHT PLAN TIME: hh+mm
    FUEL ON BOARD: nnn.n
    FUEL BOARDED: nnnnnn

    CREW PAYROLL NUMBERS

    CAPTAIN: xxxxxx
    FIRST OFFICER 1: xxxxxx
    FIRST OFFICER 2: xxxxxx
    FIRST OFFICER 3: xxxxxx
```

Figure 4-5 Automatic Flight Initialization Uplink Print Definition

5 Weather and NOTAMs

This section describes the weather and Notices to Airmen (NOTAMs) functions that the AMI implements.

5.1 Weather

The Weather function enables the flight crew to request and receive weather reports and forecasts.

5.1.1 Request

This section describes the Weather Request function.

5.1.1.1 Display Definition

The Weather Request display enables the flight crew to request weather reports and forecasts from an aircraft operator host system or other service provider. The flight crew may enter stations (airport identifiers) in either three-character International Air Transport Association (IATA) format or four-character ICAO format. Figure 5-1 shows the Weather Request display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	WEATHER REQUEST	XXXXXXXX
◆ ACTUAL	◆ FORECAST 1	
◆ FORECAST 2	◆ ACTUAL & FORECAST	
STATION 1: 0000		
STATION 2: ----		
STATION 3: ----		
STATION 4: ----		
SEND	RESET	RETURN
EXIT		
SCRATCHPAD		

Figure 5-1 Weather Request Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “WEATHER REQUEST”

Main Area

The upper portion of the display contains a group of exclusive selector buttons (ESBs) to select the type of weather request:

- ACTUAL (WXR_RQST_TYPE = SA)
- FORECAST 1 (WXR_RQST_TYPE = FC)
- FORECAST 2 (WXR_RQST_TYPE = FT)

- ACTUAL & FORECAST (WXR_RQST_TYPE = FS)

Table 5-1 describes the Weather Request display entry boxes.

Table 5-1 Weather Request Display Entry Boxes

Name	Mandatory	Length	Format	Range	Default	Parameter
STATION 1	Yes	3-4	XXXX			WXR_STA_1
STATION 2	No	3-4	XXXX			WXR_STA_2
STATION 3	No	3-4	XXXX			WXR_STA_3
STATION 4	No	3-4	XXXX			WXR_STA_4

Command Buttons

Table 5-2 describes the Weather Request display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 5-2 Weather Request Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	SEND	Send downlink	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	-	-	-	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

Review Log

When SEND is selected from this display, it becomes available for review in the WEATHER & NOTAMS category on the REVIEW menu.

5.1.1.2 Downlink Encoding Definition

When SEND is selected from this display, the following downlink encoding is used:

- ACARS SMI: WXR
- ACARS Label: 5U
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REQUEST

- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 5-3 describes the Weather Request downlink encoding definition.

Table 5-3 Weather Request Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	2	Conditionally	SA	"Weather Request Type (SA)"
3	2	Conditionally	FC	"Weather Request Type (FC)"
4	2	Conditionally	FT	"Weather Request Type (FT)"
5	2	Conditionally	FS	"Weather Request Type (FS)"
6	1	Always	<SP>	"Delimiter"
7	4	Always	WXR_STA_1	"Station 1 (XXXX)"
8	1	Conditionally	,	"Delimiter"
9	4	Conditionally	WXR_STA_2	"Station 2 (XXXX)"
10	1	Conditionally	,	"Delimiter"
11	4	Conditionally	WXR_STA_3	"Station 3 (XXXX)"
12	1	Conditionally	,	"Delimiter"
13	4	Conditionally	WXR_STA_4	"Station 4 (XXXX)"

Note: The weather station parameters and their preceding “,” (comma) delimiters are included only if they have been entered.

Sample Message

```
QU RTNBOCR
.DDLXCXA 261848
WXR
FI BO1234/AN N7873BO
DT DDL BFI 261848 M02A
- FS CDG ,EGLL
```

Note: In this sample message, the weather request type is “FS” (actual and forecast) and weather is requested for “CDG” (three-characters left-justified) and “EGLL”.

5.1.2 Uplink

This section describes the Weather Uplink function.

5.1.2.1 Uplink Decoding Definition

The decoding for the Weather Uplink is defined as follows:

- ACARS SMI: CMD
- ACARS Label: RA
- Message Identifier: ~1WXR
- Alert Level: Medium (visual “●COMM” and aural high-low chime)
- Optional Element Delimiter: “,” (comma)

Table 5-4 describes the Weather Uplink decoding definition.

Table 5-4 Weather Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	2	Yes	<none>	"Priority code (XX)"
2	7	Yes	<none>	"Ground address (XXXXXXX)"
3	5	Yes	<none>	"Message identifier (~1WXR)"
4	2	Yes	<none>	"Message version (XX)"
5	2	Yes	<none>	"Delimiter (<CR><LF>)"
6	1-16	Yes	WX_TITLE	"Message title (1-16 characters)"
7	1-3486	Yes	UPLINK_TXT_BLK	"Message text (1-3486 characters)"

Note: Element 6 is a variable-length element and accordingly should be terminated with a delimiter character. Pre-defined element delimiters are <CR>, <CR><LF>, and <ETX> and the optional element delimiter defined for this uplink decoding is “,” (comma).

Sample Message

```
QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 002A
- QURTNBOCR~1WXR01
WEATHER TITLE
WEATHER TEXT BLOCK...
```

5.1.2.2 Display Definition

Figure 5-2 shows the Weather Uplink display definition.



Figure 5-2 Weather Uplink Display Definition

Title

The title area contains:

- The current time in HHMMz format
- The title contained in the uplink (WX_TITLE)

Main Area

The main area contains the text block from the uplink. If any lines of the text block contain more than 46 characters, then that line is word-wrapped. If the entire text block cannot be displayed on a single page, then as many pages as are necessary are created and page arrows are displayed on the right side to access each page.

Command Buttons

Table 5-5 describes the Weather Uplink display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 5-5 Weather Uplink Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	PRINT	Print	Always	-
4	-	-	-	-
5	-	-	-	-
6	CANCEL	Cancel	Always	-

Review Log

When the associated uplink decoding is received, this display becomes available for review in the WEATHER & NOTAMS category on the REVIEW menu.

5.1.2.3 Print Definition

Figure 5-3 shows the Weather Uplink print definition.

TIME: hhmmz	xxxxxxxxxxxxxx	A/C REG: xxxxxxx
DATE: dd-MMM-yyyy		FLT NO: xxxxxxxxx

-----Text Block-----

Figure 5-3 Weather Uplink Print Definition

5.2 NOTAMs

The NOTAMs function enables the flight crew to request and receive NOTAMs.

5.2.1 Request

This section describes the NOTAMs Request function.

5.2.1.1 Display Definition

The NOTAMs Request display enables the flight crew to request station (airport) and/or enroute NOTAMs from the aircraft operator. The flight crew may enter separate stations for station NOTAMs or route information for enroute NOTAMs and may also enter stations (airport identifiers) in three-character IATA format or four-character ICAO format. The supplemental text fields are for entering information that may be required by the aircraft operator's host system to complete the request. Figure 5-4 shows the NOTAMs Request display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	NOTAMS REQUEST	XXXXXXXX
STATION 1: STATION 2: ENROUTE STATIONS: SUPPLEMENTAL TEXT:		
SEND	RESET	RETURN
EXIT		
SCRATCHPAD		

Figure 5-4 NOTAMs Request Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “NOTAMS REQUEST”

Main Area

Table 5-6 describes the NOTAMs Request display entry boxes.

Table 5-6 NOTAMs Request Display Entry Boxes

Name	Mand atory	Length	Format	Range	Default	Parameter
STATION 1	No	3-4	XXXX			NOTAM_STA_1
STATION 2	No	3-4	XXXX			NOTAM_STA_2

Name	Mandatory	Length	Format	Range	Default	Parameter
ENROUTE STATIONS	No	1-10	XXX...			NOTAM_ENROUTE
SUPPLEMENTAL TEXT	No	1-24	XXX...			NOTAM_REQST_1
	No	1-24	XXX...			NOTAM_REQST_2
	No	1-24	XXX...			NOTAM_REQST_3

The three text entry boxes labeled SUPPLEMENTAL TEXT each allows up to 24 characters of supplemental text entry.

Command Buttons

Table 5-7 describes the NOTAMs Request display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 5-7 NOTAMs Request Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	SEND	Send downlink	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	-	-	-	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

Review Log

When SEND is selected from this display, it becomes available for review in the WEATHER & NOTAMS category on the REVIEW menu.

5.2.1.2 Logic Unit

The AMI implements a logic unit to enable and disable the SEND button on the NOTAMs Request display as appropriate.

```
-- -----
--
-- Version C000
-- Jan 2014 SA Meisch
-- Logic fix.
-- Toggle SEND command button based on entry of one of the following:
```

```

--          Station 1, Station 2, or Enroute.
--
-- Sep 2018  SA Meisch
--          Required for NOTAM display.
--
-- Version F001  M. Matyas  29-SEP-2020
--          Updated reference from NR_SEND_COMMAND to NOTAM_SEND.
--
-- -----

if valid (NOTAM_STA_1) or valid (NOTAM_STA_2) or valid (NOTAM_ENROUTE) then
    NOTAM_SEND := TRUE;
else
    NOTAM_SEND := FALSE;
end_if;

-- -----
--
-- END

```

5.2.1.3 Downlink Encoding Definition

When SEND is selected from this display, the following downlink encoding is used:

- ACARS SMI: M38
- ACARS Label: 38
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REQUEST
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 5-8 describes the NOTAMs Request downlink encoding definition.

Table 5-8 NOTAMs Request Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	NTM	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"

#	Length	Inclusion	Content	Element Description
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	4	Always	NOTAM_STA_1	"Station 1 (XXXX)"
13	4	Always	NOTAM_STA_2	"Station 2 (XXXX)"
14	10	Always	NOTAM_ENROUTE	"Enroute Stations (XXXXXXXXXX)"
15	2	Conditionally	<CR><LF>	"Delimiter"
16	1-24	Conditionally	NOTAM_REQST_1	"Text line 1 (1-24 characters)"
17	2	Conditionally	<CR><LF>	"Delimiter"
18	1-24	Conditionally	NOTAM_REQST_2	"Text line 2 (1-24 characters)"
19	2	Conditionally	<CR><LF>	"Delimiter"
20	1-24	Conditionally	NOTAM_REQST_3	"Text line 3 (1-24 characters)"

Note: The presence of each text line and its preceding <CR><LF> delimiter in a downlink message is conditional on the validity of that text line. In other words, if text is not entered on a particular text line then that line and the preceding <CR><LF> delimiter are not present in the downlink message.

Sample Message

```
QU RTNBOCR
.DDLXCXA 221945
M38
FI BO24/AN N7873BO
DT DDL BFI 221945 M61A
- NTM01 BOE24/22221945KBFIKLGB
KLGB*****
```

Note: In this sample message, NOTAMs are requested only for "KLGB" as Station 1; no NOTAMs are requested for Station 2 or for Enroute Stations and no supplemental text is present.

```
QU RTNBOCR
.DDLXCXA 221946
M38
FI BO24/AN N7873BO
DT DDL BFI 221946 M62A
- NTM01 BOE24/22221946KBFIKLGB
LAX ****WCOAST77
TEXT LINE 1
```

Note: In this sample message, NOTAMs are requested for “LAX” (left-justified) as Station 1 and for “WCOAST77” as Enroute Stations, and one line of supplemental text is also included; no NOTAMs are requested for Station 2.

5.2.2 Uplink

This section describes the NOTAMs Uplink function.

5.2.2.1 Uplink Decoding Definition

The decoding for the NOTAMs Uplink is defined as follows:

- ACARS SMI: CMD
- ACARS Label: RA
- Message Identifier: ~1NTM
- Alert Level: Medium (visual “●COMM” and aural high-low chime)
- Optional Element Delimiter: “,” (comma)

Table 5-9 describes the NOTAMs Uplink decoding definition.

Table 5-9 NOTAMs Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	2	Yes	<none>	"Priority code (XX)"
2	7	Yes	<none>	"Ground address (XXXXXXX)"
3	5	Yes	<none>	"Message identifier (~1NTM)"
4	2	Yes	<none>	"Message version (XX)"
5	2	Yes	<none>	"Delimiter (<CR><LF>)"
6	1-16	Yes	NOTAM_UL_TITLE	"Message title (1-16 characters)"
7	1-3486	Yes	UPLINK_TXT_BLK	"Message text (1-3486 characters)"

Note: Element 6 is a variable-length element and accordingly should be terminated with a delimiter character. Pre-defined element delimiters are <CR>, <CR><LF>, and <ETX> and the optional element delimiter defined for this uplink decoding is “,” (comma).

Sample Message

```
QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 003A
- QURTNBOCR~1NTM01
NOTAM TITLE
```

NOTAM TEXT BLOCK...

5.2.2.2 Display Definition

Figure 5-5 show the NOTAMs Uplink display definition.



Figure 5-5 NOTAMs Uplink Display Definition

Title

The title area contains:

- The current time in HHMMz format
- The title contained in the uplink (NOTAM_UL_TITLE)

Main Area

The main area contains the text block from the uplink. If any lines of the text block contain more than 46 characters, then that line is word-wrapped. If the entire text block cannot be displayed on a single page, then as many pages as are necessary are created and page arrows are displayed on the right side to access each page.

Command Buttons

Table 5-10 describes the NOTAMs Uplink display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 5-10 NOTAMs Uplink Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	PRINT	Print	Always	-
4	-	-	-	-
5	-	-	-	-
6	CANCEL	Cancel	Always	-

Review Log

When the associated uplink decoding is received, this display becomes available for review in the WEATHER & NOTAMS category on the REVIEW menu.

5.2.2.3 Print Definition

Figure 5-6 shows the NOTAMs Uplink print definition.

TIME: hhmmz	xxxxxxxxxxxxxxx	A/C REG: xxxxxxx
DATE: dd-MMM-yyyy		FLT NO: xxxxxxxxxx

<-----Text Block----->

Figure 5-6 NOTAMs Uplink Print Definition

6 Messaging

This section describes the messaging functions that the AML implements.

6.1 Uplink Messaging

This section describes the uplink messaging functions that the AML implements.

6.1.1 Display Command Uplink

The Display Command Uplink function enables the aircraft operator to send the flight crew a message and control the presentation, including the message title.

6.1.1.1 Uplink Decoding Definition

The decoding for the Display Command Uplink is defined as follows:

- ACARS SMI: CMD
- ACARS Label: RA
- Message Identifier: ~1RA1
- Alert Level: Medium (visual “●COMM” and aural high-low chime)
- Optional Element Delimiter: “,” (comma) character

Table 6-1 describes the Display Command Uplink decoding definition.

Table 6-1 Display Command Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	2	Yes	<none>	"Priority code (XX)"
2	7	Yes	<none>	"Ground address (XXXXXXX)"
3	5	Yes	<none>	"Message identifier (~1RA1)"
4	2	Yes	<none>	"Message version (XX)"
5	2	Yes	<none>	"Message date (DD)"
6	4	Yes	<none>	"Message time (HHMM)"
7	2	Yes	<none>	"Delimiter (<CR><LF>)"
8	1-16	Yes	RA1_TITLE	"Message title (1-16 characters)"
9	1-3480	Yes	UPLINK_TXT_BLK	"Message text (1-3480 characters)"

Note: Element 8 is a variable-length element and accordingly should be terminated with a delimiter character. Pre-defined element delimiters are <CR>, <CR><LF>, and <ETX> and the optional element delimiter defined for this uplink decoding is “,” (comma).

Sample Message

QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 004A
- QURTNBOCR~1RA101190436
TITLE
DISPLAY COMMAND TEXT BLOCK...

6.1.1.2 Display Definition

Figure 6-1 shows the Display Command Uplink display definition.

The screenshot shows a graphical user interface for a display command uplink. At the top, there is a header bar with three sections: 'ATC' (highlighted in red), 'FLIGHT INFORMATION' (highlighted in red), and 'COMPANY'. Below this header, there are three buttons: 'REVIEW', 'MANAGER', and 'NEW MESSAGES'. The main display area is a large black rectangle. Above this rectangle, there is a row of three text fields: 'hhmmz', 'XXXXXXXXXXXXXXXXXXXXXXXXX', and 'XXXXXXXXX'. Below the black rectangle, there is a row of three buttons: 'PRINT', 'CANCEL', and 'SCRATCHPAD'. The 'SCRATCHPAD' button is highlighted in red.

Figure 6-1 Display Command Uplink Display Definition

Title

The title area contains:

- The current time in HHMMz format
- The title contained in the uplink (RA1_TITLE)

Main Area

The main area contains the text block from the uplink. If any lines of the text block contain more than 46 characters, then that line is word-wrapped. If the entire text block cannot be displayed on a single page, then as many pages as are necessary are created and page arrows are displayed on the right side to access each page.

Command Buttons

Table 6-2 describes the Display Command Uplink display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 6-2 Display Command Uplink Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	PRINT	Print	Always	-
4	-	-	-	-
5	-	-	-	-
6	CANCEL	Cancel	Always	-

Review Log

When the associated uplink decoding is received, this display becomes available for review in the COMPANY UPLINKS category on the REVIEW menu.

6.1.1.3 Print Definition

Figure 6-2 shows the Display Command Uplink print definition.

```

TIME: hhmmz                xxxxxxxxxxxxxxxx          A/C REG: xxxxxxxx
DATE: dd-MMM-yyyy          xxxxxxxxxxxxxxxx          FLT NO:  xxxxxxxxxx

<-----Text Block----->

```

Figure 6-2 Display Command Uplink Print Definition

6.1.2 E-Mail Uplink

The E-Mail Uplink function enables the aircraft operator to send the flight crew a message converted from e-mail format, including the originator's e-mail address and a subject line.

6.1.2.1 Uplink Decoding Definition

The decoding for the E-Mail Uplink is defined as follows:

- ACARS SMI: CMD
- ACARS Label: RA
- Message Identifier: ~1EML
- Alert Level: Medium (visual “●COMM” and aural high-low chime)
- Optional Element Delimiter: “,” (comma)

Table 6-3 describes the E-Mail Uplink decoding definition.

Table 6-3 E-Mail Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	2	Yes	<none>	"Priority code (XX)"
2	7	Yes	<none>	"Ground address (XXXXXXX)"
3	5	Yes	<none>	"Message identifier (~1EML)"
4	2	Yes	<none>	"Delimiter (<CR><LF>)"
5	1-24	Yes	EMAIL_UL_FROM_LOCAL	"Local name (1-24 characters)"
6	1-24	Yes	EMAIL_UL_FROM_DOMAIN	"Domain name (1-24 characters)"
7	1-24	No	EMAIL_UL_SUBJECT	"Subject (1-24 characters)"
8	1-3432	Yes	UPLINK_TXT_BLK	"Message text (1-3432 characters)"

Note: Elements 5, 6, and 7 are variable-length elements and accordingly should be terminated with a delimiter character. Pre-defined element delimiters are <CR>, <CR><LF>, and <ETX> and the optional element delimiter defined for this uplink decoding is “,” (comma).

Sample Message

```
QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 005A
- QURTNBOCR~1EML
LOCAL,DOMAIN,SUBJECT,E-MAIL UPLINK TEXT BLOCK...
```

6.1.2.2 Display Definition

Figure 6-3 shows the E-Mail Uplink display definition.

ATC

FLIGHT
INFORMATION

COMPANY

REVIEW

MANAGER

NEW MESSAGES

hhmmzE-MAIL UPLINK MESSAGEXXXXXXXXXX

FROM:XXXXXXXXXXXXXXXXXXXXXXXXXXXX

XXXXXXXXXXXXXXXXXXXXXXXXXXXX

SUBJECT:XXXXXXXXXXXXXXXXXXXXXXXXXXXX

MESSAGE:

<TEXT BLOCK>

REPLY

PRINT

CANCEL

SCRATCHPAD

Figure 6-3 E-Mail Uplink Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “E-MAIL UPLINK MESSAGE”

Main Area

Table 6-4 describes the E-Mail Uplink display parameters.

Table 6-4 E-Mail Uplink Display Parameters

Name	Length	Format	Enterable	Range	Parameter
FROM	1-24	XXX...	No		EMAIL_UL_FROM_LOCAL

Name	Length	Format	Enterable	Range	Parameter
@	1-24	XXX...	No		EMAIL_UL_FROM_DOMAIN
SUBJECT	1-24	XXX...	No		EMAIL_UL_SUBJECT

This is followed by the text block from the uplink. If any lines of the text block contain more than 46 characters, then that line is word-wrapped. If the entire text block cannot be displayed on a single page, then as many pages as are necessary are created and page arrows are displayed on the right side to access each page.

Command Buttons

Table 6-5 describes the E-Mail Uplink display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 6-5 E-Mail Uplink Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	REPLY	Show E-Mail Reply display	Always	-
2	-	-	-	-
3	PRINT	Print	Always	-
4	-	-	-	-
5	-	-	-	-
6	CANCEL	Cancel	Always	-

Review Log

When the associated uplink decoding is received, this display becomes available for review in the COMPANY UPLINKS category on the REVIEW menu.

6.1.2.3 Print Definition

Figure 6-4 shows the E-Mail Uplink print definition.

```

TIME: hhmmz           E-MAIL UPLINK MESSAGE           A/C REG: xxxxxxxx
DATE: dd-MMM-yyyy    FLT NO: xxxxxxxxxx

FROM: xxxxxxxxxxxxxxxxxxxxxxxx
    @ xxxxxxxxxxxxxxxxxxxxxxxx

SUBJECT: xxxxxxxxxxxxxxxxxxxxxxxx

MESSAGE:
<-----Text Block----->

```

Figure 6-4 E-Mail Uplink Print Definition

6.1.3 Free-Text Uplink

The Free-Text Uplink function enables the aircraft operator to send the flight crew a free-text message that includes a message title.

6.1.3.1 Uplink Decoding Definition

The decoding for the Free-Text Uplink is defined as follows:

- ACARS SMI: CMD
- ACARS Label: RA
- Message Identifier: ~1DIS
- Alert Level: Medium (visual “●COMM” and aural high-low chime)
- Optional Element Delimiter: “,” (comma) character

Table 6-6 describes the Free-Text Uplink decoding definition.

Table 6-6 Free-Text Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	2	Yes	<none>	"Priority code (XX)"
2	7	Yes	<none>	"Ground address (XXXXXXX)"
3	5	Yes	<none>	"Message identifier (~1DIS)"
4	2	Yes	<none>	"Message version (XX)"
5	2	Yes	<none>	"Message date (DD)"
6	4	Yes	<none>	"Message time (HHMM)"
7	2	Yes	<none>	"Delimiter (<CR><LF>)"
8	1-16	Yes	DISP_MSG_TITLE	"Message title (1-16 characters)"
9	1-3480	Yes	UPLINK_TXT_BLK	"Message text (1-3480 characters)"

Note: Element 8 is a variable-length element and accordingly should be terminated with a delimiter character. Pre-defined element delimiters are <CR>, <CR><LF>, and <ETX> and the optional element delimiter defined for this uplink decoding is “,” (comma).

Sample Message

```
QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 006A
- QURTNBOCR~1DIS01221953
FREE-TEXT TITLE
FREE-TEXT TEXT BLOCK...
```

6.1.3.2 Display Definition

Figure 6-5 shows the Free-Text Uplink display definition.



Figure 6-5 Free-Text Uplink Display Definition

Title

The title area contains:

- The current time in HHMMz format
- The title contained in the uplink (DISP_MSG_TITLE)

Main Area

The main area contains the text block from the uplink. If any lines of the text block contain more than 46 characters, then that line is word-wrapped. If the entire text block cannot be displayed on a single page, then as many pages as are necessary are created and page arrows are displayed on the right side to access each page.

Command Buttons

Table 6-7 describes the Free-Text Uplink display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 6-7 Free-Text Uplink Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	PRINT	Print	Always	-
4	-	-	-	-
5	-	-	-	-
6	CANCEL	Cancel	Always	-

Review Log

When the associated uplink decoding is received, this display becomes available for review in the COMPANY UPLINKS category on the REVIEW menu.

6.1.3.3 Print Definition

Figure 6-6 shows the Free-Text Uplink print definition.

Figure 6-6 Free-Text Uplink Print Definition

Table 6-8 Printer Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	1	Yes	<none>	"Period (.)"
2	7	Yes	<none>	"Ground address (XXXXXXX)"
3	1	Yes	<none>	"Space (<SP>)"
4	2	Yes	<none>	"Message date (DD)"
5	4	Yes	<none>	"Message time (HHMM)"
6	2	Yes	<none>	"Delimiter (<CR><LF>)"
7	1-3503	Yes	PRINT_TXT_BLK	"Message text (1-3503 characters)"

Sample Message

QU DDLXCXA
.RTNBOCR 221905
AGM
AN N7873BO/MA 007A
- .RTNBOCR 190523
PRINTER UPLINK TEXT BLOCK...

6.1.4.2 Print Definition

Figure 6-7 shows the Printer Uplink print definition.

TIME: hhmmz	PRINTER UPLINK	A/C REG: xxxxxxxx
DATE: dd-MM-yyyy		FLT NO: xxxxxxxxxx

<-----Text Block----->

Figure 6-7 Printer Uplink Print Definition

6.1.5 VHF Voice Contact Request

The VHF Voice Contact Request function enables the aircraft operator to send a request for the flight crew to contact the operator on a specific VHF channel.

6.1.5.1 Uplink Decoding Definition

The decoding for the VHF voice contact uplink is defined as follows:

- ACARS SMI: GVR
- ACARS Label: 54
- Message Identifier: None
- Alert Level: Medium (visual “●COMM” and aural high-low chime)
- Optional Element Delimiter: “,” (comma)

Table 6-9 describes the VHF voice contact uplink decoding definition.

Table 6-9 VHF Voice Contact Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	6	Yes	VHF_VOICE_FREQ	"VHF voice frequency (118000-136975)"
2	1-200	No	VHF_VOICE_TXT_BLK	"Message text (1-200 characters)"

The text block following the VHF voice frequency is optional and limited to 200 characters.

Sample Message

```
QU DDLXCXA
.RTNBOCR 221954
GVR
AN N7873BO/MA 008A
- 132475VHF VOICE CONTACT TEXT BLOCK...
```

Note: This sample message includes the optional text block.

6.1.5.2 Display Definition

Figure 6-8 shows the VHF Voice Contact Request display definition.

Figure 6-8 VHF Voice Contact Request Display Definition

The title area contains:

- ## Main Area

Table 6-10 VHF Voice Contact Display Parameter

NEW

Command Buttons

Table 6-11 describes the VHF Voice Contact Request display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 6-11 VHF Voice Contact Request Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	PRINT	Print	Always	-
4	-	-	-	-
5	-	-	-	-
6	CANCEL	Cancel	Always	-

Review Log

When the associated uplink decoding is received, this display becomes available for review in the COMPANY UPLINKS category on the REVIEW menu.

6.1.5.3 Print Definition

Figure 6-9 shows the VHF Voice Contact Request print definition.

TIME: hhmmz	VHF VOICE CONTACT	A/C REG: xxxxxxx
DATE: dd-MMM-yyyy		FLT NO: xxxxxxxxxx

VOICE CONTACT REQUESTED ON VHF FREQUENCY nnn.nnn MHZ

<-----Text Block----->

Figure 6-9 VHF Voice Contact Request Print Definition

6.2 Downlink Messaging

This section describes the downlink messaging functions that the AMI implements.

6.2.1 E-Mail Downlink

The E-Mail Downlink function allow the flight crew to send an ACARS message which ground systems then convert into an e-mail message.

6.2.1.1 Display Definition

Figure 6-10 shows the E-Mail Downlink display definition.

Figure 6-10 E-Mail Downlink Display Definition

The title area contains:

- ## Main Area

Table 6-12 describes the E-Mail Downlink display entry boxes.

Name	Mandatory	Length	Format	Range	Default	Parameter
TO	Yes	1-22	XXX...			EMAIL_DL_TO_LOCAL
@	Yes	1-16	XXX...			EMAIL_DL_TO_DOMAIN

Name	Mand atory	Length	Format	Range	Default	Parameter
CC	No	1-22	XXX...			EMAIL_DL_CC_1_LOCAL
@	No	1-16	XXX...			EMAIL_DL_CC_1_DOMAIN
	No	1-22	XXX...			EMAIL_DL_CC_2_LOCAL
@	No	1-16	XXX...			EMAIL_DL_CC_2_DOMAIN
	No	1-22	XXX...			EMAIL_DL_CC_3_LOCAL
@	No	1-16	XXX...			EMAIL_DL_CC_3_DOMAIN
SUBJECT	No	1-24	XXX...			EMAIL_DL_SUBJECT
MESSAGE	Yes	1-24	XXX...			EMAIL_DL_TXT_1
	No	1-24	XXX...			EMAIL_DL_TXT_2
	No	1-24	XXX...			EMAIL_DL_TXT_3
	No	1-24	XXX...			EMAIL_DL_TXT_4
	No	1-24	XXX...			EMAIL_DL_TXT_5
	No	1-24	XXX...			EMAIL_DL_TXT_6
	No	1-24	XXX...			EMAIL_DL_TXT_7
	No	1-24	XXX...			EMAIL_DL_TXT_8
	No	1-24	XXX...			EMAIL_DL_TXT_9

Note: Both the local and domain portions of the various e-mail address parameters may contain one or more periods (.). These parameters correspond to the two parts of an e-mail address. For example, to send the message to john.smith@boeing.com, "john.smith" would be entered in the TO field and "boeing.com" would be entered in the associated @ field.

Command Buttons

Table 6-13 describes the E-Mail Downlink display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 6-13 E-Mail Downlink Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	SEND	Send downlink	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	PRINT	Print	Always	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

Review Log

When SEND is selected from this display, it becomes available for review in the COMPANY DOWNLINKS category on the REVIEW menu.

6.2.1.2 Downlink Encoding Definition

When SEND is selected from this display, the following downlink encoding is used:

- ACARS SMI: EML
- ACARS Label: E1
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: MESSAGE
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 6-14 describes the E-Mail Downlink encoding definition.

Table 6-14 E-Mail Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	EML	"Message Identifier"
3	2	Always	02	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter "
12	1-22	Always	EMAIL_DL_TO_LOCAL	"To local address (1-22 characters)"
13	1	Always	,	"Delimiter"
14	1-16	Always	EMAIL_DL_TO_DOMAIN	"To domain (1-16 characters)"
15	2	Always	<CR><LF>	"Delimiter"
16	1-22	Always	EMAIL_DL_CC_1_LOCAL	"CC1 local address (1-22 characters)"
17	1	Always	,	"Delimiter"
18	1-16	Always	EMAIL_DL_CC_1_DOMAIN	"CC1 domain (1-16 characters)"
19	2	Always	<CR><LF>	"Delimiter"

#	Length	Inclusion	Content	Element Description
20	1-22	Always	EMAIL_DL_CC_2_LOCAL	"CC2 local address (1-22 characters)"
21	1	Always	,	"Delimiter"
22	1-16	Always	EMAIL_DL_CC_2_DOMAIN	"CC2 domain (1-16 characters)"
23	2	Always	<CR><LF>	"Delimiter"
24	1-22	Always	EMAIL_DL_CC_3_LOCAL	"CC3 local address (1-22 characters)"
25	1	Always	,	"Delimiter"
26	1-16	Always	EMAIL_DL_CC_3_DOMAIN	"CC3 domain (1-16 characters)"
27	2	Always	<CR><LF>	"Delimiter"
28	1-24	Always	EMAIL_DL_SUBJECT	"Subject (1-24 characters)"
29	2	Always	<CR><LF>	"Delimiter"
30	1-24	Always	EMAIL_DL_TXT_1	"Text line 1 (1-24 characters)"
31	2	Conditionally	<CR><LF>	"Delimiter"
32	1-24	Conditionally	EMAIL_DL_TXT_2	"Text line 2 (1-24 characters)"
33	2	Conditionally	<CR><LF>	"Delimiter"
34	1-24	Conditionally	EMAIL_DL_TXT_3	"Text line 3 (1-24 characters)"
35	2	Conditionally	<CR><LF>	"Delimiter"
36	1-24	Conditionally	EMAIL_DL_TXT_4	"Text line 4 (1-24 characters)"
37	2	Conditionally	<CR><LF>	"Delimiter"
38	1-24	Conditionally	EMAIL_DL_TXT_5	"Text line 5 (1-24 characters)"
39	2	Conditionally	<CR><LF>	"Delimiter"
40	1-24	Conditionally	EMAIL_DL_TXT_6	"Text line 6 (1-24 characters)"
41	2	Conditionally	<CR><LF>	"Delimiter"
42	1-24	Conditionally	EMAIL_DL_TXT_7	"Text line 7 (1-24 characters)"
43	2	Conditionally	<CR><LF>	"Delimiter"
44	1-24	Conditionally	EMAIL_DL_TXT_8	"Text line 8 (1-24 characters)"
45	2	Conditionally	<CR><LF>	"Delimiter"
46	1-24	Conditionally	EMAIL_DL_TXT_9	"Text line 9 (1-24 characters)"

Note: The presence of each text line and its preceding <CR><LF> delimiter in a downlink message is conditional on the validity of that text line. In other words, if text is not entered on a particular text line then that line and the preceding <CR><LF> delimiter are not present in the downlink message.

Note: The formats of the E-Mail Downlink and the E-Mail Reply downlink are identical.

Sample Message

```

QU RTNBOCR
.DDLXCXA 221953
EML
FI BO24/AN N7873BO
DT DDL BFI 221953 M67A
- EML02 BOE24/22221953KBFIKLGB

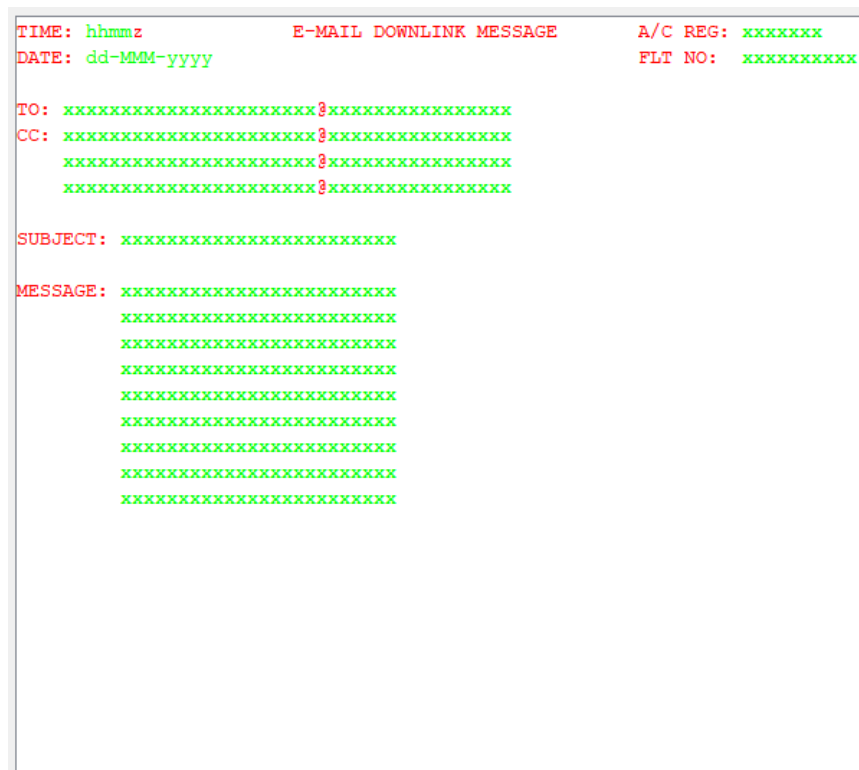
```

```
TEST.ADDRESS,BOEING.COM
*,*
*,*
*,*
SUBJECT
TEXT LINE 1
TEXT LINE 2
TEXT LINE 3
```

Note: In this sample message, no CC e-mail addresses are present.

6.2.1.3 Print Definition

Figure 6-11 shows the E-Mail Downlink print definition.



```
TIME: hhmmz          E-MAIL DOWNLINK MESSAGE      A/C REG: xxxxxxxx
DATE: dd-MMM-yyyy    FLT NO: xxxxxxxxxx

TO: xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
CC: xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
   xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
   xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
SUBJECT: xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
MESSAGE: xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
         xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
         xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
         xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
         xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
         xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
         xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
         xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
         xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
```

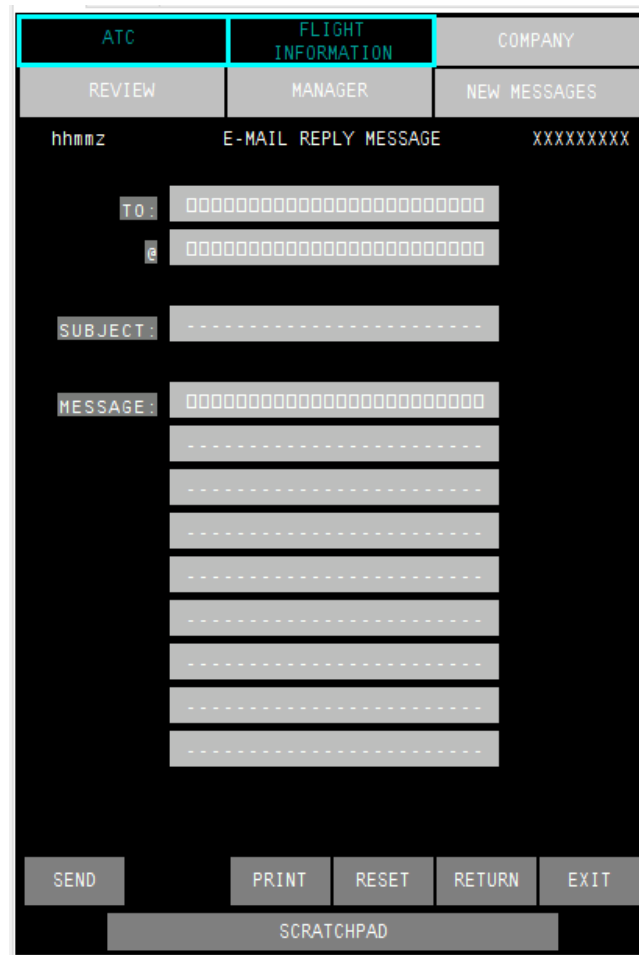
Figure 6-11 E-Mail Downlink Print Definition

6.2.2 E-Mail Reply

The E-Mail Reply function enables the flight crew to reply to an e-mail uplink by sending an ACARS message which ground systems then convert into an e-mail message.

6.2.2.1 Display Definition

Figure 6-12 shows the E-Mail Reply display definition, which is accessible from the E-Mail Uplink display described in Section 6.1.2.2.



The screenshot displays the 'E-MAIL REPLY MESSAGE' interface. At the top, there are three tabs: 'ATC' (highlighted in cyan), 'FLIGHT INFORMATION' (highlighted in cyan), and 'COMPANY'. Below these are three buttons: 'REVIEW', 'MANAGER', and 'NEW MESSAGES'. The main display area shows the time 'hhmmz' on the left, the title 'E-MAIL REPLY MESSAGE' in the center, and 'XXXXXXXXX' on the right. The form includes fields for 'TO:', 'FROM:', 'SUBJECT:', and 'MESSAGE:'. The 'MESSAGE:' field is a multi-line text area. At the bottom, there are five buttons: 'SEND', 'PRINT', 'RESET', 'RETURN', and 'EXIT', followed by a 'SCRATCHPAD' button.

Figure 6-12 E-Mail Reply Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “E-MAIL REPLY MESSAGE”

Main Area

Table 6-15 describes the E-Mail Reply display entry boxes. Of note, the values in the TO, @, and SUBJECT entry boxes default to the corresponding values in the most recent e-mail uplink message.

Table 6-15 E-Mail Reply Display Entry Boxes

Name	Mandatory	Length	Format	Default	Parameter
TO	Yes	1-22	XXX...	EMAIL_UL_SUBJECT	EMAIL_DL_REPLY_TO_LOCAL
@	Yes	1-16	XXX...	EMAIL_UL_FROM_DOMAIN	EMAIL_DL_REPLY_TO_DOMAIN
SUBJECT	No	1-24	XXX...	EMAIL_UL_FROM_LOCAL	EMAIL_DL_REPLY_SUBJECT
MESSAGE	Yes	1-24	XXX...		EMAIL_DL_REPLY_TXT_1
	No	1-24	XXX...		EMAIL_DL_REPLY_TXT_2
	No	1-24	XXX...		EMAIL_DL_REPLY_TXT_3
	No	1-24	XXX...		EMAIL_DL_REPLY_TXT_4
	No	1-24	XXX...		EMAIL_DL_REPLY_TXT_5
	No	1-24	XXX...		EMAIL_DL_REPLY_TXT_6
	No	1-24	XXX...		EMAIL_DL_REPLY_TXT_7
	No	1-24	XXX...		EMAIL_DL_REPLY_TXT_8
	No	1-24	XXX...		EMAIL_DL_REPLY_TXT_9

Note: Both the local and domain portions of the various e-mail address parameters may contain one or more periods (.). These parameters correspond to the two parts of an e-mail address. For example, to send the message to john.smith@boeing.com, "john.smith" would be entered in the TO field and "boeing.com" would be entered in the associated @ field.

Command Buttons

Table 6-16 describes the E-Mail Reply display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 6-16 E-Mail Reply Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	SEND	Send downlink	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	PRINT	Print	Always	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

Review Log

When SEND is selected from this display, it becomes available for review in the COMPANY DOWNLINKS category on the REVIEW menu.

6.2.2.2 Downlink Encoding Definition

When SEND is selected from this display, the following downlink encoding is used:

- ACARS SMI: EML
- ACARS Label: E1
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: MESSAGE
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 6-17 describes the E-Mail Reply downlink encoding definition, which is identical to the E-Mail Downlink encoding that Table 6-14 describes.

Table 6-17 E-Mail Reply Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	EML	"Message Identifier"
3	2	Always	02	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	1-24	Always	EMAIL_DL_REPLY_TO_LOCAL	"To local address (1-24 characters)"
13	1	Always	,	"Delimiter"
14	1-24	Always	EMAIL_DL_REPLY_TO_DOMAIN	"To domain (1-24 characters)"
15	2	Always	<CR><LF>	"Delimiter"
16	1	Always	*	"CC 1 local address placeholder"
17	1	Always	,	"Delimiter"
18	1	Always	*	"CC 1 domain placeholder"

#	Length	Inclusion	Content	Element Description
19	2	Always	<CR><LF>	"Delimiter"
20	1	Always	*	"CC 2 local address placeholder"
21	1	Always	,	"Delimiter"
22	1	Always	*	"CC 2 domain placeholder"
23	2	Always	<CR><LF>	"Delimiter"
24	1	Always	*	"CC 3 local address placeholder"
25	1	Always	,	"Delimiter"
26	1	Always	*	"CC 3 domain placeholder"
27	2	Always	<CR><LF>	"Delimiter"
28	1-24	Always	EMAIL_DL_REPLY_SUBJECT	"Subject (1-24 characters)"
29	2	Always	<CR><LF>	"Delimiter"
30	1-24	Always	EMAIL_DL_REPLY_TXT_1	"Text line 1 (1-24 characters)"
31	2	Conditionally	<CR><LF>	"Delimiter"
32	1-24	Conditionally	EMAIL_DL_REPLY_TXT_2	"Text line 2 (1-24 characters)"
33	2	Conditionally	<CR><LF>	"Delimiter"
34	1-24	Conditionally	EMAIL_DL_REPLY_TXT_3	"Text line 3 (1-24 characters)"
35	2	Conditionally	<CR><LF>	"Delimiter"
36	1-24	Conditionally	EMAIL_DL_REPLY_TXT_4	"Text line 4 (1-24 characters)"
37	2	Conditionally	<CR><LF>	"Delimiter"
38	1-24	Conditionally	EMAIL_DL_REPLY_TXT_5	"Text line 5 (1-24 characters)"
39	2	Conditionally	<CR><LF>	"Delimiter"
40	1-24	Conditionally	EMAIL_DL_REPLY_TXT_6	"Text line 6 (1-24 characters)"
41	2	Conditionally	<CR><LF>	"Delimiter"
42	1-24	Conditionally	EMAIL_DL_REPLY_TXT_7	"Text line 7 (1-24 characters)"
43	2	Conditionally	<CR><LF>	"Delimiter"
44	1-24	Conditionally	EMAIL_DL_REPLY_TXT_8	"Text line 8 (1-24 characters)"
45	2	Conditionally	<CR><LF>	"Delimiter"
46	1-24	Conditionally	EMAIL_DL_REPLY_TXT_9	"Text line 9 (1-24 characters)"

Note: The presence of each text line and its preceding <CR><LF> delimiter in a downlink message is conditional on the validity of that text line. In other words, if text is not entered on a particular text line then that line and the preceding <CR><LF> delimiter are not present in the downlink message.

Note: The formats of the E-Mail Downlink and the E-Mail Reply downlink are identical.

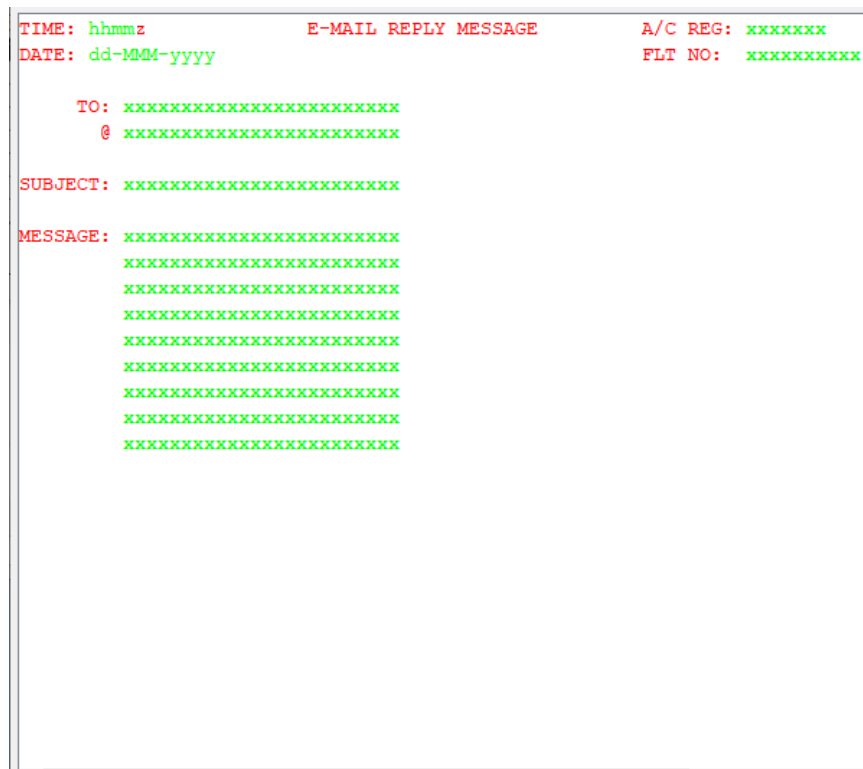
Sample Message

```
QU RTNBOCR
.DDLXCXA 221953
EML
FI BO24/AN N7873BO
DT DDL BFI 221953 M67A
```

```
- EML02      BOE24/22221953KBFIKLGB
TEST.ADDRESS,BOEING.COM
*,*
*,*
*,*
SUBJECT
TEXT LINE 1
TEXT LINE 2
TEXT LINE 3
```

6.2.2.3 Print Definition

Figure 6-13 shows the E-Mail Reply print definition.



```
TIME: hhmmz      E-MAIL REPLY MESSAGE      A/C REG: XXXXXXXX
DATE: dd-MMM-yyyy      FLT NO: XXXXXXXXXX

TO: XXXXXXXXXXXXXXXXXXXX
@ XXXXXXXXXXXXXXXXXXXX

SUBJECT: XXXXXXXXXXXXXXXXXXXX

MESSAGE: XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX
```

Figure 6-13 E-Mail Reply Print Definition

6.2.3 Message to Ground

The Message to Ground function enables the flight crew to send free-text messages to selected entities on the ground.

6.2.3.1 Display Definition

Figure 6-14 shows the Message to Ground display definition.

The screenshot shows a graphical user interface for a 'Message to Ground' display. At the top, there is a header bar with three tabs: 'ATC' (highlighted with a red border), 'FLIGHT INFORMATION', and 'COMPANY'. Below these tabs are three buttons: 'REVIEW', 'MANAGER', and 'NEW MESSAGES'. The main display area has a black background. At the top of this area, it shows 'hhmmz' on the left, 'MESSAGE TO GROUND' in the center, and 'XXXXXXXXX' on the right. Below this, there is an 'ADDRESS:' label followed by a text input field containing '0000000'. Underneath that is a 'TEXT:' label followed by a large text area with a first line containing '000000000000000000000000' and several subsequent lines of dashed lines for input. At the bottom of the screen, there is a row of five buttons: 'SEND', 'PRINT', 'RESET', 'RETURN', and 'EXIT'. Below these buttons is a wide 'SCRATCHPAD' input field.

Figure 6-14 Message to Ground Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “MESSAGE TO GROUND”

Main Area

Table 6-18 describes the Message to Ground display entry boxes.

Table 6-18 Message to Ground Display Entry Boxes

Name	Mandatory	Length	Format	Range	Default	Parameter
ADDRESS	Yes	7	XXXXXXX			AGM_ADDR
TEXT	Yes	1-24	XXX...			AGM_TEXT_1
	No	1-24	XXX...			AGM_TEXT_2
	No	1-24	XXX...			AGM_TEXT_3
	No	1-24	XXX...			AGM_TEXT_4
	No	1-24	XXX...			AGM_TEXT_5
	No	1-24	XXX...			AGM_TEXT_6
	No	1-24	XXX...			AGM_TEXT_7
	No	1-24	XXX...			AGM_TEXT_8

The ADDRESS parameter is not reset after SEND is selected, although each TEXT parameter is. The eight text entry boxes labeled TEXT, the first of which is mandatory, each allows up to 24 characters of text entry.

Command Buttons

Table 6-19 describes the Message to Ground display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 6-19 Message to Ground Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	SEND	Send downlink	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	PRINT	Print	Always	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

Review Log

When SEND is selected from this display, it becomes available for review in the COMPANY DOWNLINKS category on the REVIEW menu.

6.2.3.2 Downlink Encoding Definition

When SEND is selected from this display, the following downlink encoding is used:

- ACARS SMI: A81

- ACARS Label: 81
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: MESSAGE
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 6-20 describes the Message to Ground downlink encoding definition.

Table 6-20 Message to Ground Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	1	Always	/	"Delimiter"
3	7	Always	AGM_ADDR	"Ground Address (XXXXXXX)"
4	1	Always	.	"Delimiter"
5	1-24	Always	AGM_TEXT_1	"Text line 1 (1-24 characters)"
6	2	Conditionally	<CR><LF>	"Delimiter"
7	1-24	Conditionally	AGM_TEXT_2	"Text line 2 (1-24 characters)"
8	2	Conditionally	<CR><LF>	"Delimiter"
9	1-24	Conditionally	AGM_TEXT_3	"Text line 3 (1-24 characters)"
10	2	Conditionally	<CR><LF>	"Delimiter"
11	1-24	Conditionally	AGM_TEXT_4	"Text line 4 (1-24 characters)"
12	2	Conditionally	<CR><LF>	"Delimiter"
13	1-24	Conditionally	AGM_TEXT_5	"Text line 5 (1-24 characters)"
14	2	Conditionally	<CR><LF>	"Delimiter"
15	1-24	Conditionally	AGM_TEXT_6	"Text line 6 (1-24 characters)"
16	2	Conditionally	<CR><LF>	"Delimiter"
17	1-24	Conditionally	AGM_TEXT_7	"Text line 7 (1-24 characters)"
18	2	Conditionally	<CR><LF>	"Delimiter"
19	1-24	Conditionally	AGM_TEXT_8	"Text line 8 (1-24 characters)"

Note: The presence of each text line and its preceding <CR><LF> delimiter in a downlink message is conditional on the validity of that text line. In other words, if text is not entered on a particular text line then that line and the preceding <CR><LF> delimiter are not present in the downlink message.

Sample Message

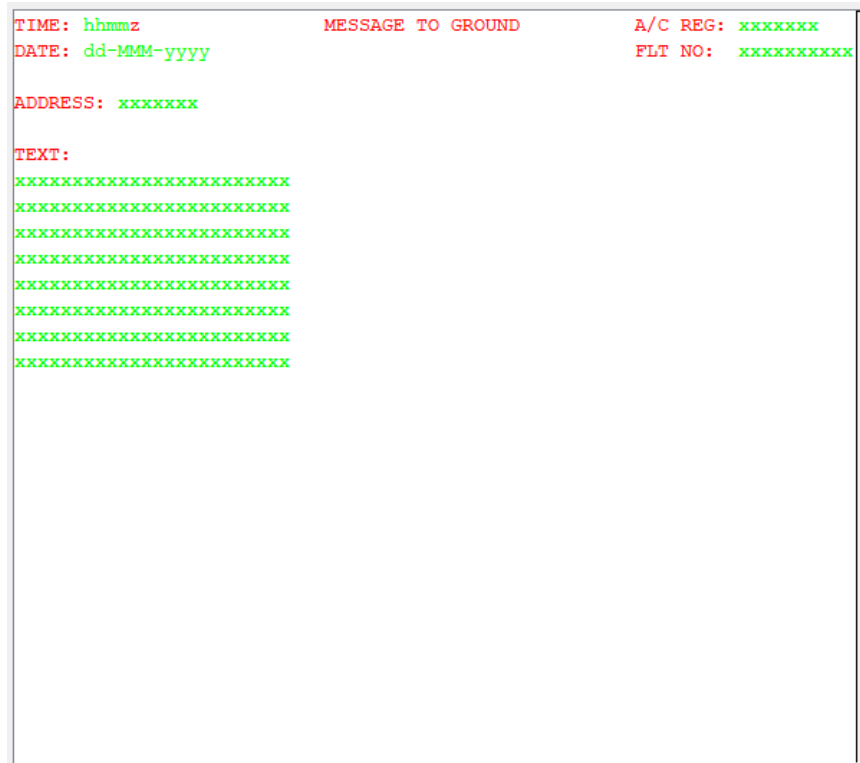
```
QU RTNBOCR FMCBOCR
.DDLXCXA 221804
A81
FI BO64/AN N7873BO
```

```
DT DDL BFI 221804 M30A
- TEXT LINE 1
TEXT LINE 2
TEXT LINE 3
```

Note: In this sample message, the Ground Address is "FMCBOCR" (which is contained in the message header and determines the message routing).

6.2.3.3 Print Definition

Figure 6-15 shows the Message to Ground print definition.



```
TIME: hhmmz          MESSAGE TO GROUND          A/C REG: xxxxxx
DATE: dd-MMM-yyyy    FLT NO: xxxxxxxx

ADDRESS: xxxxxx

TEXT:
xxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxx
```

Figure 6-15 Message to Ground Print Definition

7 Request-Response Exchanges

This section describes the request-response exchange functions that the AMI implements. These functions enable the aircraft operator to send the flight crew an uplink (which the flight crew may have requested) that requires a downlink response.

7.1 Crew Advisory

The Crew Advisory function enables the aircraft operator to send the flight crew an advisory message that requires a response from the flight crew.

7.1.1 Uplink

This section describes the Crew Advisory Uplink function.

7.1.1.1 Uplink Decoding Definition

The decoding for the Crew Advisory Uplink is defined as follows:

- ACARS SMI: CMD
- ACARS Label: RA
- Message Identifier: ~1CAA
- Alert Level: Medium (visual “●COMM” and aural high-low chime)
- Optional Element Delimiter: “,” (comma)

Table 7-1 describes the Crew Advisory Uplink decoding definition.

Table 7-1 Crew Advisory Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	2	Yes	<none>	"Priority code (XX)"
2	7	Yes	<none>	"Ground address (XXXXXXX)"
3	5	Yes	<none>	"Message identifier (~1CAA)"
4	2	Yes	<none>	"Message version (XX)"
5	2	Yes	<none>	"Message date (DD)"
6	4	Yes	<none>	"Message time (HHMM)"
7	2	Yes	<none>	"Delimiter (<CR><LF>)"
8	1-3496	Yes	UPLINK_TXT_BLK	"Message text (1-3496 characters)"

Sample Message

```
QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 009A
- QURTNBOCR~1CAA01190419
CREW ADVISORY TEXT BLOCK...
```

7.1.1.2 Display Definition

Figure 7-1 shows the Crew Advisory Uplink display definition.



Figure 7-1 Crew Advisory Uplink Display Definition

Title

The title area contains:

- The current time in HHMMz format

- “CREW ADVISORY”

Main Area

The main area contains the text block from the uplink. If any lines of the text block contain more than 46 characters, then that line is word-wrapped. If the entire text block cannot be displayed on a single page, then as many pages as are necessary are created and page arrows are displayed on the right side to access each page.

Command Buttons

Table 7-2 describes the Crew Advisory Uplink display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 7-2 Crew Advisory Uplink Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	ACCEPT	Send ACCEPT response	Always	When the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	PRINT	Print	Always	-
4	-	-	-	-
5	REJECT REASONS	Show RESPONSE REASONS display	Always	When the downlink queue is full or datalink is NO COMM
6	REJECT	Send REJECT response	Always	When the downlink queue is full or datalink is NO COMM

Review Log

When the associated uplink decoding is received, this display becomes available for review in the COMPANY UPLINKS category on the REVIEW menu.

7.1.1.3 Print Definition

Figure 7-2 shows the Crew Advisory Uplink print definition.

TIME: hhmmz DATE: dd-MMM-yyyy	CREW ADVISORY	A/C REG: xxxxxxxx FLT NO: xxxxxxxxxxxx
----------------------------------	---------------	---

Text Block

Figure 7-2 Crew Advisory Uplink Print Definition

7.1.2 Response

This section describes the Crew Advisory Response function.

7.1.2.1 Accept Downlink Encoding Definition

When ACCEPT is selected, the following downlink is sent:

- ACARS SMI: M40
- ACARS Label: 40
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_CREW
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 7-3 describes the Crew Advisory Accept downlink encoding definition.

Table 7-3 Crew Advisory Accept Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	CAA	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	1	Always	A	"Acceptance Flag"

Sample Message

```
QU RTNBOCR
.DDLXCXA 222013
M40
FI BO24/AN N7873BO
DT DDL BFI 222013 M72A
- CAA01      BOE24/22222013KBFIKLGB
A
```

7.1.2.2 Reject Downlink Encoding Definition

When REJECT is selected, the following downlink is sent:

- ACARS SMI: M40
- ACARS Label: 40
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_CREW
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 7-4 describes the Crew Advisory Reject downlink encoding definition.

Table 7-4 Crew Advisory Reject Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	CAA	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	1	Always	R	"Rejection Flag"

Sample Message

```
QU RTNBOCR
.DDLXCXA 222013
M40
FI BO24/AN N7873BO
DT DDL BFI 222013 M72A
- CAA01      BOE24/22222013KBFIKLGB
R
```

7.1.2.3 Reject with Reason

The Crew Advisory Reject with Reason display is displayed when REJECT REASONS is selected from the Crew Advisory Uplink display, which enables the flight crew to compose a response indicating the reason for the rejection.

7.1.2.3.1 Display Definition

Figure 7-3 shows the Crew Advisory Reject with Reason display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	CREW ADVISORY	XXXXXXXX
REJECTION REASON:		
◆ MESSAGE UNCLEAR		
◆ UNABLE TO COMPLY		
◆ CHECK ITEM NUMBER:	□□	
◆ CHECK LINE NUMBER:	□□	
◆ OTHER:	□□□□□□□□□□□□□□□□□□□□	
<TEXT BLOCK>		
REJECT		
SCRATCHPAD		

Figure 7-3 Crew Advisory Reject with Reason Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “CREW ADVISORY RESPONSE REASONS”

Main Area

This contains a group of four ESBs:

- MESSAGE UNCLEAR (CAA_REASON = 1)
- UNABLE TO COMPLY (CAA_REASON = 2)
- CHECK ITEM NUMBER (CAA_REASON = 3)
- CHECK LINE NUMBER (CAA_REASON = 4)

- OTHER (CAA_REASON = 5)

There are also three entry boxes that become visible only when the associated ESB has been selected. Table 7-5 describes the Crew Advisory Reject with Reason display entry boxes.

Table 7-5 Crew Advisory Reject with Reason Display Entry Boxes

Name	Mandatory	Length	Format	Range	Display	Parameter
CHECK ITEM NUMBER	Yes	1-2	XX		If CHECK ITEM NUMBER selected	CAA_CHK_ITEM_NO
CHECK LINE NUMBER	Yes	1-2	XX		If CHECK LINE NUMBER selected	CAA_CHK_LINE_NO
OTHER	Yes	1-24	XXX...		If OTHER selected	CAA_OTHER

Below this (and extending to multiple pages, as necessary) is displayed the crew advisory uplink that is being rejected.

Command Buttons

Table 7-6 describes the Crew Advisory Reject with Reason display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 7-6 Crew Advisory Reject with Reason Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-
4	-	-	-	-
5	-	-	-	-
6	REJECT	Reject	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM

7.1.2.3.2 Downlink Encoding Definition

When REJECT is selected from this display, the following downlink encoding is used:

- ACARS SMI: M40
- ACARS Label: 40
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_CREW

- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 7-7 describes the Crew Advisory Reject with Reason downlink encoding definition.

Table 7-7 Crew Advisory Reject with Reason Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	CAA	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	1	Always	X	"Rejection with Reasons Flag"
13	1	Always	CAA_REASON	"Rejection Reason Code (N)"
14	2	Conditionally	CAA_CHK_ITEM_NO	"Check Item Number (XX)"
15	2	Conditionally	CAA_CHK_LINE_NO	"Check Line Number (XX)"
16	1-24	Conditionally	CAA_OTHER	"Other Reason Text (1-24 characters)"
17	2	Always	<CR><LF>	"Delimiter"
18	1-3250	Always	UPLINK_TXT_BLK	"Uplink Text (1-3250 characters)"

Sample Message

```

QU RTNBOCR
.DDLXCXA 222013
M40
FI BO24/AN N7873BO
DT DDL BFI 222013 M72A
- CAA01      BOE24/22222013KBFIKLGB
X2
CREW ADVISORY TEXT BLOCK...
```

Note: In this sample message, the rejection reason code is "2" (unable to comply).

7.2 Flight Release

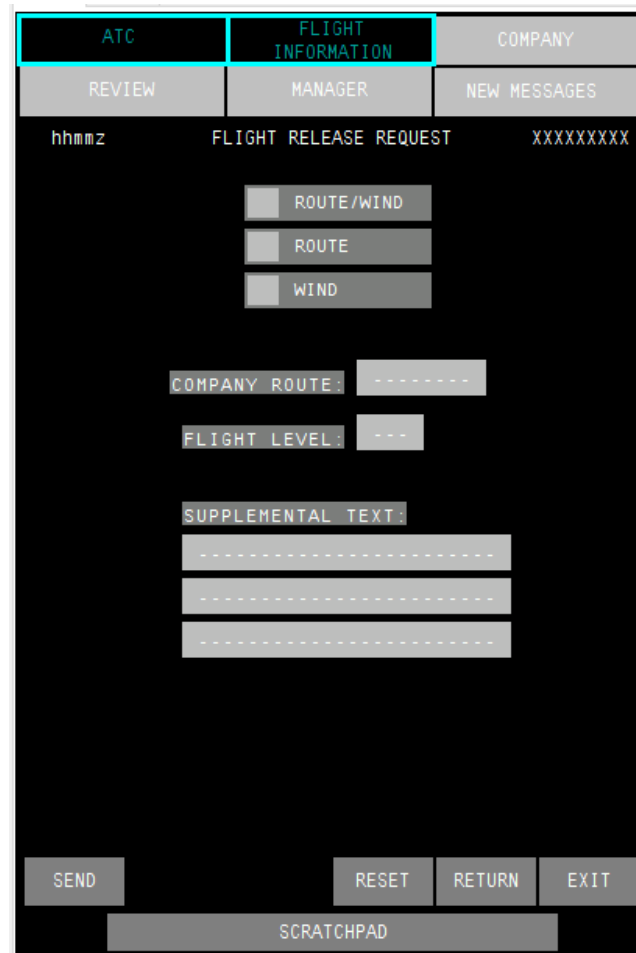
The Flight Release function enables the flight crew to request flight release information, the aircraft operator to send that information to the flight crew, and the flight crew to respond accordingly. The aircraft operator may also send the flight release information without a flight crew request.

7.2.1 Request

This section describes the Flight Release Request function.

7.2.1.1 Display Definition

Figure 7-4 shows the Flight Release Request display definition.



The image shows a graphical user interface for a 'Flight Release Request' function. At the top, there is a header bar with three sections: 'ATC' (highlighted with a red border), 'FLIGHT INFORMATION' (highlighted with a red border), and 'COMPANY'. Below this header, there are three buttons: 'REVIEW', 'MANAGER', and 'NEW MESSAGES'. The main display area has a black background with white text. It starts with 'hhmmz' on the left, 'FLIGHT RELEASE REQUEST' in the center, and 'XXXXXXXXX' on the right. Below this, there are three buttons: 'ROUTE/WIND', 'ROUTE', and 'WIND'. Further down, there are two input fields: 'COMPANY ROUTE:' followed by a field with seven dashes, and 'FLIGHT LEVEL:' followed by a field with three dashes. Below these is a section labeled 'SUPPLEMENTAL TEXT:' followed by three lines of input fields, each containing a series of dashes. At the bottom of the display, there are four buttons: 'SEND', 'RESET', 'RETURN', and 'EXIT'. Below these buttons is a wide 'SCRATCHPAD' area.

Figure 7-4 Flight Release Request Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “FLIGHT RELEASE REQUEST”

Main Area

The upper part of this display definition consists of a group of three non-exclusive selector buttons (NESBs) labeled:

- ROUTE/WIND (FTREL_RTE_WIND = Y if selected, N if not selected)
- ROUTE (FTREL_ROUTE = Y if selected, N if not selected)
- WIND (FTREL_WIND = Y if selected, N if not selected)

Table 7-8 describes the Flight Release Request display entry boxes.

Table 7-8 Flight Release Request Display Entry Boxes

Name	Mandatory	Length	Format	Range	Default	Parameter
COMPANY ROUTE	No	1-8	XXX...			FTREL_CO_RTE
FLIGHT LEVEL	No	3	NNN	0-490		FTREL_FLT_LVL
SUPPLEMENTAL TEXT	No	1-24	XXX...			FTREL_REQST_1
	No	1-24	XXX...			FTREL_REQST_2
	No	1-24	XXX...			FTREL_REQST_3

The three text entry boxes labeled SUPPLEMENTAL TEXT each allows up to 24 characters of supplemental text entry.

Command Buttons

Table 7-9 describes the Flight Release Request display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 7-9 Flight Release Request Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	SEND	Send downlink	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	-	-	-	-

Position	Legend	Function	Displayed	Displayed as Inhibited
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

Review Log

When SEND is selected from this display, it becomes available for review in the COMPANY DOWNLINKS category on the REVIEW menu.

7.2.1.2 Downlink Encoding Definition

When SEND is selected from this display, the following downlink encoding is used:

- ACARS SMI: M35
- ACARS Label: 35
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REQUEST
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 7-10 describes the Flight Release Request downlink encoding definition.

Table 7-10 Flight Release Request Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	FRL	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	1	Always	FTREL_RTE_WIND	"Route/Wind Selection (X)"
13	1	Always	FTREL_ROUTE	"Route Selection (X)"
14	1	Always	FTREL_WIND	"Wind Selection (X)"

#	Length	Inclusion	Content	Element Description
15	8	Always	FTREL_CO RTE	"Company Route (XXXXXXXX)"
16	3	Always	FTREL_FLT_LVL	"Flight Level (NNN)"
17	2	Conditionally	<CR><LF>	"Delimiter"
18	1-24	Conditionally	FTREL_REQST_1	"Text line 1 (1-24 characters)"
19	2	Conditionally	<CR><LF>	"Delimiter"
20	1-24	Conditionally	FTREL_REQST_2	"Text line 2 (1-24 characters)"
21	2	Conditionally	<CR><LF>	"Delimiter"
22	1-24	Conditionally	FTREL_REQST_3	"Text line 3 (1-24 characters)"

Note: The presence of each text line and its preceding <CR><LF> delimiter in a downlink message is conditional on the validity of that text line. In other words, if text is not entered on a particular text line then that line and the preceding <CR><LF> delimiter are not present in the downlink message.

Sample Message

```
QU RTNBOCR
.DDLXCXA 221954
M35
FI BO24/AN N7873BO
DT DDL BFI 221954 M68A
- FRL01 BOE24/19190549KBFIPHOG
Y**ALOHA50 390
```

Note: In this sample message, the flight release request is "Y**" (for combined route and winds), the company route is "ALOHA50" (left-justified), and the flight level is "390" (FL390).

7.2.2 Uplink

This section describes the Flight Release Uplink function.

7.2.2.1 Uplink Decoding Definition

The decoding for the Flight Release Uplink is defined as follows:

- ACARS SMI: CMD
- ACARS Label: RA
- Message Identifier: ~1FRL
- Alert Level: Medium (visual "●COMM" and aural high-low chime)
- Optional Element Delimiter: "," (comma)

Table 7-11 describes the Flight Release Uplink decoding definition.

Table 7-11 Flight Release Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	2	Yes	<none>	"Priority code (XX)"
2	7	Yes	<none>	"Ground address (XXXXXXX)"
3	5	Yes	<none>	"Message identifier (~1FRL)"
4	2	Yes	<none>	"Message version (XX)"
5	2	Yes	<none>	"Message date (DD)"
6	4	Yes	<none>	"Message time (HHMM)"
7	2	Yes	<none>	"Delimiter (<CR><LF>)"
8	1-3496	Yes	UPLINK_TXT_BLK	"Message text (1-3496 characters)"

Sample Message

```
QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 010A
- QURTNBOCR~1FRL01190448
FLIGHT RELEASE TEXT BLOCK...
```

7.2.2.2 Display Definition

Figure 7-5 shows the Flight Release Uplink display definition.



Figure 7-5 Flight Release Uplink Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “FLIGHT RELEASE”

Main Area

The main area contains the text block from the uplink. If any lines of the text block contain more than 46 characters, then that line is word-wrapped. If the entire text block cannot be displayed on a single page, then as many pages as are necessary are created and page arrows are displayed on the right side to access each page.

Command Buttons

Table 7-12 describes the Flight Release Uplink display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 7-12 Flight Release Uplink Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	ACCEPT	Send ACCEPT downlink	Always	When the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	PRINT	Print	Always	-
4	-	-	-	-
5	REJECT REASONS	Show RESPONSE REASONS display	Always	When the downlink queue is full or datalink is NO COMM
6	REJECT	Send REJECT downlink	Always	When the downlink queue is full or datalink is NO COMM

Review Log

When the associated uplink decoding is received, this display becomes available for review in the COMPANY UPLINKS category on the REVIEW menu.

7.2.2.3 Print Definition

Figure 7-6 shows the Flight Release Uplink print definition.

TIME: hhmmz	FLIGHT RELEASE	A/C REG: xxxxxxx
DATE: dd-MMM-yyyy		FLT NO: xxxxxxxxxx

<-----Text Block----->

Figure 7-6 Flight Release Uplink Print Definition

7.2.3 Response

This section describes the Flight Release Response function.

7.2.3.1 Accept Downlink Encoding Definition

When ACCEPT is selected, the following downlink is sent:

- ACARS SMI: M45
- ACARS Label: 45
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_CREW
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 7-13 describes the Flight Release Accept downlink encoding definition .

Table 7-13 Flight Release Accept Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	FRL	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	1	Always	A	"Acceptance Flag"

Sample Message

```
QU RTNBOCR
.DDLXCXA 221916
M45
FI BO24/AN N7873BO
DT DDL BFI 221916 M51A
- FRL01      BOE24/22221916KBFIKLGB
A
```

7.2.3.2 Reject Downlink Encoding Definition

When REJECT is selected, the following downlink is sent:

- ACARS SMI: M45
- ACARS Label: 45
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_CREW
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 7-14 describes the Flight Release Reject downlink encoding definition.

Table 7-14 Flight Release Reject Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	FRL	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	1	Always	R	"Rejection Flag"

Sample Message

```
QU RTNBOCR
.DDLXCXA 221916
M45
FI BO24/AN N7873BO
DT DDL BFI 221916 M51A
- FRL01      BOE24/22221916KBFIKLGB
R
```

7.2.3.3 Reject with Reason

The Flight Release Reject with Reason display is displayed when REJECT REASONS is selected from the Flight Release Uplink display, which enables the flight crew to compose a response indicating the reason for the rejection.

7.2.3.3.1 Display Definition

Figure 7-7 shows the Flight Release Reject with Reason display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	FLIGHT RELEASE	XXXXXXXXX
REJECTION REASON:		
☐ UNABLE TO COMPLY		
☐ CHECK ITEM NUMBER:		
☐ CHECK LINE NUMBER:		
☐ OTHER:		
<TEXT BLOCK>		
SCRATCHPAD		
		REJECT

Figure 7-7 Flight Release Reject with Reason Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “FLIGHT RELEASE RESPONSE REASONS”

Main Area

This contains a group of four ESBs:

- UNABLE TO COMPLY (FRL_REASON = 1)
- CHECK ITEM NUMBER (FRL_REASON = 2)
- CHECK LINE NUMBER (FRL_REASON = 3)
- OTHER (FRL_REASON = 4)

There are also three entry boxes that become visible only when the associated ESB has been selected. Table 7-15 describes the Flight Release Reject with Reason display entry boxes.

Table 7-15 Flight Release Reject with Reason Display Entry Boxes

Name	Mandatory	Length	Format	Range	Display	Parameter
CHECK ITEM NUMBER	Yes	1-2	XX		If CHECK ITEM NUMBER selected	FRL_CHK_ITEM_NO
CHECK LINE NUMBER	Yes	1-2	XX		If CHECK LINE NUMBER selected	FRL_CHK_LINE_NO
OTHER	Yes	1-24	XXX...		If OTHER selected	FRL_OTHER

Below this (and extending to multiple pages, as necessary) is displayed the flight release uplink that is being rejected.

Command Buttons

Table 7-16 describes the Flight Release Reject with Reason display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 7-16 Flight Release Reject with Reason Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-
4	-	-	-	-
5	-	-	-	-
6	REJECT	Reject	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM

7.2.3.3.2 Downlink Encoding Definition

When REJECT is selected from this display, the following downlink encoding is used:

- ACARS SMI: M45
- ACARS Label: 45
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_CREW
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 7-17 describes the Flight Release Reject with Reason downlink encoding definition.

Table 7-17 Flight Release Reject with Reason Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	FRL	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	1	Always	X	"Rejection with Reasons Flag"
13	1	Always	FRL_REASON	"Rejection Reason Code (N)"
14	2	Conditionally	FRL_CHK_ITEM_NO	"Check Item Number (XX)"
15	2	Conditionally	FRL_CHK_LINE_NO	"Check Line Number (XX)"
16	1-24	Conditionally	FRL_OTHER	"Other Reason Text (1-24 characters)"
17	2	Always	<CR><LF>	"Delimiter"
18	1-3250	Always	UPLINK_TXT_BLK	"Uplink Text (1-3250 characters)"

Sample Message

```
QU RTNBOCR
.DDLXCXA 221916
M45
FI BO24/AN N7873BO
DT DDL BFI 221916 M51A
- FRL01      BOE24/22221916KBFIKLGB
X311
FLIGHT RELEASE TEXT BLOCK...
```

Note: In this sample message, the rejection reason code is "3" (check line number) and the line number is "11".

7.3 Gate Assignment

The Gate Assignment function enables the flight crew to request gate assignment information, the aircraft operator to send that information to the flight crew, and the flight crew to respond

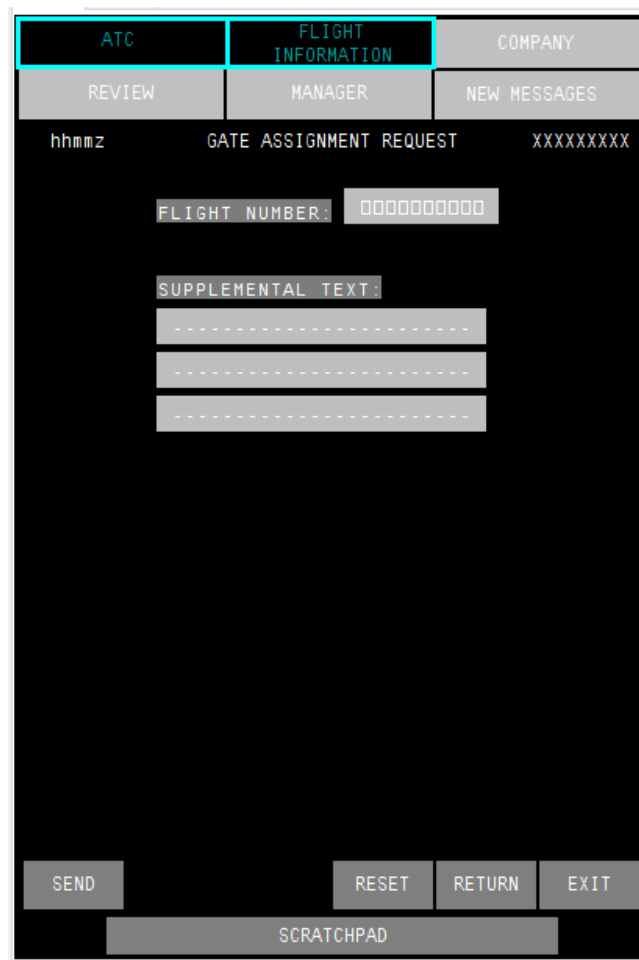
accordingly. The aircraft operator may also send the gate assignment information without a flight crew request.

7.3.1 Request

This section describes the Gate Assignment Request function.

7.3.1.1 Display Definition

Figure 7-8 shows the Gate Assignment Request display definition.



The image shows a graphical user interface for a 'GATE ASSIGNMENT REQUEST' function. At the top, there is a header bar with three tabs: 'ATC' (highlighted in red), 'FLIGHT INFORMATION' (highlighted in red), and 'COMPANY'. Below the tabs are three buttons: 'REVIEW', 'MANAGER', and 'NEW MESSAGES'. The main display area has a black background. At the top of this area, the text 'hhmmz' is on the left, 'GATE ASSIGNMENT REQUEST' is in the center, and 'XXXXXXXXX' is on the right. Below this, there is a 'FLIGHT NUMBER:' label followed by a text input field containing '0000000000'. Underneath that is a 'SUPPLEMENTAL TEXT:' label followed by three stacked text input fields, each containing a series of dashes. At the bottom of the display, there are four buttons: 'SEND', 'RESET', 'RETURN', and 'EXIT'. Below these buttons is a wide 'SCRATCHPAD' input field.

Figure 7-8 Gate Assignment Request Display Definition

Title

The title area contains:

- The current time in HHMMz format

- “GATE ASSIGNMENT REQUEST”

Main Area

Table 7-18 describes the Gate Assignment Request display entry boxes.

Table 7-18 Gate Assignment Request Display Entry Boxes

Name	Mandatory	Length	Format	Range	Default	Parameter
FLIGHT NUMBER	Yes	1-10	XXX...		ENTERED_FLTNO	GATE_RQST_FLTNO
SUPPLEMENTAL TEXT	No	1-24	XXX...			GATE_REQUEST_1
	No	1-24	XXX...			GATE_REQUEST_2
	No	1-24	XXX...			GATE_REQUEST_3

The three text entry boxes labeled SUPPLEMENTAL TEXT each allows up to 24 characters of supplemental text entry.

Command Buttons

Table 7-19 describes the Gate Assignment Request display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 7-19 Gate Assignment Request Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	SEND	Send downlink	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	-	-	-	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

Review Log

When SEND is selected from this display, it becomes available for review in the COMPANY DOWNLINKS category on the REVIEW menu.

7.3.1.2 Downlink Encoding Definition

When SEND is selected from this display, the following downlink encoding is used:

- ACARS SMI: M36

- ACARS Label: 36
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REQUEST
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 7-20 describes the Gate Assignment Request downlink encoding definition.

Table 7-20 Gate Assignment Request Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	GTA	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	10	Always	GATE_RQST_FLTNO	"Flight Number (XXXXXXXXXX)"
13	2	Conditionally	<CR><LF>	"Delimiter"
14	1-24	Conditionally	GATE_REQUEST_1	"Text line 1 (1-24 characters)"
15	2	Conditionally	<CR><LF>	"Delimiter"
16	1-24	Conditionally	GATE_REQUEST_2	"Text line 2 (1-24 characters)"
17	2	Conditionally	<CR><LF>	"Delimiter"
18	1-24	Conditionally	GATE_REQUEST_3	"Text line 3 (1-24 characters)"

Note: The presence of each text line and its preceding <CR><LF> delimiter in a downlink message is conditional on the validity of that text line. In other words, if text is not entered on a particular text line then that line and the preceding <CR><LF> delimiter are not present in the downlink message.

Sample Message

```
QU RTNBOCR
.DDLXCXA 221949
M36
FI BO24/AN N7873BO
DT DDL BFI 221949 M64A
```

- GTA01 BOE24/22221949KBFIKLGB
 BOE24

Note: In this sample message, the flight number is “BOE24” (right-justified).

7.3.2 Uplink

This section describes the Gate Assignment Uplink function

7.3.2.1 Uplink Decoding Definition

The decoding for the Gate Assignment Uplink is defined as follows:

- ACARS SMI: CMD
- ACARS Label: RA
- Message Identifier: ~1GTA
- Alert Level: Medium (visual “●COMM” and aural high-low chime)
- Optional Element Delimiter: “,” (comma)

Table 7-21 describes the Gate Assignment Uplink decoding definition.

Table 7-21 Gate Assignment Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	2	Yes	<none>	"Priority code (XX)"
2	7	Yes	<none>	"Ground address (XXXXXXX)"
3	5	Yes	<none>	"Message identifier (~1GTA)"
4	2	Yes	<none>	"Message version (XX)"
5	2	Yes	<none>	"Message date (DD)"
6	4	Yes	<none>	"Message time (HHMM)"
7	2	Yes	<none>	"Delimiter (<CR><LF>)"
8	1-3496	Yes	UPLINK_TXT_BLK	"Message text (1-3496 characters)"

Sample Message

```
QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 011A
- QURTNBOCR~1GTA01190453
GATE ASSIGNMENT TEXT BLOCK...
```

7.3.2.2 Display Definition

Figure 7-9 shows the Gate Assignment Uplink display definition.

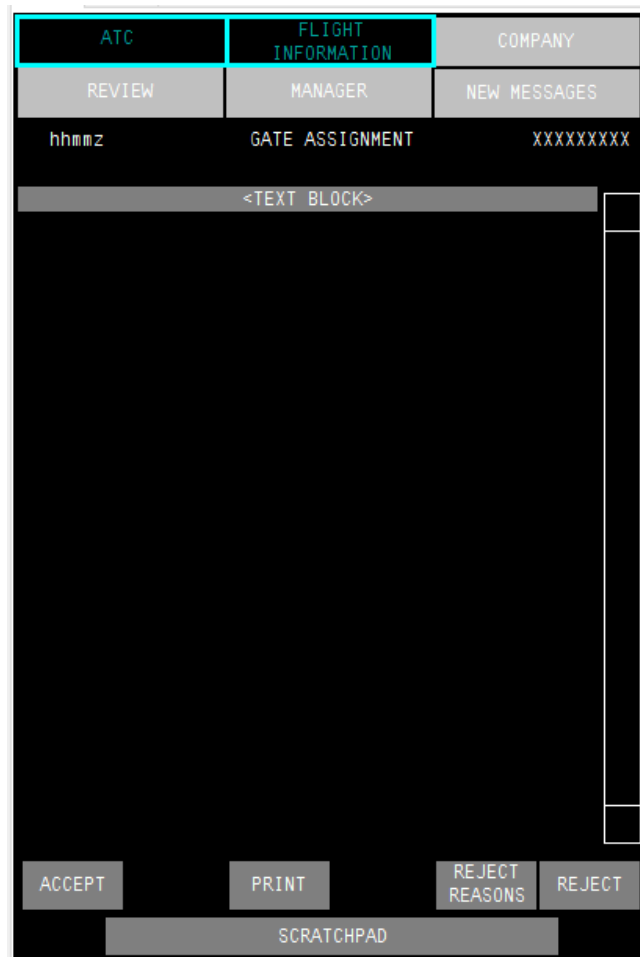


Figure 7-9 Gate Assignment Uplink Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “GATE ASSIGNMENT”

Main Area

The main area contains the text block from the uplink. If any lines of the text block contain more than 46 characters, then that line is word-wrapped. If the entire text block cannot be displayed

on a single page, then as many pages as are necessary are created and page arrows are displayed on the right side to access each page.

Command Buttons

Table 7-22 describes the Gate Assignment Uplink display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 7-22 Gate Assignment Uplink Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	ACCEPT	Send ACCEPT response	Always	When the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	PRINT	Print	Always	-
4	-	-	-	-
5	REJECT REASONS	Show RESPONSE REASONS display	Always	When the downlink queue is full or datalink is NO COMM
6	REJECT	Send REJECT response	Always	When the downlink queue is full or datalink is NO COMM

Review Log

When the associated uplink decoding is received, this display becomes available for review in the COMPANY UPLINKS category on the REVIEW menu.

7.3.2.3 Print Definition

Figure 7-10 shows the Gate Assignment Uplink print definition.

TIME: hhmmz DATE: dd-MM-yyyy	GATE ASSIGNMENT	A/C REG: xxxxxxx FLT NO: xxxxxxxxxx
---------------------------------	-----------------	--

--Text Block--

Figure 7-10 Gate Assignment Uplink Print Definition

7.3.3 Response

This section describes the Gate Assignment Response function.

7.3.3.1 Accept Downlink Encoding Definition

When ACCEPT is selected, the following downlink is sent:

- ACARS SMI: M46
- ACARS Label: 46
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_CREW
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 7-23 describes the Gate Assignment Accept downlink encoding definition.

Table 7-23 Gate Assignment Accept Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	GTA	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	1	Always	A	"Acceptance Flag"

Sample Message

```
QU RTNBOCR
.DDLXCXA 221922
M46
FI BO24/AN N7873BO
DT DDL BFI 221922 M57A
- GTA01      BOE24/22221922KBFIKLGB
A
```

7.3.3.2 Reject Downlink Encoding Definition

When REJECT is selected, the following downlink is sent:

- ACARS SMI: M46
- ACARS Label: 46
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_CREW
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 7-24 describes the Gate Assignment Reject downlink encoding definition.

Table 7-24 Gate Assignment Reject Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	GTA	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	1	Always	R	"Rejection Flag"

Sample Message

```
QU RTNBOCR
.DDLXCXA 221922
M46
FI BO24/AN N7873BO
DT DDL BFI 221922 M57A
- GTA01      BOE24/22221922KBFIKLGB
R
```

7.3.3.3 Reject with Reason

The Gate Assignment Reject with Reason display is displayed when REJECT REASONS is selected from the Gate Assignment Uplink display, which enables the flight crew to compose a response indicating the reason for the rejection.

7.3.3.3.1 Display Definition

Figure 7-11 shows the Gate Assignment Reject with Reason display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES

hhmmz

GATE ASSIGNMENT

XXXXXXXXXX

REJECTION REASON:

◆ GATE OCCUPIED

◆ NO SUCH GATE

◆ GATE BLOCKED

◆ OTHER:

<TEXT BLOCK>

REJECT

SCRATCHPAD

Figure 7-11 Gate Assignment Reject with Reason Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “GATE ASSIGNMENT RESPONSE REASONS”

Main Area

This contains a group of four ESBs:

- GATE OCCUPIED (GTA_REASON = 1)
- NO SUCH GATE (GTA_REASON = 2)
- GATE BLOCKED (GTA_REASON = 3)
- OTHER (GTA_REASON = 4)

There is also an entry box that becomes visible only when the associated ESB has been selected. Table 7-25 describes the Gate Assignment Reject with Reason display entry box.

Table 7-25 Gate Assignment Reject with Reason Display Entry Box

Name	Mandatory	Length	Format	Range	Display	Parameter
OTHER	Yes	1-24	XXX...		If OTHER selected	GTA_OTHER

Below this (and extending to multiple pages, as necessary) is displayed the gate assignment uplink that is being rejected.

Command Buttons

Table 7-26 describes the Gate Assignment Reject with Reason display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 7-26 Gate Assignment Reject with Reason Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-
4	-	-	-	-
5	-	-	-	-
6	REJECT	Reject	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM

7.3.3.3.2 Downlink Encoding Definition

When REJECT is selected from this display, the following downlink encoding is used:

- ACARS SMI: M46
- ACARS Label: 46
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_CREW
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 7-27 describes the Gate Assignment Reject with Reason downlink encoding definition.

Table 7-27 Gate Assignment Reject with Reason Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	GTA	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	1	Always	X	"Rejection with Reasons Flag"
13	1	Always	GTA_REASON	"Rejection Reason Code (N)"
14	1-24	Conditionally	GTA_OTHER	"Other Reason Text (1-24 characters)"
15	2	Always	<CR><LF>	"Delimiter"
16	1-3250	Always	UPLINK_TXT_BLK	"Uplink Text (1-3250 characters)"

Sample Message

```
QU RTNBOCR
.DDLXCXA 221922
M46
FI BO24/AN N7873BO
DT DDL BFI 221922 M57A
- GTA01      BOE24/22221922KBFIKLGB
X4NO RAMP PERSONNEL
GATE ASSIGNMENT TEXT BLOCK...
```

Note: In this sample message, the rejection reason code is "4" (other) and the other reason text is "NO RAMP PERSONNEL".

7.4 Loadsheets

The Loadsheets function enables the aircraft operator to send the flight crew an uplink message containing loadsheet information. The information is partially in a free-text format and requires a response from the flight crew.

In order to remove a loadsheet uplink message from NEW MESSAGES, the flight crew must respond to it by selecting either ACCEPT LOADSHEET or REJECT LOADSHEET, entering valid data in the captain ID and code entry boxes, and then selecting SEND.

7.4.1 Uplink

This section describes the Loadsheet Uplink function.

7.4.1.1 Uplink Decoding Definition

The decoding for the Loadsheet Uplink is defined as follows:

- ACARS SMI: M1L
- ACARS Label: 1L
- Message Identifier: LDS
- Alert Level: Medium (visual “●COMM” and aural high-low chime)
- Optional Element Delimiter: “/” (forward slash)

Table 7-28 describes the Loadsheet Uplink decoding definition.

Table 7-28 Loadsheet Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	3	Yes	<none>	"Message identifier (LDS)"
2	1	Yes	<none>	"Delimiter (/)"
3	3-10	Yes	LDS_FLIGHTNO	"Flight number (3-10 characters)"
4	2-11	Yes	LDS_DATE	"Date (2-11 characters)"
5	5-6	No	LDS_TYPE	"Type (PRELIM or FINAL)"
6	1-8	No	LDS_EDNO	"Edition (1-8 characters)"
7	1-3481	Yes	UPLINK_TXT_BLK	"Message text (1-3481 characters)"

Note: Elements 3 through 6 are variable-length elements and accordingly should be terminated with a delimiter character. Pre-defined element delimiters are <CR>, <CR><LF>, and <ETX> and the optional element delimiter defined for this uplink decoding is “/” (forward slash).

Sample Message

```
QU DDLXCXA
.RTNBOCR 221954
M1L
AN N7873BO/MA 012A
- LDS/BOE718/22-FEB-2019/PRELIM/V1
TEXT BLOCK LINE 01
TEXT BLOCK LINE 02
TEXT BLOCK LINE 03
TEXT BLOCK LINE 04
TEXT BLOCK LINE 05
```

```
TEXT BLOCK LINE 06
TEXT BLOCK LINE 07
TEXT BLOCK LINE 08
TEXT BLOCK LINE 09
TEXT BLOCK LINE 10
```

7.4.1.2 Display Definition

Figure 7-12 shows the Loadsheet display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES

hmmz LOADSHEET XXXXXXXXX

FLIGHT NUMBER: XXXXXXXXX

DATE: XXXXXXXXXXXX

TYPE: XXXXXX

EDITION: XXXXXXXX


ACCEPT
LOADSHEET


REJECT
LOADSHEET

CAPTAIN: 000000000000000000000000 000000

LOADSHEET:

<TEXT BLOCK>

SEND

PRINT

RESET

RETURN

EXIT

SCRATCHPAD

Figure 7-12 Loadsheet Display Definition

Title

The title area contains:

- The current time in HHMMz format
- "LOADSHEET"

Main Area

Table 7-29 describes the Loadsheet display parameters.

Table 7-29 Loadsheet Display Parameters

Name	Length	Format	Enterable	Range	Parameter
FLIGHT NUMBER	3-10	XXXXXXXXXX	No		LDS_FLIGHTNO
DATE	2-11	XXXXXXXX	No		LDS_DATE
TYPE	5-6	PRELIM or FINAL	No		LDS_TYPE
EDITION	1-8	XXXXXXXX	No		LDS_EDNO

This is followed by a group of two ESBs:

- ACCEPT LOADSHEET (LDS_CONFIRM_REVIEW = 1)
- REJECT LOADSHEET (LDS_CONFIRM_REVIEW = 2)

Table 7-30 describes the Loadsheet display entry boxes.

Table 7-30 Loadsheet Display Entry Boxes

Name	Mandatory	Length	Format	Range	Default	Parameter
CAPTAIN	Yes	1-24	XXX...			LDS_CAPTAIN_ID
	Yes	1-6	XXX...			LDS_CAPTAIN_CODE

This is followed by the text block from the uplink. If any lines of the text block contain more than 46 characters, then that line is word-wrapped. If the entire text block cannot be displayed on a single page, then as many pages as are necessary are created and page arrows are displayed on the right side to access each page.

Command Buttons

Table 7-31 describes the Loadsheet display command buttons.

Table 7-31 Loadsheet Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	SEND	Send downlink	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	PRINT	Print	Always	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

Review Log

When the associated uplink decoding is received, this display becomes available for review in the LOADSHEET category on the REVIEW menu.

7.4.1.3 Print Definition

Figure 7-13 shows the Loadsheet print definition.

```
TIME: hhmmz                      LOADSHEET                      A/C REG: xxxxxx
DATE: dd-MMM-yyyy                FLT NO: xxxxxxxxxx

FLIGHT NUMBER: xxxxxxxxxx
    DATE: xxxxxxxxxx
    TYPE: xxxxxx
    EDITION: xxxxxxxx

    ACCEPTED  REJECTED

    CAPTAIN ID: xxxxxxxxxxxxxxxxxxxxxxxx

LOADSHEET:
<-----Text Block----->
```

Figure 7-13 Loadsheet Print Definition

7.4.2 Response

This section describes the Loadsheet Response function.

7.4.2.1 Downlink Encoding Definition

When SEND is selected from this display, the following downlink encoding is used:

- ACARS SMI: M1L
- ACARS Label: 1L
- Allowable Subnetworks: All

- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_CREW
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 7-32 describes the Loadsheet Response downlink encoding definition.

Table 7-32 Loadsheet Response Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3-10	Always	LDS_FLIGHTNO	"Flight Number (3-10 characters)"
3	1	Always	/	"Delimiter"
4	2-11	Always	LDS_DATE	"Date (2-11 characters)"
5	1	Always	/	"Delimiter"
6	5-6	Always	LDS_TYPE	"Type (PRELIM or FINAL)"
7	1	Always	/	"Delimiter"
8	1-8	Always	LDS_EDNO	"Edition (1-8 characters)"
9	2	Always	<CR><LF>	"Delimiter"
10	6	Conditionally	ACCEPT	"ACCEPT Response"
11	6	Conditionally	REJECT	"REJECT Response"
12	2	Always	<CR><LF>	"Delimiter"
13	1-24	Always	LDS_CAPTAIN_ID	"Captain ID (1-24 characters)"
14	1	Always	/	"Delimiter"
15	1-6	Always	LDS_CAPTAIN_CODE	"Captain Code (1-6 characters)"

Sample Message

```

QU RTNBOCR
.DDLXCXA 221626
M1L
FI BO718/AN N7873BO
DT DDL BFI 221626 M01A
- BOE718/22-FEB-2019/PRELIM/V1
ACCEPT
SMITH/789

```

7.5 Weight and Balance

The Weight and Balance function enables the flight crew to request weight and balance information, the aircraft operator to send that information to the flight crew, and the flight crew to

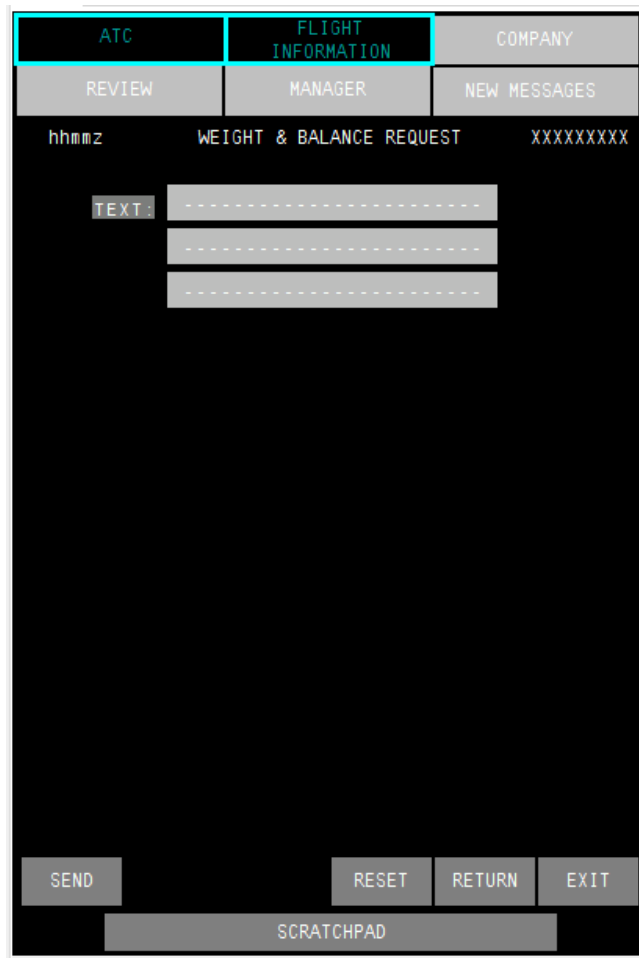
respond accordingly. The aircraft operator may also send the weight and balance information without a flight crew request.

7.5.1 Request

This section describes the Weight and Balance Request function.

7.5.1.1 Display Definition

Figure 7-14 shows the Weight and Balance Request display definition.



The figure shows a graphical user interface for a 'Weight & Balance Request' function. At the top, there are three tabs: 'ATC' (highlighted with a red border), 'FLIGHT INFORMATION' (highlighted with a red border), and 'COMPANY'. Below these are three buttons: 'REVIEW', 'MANAGER', and 'NEW MESSAGES'. The main display area has a title bar with 'hhmmz' on the left, 'WEIGHT & BALANCE REQUEST' in the center, and 'XXXXXXXX' on the right. Below the title bar, there is a 'TEXT' label followed by three horizontal input fields. At the bottom of the screen, there are four buttons: 'SEND', 'RESET', 'RETURN', and 'EXIT'. Below these buttons is a 'SCRATCHPAD' label.

Figure 7-14 Weight and Balance Request Display Definition

Title

The title area contains:

- The current time in HHMMz format

- “WEIGHT & BALANCE REQUEST”

Main Area

The main area contains a group of three text entry boxes labeled TEXT, each of which allows up to 24 characters of text entry. Table 7-33 describes the Weight and Balance Request display entry boxes.

Table 7-33 Weight and Balance Request Display Entry Boxes

Name	Mandatory	Length	Format	Range	Default	Parameter
TEXT	No	1-24	XXX...			WTBAL_RQST_TXT_1
	No	1-24	XXX...			WTBAL_RQST_TXT_2
	No	1-24	XXX...			WTBAL_RQST_TXT_3

Command Buttons

Table 7-34 describes the Weight and Balance Request display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 7-34 Weight and Balance Request Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	SEND	Send downlink	Always	When the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	-	-	-	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

Review Log

When SEND is selected from this display, it becomes available for review in the COMPANY DOWNLINKS category on the REVIEW menu.

7.5.1.2 Downlink Encoding Definition

When SEND is selected from this display, the following downlink encoding is used:

- ACARS SMI: M31
- ACARS Label: 31
- Allowable Subnetworks: All

- Downlink Priority: Flight Regularity
- Downlink Buffer: REQUEST
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 7-35 describes the Weight and Balance Request downlink encoding definition.

Table 7-35 Weight and Balance Request Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	WAB	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Conditionally	<CR><LF>	"Delimiter"
12	1-24	Conditionally	WTBAL_RQST_TXT_1	"Text line 1 (1-24 characters)"
13	2	Conditionally	<CR><LF>	"Delimiter"
14	1-24	Conditionally	WTBAL_RQST_TXT_2	"Text line 2 (1-24 characters)"
15	2	Conditionally	<CR><LF>	"Delimiter"
16	1-24	Conditionally	WTBAL_RQST_TXT_3	"Text line 3 (1-24 characters)"

Note: The presence of each text line and its preceding <CR><LF> delimiter in a downlink message is conditional on the validity of that text line. In other words, if text is not entered on a particular text line then that line and the preceding <CR><LF> delimiter are not present in the downlink message.

Sample Message

```
QU RTNBOCR
.DDLXCXA 221937
M31
FI BO24/AN N7873BO
DT DDL BFI 221937 M58A
- WAB01 BOE24/22221937KBFIKLGB
```

Note: In this sample message, no supplemental text is present.

7.5.2 Uplink

This section describes the Weight and Balance Uplink function.

7.5.2.1 Uplink Decoding Definition

The decoding for the Weight and Balance Uplink is defined as follows:

- ACARS SMI: CMD
- ACARS Label: RA
- Message Identifier: ~1WAB
- Alert Level: Medium (visual “●COMM” and aural high-low chime)
- Optional Element Delimiter: “,” (comma)

Table 7-36 describes the Weight and Balance Uplink decoding definition.

Table 7-36 Weight and Balance Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	2	Yes	<none>	"Priority code (XX)"
2	7	Yes	<none>	"Ground address (XXXXXXX)"
3	5	Yes	<none>	"Message identifier (~1WAB)"
4	2	Yes	<none>	"Message version (XX)"
5	2	Yes	<none>	"Message date (DD)"
6	4	Yes	<none>	"Message time (HHMM)"
7	2	Yes	<none>	"Delimiter (<CR><LF>)"
8	1-3496	Yes	UPLINK_TXT_BLK	"Message text (1-3496 characters)"

Sample Message

```
QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 013A
- QURTNBOCR~1WAB01190526
WEIGHT & BALANCE TEXT BLOCK...
```

7.5.2.2 Display Definition

Figure 7-15 shows the Weight and Balance Uplink display definition.



Figure 7-15 Weight and Balance Uplink Display Definition

Title

The title area contains:

- The current time in HHMMz format
- "WEIGHT & BALANCE"

Main Area

The main area contains the text block from the uplink. If any lines of the text block contain more than 46 characters, then that line is word-wrapped. If the entire text block cannot be displayed on a single page, then as many pages as are necessary are created and page arrows are displayed on the right side to access each page.

Command Buttons

Table 7-37 describes the Weight and Balance Uplink display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 7-37 Weight and Balance Uplink Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	ACCEPT	Send ACCEPT downlink	Always	When the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	PRINT	Print	Always	-
4	-	-	-	-
5	REJECT REASONS	Show RESPONSE REASONS display	Always	When the downlink queue is full or datalink is NO COMM
6	REJECT	Send REJECT downlink	Always	When the downlink queue is full or datalink is NO COMM

Review Log

When the associated uplink decoding is received, this display becomes available for review in the COMPANY UPLINKS category on the REVIEW menu.

7.5.2.3 Print Definition

Figure 7-16 shows the Weight and Balance Uplink print definition.

TIME: hhmmz	WEIGHT & BALANCE	A/C REG: xxxxxxxx
DATE: dd-MMM-yyyy		FLT NO: xxxxxxxxxxxx
--Text Block--		

Figure 7-16 Weight and Balance Uplink Print Definition

7.5.3 Response

This section describes the Weight and Balance Response function.

7.5.3.1 Accept Downlink Encoding Definition

When ACCEPT is selected, the following downlink is sent:

- ACARS SMI: M41
- ACARS Label: 41
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_CREW
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 7-38 describes the Weight and Balance Accept downlink encoding definition.

Table 7-38 Weight and Balance Accept Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	WAB	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	1	Always	A	"Acceptance Flag"

Sample Message

```
QU RTNBOCR
.DDLXCXA 221834
M41
FI BO64/AN N7873BO
DT DDL BFI 221834 M39A
- WAB01      BOE64/22221834KBFIPHOG
A
```

7.5.3.2 Reject Downlink Encoding Definition

When REJECT is selected, the following downlink is sent:

- ACARS SMI: M41
- ACARS Label: 41
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_CREW
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 7-39 describes the Weight and Balance Reject downlink encoding definition.

Table 7-39 Weight and Balance Reject Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	WAB	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	1	Always	R	"Rejection Flag"

Sample Message

```
QU RTNBOCR
.DDLXCXA 221834
M41
FI BO64/AN N7873BO
DT DDL BFI 221834 M69A
- WAB01      BOE64/22221834KBFIPHOG
R
```

7.5.3.3 Reject with Reason

The Weight and Balance Reject with Reason display is displayed when REJECT REASONS is selected from the Weight and Balance Uplink display, which enables the flight crew to compose a response indicating the reason for the rejection.

7.5.3.3.1 Display Definition

Figure 7-17 shows the Weight and Balance Reject with Reason display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	WEIGHT & BALANCE	XXXXXXXXX
REJECTION REASON:		
◆ PASSENGER COUNT DISCREPANCY		
◆ FUEL ORDER DISCREPANCY		
◆ CREW COUNT DISCREPANCY		
◆ CHECK ITEM NUMBER:	□□	
◆ CHECK LINE NUMBER:	□□	
◆ OTHER:	□□□□□□□□□□□□□□□□□□□□	
<TEXT BLOCK>		
REJECT		
SCRATCHPAD		

Figure 7-17 Weight and Balance Reject with Reason Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “WEIGHT & BALANCE RESPONSE REASONS”

Main Area

This contains a group of six ESBs:

- PASSENGER COUNT DISCREPANCY (WAB_REASON = 1)
- FUEL ORDER DISCREPANCY (WAB_REASON = 2)
- CREW COUNT DISCREPANCY (WAB_REASON = 3)
- CHECK ITEM NUMBER (WAB_REASON = 4)

- CHECK LINE NUMBER (WAB_REASON = 5)
- OTHER (WAB_REASON = 6)

There are also three entry boxes that become visible only when the associated ESB is selected. Table 7-40 describes the Weight and Balance Reject with Reason display entry boxes.

Table 7-40 Weight and Balance Reject with Reason Display Entry Boxes

Name	Mandatory	Length	Format	Range	Display	Parameter
CHECK ITEM NUMBER	Yes	1-2	XX		If CHECK ITEM NUMBER selected	WAB_CHK_ITEM_NO
CHECK LINE NUMBER	Yes	1-2	XX		If CHECK LINE NUMBER selected	WAB_CHK_LINE_NO
OTHER	Yes	1-24	XXX...		If OTHER selected	WAB_OTHER

Below this (and extending to multiple pages, as necessary) is displayed the weight and balance uplink that is being rejected.

Command Buttons

Table 7-41 describes the Weight and Balance Reject with Reason display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 7-41 Weight and Balance Reject with Reason Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-
4	-	-	-	-
5	-	-	-	-
6	REJECT	Reject	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM

7.5.3.3.2 Downlink Encoding Definition

When REJECT is selected from this display, the following downlink encoding is used:

- ACARS SMI: M41
- ACARS Label: 41
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity

- Downlink Buffer: RESPONSE_CREW
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 7-42 describes the Weight and Balance Reject with Reason downlink encoding definition.

Table 7-42 Weight and Balance Reject with Reason Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	WAB	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	1	Always	X	"Rejection with Reasons Flag"
13	1	Always	WAB_REASON	"Rejection Reason Code (N)"
14	2	Conditionally	WAB_CHK_ITEM_NO	"Check Item Number (XX)"
15	2	Conditionally	WAB_CHK_LINE_NO	"Check Line Number (XX)"
16	1-24	Conditionally	WAB_OTHER	"Other Reason Text (1-24 characters)"
17	2	Always	<CR><LF>	"Delimiter"
18	1-3250	Always	UPLINK_TXT_BLK	"Uplink Text (1-3250 characters)"

Sample Message

```

QU RTNBOCR
.DDLXCXA 221916
M41
FI BO64/AN N7873BO
DT DDL BFI 221916 M51A
- WAB01      BOE64/22221916KBFIPHOG
X432
WEIGHT & BALANCE TEXT BLOCK...
```

Note: In this sample message, the rejection reason code is "4" (check item number) and the item number that the flight crew entered is "32".

8 Manual Reports

This section describes the manual (flight crew-initiated) report functions that the AMI implements.

8.1 Arrival Report

The Arrival Report function enables the flight crew to send additional information that is not included in the automatic On and In reports, such as the pilot who performed the landing, the landing category, and autoland information. The data on this display is reset after SEND is selected.

8.1.1 Display Definition

Figure 8-1 shows the Arrival Report page 1 of 2 display definition.

ATC	FLIGHT INFORMATION	COMPANY	
REVIEW	MANAGER	NEW MESSAGES	
hhmmz	ARRIVAL REPORT	XXXXXXXXX	
ON TIME: 00:00	IN TIME: 00:00		
FUEL ON BOARD: - - - . -			
LANDING PILOT:			
◆ CAPT	◆ F/O 1		
◆ F/O 2	◆ F/O 3		
LANDING CATEGORY:			
◆ I	◆ II		
◆ IIIA	◆ IIIB		◆ IIIC
AUTOLAND:			
◆ YES	◆ NO		
◆ SATISFACTORY			
◆ UNSATISFACTORY			
SEND	RESET	RETURN	
SCRATCHPAD			

Figure 8-1 Arrival Report Page 1 of 2 Display Definition

Figure 8-2 shows the Arrival Report page 2 of 2 display definition.

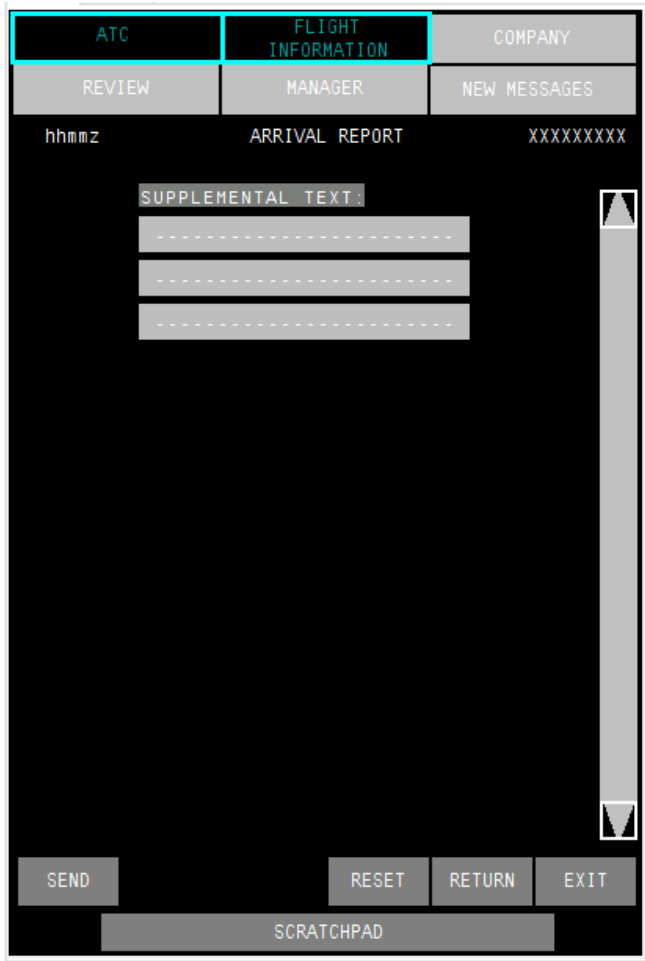


Figure 8-2 Arrival Report Page 2 of 2 Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “ARRIVAL REPORT”

Main Area

Table 8-1 describes the Arrival Report display entry boxes.

Table 8-1 Arrival Report Display Entry Boxes

Name	Mand atory	Length	Format	Range	Default	Parameter
ON TIME	Yes	4	HHMM		ON_TIME	ARR_ON_TIME
IN TIME	Yes	4	HHMM		IN_TIME	ARR_IN_TIME

Name	Mandatory	Length	Format	Range	Default	Parameter
FUEL ON BOARD	No	1-5	NNN.N	0.0-280.0	IN_FOB	ARR_FOB
SUPPLEMENTAL TEXT	No	1-24	XXX...			ARR_SUPLMT_TXT_1
	No	1-24	XXX...			ARR_SUPLMT_TXT_2
	No	1-24	XXX...			ARR_SUPLMT_TXT_3

The first three entry boxes are followed by a group of ESBs labeled LANDING PILOT to select which flight crew member performed the landing:

- CAPT (ARR_LNDG_PILOT = 1)
- F/O 1 (ARR_LNDG_PILOT = 2)
- F/O 2 (ARR_LNDG_PILOT = 3)
- F/O 3 (ARR_LNDG_PILOT = 4)

Those ESBs are followed by another group of ESBs labeled LANDING CATEGORY to select the landing category:

- I (ARR_LAND_CAT_CND = 01)
- II (ARR_LAND_CAT_CND = 02)
- IIIA (ARR_LAND_CAT_CND = 3A)
- IIIB (ARR_LAND_CAT_CND = 3B)
- IIIC (ARR_LAND_CAT_CND = 3C)

Those ESBs are followed by another group of ESBs labeled AUTOLAND to indicate whether an autoland was performed:

- YES
- NO (ARR_AUTOLAND = NO)

If the “YES” ESB is selected, then another group of ESBs is displayed:

- SATISFACTORY (ARR_AUTOLND_QUAL = YS)
- UNSATISFACTORY (ARR_AUTOLND_QUAL = YU)

Page 2 of the ARRIVAL REPORT display contains a group of three text entry boxes labeled SUPPLEMENTAL TEXT, each of which allows up to 24 characters of supplemental text entry.

Command Buttons

Table 8-2 describes the Arrival Report display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 8-2 Arrival Report Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	SEND	Send downlink	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	-	-	-	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

Review Log

When SEND is selected from this display, it becomes available for review in the COMPANY DOWNLINKS category on the REVIEW menu.

8.1.2 Downlink Encoding Definition

When SEND is selected from this display, the following downlink encoding is used:

- ACARS SMI: M28
- ACARS Label: 28
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REPORT_CREW
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 8-3 describes the Arrival Report downlink encoding definition.

Table 8-3 Arrival Report Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	LND	"Message Identifier"
3	2	Always	02	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"

#	Length	Inclusion	Content	Element Description
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	4	Always	ARR_ON_TIME	"On Time (HHMM)"
13	4	Always	ARR_IN_TIME	"In Time (HHMM)"
14	4	Always	ARR_FOB	"Fuel On Board / 100 (NNNN)"
15	2	Conditionally	01	"Landing Category"
16	2	Conditionally	02	"Landing Category"
17	2	Conditionally	3A	"Landing Category"
18	2	Conditionally	3B	"Landing Category"
19	2	Conditionally	3C	"Landing Category"
20	1	Always	ARR_LNDG_PILOT	"Landing Pilot (N)"
21	0	Conditionally	BEGIN_GROUP	
22	2	Conditionally	YS	"Autoland Satisfactory"
23	2	Conditionally	YU	"Autoland Unsatisfactory"
24	0	<n/a>	END_GROUP	
25	2	Conditionally	NO	"No Autoland"
26	2	Conditionally	<CR><LF>	"Delimiter"
27	1-24	Conditionally	ARR_SUPLMT_TXT_1	"Text line 1 (1-24 characters)"
28	2	Conditionally	<CR><LF>	"Delimiter"
29	1-24	Conditionally	ARR_SUPLMT_TXT_2	"Text line 2 (1-24 characters)"
30	2	Conditionally	<CR><LF>	"Delimiter"
31	1-24	Conditionally	ARR_SUPLMT_TXT_3	"Text line 3 (1-24 characters)"

Note: The presence of each text line and its preceding <CR><LF> delimiter in a downlink message is conditional on the validity of that text line. In other words, if text is not entered on a particular text line then that line and the preceding <CR><LF> delimiter are not present in the downlink message.

Sample Message

```

QU RTNBOCR
.DDLXCXA 222016
M28
FI BO24/AN N7873BO
DT DDL BFI 222016 M74A
- LND02 BOE24/22222016KBFIKLGB
1425143604583B4YU
TEXT LINE 1

```

Note: In this sample message, the fuel on board is “0458” (45,800), the landing category is “3B”, the landing pilot is “4” (F/O 3), and the autoland indication is “YU” (yes, unsatisfactory).

8.2 ATC Datalink Problem Report

The ATC Datalink Problem Report function enables the flight crew to send a problem report in a format suitable for submission to the appropriate reporting agency that monitors ATC datalink operations in the airspace in question.

8.2.1 Display Definition

Figure 8-3 shows the ATC Datalink Problem Report page 1 of 2 display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz ATC DATALINK PROBLEM REPORT XXXXXXXXX		
FLIGHT NUMBER: XXXXXXXXX		▲ ▼
AIRCRAFT REGISTRATION: XXXXXXX		
AIRCRAFT TYPE: XXXX		
DATE UTC: □□		
TIME UTC: □□:□□		
ACTIVE CENTER: -----		
NEXT CENTER: -----		
POSITION: -----		
ORIGINATOR: -----		
DESCRIPTION: □□□□□□□□□□□□□□□□□□□□		

SEND	PRINT	RESET
RETURN		
EXIT		
SCRATCHPAD		

Figure 8-3 ATC Datalink Problem Report Page 1 of 2 Display Definition

Figure 8-4 shows the ATC Datalink Problem Report page 2 of 2 display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	ATC DATALINK PROBLEM REPORT	XXXXXXXX
ADDITIONAL		
DESCRIPTION:		
		
		
		
		
		
		
		
		
		
		
		
SEND	PRINT	RESET
	RETURN	EXIT
SCRATCHPAD		

Figure 8-4 ATC Datalink Problem Report Page 2 of 2 Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “ATC DATALINK PROBLEM REPORT”

Main Area

Table 8-4 describes the ATC Datalink Problem Report display parameters.

Table 8-4 ATC Datalink Problem Report Display Parameters

Name	Length	Format	Range	Parameter
FLIGHT NUMBER	1-10	XXXXXXXX		ENTERED_FLTNO
AIRCRAFT REGISTRATION	1-7	XXXXXXXX		REGISTRATION_NO

Name	Length	Format	Range	Parameter
AIRCRAFT TYPE	4	XXXX		AIRCRAFT_TYPE

CMF determines the aircraft type (ICAO aircraft type designator) as follows:

- The first character is the airplane manufacturer, namely “B” (Boeing).
- The second and third characters are the major model, namely “78” (787).
- The fourth character is the minor model, “8” (787-8), “9” (787-9), or “X” (787-10).
- Any undefined characters are shown as “*” (asterisk).

Table 8-5 describes the ATC Datalink Problem Report display entry boxes.

Table 8-5 ATC Datalink Problem Report Display Entry Boxes

Name	Mand atory	Length	Format	Range	Default	Parameter
DATE UTC	Yes	2	DD	1-31		PR.UTC_DATE
TIME UTC	Yes	4	HHMM			PR.UTC_TIME
ACTIVE CENTER	No	1-8	XXX...			PR.ACT_CTR
NEXT CENTER	No	1-8	XXX...			PR.NEXT_CTR
POSITION	No	1-15	XXX...			PR.POSITION
ORIGINATOR	No	1-24	XXX...			PR.ORIGINATOR
DESCRIPTION	Yes	1-24	XXX...			PR.TEXT_1
	No	1-24	XXX...			PR.TEXT_2
	No	1-24	XXX...			PR.TEXT_3
	No	1-24	XXX...			PR.TEXT_4
ADDITIONAL DESCRIPTION	No	1-24	XXX...			PR.TEXT_5
	No	1-24	XXX...			PR.TEXT_6
	No	1-24	XXX...			PR.TEXT_7
	No	1-24	XXX...			PR.TEXT_8
	No	1-24	XXX...			PR.TEXT_9
	No	1-24	XXX...			PR.TEXT_10
	No	1-24	XXX...			PR.TEXT_11
	No	1-24	XXX...			PR.TEXT_12
	No	1-24	XXX...			PR.TEXT_13
	No	1-24	XXX...			PR.TEXT_14
	No	1-24	XXX...			PR.TEXT_15
	No	1-24	XXX...			PR.TEXT_16
	No	1-24	XXX...			PR.TEXT_17
	No	1-24	XXX...			PR.TEXT_18

Command Buttons

Table 8-6 describes the ATC Datalink Problem Report display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 8-6 ATC Datalink Problem Report Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	SEND	Send downlink	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	PRINT	Print	Always	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

Review Log

When SEND is selected from this display, it becomes available for review in the COMPANY DOWNLINKS category on the REVIEW menu.

8.2.2 Downlink Encoding Definition

When SEND is selected from this display, the following downlink encoding is used:

- ACARS SMI: A82
- ACARS Label: 82
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REPORT_CREW
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 8-7 describes the ATC Datalink Problem Report downlink encoding definition.

Table 8-7 ATC Datalink Problem Report Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	CRA	"Message Identifier"
3	2	Always	02	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"

#	Length	Inclusion	Content	Element Description
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	1-7	Always	REGISTRATION_NO	"Aircraft Registration (1-7 characters)"
13	1	Always	,	"Delimiter"
14	4	Always	AIRCRAFT_TYPE	"Aircraft Type (XXXX)"
15	1	Always	,	"Delimiter"
16	2	Always	PR.UTC_DATE	"PR Date (DD)"
17	1	Always	,	"Delimiter"
18	4	Always	PR.UTC_TIME	"PR Time (HHMM)"
19	1	Always	,	"Delimiter"
20	1-8	Always	PR_ACT_CTR	"Active Center (1-8 characters)"
21	1	Always	,	"Delimiter"
22	1-8	Always	PR_NEXT_CTR	"Next Center (1-8 characters)"
23	1	Always	,	"Delimiter"
24	1-13	Always	PR_POSITION	"PR Position (1-13 characters)"
25	1	Always	,	"Delimiter"
26	1-24	Always	PR_ORIGINATOR	"PR Originator (1-24 characters)"
27	2	Always	<CR><LF>	"Delimiter"
28	1-24	Always	PR_TEXT_1	"Text line 1 (1-24 characters)"
29	2	Conditionally	<CR><LF>	"Delimiter"
30	1-24	Conditionally	PR_TEXT_2	"Text line 2 (1-24 characters)"
31	2	Conditionally	<CR><LF>	"Delimiter"
32	1-24	Conditionally	PR_TEXT_3	"Text line 3 (1-24 characters)"
33	2	Conditionally	<CR><LF>	"Delimiter"
34	1-24	Conditionally	PR_TEXT_4	"Text line 4 (1-24 characters)"
35	2	Conditionally	<CR><LF>	"Delimiter"
36	1-24	Conditionally	PR_TEXT_5	"Text line 5 (1-24 characters)"
37	2	Conditionally	<CR><LF>	"Delimiter"
38	1-24	Conditionally	PR_TEXT_6	"Text line 6 (1-24 characters)"
39	2	Conditionally	<CR><LF>	"Delimiter"
40	1-24	Conditionally	PR_TEXT_7	"Text line 7 (1-24 characters)"
41	2	Conditionally	<CR><LF>	"Delimiter"
42	1-24	Conditionally	PR_TEXT_8	"Text line 8 (1-24 characters)"
43	2	Conditionally	<CR><LF>	"Delimiter"
44	1-24	Conditionally	PR_TEXT_9	"Text line 9 (1-24 characters)"
45	2	Conditionally	<CR><LF>	"Delimiter"

#	Length	Inclusion	Content	Element Description
46	1-24	Conditionally	PR_TEXT_10	"Text line 10 (1-24 characters)"
47	2	Conditionally	<CR><LF>	"Delimiter"
48	1-24	Conditionally	PR_TEXT_11	"Text line 11 (1-24 characters)"
49	2	Conditionally	<CR><LF>	"Delimiter"
50	1-24	Conditionally	PR_TEXT_12	"Text line 12 (1-24 characters)"
51	2	Conditionally	<CR><LF>	"Delimiter"
52	1-24	Conditionally	PR_TEXT_13	"Text line 13 (1-24 characters)"
53	2	Conditionally	<CR><LF>	"Delimiter"
54	1-24	Conditionally	PR_TEXT_14	"Text line 14 (1-24 characters)"
55	2	Conditionally	<CR><LF>	"Delimiter"
56	1-24	Conditionally	PR_TEXT_15	"Text line 15 (1-24 characters)"
57	2	Conditionally	<CR><LF>	"Delimiter"
58	1-24	Conditionally	PR_TEXT_16	"Text line 16 (1-24 characters)"
59	2	Conditionally	<CR><LF>	"Delimiter"
60	1-24	Conditionally	PR_TEXT_17	"Text line 17 (1-24 characters)"
61	2	Conditionally	<CR><LF>	"Delimiter"
62	1-24	Conditionally	PR_TEXT_18	"Text line 18 (1-24 characters)"

Note: The presence of each text line and its preceding <CR><LF> delimiter in a downlink message is conditional on the validity of that text line. In other words, if text is not entered on a particular text line then that line and the preceding <CR><LF> delimiter are not present in the downlink message.

Note: Although the ARINC Specification 620 definition of a label 82 downlink message allows a supplementary address to be included, the baseline AMI does not include one.

Sample Message

```

QU RTNBOCR
.DDLXCXA 222015
A82
FI BO24/AN N7873BO
DT DDL BFI 222015 M73A
- CRA02 BOE24/22222015KBFIKLGB
N7873BO,B78X,23,1221,KZSE,KZOA,80NM SOUTH OF SEA,SMITH
TEXT LINE 1
TEXT LINE 2
TEXT LINE 3

```

8.2.3 Print Definition

Figure 8-5 shows the ATC Datalink Problem Report print definition.

[illegible]

Figure 8-5 ATC Datalink Problem Report Print Definition

8.3 Delay Report

The Delay Report function enables the flight crew to inform the aircraft operator of a departure, takeoff, enroute, or arrival delay. The report includes data pertinent to the type of delay and the reason for the delay.

8.3.1 Display Definition

The Delay Report display definition depends on the selected delay type. AMI logic automatically selects the delay type as follows:

- The logic selects the departure delay type if:
 - CMF initializes when the airplane is on the ground, all doors are closed, and the parking brake is set,
 - CMF initializes when the airplane is on the ground and not all doors are closed, or
 - the OOOI state changes to In.
- The logic selects the takeoff delay type if:
 - CMF initializes when the airplane is on the ground, all doors are closed, and the parking brake is released, or

- the OOOI state changes to Out.
- The logic selects the enroute delay type if:
 - CMF initializes when the airplane is in the air, or
 - the OOOI state changes to Off.
- The logic selects the arrival delay type if the OOOI state changes to On.
- If none of these conditions apply, then the logic makes no automatic selection.

The flight crew may manually select any delay type at any time.

Figure 8-6 shows the Delay Report page 1 of 2 display definition when the departure delay type is selected, including relevant information and delay reasons.

The screenshot shows a cockpit display titled "DELAY REPORT" with a time of "1636z". At the top are three tabs: "ATC" (blue), "FLIGHT INFORMATION" (blue), and "COMPANY" (green). Below these are three buttons: "REVIEW" (blue), "MANAGER" (blue), and "NEW MESSAGES" (blue). The main display area is black with white text. It shows "DELAY TYPE:" with four options: "DEPARTURE" (selected with a green diamond), "ENROUTE" (grey diamond), "TAKEOFF" (grey diamond), and "ARRIVAL" (grey diamond). Below this are "ESTIMATED OUT TIME: 00:00" and "FUEL ON BOARD: 198.7". Under "DEPARTURE DELAY REASON:", there are eight options arranged in two columns: "AIRPORT CONGESTION", "MECHANICAL PROBLEM", "LATE CREW", "LATE AIRCRAFT", "OTHER:" (with a text input field), "LATE PASSENGER", "LOADING", "ATC HOLD", and "WEATHER". At the bottom are three buttons: "RESET", "RETURN", and "EXIT". A vertical scrollbar is on the right side of the display.

Figure 8-6 Delay Report Page 1 of 2 Display Definition – Departure Delay Type

Figure 8-7 shows the Delay Report page 1 of 2 display definition when the takeoff delay type is selected, including relevant information and delay reasons.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES

1637z DELAY REPORT

DELAY TYPE:

◆ DEPARTURE	◆ TAKEOFF
◆ ENROUTE	◆ ARRIVAL

ESTIMATED OFF TIME: 00:00

FUEL ON BOARD: 198.7

TAKEOFF DELAY REASON:

◆ AIRPORT CONGESTION	◆ ATC HOLD
◆ MECHANICAL PROBLEM	◆ WEATHER
◆ OTHER: -----	

RESET RETURN EXIT

Figure 8-7 Delay Report Page 1 of 2 Display Definition – Takeoff Delay Type

Figure 8-8 shows the Delay Report page 1 of 2 display definition when the enroute delay type is selected, including relevant information and delay reasons.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES

1637z DELAY REPORT

DELAY TYPE:

☐ DEPARTURE	☐ TAKEOFF
☒ ENROUTE	☐ ARRIVAL

ESTIMATED ON TIME: 16:36

FUEL ON BOARD: 198.7

ENROUTE DELAY REASON:

☐ ATC HOLD	☐ PERF LIMITS
☐ ATC REROUTE	☐ FLIGHT LEVEL
☐ WEATHER/HEADWINDS	
☐ OTHER: _____	

RESET RETURN EXIT

Figure 8-8 Delay Report Page 1 of 2 Display Definition – Enroute Delay Type

Figure 8-9 shows the Delay Report page 1 of 2 display definition when the arrival delay type is selected, including relevant information and delay reasons.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES

1637z DELAY REPORT

DELAY TYPE:

◆ DEPARTURE	◆ TAKEOFF
◆ ENROUTE	◆ ARRIVAL

ESTIMATED IN TIME:

FUEL ON BOARD:

ARRIVAL DELAY REASON:

◆ AIRPORT CONGESTION	◆ ATC HOLD
◆ MECHANICAL PROBLEM	◆ WEATHER
◆ GATE OCCUPIED	◆ GATE BLOCKED
◆ GROUND CREW	
◆ OTHER: -----	

RESET RETURN EXIT

Figure 8-9 Delay Report Page 1 of 2 Display Definition – Arrival Delay Type

Figure 8-10 shows the Delay Report page 2 of 2 display definition. This page includes a group of three text entry boxes labeled SUPPLEMENTAL TEXT, each of which allows up to 24 characters of supplemental text entry.

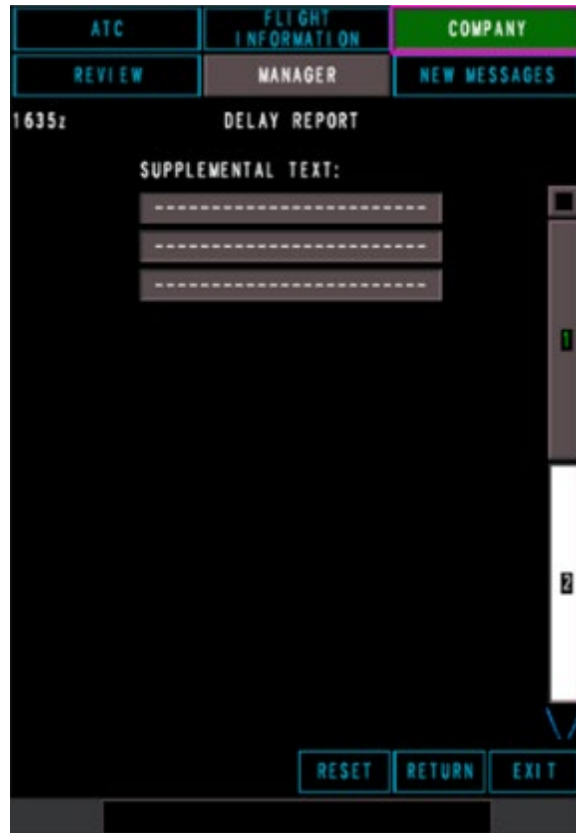


Figure 8-10 Delay Report Page 2 of 2 Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “DELAY REPORT”

Main Area

Table 8-8 describes the Delay Report display delay type ESBs.

Table 8-8 Delay Report Display Delay Type ESBs

Button Title	Result of Selection
DEPARTURE	Lower section of display contains prompts for relevant information and ESBs defined for departure delay reasons
TAKEOFF	Lower section of display contains prompts for relevant information and ESBs defined for takeoff delay reasons
ENROUTE	Lower section of display contains prompts for relevant information and ESBs defined for enroute delay reasons

Button Title	Result of Selection
ARRIVAL	Lower section of display contains prompts for relevant information and ESBs defined for arrival delay reasons

Departure Delay Type Selected

Table 8-9 describes the Delay Report display estimated out time entry box that the AMI displays when the departure delay type is selected.

Table 8-9 Delay Report Display Estimated Out Time Entry Box

Name	Mandatory	Length	Format	Range	Default	Parameter
ESTIMATED OUT TIME	Yes	4	HHMM			EST_OUT_TIME

When the departure delay type is selected the AMI displays the DEPARTURE DELAY REASON ESB group, which is composed of the following selections:

- AIRPORT CONGESTION
- MECHANICAL PROBLEM
- LATE CREW
- LATE AIRCRAFT
- LATE PASSENGER
- LOADING
- ATC HOLD
- WEATHER
- OTHER (with associated entry box for up to 24 characters)

Takeoff Delay Type Selected

Table 8-10 describes the Delay Report display estimated off time entry box that the AMI displays when the takeoff delay type is selected.

Table 8-10 Delay Report Display Estimated Off Time Entry Box

Name	Mandatory	Length	Format	Range	Default	Parameter
ESTIMATED OFF TIME	Yes	4	HHMM			EST_OFF_TIME

When the takeoff delay type is selected the AMI displays the TAKEOFF DELAY REASON ESB group, which is composed of the following selections:

- AIRPORT CONGESTION
- MECHANICAL PROBLEM
- ATC HOLD
- WEATHER
- OTHER (with associated entry box for up to 24 characters)

Enroute Delay Type Selected

Table 8-11 describes the Delay Report display estimated on time entry box that the AMI displays when the enroute delay type is selected.

Table 8-11 Delay Report Display Estimated On Time Entry Box

Name	Mandatory	Length	Format	Range	Default	Parameter
ESTIMATED ON TIME	Yes	4	HHMM		FMF_DEST_ETA	EST_ON_TIME

When the enroute delay type is selected the AMI displays the ENROUTE DELAY REASON ESB group, which is composed of the following selections:

- ATC HOLD
- ATC REROUTE
- WEATHER/HEADWINDS
- PERF LIMITS
- FLIGHT LEVEL
- OTHER (with associated entry box for up to 24 characters)

Arrival Delay Type Selected

Table 8-12 describes the Delay Report display estimated in time entry box that the AMI displays when the arrival delay type is selected.

Table 8-12 Delay Report Display Estimated In Time Entry Box

Name	Mandatory	Length	Format	Range	Default	Parameter
ESTIMATED IN TIME	Yes	4	HHMM			EST_IN_TIME

When the arrival delay type is selected the AMI displays the ARRIVAL DELAY REASON ESB group, which is composed of the following selections:

- AIRPORT CONGESTION
- MECHANICAL PROBLEM
- GATE OCCUPIED
- GROUND CREW
- ATC HOLD
- WEATHER
- GATE BLOCKED
- OTHER (with associated entry box for up to 24 characters)

Any Delay Type Selected

Table 8-13 describes the Delay Report display entry boxes that the AMI displays when any delay type is selected.

Table 8-13 Delay Report Display Entry Boxes

Name	Mandatory	Length	Format	Range	Default	Parameter
FUEL ON BOARD	No	1-5	NNN.N	0.0-280.0	FUEL_ON_BOARD	DLY_FUEL_ON_BD
SUPPLEMENTAL TEXT	No	1-24	XXX...			DLY_SUPLMT_TXT_1
	No	1-24	XXX...			DLY_SUPLMT_TXT_2
	No	1-24	XXX...			DLY_SUPLMT_TXT_3

Command Buttons

Table 8-14 describes the Delay Report display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 8-14 Delay Report Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	SEND	Send downlink	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	-	-	-	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

Review Log

When SEND is selected from this display, it becomes available for review in the COMPANY DOWNLINKS category on the REVIEW menu.

8.3.2 Downlink Encoding Definition

When SEND is selected from this display, the following downlink encoding is used:

- ACARS SMI: M21
- ACARS Label: 21
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REPORT_CREW
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 8-15 describes the Delay Report downlink encoding definition.

Table 8-15 Delay Report Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Conditionally	DPD	"Message Identifier"
3	3	Conditionally	DTO	"Message Identifier"
4	3	Conditionally	DNR	"Message Identifier"
5	3	Conditionally	DAR	"Message Identifier"
6	2	Always	02	"Message Version"
7	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
8	1	Always	/	"Delimiter"
9	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
10	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
11	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
12	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
13	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
14	2	Always	<CR><LF>	"Delimiter"
15	0	Conditionally	BEGIN_GROUP	
16	4	Always	EST_OUT_TIME	"Estimated Out Time (HHMM)"
17	2	Conditionally	AR	"Depart - Airport Congestion"
18	2	Conditionally	ME	"Depart - Mechanical Problem"
19	2	Conditionally	CR	"Depart - Late Crew"

#	Length	Inclusion	Content	Element Description
20	2	Conditionally	AC	"Depart - Late Aircraft"
21	2	Conditionally	PL	"Depart - Late Passenger"
22	2	Conditionally	LD	"Depart - Loading"
23	2	Conditionally	AT	"Depart - ATC Hold"
24	2	Conditionally	WX	"Depart - Weather"
25	0	Conditionally	BEGIN_SUBGROUP	
26	2	Always	OC	"Depart - Other"
27	1-24	Always	DLY_DEPT_TEXT	"Depart - Text (1-24 characters)"
28	0	<n/a>	END_SUBGROUP	
29	0	<n/a>	END_GROUP	
30	0	Conditionally	BEGIN_GROUP	
31	4	Always	EST_OFF_TIME	"Estimated Off Time (HHMM)"
32	2	Conditionally	AR	"Takeoff - Airport Congestion"
33	2	Conditionally	ME	"Takeoff - Mechanical Problem"
34	2	Conditionally	AT	"Takeoff - ATC Hold"
35	2	Conditionally	WX	"Takeoff - Weather"
36	0	Conditionally	BEGIN_SUBGROUP	
37	2	Always	OC	"Takeoff - Other"
38	1-24	Always	DLY_TAKEOFF_TEXT	"Takeoff - Text (1-24 characters)"
39	0	<n/a>	END_SUBGROUP	
40	0	<n/a>	END_GROUP	
41	0	Conditionally	BEGIN_GROUP	
42	4	Always	EST_ON_TIME	"Estimated On Time (HHMM)"
43	2	Conditionally	AT	"Enroute - ATC Hold"
44	2	Conditionally	TR	"Enroute - ATC Reroute"
45	2	Conditionally	WW	"Enroute - Weather/Headwinds"
46	2	Conditionally	PF	"Enroute - Performance"
47	2	Conditionally	FL	"Enroute - Flight Level"
48	0	Conditionally	BEGIN_SUBGROUP	
49	2	Always	OC	"Enroute - Other"
50	1-24	Always	DLY_ENROUTE_TEXT	"Enroute - Text (1-24 characters)"
51	0	<n/a>	END_SUBGROUP	
52	0	<n/a>	END_GROUP	
53	0	Conditionally	BEGIN_GROUP	
54	4	Always	EST_IN_TIME	"Estimated In Time (HHMM)"
55	2	Conditionally	AR	"Arrival - Airport Congestion"
56	2	Conditionally	AT	"Arrival - ATC Hold"
57	2	Conditionally	ME	"Arrival - Mechanical Problem"
58	2	Conditionally	WX	"Arrival - Weather"
59	2	Conditionally	GO	"Arrival - Gate Occupied"
60	2	Conditionally	GB	"Arrival - Gate Blocked"

#	Length	Inclusion	Content	Element Description
61	2	Conditionally	CR	"Arrival - Ground Crew"
62	0	Conditionally	BEGIN_SUBGROUP	
63	2	Always	OC	"Arrival - Other"
64	1-24	Always	DLY_ARRIVAL_TEXT	"Arrival - Text (1-24 characters)"
65	0	<n/a>	END_SUBGROUP	
66	0	<n/a>	END_GROUP	
67	2	Always	<CR><LF>	"Delimiter"
68	4	Always	DLY_FUEL_ON_BD	"Fuel On Board / 100 (NNNN)"
69	2	Conditionally	<CR><LF>	"Delimiter"
70	1-24	Conditionally	DLY_SUPLMT_TXT_1	"Text line 1 (1-24 characters)"
71	2	Conditionally	<CR><LF>	"Delimiter"
72	1-24	Conditionally	DLY_SUPLMT_TXT_2	"Text line 2 (1-24 characters)"
73	2	Conditionally	<CR><LF>	"Delimiter"
74	1-24	Conditionally	DLY_SUPLMT_TXT_3	"Text line 3 (1-24 characters)"

Note: The presence of each text line and its preceding <CR><LF> delimiter in a downlink message is conditional on the validity of that text line. In other words, if text is not entered on a particular text line then that line and the preceding <CR><LF> delimiter are not present in the downlink message.

Table 8-16 describes the Delay Report downlink encoding message identifiers.

Table 8-16 Delay Report Downlink Encoding Message Identifiers

Delay Type	Message Identifier
Departure	DPD
Takeoff	DTO
Enroute	DNR
Arrival	DAR

Table 8-17 describes the Delay Report downlink encoding departure delay reasons.

Table 8-17 Delay Report Downlink Encoding Departure Delay Reasons

Delay Reason	Code
AIRPORT CONGESTION	AR
MECHANICAL PROBLEM	ME
LATE CREW	CR
LATE AIRCRAFT	AC
LATE PASSENGER	PL
LOADING	LD
ATC HOLD	AT
WEATHER	WX
OTHER	OC

Table 8-18 describes the Delay Report downlink encoding takeoff delay reasons.

Table 8-18 Delay Report Downlink Encoding Takeoff Delay Reasons

Delay Reason Selected	Code in Field 13
AIRPORT CONGESTION	AR
MECHANICAL PROBLEM	ME
ATC HOLD	AT
WEATHER	WX
OTHER	OC

Table 8-19 describes the Delay Report downlink encoding enroute delay reasons.

Table 8-19 Delay Report Downlink Encoding Enroute Delay Reasons

Delay Reason Selected	Code in Field 13
ATC HOLD	AT
ATC REROUTE	TR
WEATHER/HEADWINDS	WW
PERF LIMITS	PF
FLIGHT LEVEL	FL
OTHER	OC

Table 8-20 describes the Delay Report downlink encoding arrival delay reasons.

Table 8-20 Delay Report Downlink Encoding Arrival Delay Reasons

Delay Reason Selected	Code in Field 13
AIRPORT CONGESTION	AR
MECHANICAL PROBLEM	ME
GATE OCCUPIED	GO
GROUND CREW	CR
ATC HOLD	AT
WEATHER	WX
GATE BLOCKED	GB
OTHER	OC

If the flight crew selected the “OTHER” delay reason, then the text that they entered in the associated entry box (up to 24 characters) follows the “OC” reason code.

Sample Messages

```
QU RTNBOCR
.DDLXCXA 221440
M21
```

```
FI BO428/AN N7873BO
DT DDL BFI 221440 M00A
-   DPD02      BOE428/22221440KBFIKBFI
1450AT
0450
```

Note: In sample message above, the message identifier is “DPD” (departure delay), the estimated Off time is “1450” (14:50), the delay reason is “AT” (ATC hold blocked), the fuel on board is “0450” (45,000), and no supplemental text is present.

```
QU RTNBOCR
.DDLXCXA 221956
M21
FI BO24/AN N7873BO
DT DDL BFI 221956 M70A
-   DAR02      BOE24/22221956KBFIKLGB
1258GB
0321
TEXT LINE 1
TEXT LINE 2
TEXT LINE 3
```

Note: In sample message above, the message identifier is “DAR” (arrival delay), the estimated In time is “1258” (12:58), the delay reason is “GB” (gate blocked), and the fuel on board is “0321” (32,100).

8.4 Departure Report

The Departure Report function enables the flight crew to send additional information that is not included in the automatic Out and Off reports, such as the pilot who performed the takeoff, and also thrust and flap settings.

8.4.1 Display Definition

Figure 8-11 shows the Departure Report display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	DEPARTURE REPORT	XXXXXXXXX
OUT TIME: 00:00	OFF TIME: 00:00	
FUEL ON BOARD: - - - -	TAKEOFF FLAPS: - -	
TAKEOFF N1: - - - - %	TAKEOFF TPR: - - -	
TAKEOFF PILOT:		
◆ CAPT	◆ F/O 1	
◆ F/O 2	◆ F/O 3	
SUPPLEMENTAL TEXT:		

SEND	RESET	RETURN
EXIT		
SCRATCHPAD		

Figure 8-11 Departure Report Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “DEPARTURE REPORT”

Main Area

Table 8-21 describes the Departure Report display entry boxes.

Table 8-21 Departure Report Display Entry Boxes

Name	Mand atory	Length	Format	Range	Default	Parameter
OUT TIME	Yes	4	HHMM		OUT_TIME	DEPT_TIME
OFF TIME	Yes	4	HHMM		OFF_TIME	TKO_TIME

Name	Mandatory	Length	Format	Range	Default	Parameter
FUEL ON BOARD	No	1-5	NNN.N	0.0-280.0	OFF_FOB	TKO_FUEL
TAKEOFF FLAPS	No	1-2	NN	0-30		TKO_FLAP
TAKEOFF N1	No	2-5	NNN.N	80.0-150.0		TKO_N1
TAKEOFF TPR	No	2-3	NNN	50-150		TKO_TPR
SUPPLEMENTAL TEXT	No	1-24	XXX...			TKO_SUPLMT_TXT_1
	No	1-24	XXX...			TKO_SUPLMT_TXT_2
	No	1-24	XXX...			TOK_SUPLMT_TXT_3

The “TAKEOFF TPR” label and entry box are only displayed on airplanes with Rolls Royce engines. On airplanes with General Electric (GE) engines, the label and entry box are invisible and the associated field in the downlink encoding contains “*”.

The first five or six entry boxes are followed by a group of ESBs labeled TAKEOFF PILOT to select which flight crew member performed the takeoff:

- CAPT (TKO_PILOT = 1)
- F/O 1 (TKO_PILOT = 2)
- F/O 2 (TKO_PILOT = 3)
- F/O 3 (TKO_PILOT = 4)

Those ESBs are followed by a group of three text entry boxes labeled SUPPLEMENTAL TEXT, each of which allows up to 24 characters of supplemental text entry.

Command Buttons

Table 8-22 describes the Departure Report display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 8-22 Departure Report Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	SEND	Send downlink	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	-	-	-	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

Review Log

When SEND is selected from this display, it becomes available for review in the COMPANY DOWNLINKS category on the REVIEW menu.

8.4.2 Downlink Encoding Definition

When SEND is selected from this display, the following downlink encoding is used:

- ACARS SMI: M24
- ACARS Label: 24
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REPORT_CREW
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 8-23 describes the Departure Report downlink encoding definition.

Table 8-23 Departure Report Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	TKO	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	4	Always	DEPT_TIME	"Out Time (HHMM)"
13	4	Always	TKO_TIME	"Off Time (HHMM)"
14	4	Always	TKO_FUEL	"Off Fuel On Board / 100 (NNNN)"
15	1	Always	TKO_PILOT	"Takeoff Pilot (N)"
16	2	Always	TKO_FLAP	"Takeoff Flaps (NN)"
17	3	Always	TKO_TPR	"Takeoff TPR (NNN)"
18	4	Always	TKO_N1	"Takeoff N1 x 10 (NNNN)"
19	2	Conditionally	<CR><LF>	"Delimiter"

#	Length	Inclusion	Content	Element Description
20	1-24	Conditionally	TKO_SUPLMT_TXT_1	"Text line 1 (1-24 characters)"
21	2	Conditionally	<CR><LF>	"Delimiter"
22	1-24	Conditionally	TKO_SUPLMT_TXT_2	"Text line 2 (1-24 characters)"
23	2	Conditionally	<CR><LF>	"Delimiter"
24	1-24	Conditionally	TKO_SUPLMT_TXT_3	"Text line 3 (1-24 characters)"

Note: The presence of each text line and its preceding <CR><LF> delimiter in a downlink message is conditional on the validity of that text line. In other words, if text is not entered on a particular text line then that line and the preceding <CR><LF> delimiter are not present in the downlink message.

Sample Message

```
QU RTNBOCR
.DDLXCXA 221955
M24
FI BO24/AN N7873BO
DT DDL BFI 221955 M69A
- TKO01 BOE24/22221955KBFIKLGB
1456150606983200990999
TEXT LINE 1
```

Note: In this sample message, the fuel on board is "0698" (69,800), the takeoff pilot is "3" (F/O 2), the takeoff flap setting is "20", the takeoff TPR is "099" (99), and the takeoff N1 is "0999" (99.9%).

8.5 Manual Diversion Report

The Manual Diversion Report function enables the flight crew to notify the aircraft operator of a diversion. The report includes the diversion station (in either three-character IATA format or four-character ICAO format), estimated time of arrival (ETA) at the diversion station, estimated fuel at the diversion station, fuel on board, reason for diverting, and optional supplemental text.

8.5.1 Display Definition

Figure 8-12 shows the Manual Diversion Report display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	DIVERSION REPORT	XXXXXXXXX
DIVERSION STATION:		0000
ETA AT DIVERSION STATION:		--:--
FUEL AT DIVERSION STATION:		----
FUEL ON BOARD:		----
DIVERSION REASON:		
☐ FUEL	☐ MEDICAL	
☐ TECHNICAL	☐ SECURITY	
☐ ENROUTE WEATHER	☐ DESTINATION WEATHER	
☐ OTHER:	-----	
SUPPLEMENTAL TEXT:		

SEND	RESET	RETURN EXIT
SCRATCHPAD		

Figure 8-12 Manual Diversion Report Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “DIVERSION REPORT”

Main Area

Table 8-24 describes the Manual Diversion Report display entry boxes.

Table 8-24 Manual Diversion Report Display Entry Boxes

Name	Mand atory	Length	Format	Range	Default	Parameter
DIVERSION STATION	Yes	3-4	XXXX		FMF_ACTRTEDEST	DIV_DEST

Name	Mandatory	Length	Format	Range	Default	Parameter
ETA AT DIVERSION STATION	No	4	HHMM		FMF_DEST_ETA	DIV_DEST_ETA
FUEL AT DIVERSION STATION	No	1-5	NNN.N	0.0-280.0	FMF_FUEL_AT_DEST	DIV_DEST_FUEL
FUEL ON BOARD	No	1-5	NNN.N	0.0-280.0	FUEL_ON_BOARD	DIV_FUEL_ON_BOARD
SUPPLEMENTAL TEXT	No	1-24	XXX...			DIV_TXT_1
	No	1-24	XXX...			DIV_TXT_2
	No	1-24	XXX...			DIV_TXT_3

The first four entry boxes are followed by a group of ESBs labeled SELECT DIVERSION REASON:

- FUEL (DIV_CODE = 1)
- MEDICAL (DIV_CODE = 3)
- TECHNICAL (DIV_CODE = 2)
- SECURITY (DIV_CODE = 8)
- ENROUTE WEATHER (DIV_CODE = 6)
- DESTINATION WEATHER (DIV_CODE = 7)
- OTHER (DIV_CODE = 5) (with associated entry box for up to 24 characters)

The ESBs are followed by a group of three text entry boxes labeled SUPPLEMENTAL TEXT, each of which allows up to 24 characters of supplemental text entry.

Command Buttons

Table 8-25 describes the Manual Diversion Report display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 8-25 Manual Diversion Report Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	SEND	Send downlink	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	-	-	-	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

Review Log

When SEND is selected from this display, it becomes available for review in the COMPANY DOWNLINKS category on the REVIEW menu.

8.5.2 Downlink Encoding Definition

When SEND is selected from this display, the following downlink encoding is used:

- ACARS SMI: M25
- ACARS Label: 25
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REPORT_CREW
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 8-26 describes the Manual Diversion Report downlink encoding definition.

Table 8-26 Manual Diversion Report Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	DIV	"Message Identifier"
3	2	Always	02	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	4	Always	DIV_DEST	"Divert station (XXXX)"
13	4	Always	DIV_DEST_ETA	"Divert station ETA (HHMM)"
14	4	Always	DIV_DEST_FUEL	"Divert station fuel / 100 (NNNN)"
15	4	Always	DIV_FUEL_ON_BOARD	"Fuel on board / 100 (NNNN)"
16	1	Always	DIV_REASON	"Divert reason (N)"
17	1-24	Conditionally	DIV_REASON_OTHER	"Other reason text (1-24 characters)"
18	2	Conditionally	<CR><LF>	"Delimiter"
19	1-24	Conditionally	DIV_TXT_1	"Text line 1 (1-24 characters)"

#	Length	Inclusion	Content	Element Description
20	2	Conditionally	<CR><LF>	"Delimiter"
21	1-24	Conditionally	DIV_TXT_2	"Text line 2 (1-24 characters)"
22	2	Conditionally	<CR><LF>	"Delimiter"
23	1-24	Conditionally	DIV_TXT_3	"Text line 3 (1-24 characters)"

Note: The presence of each text line and its preceding <CR><LF> delimiter in a downlink message is conditional on the validity of that text line. In other words, if text is not entered on a particular text line then that line and the preceding <CR><LF> delimiter are not present in the downlink message.

Note: The formats of the Manual Diversion Report downlink and the Automatic Diversion Report downlink are identical, although the Automatic Diversion Report downlink never includes other reason text or supplemental text that are optional in the Manual Diversion Report downlink.

Sample Message

```
QU RTNBOCR
.DDLXCXA 221954
M25
FI BO24/AN N7873BO
DT DDL BFI 221954 M68A
- DIV02      BOE24/22221954KBFIKSFO
KSJC2046032103987
```

Note: In this sample message, the diversion station is "KSJC", the ETA at the diversion station is "2046" (20:46), the estimated fuel at the diversion station is "0321" (32,100), the fuel on board is "0398" (39,800), the diversion reason is "7" (destination weather), and no supplemental text is present.

8.6 ETA Report

The ETA Report function enables the flight crew to inform the aircraft operator of the estimated time of arrival (ETA) at the destination airport.

8.6.1 Display Definition

Figure 8-13 shows the ETA Report display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	ETA REPORT	XXXXXXXX
ESTIMATED TIME OF ARRIVAL:		00:00
FUEL ON BOARD:		
SUPPLEMENTAL TEXT:		
.....		
.....		
.....		
SEND	RESET	RETURN
EXIT		
SCRATCHPAD		

Figure 8-13 ETA Report Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “ETA REPORT”

Main Area

Table 8-27 describes the ETA Report display entry boxes.

Table 8-27 ETA Report Display Entry Boxes

Name	Mand atory	Length	Format	Range	Default	Parameter
ESTIMATED TIME OF ARRIVAL	Yes	4	HHMM		FMF_DEST_ETA	ETA_ETA

Name	Mandatory	Length	Format	Range	Default	Parameter
FUEL ON BOARD	No	1-5	NNN.N	0.0-280.0	FUEL_ON_BOARD	ETA_FUEL_ON_BD
SUPPLEMENTAL TEXT	No	1-24	XXX...			ETA_SUPLMT_TXT_1
	No	1-24	XXX...			ETA_SUPLMT_TXT_2
	No	1-24	XXX...			ETA_SUPLMT_TXT_3

The three text entry boxes labeled SUPPLEMENTAL TEXT each allows up to 24 characters of supplemental text entry.

Command Buttons

Table 8-28 describes the ETA REPORT display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 8-28 ETA Report Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	SEND	Send downlink	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	-	-	-	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

Review Log

When SEND is selected from this display, it becomes available for review in the COMPANY DOWNLINKS category on the REVIEW menu.

8.6.2 Downlink Encoding Definition

When SEND is selected from this display, the following downlink encoding is used:

- ACARS SMI: M26
- ACARS Label: 26
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REPORT_CREW
- Delete at System Reset: Yes

- Delete on Buffer Full: Yes

Table 8-29 describes the ETA Report downlink encoding definition.

Table 8-29 ETA Report Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	ETA	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	4	Always	ETA_ETA	"Destination ETA (HHMM)"
13	4	Always	ETA_FUEL_ON_BD	"Fuel on Board / 100 (NNNN)"
14	2	Conditionally	<CR><LF>	"Delimiter"
15	1-24	Conditionally	ETA_SUPLMT_TXT_1	"Text line 1 (1-24 characters)"
16	2	Conditionally	<CR><LF>	"Delimiter"
17	1-24	Conditionally	ETA_SUPLMT_TXT_2	"Text line 2 (1-24 characters)"
18	2	Conditionally	<CR><LF>	"Delimiter"
19	1-24	Conditionally	ETA_SUPLMT_TXT_3	"Text line 3 (1-24 characters)"

Note: The presence of each text line and its preceding <CR><LF> delimiter in a downlink message is conditional on the validity of that text line. In other words, if text is not entered on a particular text line then that line and the preceding <CR><LF> delimiter are not present in the downlink message.

Sample Message

```

QU RTNBOCR
.DDLXCXA 221950
M26
FI BO24/AN N7873BO
DT DDL BFI 221950 M65A
- ETA01 BOE24/22221950KBFIKLGB
21480328
TEXT LINE 1

```

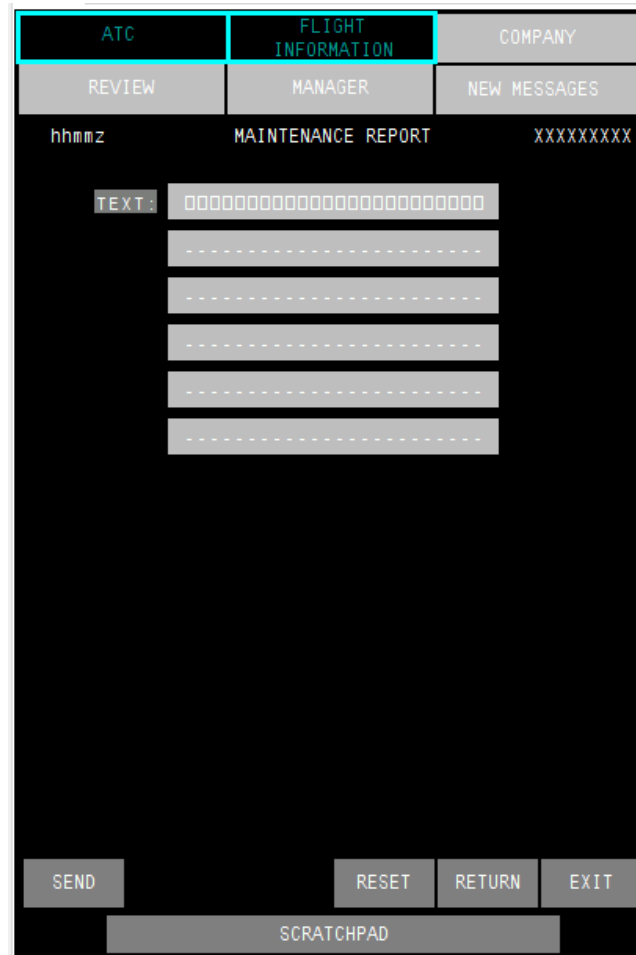
Note: In this sample message, the ETA is "2148" (21:48) and the fuel on board is "0328" (32,800).

8.7 Maintenance Report

The Maintenance Report function enables the flight crew to enter text regarding maintenance problems for transmission to the ground.

8.7.1 Display Definition

Figure 8-14 shows the Maintenance Report display definition.



The figure shows a graphical user interface for a Maintenance Report. At the top, there is a header bar with three sections: 'ATC' (highlighted with a red border), 'FLIGHT INFORMATION' (highlighted with a red border), and 'COMPANY'. Below this header, there are three buttons: 'REVIEW', 'MANAGER', and 'NEW MESSAGES'. The main display area has a black background. At the top of this area, it shows 'hhmmz' on the left, 'MAINTENANCE REPORT' in the center, and 'XXXXXXXX' on the right. Below this, there is a 'TEXT:' label followed by a series of input fields. The first field is a long horizontal bar, followed by five shorter horizontal bars, each containing a dashed line. At the bottom of the display, there are four buttons: 'SEND', 'RESET', 'RETURN', and 'EXIT'. Below these buttons is a wide 'SCRATCHPAD' area.

Figure 8-14 Maintenance Report Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “MAINTENANCE REPORT”

Main Area

The main area of this display definition consists of a group of six text entry boxes labeled TEXT, each of which allows up to 24 characters of text entry.

Table 8-30 describes the Maintenance Report display entry boxes.

Table 8-30 Maintenance Report Display Entry Boxes

Name	Mandatory	Length	Format	Range	Default	Parameter
TEXT	Yes	1-24	XXX...			MAINT_TEXT_1
	No	1-24	XXX...			MAINT_TEXT_2
	No	1-24	XXX...			MAINT_TEXT_3
	No	1-24	XXX...			MAINT_TEXT_4
	No	1-24	XXX...			MAINT_TEXT_5
	No	1-24	XXX...			MAINT_TEXT_6

Command Buttons

Table 8-31 describes the Maintenance Report display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 8-31 Maintenance Report Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	SEND	Send downlink	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	-	-	-	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

Review Log

When SEND is selected from this display, it becomes available for review in the COMPANY DOWNLINKS category on the REVIEW menu.

8.7.2 Downlink Encoding Definition

When SEND is selected from this display, the following downlink encoding is used:

- ACARS SMI: M2A

- ACARS Label: 2A
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REPORT_CREW
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 8-32 describes the Maintenance Report downlink encoding definition.

Table 8-32 Maintenance Report Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	MNT	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	1-24	Always	MAINT_TEXT_1	"Text line 1 (1-24 characters)"
13	2	Conditionally	<CR><LF>	"Delimiter"
14	1-24	Conditionally	MAINT_TEXT_2	"Text line 2 (1-24 characters)"
15	2	Conditionally	<CR><LF>	"Delimiter"
16	1-24	Conditionally	MAINT_TEXT_3	"Text line 3 (1-24 characters)"
17	2	Conditionally	<CR><LF>	"Delimiter"
18	1-24	Conditionally	MAINT_TEXT_4	"Text line 4 (1-24 characters)"
19	2	Conditionally	<CR><LF>	"Delimiter"
20	1-24	Conditionally	MAINT_TEXT_5	"Text line 5 (1-24 characters)"
21	2	Conditionally	<CR><LF>	"Delimiter"
22	1-24	Conditionally	MAINT_TEXT_6	"Text line 6 (1-24 characters)"

Note: The presence of each text line and its preceding <CR><LF> delimiter in a downlink message is conditional on the validity of that text line. In other words, if text is not entered on a particular text line then that line and the preceding <CR><LF> delimiter are not present in the downlink message.

Sample Message

```
QU RTNBOCR
.DDLXCXA 221947
M2A
FI BO24/AN N7873BO
DT DDL BFI 221947 M63A
- MNT01      BOE24/22221947KBFIKLGB
TEXT LINE 1
TEXT LINE 2
TEXT LINE 3
```

8.8 Miscellaneous Code Report

The Miscellaneous Code Report function enables the aircraft operator to support new or otherwise unsupported functions. When a need arises for a new function, an alphanumeric code is assigned to the function and flight crews are advised of the purpose and data entry requirements for the new application. To send the message for the function, the flight crew enters the assigned miscellaneous code and the required data in the text fields.

8.8.1 Display Definition

Figure 8-15 shows the Miscellaneous Code Report display definition.

The image shows a graphical user interface for a 'Miscellaneous Code Report'. At the top, there are three tabs: 'ATC' (highlighted in red), 'FLIGHT INFORMATION' (highlighted in red), and 'COMPANY'. Below these are three buttons: 'REVIEW', 'MANAGER', and 'NEW MESSAGES'. The main display area has a black background. At the top of this area, it shows 'hhmmz' on the left, 'MISCELLANEOUS CODE REPORT' in the center, and 'XXXXXXXXX' on the right. Below this, there is a label 'MISCELLANEOUS CODE:' followed by a two-digit input field showing '00'. Underneath is a label 'SUPPLEMENTAL TEXT:' followed by six horizontal input lines, each with a dashed line pattern. At the bottom of the screen, there are four buttons: 'SEND', 'RESET', 'RETURN', and 'EXIT'. Below these buttons is a wide 'SCRATCHPAD' area.

Figure 8-15 Miscellaneous Code Report Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “MISCELLANEOUS CODE REPORT”

Main Area

Table 8-33 describes the Miscellaneous Code Report display entry boxes.

Table 8-33 Miscellaneous Code Report Display Entry Boxes

Name	Mandatory	Length	Format	Range	Default	Parameter
MISCELLANEOUS CODE	Yes	1-2	XX			MISC_CODE

Name	Mandatory	Length	Format	Range	Default	Parameter
SUPPLEMENTAL TEXT	No	1-24	XXX...			MISC_TEXT_1
	No	1-24	XXX...			MISC_TEXT_2
	No	1-24	XXX...			MISC_TEXT_3
	No	1-24	XXX...			MISC_TEXT_4
	No	1-24	XXX...			MISC_TEXT_5
	No	1-24	XXX...			MISC_TEXT_6

The six text entry boxes labeled SUPPLEMENTAL TEXT each allows up to 24 characters of supplemental text entry.

Command Buttons

Table 8-34 describes the Miscellaneous Code Report display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 8-34 Miscellaneous Code Report Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	SEND	Send downlink	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	-	-	-	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

Review Log

When SEND is selected from this display, it becomes available for review in the COMPANY DOWNLINKS category on the REVIEW menu.

8.8.2 Downlink Encoding Definition

When SEND is selected from this display, the following downlink encoding is used:

- ACARS SMI: M29
- ACARS Label: 29
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REPORT_CREW

- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 8-35 describes the Miscellaneous Code Report downlink encoding definition.

Table 8-35 Miscellaneous Code Report Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	MSC	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	2	Always	MISC_CODE	"Miscellaneous Code (XX)"
13	2	Conditionally	<CR><LF>	"Delimiter"
14	1-24	Conditionally	MISC_TEXT_1	"Text line 1 (1-24 characters)"
15	2	Conditionally	<CR><LF>	"Delimiter"
16	1-24	Conditionally	MISC_TEXT_2	"Text line 2 (1-24 characters)"
17	2	Conditionally	<CR><LF>	"Delimiter"
18	1-24	Conditionally	MISC_TEXT_3	"Text line 3 (1-24 characters)"
19	2	Conditionally	<CR><LF>	"Delimiter"
20	1-24	Conditionally	MISC_TEXT_4	"Text line 4 (1-24 characters)"
21	2	Conditionally	<CR><LF>	"Delimiter"
22	1-24	Conditionally	MISC_TEXT_5	"Text line 5 (1-24 characters)"
23	2	Conditionally	<CR><LF>	"Delimiter"
24	1-24	Conditionally	MISC_TEXT_6	"Text line 6 (1-24 characters)"

Note: The presence of each text line and its preceding <CR><LF> delimiter in a downlink message is conditional on the validity of that text line. In other words, if text is not entered on a particular text line then that line and the preceding <CR><LF> delimiter are not present in the downlink message.

Sample Message

```
QU RTNBOCR
.QXSXMXS 221853
M29
FI BO0024/AN N7873BO
```

```
DT QXS BFI7 221853 M45A
- MSC01      BOE24/22221853KBFIKLGB
A1
TEXT LINE 1
TEXT LINE 2
TEXT LINE 3
```

Note: In this sample message, the miscellaneous code is “A1”.

8.9 Situation Report

The Situation Report function enables the flight crew to notify the aircraft operator of a hijacking or other emergency situation. The flight crew enters text into the text fields in a predefined format or in free-text form and then CMF and the DSPs give the message high priority.

This function also enables the flight crew to manually set the Automatic Emergency Report function to the Armed, On, or Off state. When the flight crew sets that function to the On state, CMF immediately sends a message containing key airplane information and periodically sends that message thereafter.

Note: The logic units that Section 9.4.1 and Section 9.4.2 describe have the ability to automatically initiate automatic emergency reports.

8.9.1 Display Definition

Figure 8-16 shows the Situation Report display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	SITUATION REPORT	XXXXXXXX
TEXT:		
AUTOMATIC EMERGENCY REPORTS ON AUTOMATIC EMERGENCY REPORTS ARMED AUTOMATIC EMERGENCY REPORTS OFF		
SEND	RESET	RETURN
EXIT		
SCRATCHPAD		

Figure 8-16 Situation Report Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “SITUATION REPORT”

Main Area

Table 8-36 describes the Situation Report display entry boxes.

Table 8-36 Situation Report Display Entry Boxes

Name	Mandatory	Length	Format	Range	Default	Parameter
TEXT	Yes	1-24	XXX...			EMER_SITU_TXT_1
	No	1-24	XXX...			EMER_SITU_TXT_2

Name	Mandatory	Length	Format	Range	Default	Parameter
	No	1-24	XXX...			EMER_SITU_TXT_3

The three text entry boxes labeled TEXT, the first of which is mandatory, each allows up to 24 characters of supplemental text entry.

Additionally, this display contains a group of ESBs to control the state of the Automatic Emergency Report function:

- AUTOMATIC EMERGENCY REPORTS ON (EMERGENCY_SELECTION = 1)
- AUTOMATIC EMERGENCY REPORTS ARMED (EMERGENCY_SELECTION = 2)
- AUTOMATIC EMERGENCY REPORTS OFF (EMERGENCY_SELECTION = 3)

Selection of AUTOMATIC EMERGENCY REPORTS ON causes CMF to immediately begin sending automatic emergency reports, while selection of SEND causes CMF to send the situation report. Section 9.4.3 describes the Automatic Emergency Report downlink encoding definition.

Command Buttons

Table 8-37 describes the Situation Report display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 8-37 Situation Report Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	SEND	Send downlink	When all mandatory data is entered or selected	When all mandatory data is entered or selected but the downlink queue is full or datalink is NO COMM
2	-	-	-	-
3	-	-	-	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

Review Log

When SEND is selected from this display, it becomes available for review in the COMPANY DOWNLINKS category on the REVIEW menu.

8.9.2 Downlink Encoding Definition

When SEND is selected from this display, the following downlink encoding is used:

- ACARS SMI: HJK
- ACARS Label: 00
- Allowable Subnetworks: All
- Downlink Priority: Emergency
- Downlink Buffer: REPORT_CREW
- Delete at System Reset: No
- Delete on Buffer Full: No

Table 8-38 describes the Situation Report downlink encoding definition.

Table 8-38 Situation Report Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	SIT	"Message Identifier"
3	2	Always	02	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	1-24	Always	EMER_SITU_TXT_1	"Text line 1 (1-24 characters)"
13	2	Conditionally	<CR><LF>	"Delimiter"
14	1-24	Conditionally	EMER_SITU_TXT_2	"Text line 2 (1-24 characters)"
15	2	Conditionally	<CR><LF>	"Delimiter"
16	1-24	Conditionally	EMER_SITU_TXT_3	"Text line 3 (1-24 characters)"

Note: The presence of each text line and its preceding <CR><LF> delimiter in a downlink message is conditional on the validity of that text line. In other words, if text is not entered on a particular text line then that line and the preceding <CR><LF> delimiter are not present in the downlink message.

Sample Message

```
QU RTNBOCR
.DDLXCXA 221938
HJK
FI BO24/AN N7873BO
DT DDL BFI 221938 M59A
```

- SIT02 BOE24/22221938KBFIKLGB
TEXT LINE 1

9 Automatic Reports

This section describes the automatic report functions that the AMI implements.

9.1 AHM Flight Phase Report

This section describes the Airplane Health Management (AHM) Flight Phase Report function that the AMI implements. This function should not be modified in order to ensure that Boeing AHM receives necessary flight phase reports.

Note: There is no display definition for this function, although the SENSOR STATUS display described in Section 12.6.1 shows the flight phase and the CMF PARAMETERS display described in Section 12.6.2.1.1 allows AHM flight phase reports to be enabled or disabled.

9.1.1 Logic Unit

The AMI implements a logic unit to detect flight phase changes (including OOOI, diversion, and flight initialization events) and generate the AHM Flight Phase Report downlink encoding.

```
-- -----
--
-- Revision history
--
-- 17-OCT-2010  Original version
--
-- VERSION C000 --- NEW VERSIONING --- NO CHANGE IN LOGIC
--
-- Version D001  M. Matyas  24-OCT-2017
--   Updated "AHS" references to "AHM".
--   Updated references to AHS_IDENT parameter to AHM_FLIGHT_PHASE_IDENT.
--   Updated references to FLIGHT_PHASE parameter to AHM_FLIGHT_PHASE_ENABLE.
--   No change to logic.
--
-- Version F001  M. Matyas  29-SEP-2020
--   Added logic to send AHM flight phase reports at OOOI events, diversion
--   event, and flight initialization event.
--
-- -----
--
-- When AHM flight phase reports are enabled, CMF creates and sends AHM flight
-- phase reports at the following flight phase transition points and also at
-- at OOOI and diversion events:
--
```

```

-- Report Identifier: POES      Transition from PO (1) to ES (2)
-- Report Identifier: ESTA      Transition from ES (2) to TA (3)
-- Report Identifier: TAES      Transition from TA (3) to ES (2)
-- Report Identifier: TATO      Transition from TA (3) to TO (4)
-- Report Identifier: TOIC      Transition from TO (4) to IC (5)
-- Report Identifier: ICCL      Transition from IC (5) to CL (6)
-- Report Identifier: CLER      Transition from CL (6) to ER (7)
-- Report Identifier: ERCL      Transition from ER (7) to CL (6)
-- Report Identifier: ERDC      Transition from ER (7) to DC (8)
-- Report Identifier: DCAP      Transition from DC (8) to AP (9)
-- Report Identifier: APGA      Transition from AP (9) to GA (10)
-- Report Identifier: APFL      Transition from AP (9) to FL (11)
-- Report Identifier: FLRO      Transition from FL (11) to RO (12)
-- Report Identifier: ECER      Transition from DC (8) to ER (7)
-- Report Identifier: ROTI      Transition from RO (12) to TI (13)
-- Report Identifier: TISD      Transition from TI (13) to SD (14)
-- Report Identifier: SDPO      Transition from SD (14) to PO (1)
-- Report identifier: OUT        Out event
-- Report identifier: OFF        Off event
-- Report identifier: ON         On event
-- Report identifier: IN         In event
-- Report identifier: RTG        Return to gate event
-- Report identifier: DVRT       Diversion event
-- Report identifier: INIT       Flight initialization event
--
-- Note: The number next to the flight phase abbreviation is the integer
-- representation of the flight phase.
--
-- -----

```

```

set_default (CMCF_FLIGHT_PHASE_CURRENT);
set_default (AHM_FLIGHT_PHASE_IDENT);

if AHM_FLIGHT_PHASE_ENABLE = TRUE then

    if valid (CMCF_FLIGHT_PHASE) then
        CMCF_FLIGHT_PHASE_CURRENT := CMCF_FLIGHT_PHASE;
    end_if;

    if (CMCF_FLIGHT_PHASE_CURRENT changed from (1) to (2)) then
        AHM_FLIGHT_PHASE_IDENT := "POES";
        send(BCG_AHM_FLIGHT_PHASE);

    else_if (CMCF_FLIGHT_PHASE_CURRENT changed from (2) to (3)) then
        AHM_FLIGHT_PHASE_IDENT := "ESTA";
        send(BCG_AHM_FLIGHT_PHASE);
    end_if;
end_if;

```

```
else_if (CMCF_FLIGHT_PHASE_CURRENT changed from (3) to (2)) then
    AHM_FLIGHT_PHASE_IDENT := "TAES";
    send(BCG_AHM_FLIGHT_PHASE);

else_if (CMCF_FLIGHT_PHASE_CURRENT changed from (3) to (4)) then
    AHM_FLIGHT_PHASE_IDENT := "TATO";
    send(BCG_AHM_FLIGHT_PHASE);

else_if (CMCF_FLIGHT_PHASE_CURRENT changed from (4) to (5)) then
    AHM_FLIGHT_PHASE_IDENT := "TOIC";
    send(BCG_AHM_FLIGHT_PHASE);

else_if (CMCF_FLIGHT_PHASE_CURRENT changed from (5) to (6)) then
    AHM_FLIGHT_PHASE_IDENT := "ICCL";
    send(BCG_AHM_FLIGHT_PHASE);

else_if (CMCF_FLIGHT_PHASE_CURRENT changed from (6) to (7)) then
    AHM_FLIGHT_PHASE_IDENT := "CLER";
    send(BCG_AHM_FLIGHT_PHASE);

else_if (CMCF_FLIGHT_PHASE_CURRENT changed from (7) to (6)) then
    AHM_FLIGHT_PHASE_IDENT := "ERCL";
    send(BCG_AHM_FLIGHT_PHASE);

else_if (CMCF_FLIGHT_PHASE_CURRENT changed from (7) to (8)) then
    AHM_FLIGHT_PHASE_IDENT := "ERDC";
    send(BCG_AHM_FLIGHT_PHASE);

else_if (CMCF_FLIGHT_PHASE_CURRENT changed from (8) to (9)) then
    AHM_FLIGHT_PHASE_IDENT := "DCAP";
    send(BCG_AHM_FLIGHT_PHASE);

else_if (CMCF_FLIGHT_PHASE_CURRENT changed from (9) to (10)) then
    AHM_FLIGHT_PHASE_IDENT := "APGA";
    send(BCG_AHM_FLIGHT_PHASE);

else_if (CMCF_FLIGHT_PHASE_CURRENT changed from (9) to (11)) then
    AHM_FLIGHT_PHASE_IDENT := "APFL";
    send(BCG_AHM_FLIGHT_PHASE);

else_if (CMCF_FLIGHT_PHASE_CURRENT changed from (11) to (12)) then
    AHM_FLIGHT_PHASE_IDENT := "FLRO";
    send(BCG_AHM_FLIGHT_PHASE);

else_if (CMCF_FLIGHT_PHASE_CURRENT changed from (8) to (7)) then
```

```

    AHM_FLIGHT_PHASE_IDENT := "ECER";
    send (BCG_AHM_FLIGHT_PHASE);

else_if (CMCF_FLIGHT_PHASE_CURRENT changed from (12) to (13)) then
    AHM_FLIGHT_PHASE_IDENT := "ROTI";
    send (BCG_AHM_FLIGHT_PHASE);

else_if (CMCF_FLIGHT_PHASE_CURRENT changed from (13) to (14)) then
    AHM_FLIGHT_PHASE_IDENT := "TISD";
    send (BCG_AHM_FLIGHT_PHASE);

else_if (CMCF_FLIGHT_PHASE_CURRENT changed from (14) to (1)) then
    AHM_FLIGHT_PHASE_IDENT := "SDPO";
    send (BCG_AHM_FLIGHT_PHASE);

else_if OOOI_STATE changed to (OUT) then
    AHM_FLIGHT_PHASE_IDENT := "OUT";
    send (BCG_AHM_FLIGHT_PHASE);

else_if OOOI_STATE changed to (OFF) then
    AHM_FLIGHT_PHASE_IDENT := "OFF";
    send (BCG_AHM_FLIGHT_PHASE);

else_if OOOI_STATE changed to (ON) then
    AHM_FLIGHT_PHASE_IDENT := "ON";
    send (BCG_AHM_FLIGHT_PHASE);

else_if OOOI_STATE changed to (IN) then
    if RETURN_TO_GATE then
        AHM_FLIGHT_PHASE_IDENT := "RTG";
        send (BCG_AHM_FLIGHT_PHASE);
    else
        AHM_FLIGHT_PHASE_IDENT := "IN";
        send (BCG_AHM_FLIGHT_PHASE);
    end_if;

else_if AIRCRAFT_IN_AIR
and valid (FMF_ACTRTEDEST)
and (AHM_DVRT_PREV_DEST != FMF_ACTRTEDEST) then
    AHM_DVRT_FLAG := TRUE;

else_if not AHM_INIT_FLAG then
    if valid (IRU_RDC_PITCH_ANGLE, FMF_ACTRTEORIGIN, FMF_ACTRTEDEST,
    FMF_ORIGIN_RWY_ID, FMF_CG) then
        AHM_INIT_FLAG := TRUE;
        AHM_FLIGHT_PHASE_IDENT := "INIT";

```

```

        send (BCG_AHM_FLIGHT_PHASE);
    end_if;

end_if;

if for 5 seconds (AHM_DVRT_FLAG) then    -- Give FMF time to update params
    AHM_FLIGHT_PHASE_IDENT := "DVRT";
    send (BCG_AHM_FLIGHT_PHASE);
    set_default (AHM_DVRT_FLAG);
end_if;

end_if;

AHM_DVRT_PREV_DEST := FMF_ACTRTEDEST;

-----
--
-- END

```

9.1.2 Downlink Encoding Definition

If the AMI detects a flight phase transition (e.g., a flight phase transition from Power On to Engine Start) and AHM flight phase reports are enabled, then it sends the following downlink encoding:

- ACARS SMI: M49
- ACARS Label: 49
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REPORT_AUTO
- Delete at System Reset: No
- Delete on Buffer Full: Yes

Table 9-1 describes the AHM Flight Phase Report downlink encoding definition.

Table 9-1 AHM Flight Phase Report Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	2	Always	01	"Message Version"
3	4	Always	AHM_FLIGHT_PHASE_IDENT	"Message Identifier (YYZZ)"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"

#	Length	Inclusion	Content	Element Description
5	1	Always	/	"Delimiter"
6	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
7	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
8	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
9	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
10	2	Always	<CR><LF>	"Delimiter"
11	6	Always	NAV_ALTITUDE	"Altitude (±NNNNN)"
12	5	Always	ADRF_CAS_VOTED	"CAS (NNN.N)"
13	6	Always	ADRF_TAT_VOTED	"TAT (±NNN.N)"
14	4	Always	FUEL_ON_BOARD	"Fuel on board / 100 (NNNN)"
15	4	Always	FMF_FUEL_AT_DEST	"Fuel at destination / 100 (NNNN)"
16	4	Always	FMF_DEST_ETA	"ETA at destination (HHMM)"

"YY" is the flight phase from which the transition is occurring and "ZZ" is the flight phase to which the transition is occurring. The following flight phase message identifiers are possible:

- "POES" (transition from Power On to Engine Start)
- "ESTA" (transition from Engine Start to Taxi Out)
- "TAES" (transition from Taxi Out to Engine Start)
- "TATO" (transition from Taxi Out to Takeoff)
- "TOIC" (transition from Takeoff to Initial Climb)
- "ICCL" (transition from Initial Climb to Climb)
- "CLER" (transition from Climb to Enroute)
- "ERCL" (transition from Enroute to Climb)
- "ERDC" (transition from Enroute to Descent)
- "DCAP" (transition from Descent to Approach)
- "APGA" (transition from Approach to Go Around)
- "APFL" (transition from Approach to Flare)
- "FLRO" (transition from Flare to Rollout)
- "DCER" (transition from Descent to Enroute)
- "ROTI" (transition from Rollout to Taxi In)
- "TISD" (transition from Taxi In to Engine Shutdown)
- "SDPO" (transition from Engine Shutdown to Power On)

Additional flight phase message identifiers (which are always four characters, potentially including space characters) include the following:

- “OUT ” (Out event)
- “OFF ” (Off event)
- “ON ” (On event)
- “IN ” (In event)
- “RTG ” (Return to Gate event)
- “DVRT” (diversion event, which occurs if the FMF destination airport changes when the airplane is airborne)
- “INIT” (flight initialization event, which occurs when the IRU pitch angle, FMF origin airport, FMF destination airport, FMF origin airport runway, and FMF center-of-gravity are all valid for the first time during a flight)

Sample Message

```
QU RTNBOCR
.DDLXCXA 070034
M49
FI BO1234/AN N7873BO
DT DDL BFI 070034 M16A
- 01ERDC BOE1234/070034KBFIKBFI
+10999293.8+ 7.7 607 5940043
```

Note: In this sample message, the message identifier is “ERDC” (transition from the Enroute flight phase to the Descent flight phase), the altitude is “+10999” (10,999 feet), the CAS is 293.8 (293.8 KCAS), the TAT is “+ 7.7” (7.7°C) , the fuel on board is “ 607” (60,700), the predicted fuel at the destination is “ 594” (59,400), and the predicted ETA at the destination is “0043” (00:43).

9.1.3 Uplink Decoding Definition

An aircraft operator may send the AHM Flight Phase Report Configuration Uplink to enable or disable AHM flight phase reports. CMF stores the value contained in this uplink in non-volatile memory (NVM) in order to retain it across power cycles.

The decoding for the AHM Flight Phase Report Configuration Uplink is defined as follows:

- ACARS SMI: CMD
- ACARS Label: RA
- Message Identifier: ~1AHM

- Alert Level: None
- Optional Element Delimiter: “,” (comma)

Table 9-2 describes the AHM Flight Phase Report Configuration Uplink decoding definition.

Table 9-2 AHM Flight Phase Report Configuration Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	2	Yes	<none>	"Priority code (XX)"
2	7	Yes	<none>	"Ground address (XXXXXXX)"
3	5	Yes	<none>	"Message identifier (~1AHM)"
4	2	Yes	<none>	"Delimiter (<CR><LF>)"
5	1	Yes	AHM_FLIGHT_PHASE_ENABLE	"Enable (1) or disable (0)"

Sample Message

```
QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 014A
- QURTNBOCR~1AHM
1
```

Note: This sample message would enable AHM flight phase reports.

9.2 AoIP Link Advisory

This section describes the ACARS-over-IP (AoIP) Link Advisory function that the AMI implements. This function is intended to emulate the media advisory function for traditional ACARS media (which is implemented in CMF operational software) by sending an AoIP link advisory when the AMI detects an AoIP link status change. This function also includes an uplink decoding definition to enable or disable the advisories.

Note: There is no display definition for this function, although the CMF PARAMETERS display described in Section 12.6.2.1.1 allows AoIP advisories to be enabled or disabled.

Note: If AoIP is not enabled – either because the Option Selection Software (OSS) does not enable the CMF AoIP software option or the AMI does not define AoIP as a subnetwork preference – then the AMI does not generate AoIP link advisories because CMF sets the AoIP link status to Failed.

9.2.1 Logic Unit

The AMI implements a logic unit to generate the AoIP Link Advisory downlink encoding.

```

-- -----
--
-- Version D001      M. Matyas   24-OCT-2017
--   Developed logic to detect an AoIP link status change and send a
--   corresponding advisory downlink.
--
-- Version E001      M. Matyas   05-FEB-2019
--   Enhanced logic to consider whether AoIP link advisories are enabled.
--
-- -----

if AOIP_LINK_ADV_ENABLE and AOIP_LINK_STATUS changed then
  AOIP_LINK_ADV_STATUS  := AOIP_LINK_STATUS;
  AOIP_LINK_ADV_TIME    := CURRENT_TIME;
  send (BCG_AOIP_LINK_ADV);
end_if;

-- -----
-- END
-- -----

```

9.2.2 Downlink Encoding Definition

The encoding for the AoIP Link Advisory downlink is defined as follows:

- ACARS SMI: M19
- ACARS Label: 19
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REPORT_AUTO
- Delete at System Reset: Yes
- Delete on Buffer Full: Yes

Table 9-3 describes the AoIP Link Advisory downlink encoding definition.

Table 9-3 AoIP Link Advisory Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	AOI	"Message Identifier"
3	2	Always	02	"Message Version"

#	Length	Inclusion	Content	Element Description
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	1	Conditionally	A	"Available flag"
13	1	Conditionally	N	"Not Available flag"
14	1	Conditionally	F	"Failed flag"
15	1	Conditionally	*	"Invalid flag"
16	6	Always	AOIP_LINK_ADV_TIME	"Timestamp (HHMMSS)"

Sample Message

```

QU RTNBOCR
.DDLXCXA 061508
M19
FI BO1234/AN N7873BO
DT DDL BFI 061508 M01A
- AOI02 BOE1234/06061508KBFIKBFI
N150802

```

9.2.3 Uplink Decoding Definition

An aircraft operator may send the AoIP Link Advisory Configuration Uplink to enable or disable AoIP link advisory downlinks. CMF stores the value contained in this uplink in NVM in order to retain it across power cycles.

The decoding for the AoIP Link Advisory Configuration Uplink is defined as follows:

- ACARS SMI: CMD
- ACARS Label: RA
- Message Identifier: ~1AOI
- Alert Level: None
- Optional Element Delimiter: “,” (comma)

Table 9-4 describes the AoIP Link Advisory Configuration Uplink decoding definition.

Table 9-4 AoIP Link Advisory Configuration Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	2	Yes	<none>	"Priority code (XX)"
2	7	Yes	<none>	"Ground address (XXXXXXX)"
3	5	Yes	<none>	"Message identifier (~1AOI)"
4	2	Yes	<none>	"Delimiter (<CR><LF>)"
5	1	Yes	AOIP_LINK_ADV_ENABLE	"Enable (1) or disable (0)"

Sample Message

```
QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 015A
- QURTNBOCR~1AOI
0
```

Note: This sample message would disable AoIP link advisories.

9.3 Automatic Diversion Report

This section describes the Automatic Diversion Report function that the AML implements. This function automatically generates a diversion report if the FMF destination airport changes when the airplane is airborne.

9.3.1 Logic Unit

The AMI implements a logic unit to generate the Automatic Diversion Report downlink encoding.

```
-- -----  
--  
-- Version F001 M. Matyas 29-SEP-2020  
-- Logic to detect diversion event and send automatic diversion report.  
--  
-- -----  
  
if AIRCRAFT_IN_AIR  
and valid (FMF_ACTRTEDEST)  
and (DIV_PREV_DEST != FMF_ACTRTEDEST) then  
    DIV_AUTO_FLAG := TRUE;  
end_if;  
  
if for 5 seconds (DIV_AUTO_FLAG) then    -- Give FMF time to update params  
    set_default (DIV_DEST);  
    set_default (DIV_DEST_ETA);  
    set_default (DIV_DEST_FUEL);  
    set_default (DIV_FUEL_ON_BOARD);  
    send (BCG_DIVERT_AUTO);  
    set_default (DIV_AUTO_FLAG);  
end_if;  
  
DIV_PREV_DEST := FMF_ACTRTEDEST;  
  
-- -----  
--  
-- END
```

9.3.2 Downlink Encoding Definition

The encoding for the Automatic Diversion Report downlink is defined as follows:

- ACARS SMI: M25
- ACARS Label: 25
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REPORT_AUTO
- Delete at System Reset: Yes

- Delete on Buffer Full: Yes

Table 9-5 describes the Automatic Diversion Report downlink encoding definition.

Table 9-5 Automatic Diversion Report Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	DIV	"Message Identifier"
3	2	Always	02	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	4	Always	DIV_DEST	"Divert station (XXXX)"
13	4	Always	DIV_DEST_ETA	"Divert station ETA (HHMM)"
14	4	Always	DIV_DEST_FUEL	"Divert station fuel / 100 (NNNN)"
15	4	Always	DIV_FUEL_ON_BOARD	"Fuel on board / 100 (NNNN)"
16	1	Always	A	"Divert reason (A = auto)"

Note: The formats of the Manual Diversion Report downlink and the Automatic Diversion Report downlink are identical, although the Automatic Diversion Report downlink never includes other reason text or supplemental text that are optional in the Manual Diversion Report downlink.

Sample Message

```
QU RTNBOCR
.DDLXCXA 221541
M25
FI BO24/AN N7873BO
DT DDL BFI 221541 M10A
- DIV02 BOE24/22221541KBFIPHOG
PHNL143203210398A
```

Note: In this sample message, the diversion station is "PHNL", the ETA at the diversion station is "1432" (14:32), the estimated fuel at the diversion station is "0321" (32,100), the fuel on board is "0398" (39,800), and the diversion reason is "A" (automatic).

9.4 Automatic Emergency Report

This section describes the Automatic Emergency Report function that the AMI implements. This function includes the uplink decoding definition described in Section 11.1.1 that the aircraft operator may send to set the function to the Armed, Off, or On state as well as the uplink decoding definition described in Section 9.4.4 that the aircraft operator may send to set the thresholds for detecting an emergency condition.

Note: The flight crew may also manually set the Automatic Emergency Report function to the Armed, Off, or On state on the Situation Report display that Section 8.9.1 describes.

With the default settings, the AMI enables the automatic emergency report function (i.e., potentially sends automatic emergency reports) under the following conditions:

- radio altimeter height is greater than 2,000 feet; and
- FMF origin airport, destination airport, and flight number are valid; and
- CAS is greater than 100 knots; and
- pressure altitude is greater than 10,000 feet.

With the default settings, the AMI disables the automatic emergency function (i.e., does not potentially send automatic emergency reports) under the following conditions:

- radio altimeter height is less than 500 feet; or
- FMF destination airport is invalid, FMF flight number is invalid, and airplane is on the ground.

With the default settings, the AMI begins sending automatic emergency reports under the following conditions:

- Altitude rate (vertical speed)
 - Climb rate: greater than 7,500 feet per minute
 - Descent rate: greater than 7,500 feet per minute
- Speed
 - CAS: less than 100 knots CAS
 - V_{MO} : greater than 390 knots CAS
 - M_{MO} : greater than 0.95 Mach
- Attitude
 - Pitch angle: greater than 20 degrees (up or down)
 - Pitch rate: greater than 10 degrees per second
 - Yaw rate: greater than 10 degrees per second
 - Vertical acceleration: greater than 2.5g

-
- Engines
 - N1: both engines less than 5% N1
 - TPR: both engines than 5% TPR
 - Fuel control switches: both set to CUTOFF
 - Fuel: less than 5,500 pounds

With the default settings, if the AMI begins sending automatic emergency reports then it sends a report every 60 seconds for up to 60 minutes.

9.4.1 Detection Logic Unit

The AMI implements a logic unit to automatically detect an emergency condition.

```

-----
-- This Logic Unit determines if an emergency situation exists.  --
-- This provides automatic determination of an emergency state  --
-- based on whether certain state parameters exceed a          --
-- predetermined limit.                                         --
--                                                              --
-- G.Sandell  7-02-2009  Initial development                    --
--                                                              --
-- G.Sandell  4-10-2013  Changed logic for ON/ARM/OFF           --
--                      Added check on climb rate              --
--                      Changed enabling logic                  --
--                                                              --
-- G.Sandell  5-1-2013   Allowed for LRRA invalid above 7500   --
--                                                              --
-- S.Meisch   2014-01-22 REVERSION TO C000 - NO CHANGE         --
--                                                              --
-- Version C001  G.Sandell  18-APR-2014                         --
--    Changed disabling                                         --
--    Changed filtering on des rate to 5 sec                    --
--    Changed EO logic                                           --
--    Separated CLB AND DES rate                                 --
--                                                              --
-- Version D001  M. Matyas  01-NOV-2017                         --
--    Added logic to determine overall vertical rate emergency --
--    state.                                                      --
--                                                              --
-- Version D002  S. Meisch  AUG-2018                           --
--    Added logic to determine overall pitch rate emergency    --
--    state.                                                      --
--

```

```
-- Version F001 M. Matyas 29-SEP-2020
-- Updated references from ACARS_10_CHAR_FLIGHT_NUMBER to
-- FMF_CMF_FLIGHT_NUMBER.
-- Updated reference from CUR_FUELREMAIN_WT to FMF_FUEL_REMAINING_WEIGHT.
--
```

```
-----
-- Input Variables (ACPs and Existing UDPs) --
-- AIRCRAFT_IN_AIR (Boolean) --
-- ADRF_ALTRATE_VOTED --
-- ADRF_CAS_VOTED --
-- ADRF_MACH_VOTED --
-- ADRF_PRESSALT_VOTED --
-- ADRF_TAT_VOTED --
-- FMF_FUEL_REMAINING_WEIGHT --
-- EICAS_ENG_N1_SPD_L --
-- EICAS_ENG_N1_SPD_R --
-- FUEL_SW_L_CUTOFF --
-- FUEL_SW_R_CUTOFF --
-- ENG_TPR_ACTUAL_L --
-- ENG_TPR_ACTUAL_R --
-- IRU_RDC_BODY_PITCH_RATE --
-- IRU_RDC_BODY_YAW_RATE --
-- IRU_RDC_INERTIAL_VERT_SPEED --
-- IRU_RDC_PITCH_ANGLE --
-- IRU_RDC_VERT_ACCELERATION --
-- METRIC_ENABLE_ON --
-- TOTAL_FILTERED_FUEL_MASS --
-- TIME_SINCE_POWER_ON --
-- FMF_ACTRTEORIGIN --
-- FMF_ACTRTEDEST --
-- FMF_CMF_FLIGHT_NUMBER --
-- LRRR_RADIO_HEIGHT --
--
-- UDPs --
-- EMERGENCY_DESRATE_THRESHOLD --
-- EMERGENCY_CLBRATE_THRESHOLD --
-- EMERGENCY_MMO_THRESHOLD --
-- EMERGENCY_VMO_THRESHOLD --
-- EMERGENCY_LOW_SPEED_THRESHOLD --
-- EMERGENCY_PITCH_THRESHOLD --
-- EMERGENCY_PITCH_RATE_THRESHOLD --
-- EMERGENCY_YAW_RATE_THRESHOLD --
-- EMERGENCY_VERT_ACCEL_THRESHOLD --
-- EMERGENCY_FUEL_THRESHOLD --
-- EMERGENCY_TPR_THRESHOLD --
-- EMERGENCY_N1_THRESHOLD --
```

```

--      EMERGENCY_MIN_ALTITUDE                                --
--      EMERGENCY_ENABLED                                      --
--      RADIO_ALT_REACHED                                     --
--      VMO                                                    --
--      MMO                                                    --
--
-- Internal/Output Parameters                                  --
--      LAST_EMERGENCY_STATE                                   --
--      CUR_EMERGENCY_STATE                                    --
--      EMERGENCY_STATE_CLB                                   --
--      EMERGENCY_STATE_DES                                   --
--      EMERGENCY_STATE_RATE                                   --
--      EMERGENCY_STATE_EO                                    --
--      ENGINE_OUT_LEFT                                       --
--      ENGINE_OUT_RIGHT                                       --
--      EMERGENCY_STATE_SPD                                    --
--      EMERGENCY_STATE_FUEL                                   --
--      EMERGENCY_STATE_ATTITUDE                              --
--      OVER_CAS                                               --
--      OVER_MACH                                              --
--      UNDER_CAS                                             --
--      OVER_PITCH_RATE                                        --
--      OVER_YAW_RATE                                          --
--      OVER_PITCH                                             --
--      OVER_ACCEL                                             --
--
-----
-- Initialize Parameters                                     --
-----

LAST_EMERGENCY_STATE := CUR_EMERGENCY_STATE;
set_default (CUR_EMERGENCY_STATE);
-----

-- Determine if reporting is enabled                         --
-----

If (LRRA_RADIO_HEIGHT > 2000) then
    RADIO_ALT_REACHED := TRUE;
end_if;

If valid (FMF_ACTRTEORIGIN) and
    valid (FMF_ACTRTEDEST) and
    valid (FMF_CMF_FLIGHT_NUMBER) and
    RADIO_ALT_REACHED and
    (ADRF_CAS_VOTED > EMERGENCY_LOW_SPEED_THRESHOLD) and
    (ADRF_PRESSALT_VOTED > EMERGENCY_MIN_ALTITUDE) then
    EMERGENCY_ENABLED := TRUE;
end_if;

```



```

-----
-- Determine when reporting is disabled                                --
-----

If (not valid (FMF_ACTRTEDEST) and
    not valid (FMF_CMF_FLIGHT_NUMBER)
    and not (AIRCRAFT_IN_AIR)) or
    (LRRR_RADIO_HEIGHT < 500) then
    EMERGENCY_ENABLED := FALSE;
    RADIO_ALT_REACHED := FALSE;
end_if;
-----

-----
-- Look for Vertical Rate                                            --
-----

If EMERGENCY_ENABLED then
    if valid (ADRF_ALTRATE_VOTED) then
        if for 5 seconds (-ADRF_ALTRATE_VOTED >
            (EMERGENCY_DESRATE_THRESHOLD + 500)) then
            EMERGENCY_STATE_DES := TRUE;
        end_if;
    end_if;

    if valid (IRU_RDC_INERTIAL_VERT_SPEED) then
        if for 5 seconds (-IRU_RDC_INERTIAL_VERT_SPEED >
            EMERGENCY_DESRATE_THRESHOLD) then
            EMERGENCY_STATE_DES := TRUE;
        end_if;
    end_if;
end_if;

If EMERGENCY_ENABLED then
    if valid (ADRF_ALTRATE_VOTED) then
        if for 5 seconds (ADRF_ALTRATE_VOTED >
            (EMERGENCY_CLBRATE_THRESHOLD + 500)) then
            EMERGENCY_STATE_CLB := TRUE;
        end_if;
    end_if;

    if valid (IRU_RDC_INERTIAL_VERT_SPEED) then
        if for 5 seconds (IRU_RDC_INERTIAL_VERT_SPEED >
            EMERGENCY_CLBRATE_THRESHOLD) then
            EMERGENCY_STATE_CLB := TRUE;
        end_if;
    end_if;
end_if;

```

```

end_if;

if EMERGENCY_STATE_DES or EMERGENCY_STATE_CLB then
    EMERGENCY_STATE_VERT_RATE := TRUE;
end_if;

-----

-----

-- Look for Multiple Engine Out --
-----

If EMERGENCY_ENABLED and AIRCRAFT_IN_AIR then
    if for 5 seconds ((valid (ENG_TPR_ACTUAL_L) and
        ENG_TPR_ACTUAL_L < EMERGENCY_TPR_THRESHOLD) or
        (valid (EICAS_ENG_N1_SPD_L) and
        EICAS_ENG_N1_SPD_L < EMERGENCY_N1_THRESHOLD) or
        FUEL_SW_L_CUTOFF or
        (not valid (ENG_TPR_ACTUAL_L) and not valid (EICAS_ENG_N1_SPD_L))) then
        ENGINE_OUT_LEFT := TRUE;
    end_if;

    if for 5 seconds (((valid (ENG_TPR_ACTUAL_L) and
        ENG_TPR_ACTUAL_L > EMERGENCY_TPR_THRESHOLD + 5) or
        (valid (EICAS_ENG_N1_SPD_L) and
        EICAS_ENG_N1_SPD_L > EMERGENCY_N1_THRESHOLD + 5)) and
        not (FUEL_SW_L_CUTOFF)) then
        ENGINE_OUT_LEFT := FALSE;
    end_if;

    if for 5 seconds ((valid (ENG_TPR_ACTUAL_R) and
        ENG_TPR_ACTUAL_R < EMERGENCY_TPR_THRESHOLD) or
        (valid (EICAS_ENG_N1_SPD_R) and
        EICAS_ENG_N1_SPD_R < EMERGENCY_N1_THRESHOLD) or
        FUEL_SW_R_CUTOFF or
        (not valid (ENG_TPR_ACTUAL_R) and not valid (EICAS_ENG_N1_SPD_R))) then
        ENGINE_OUT_RIGHT := TRUE;
    end_if;

    if for 5 seconds (((valid (ENG_TPR_ACTUAL_R) and
        ENG_TPR_ACTUAL_R > EMERGENCY_TPR_THRESHOLD + 5) or
        (valid (EICAS_ENG_N1_SPD_R) and
        EICAS_ENG_N1_SPD_R > EMERGENCY_N1_THRESHOLD + 5)) and
        not (FUEL_SW_R_CUTOFF)) then
        ENGINE_OUT_RIGHT := FALSE;
    end_if;

```

```

end_if;

If ENGINE_OUT_LEFT and ENGINE_OUT_RIGHT then
    EMERGENCY_STATE_EO := TRUE;
end_if;

-----

-----

-- Look for Excessive Airspeed (High or Low) --
-----

If EMERGENCY_ENABLED and AIRCRAFT_IN_AIR then
    if valid (ADRF_CAS_VOTED) and
        ADRF_CAS_VOTED > (VMO + EMERGENCY_VMO_THRESHOLD) then
        OVER_CAS := TRUE;
    end_if;
    if valid (ADRF_MACH_VOTED) and
        ADRF_MACH_VOTED > (MMO + EMERGENCY_MMO_THRESHOLD) then
        OVER_MACH := TRUE;
    end_if;
    if valid (ADRF_CAS_VOTED) and
        ADRF_CAS_VOTED < EMERGENCY_LOW_SPEED_THRESHOLD then
        UNDER_CAS := TRUE;
    end_if;
end_if;

If OVER_CAS or OVER_MACH or UNDER_CAS then
    EMERGENCY_STATE_SPD := TRUE;
end_if;

-----

-----

-- Look for Unusual Attitude or Maneuver --
-----

If EMERGENCY_ENABLED and AIRCRAFT_IN_AIR then
    if valid (IRU_RDC_BODY_PITCH_RATE) and
        (IRU_RDC_BODY_PITCH_RATE > EMERGENCY_PITCH_RATE_THRESHOLD or
        IRU_RDC_BODY_PITCH_RATE < -EMERGENCY_PITCH_RATE_THRESHOLD) then
        OVER_PITCH_RATE := TRUE;
    end_if;
    if valid (IRU_RDC_BODY_YAW_RATE) and
        (IRU_RDC_BODY_YAW_RATE > EMERGENCY_YAW_RATE_THRESHOLD or
        IRU_RDC_BODY_YAW_RATE < -EMERGENCY_YAW_RATE_THRESHOLD) then
        OVER_YAW_RATE := TRUE;
    end_if;

```

```

    if valid (IRU_RDC_PITCH_ANGLE) and
        IRU_RDC_PITCH_ANGLE > EMERGENCY_PITCH_THRESHOLD or
        IRU_RDC_PITCH_ANGLE < -EMERGENCY_PITCH_THRESHOLD then
        OVER_PITCH := TRUE;
    end_if;
    if valid (IRU_RDC_VERT_ACCELERATION) and
        IRU_RDC_VERT_ACCELERATION > EMERGENCY_VERT_ACCEL_THRESHOLD then
        OVER_ACCEL := TRUE;
    end_if;
end_if;

If OVER_PITCH_RATE or OVER_YAW_RATE or OVER_PITCH or OVER_ACCEL then
    EMERGENCY_STATE_ATTITUDE := TRUE;
end_if;

-----

-----
-- Look for Emergency/Hijack Transponder Code Set                --
-----
--
--
-- Transponder code not available on 787.                          --
--
-----

-----
-- Look for Low Fuel Situation                                     --
-----

If EMERGENCY_ENABLED and AIRCRAFT_IN_AIR then
    if for 10 seconds
        (FMF_FUEL_REMAINING_WEIGHT < EMERGENCY_FUEL_THRESHOLD) then
            EMERGENCY_STATE_FUEL := TRUE;
        end_if;
    if for 10 seconds
        (TOTAL_FILTERED_FUEL_MASS < EMERGENCY_FUEL_THRESHOLD) then
            EMERGENCY_STATE_FUEL := TRUE;
        end_if;
    end_if;
end_if;

-----

-----
-- Look for Depressurization (Cabin Altitude)                    --
-----
--
-- Cabin altitude not available on 787.                            --

```

```

--
-----

-----
-- Determine final emergency state
-----

CUR_EMERGENCY_STATE := EMERGENCY_STATE_VERT_RATE or
EMERGENCY_STATE_EO or EMERGENCY_STATE_SPD or EMERGENCY_STATE_FUEL or
EMERGENCY_STATE_ATTITUDE;
-----

```

9.4.2 Reporting Logic Unit

The AMI implements a logic unit to generate the Automatic Emergency Report downlink encoding.

```

-----
-- This Logic Unit sends reports in emergency situations.
--
-- G.Sandell 7-02-2009 Initial development
--
-- G.Sandell 4-10-2013 Changed ON/OFF to ON/ARM/OFF
--
-- VERSION C000 NEW VERSIONING -- no change since B000
--
-- VERSION C001 G Sandell 4-18-2014
-- Changed for separating CLB and DES rate
--
-- Version D001 M Matyas 01-NOV-2017
-- Added logic to reset overall vertical rate emergency state.
--
-- Version D002 S Meisch 2018SEP
-- Changed downlink report name to: BCG_EMERG_AUTO
--
-- Version E001 M Matyas 21-FEB-2019
-- Changed encoding name to BCG_EMRG for consistency.
--
-----

--
-- Input Parameters
-- EMERGENCY_STATE_CLB
-- EMERGENCY_STATE_DES
-- EMERGENCY_STATE_EO
-- EMERGENCY_STATE_SPD
--

```

```

--      EMERGENCY_STATE_FUEL                      --
--      EMERGENCY_STATE_ATTITUDE                  --
--      EMERGENCY_SELECTION                      --
--      EMERGENCY_FREQUENCY                      --
--      MAX_EMERGENCY_RPT                      --
--      CUR_EMERGENCY_STATE                      --
--
-- Internal/Output Parameters                      --
--      LAST_EMERGENCY_SELECTION                  --
--      EMERGENCY_COUNTER                      --
--      NEW_EMERGENCY                          --
--      END_EMERGENCY                          --
--      MANUAL_EMERGENCY                      --
--      LAST_EMERGENCY_STATE                      --
--
-----
-- Send reports for continuing emergency situations --
-----

if (EMERGENCY_SELECTION = 1) and (LAST_EMERGENCY_SELECTION = 1) then
    EMERGENCY_COUNTER := EMERGENCY_COUNTER + 1;
    if EMERGENCY_COUNTER = EMERGENCY_FREQUENCY * (EMERGENCY_COUNTER/
    EMERGENCY_FREQUENCY) then
        send (BCG_EMRG);
    end_if;
end_if;

-----

-----
-- Look for manually-initiated start                --
-----

if (EMERGENCY_SELECTION = 1) and (LAST_EMERGENCY_SELECTION !=1) then
    NEW_EMERGENCY := TRUE;
    MANUAL_EMERGENCY := TRUE;
end_if;

-----

-----
-- Look for auto-initiated start                    --
-----

if (EMERGENCY_SELECTION =2) and CUR_EMERGENCY_STATE and
    not (LAST_EMERGENCY_STATE) then
    NEW_EMERGENCY := TRUE;
    MANUAL_EMERGENCY := FALSE;
end_if;

```

```

-----

-----
-- Set up reporting process and send initial report      --
-----

if NEW_EMERGENCY then
    send (BCG_EMRG);
    EMERGENCY_COUNTER := 1;
    NEW_EMERGENCY := FALSE;
    EMERGENCY_SELECTION := 1;
end_if;

-----

-----
-- Look for manual cancellation                          --
-----

if (LAST_EMERGENCY_SELECTION = 1) and (EMERGENCY_SELECTION != 1) then
    END_EMERGENCY := TRUE;
end_if;

-----

-----
-- End reporting                                         --
-----

if END_EMERGENCY or EMERGENCY_COUNTER >= MAX_EMERGENCY_RPT*60 then
    END_EMERGENCY := FALSE;
    if (EMERGENCY_SELECTION = 1) then
        EMERGENCY_SELECTION := 2;
    end_if;
    EMERGENCY_COUNTER := MAX_EMERGENCY_RPT*60 + 10;
-- And reset all the variables                          --
    EMERGENCY_STATE_CLB := FALSE;
    EMERGENCY_STATE_DES := FALSE;
    EMERGENCY_STATE_VERT_RATE := FALSE;
    EMERGENCY_STATE_EO := FALSE;
    ENGINE_OUT_LEFT := FALSE;
    ENGINE_OUT_RIGHT := FALSE;
    EMERGENCY_STATE_SPD := FALSE;
    EMERGENCY_STATE_FUEL := FALSE;
    EMERGENCY_STATE_ATTITUDE := FALSE;
    OVER_CAS := FALSE;
    OVER_MACH := FALSE;
    UNDER_CAS := FALSE;
    OVER_PITCH_RATE := FALSE;
    OVER_YAW_RATE := FALSE;
    OVER_PITCH := FALSE;

```

```

OVER_ACCEL := FALSE;
MANUAL_EMERGENCY := FALSE;
CUR_EMERGENCY_STATE := FALSE;
end_if;

-----

-----

-- Clean up for the next cycle --

-----

LAST_EMERGENCY_SELECTION := EMERGENCY_SELECTION;
-----

```

9.4.3 Downlink Encoding Definition

The encoding for the Automatic Emergency Report downlink is defined as follows:

- ACARS SMI: SIT
- ACARS Label: 99
- Allowable Subnetworks: All
- Downlink Priority: Emergency
- Downlink Buffer: REPORT_AUTO
- Delete at System Reset: No
- Delete on Buffer Full: No

Table 9-6 describes the Automatic Emergency Report downlink encoding definition.

Table 9-6 Automatic Emergency Report Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	EMR	"Message Identifier"
3	2	Always	06	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	13	Always	AIRCRAFT_POSITION	"Position (XDDMMTXDDMMT) "

#	Length	Inclusion	Content	Element Description
13	1	Always	,	"Delimiter"
14	1-5	Always	NAV_ALTITUDE	"Altitude (NNNNN)"
15	1	Always	,	"Delimiter"
16	6	Always	CURRENT_TIME	"Timestamp (HHMMSS)"
17	2	Always	<CR><LF>	"Delimiter"
18	4	Always	NAV_MAG_HDG	"Magnetic Heading (±NNN)"
19	1	Always	,	"Delimiter"
20	4	Always	NAV_MAG_TRACK	"Magnetic Track (±NNN)"
21	1	Always	,	"Delimiter"
22	4	Always	NAV_TRUE_HDG	"True Heading (±NNN)"
23	1	Always	,	"Delimiter"
24	4	Always	NAV_TRUE_TRACK	"True Track (±NNN)"
25	1	Always	,	"Delimiter"
26	4	Always	NAV_WIND_TRUE_BRG	"True Wind Bearing (±NNN)"
27	1	Always	,	"Delimiter"
28	1-4	Always	NAV_WIND_MAG	"Wind Speed (NNNN)"
29	1	Always	,	"Delimiter"
30	1-4	Always	NAV_GROUNDSPEED	"Groundspeed (NNNN)"
31	1	Always	,	"Delimiter"
32	4	Always	FMF_DEST_ETA	"Destination ETA (HHMM)"
33	2	Always	<CR><LF>	"Delimiter"
34	1-5	Always	ADRF_PRESSALT_VOTED	"Pressure Altitude (NNNNN)"
35	1	Always	,	"Delimiter"
36	2-6	Always	ADRF_ALTRATE_VOTED	"Altitude Rate" (±NNNNN)
37	1	Always	,	"Delimiter"
38	1-3	Always	ADRF_CAS_VOTED	"CAS (NNN)"
39	1	Always	,	"Delimiter"
40	5	Always	ADRF_MACH_VOTED	"Mach (N.NNN)"
41	1	Always	,	"Delimiter"
42	2-4	Always	ADRF_SAT_VOTED	"SAT (±NNN)"
43	2	Always	<CR><LF>	"Delimiter"
44	2-6	Always	IRU_RDC_INERTIAL_VERT_SPEED	"Vertical Speed (±NNNNN)"
45	1	Always	,	"Delimiter"
46	4	Always	IRU_RDC_VERT_ACCELERATION	"Vertical Acceleration (±N.N)"
47	1	Always	,	"Delimiter"
48	4-6	Always	IRU_RDC_PITCH_ANGLE	"Pitch Angle (±NNN.N)"
49	1	Always	,	"Delimiter"
50	4-6	Always	IRU_RDC_BODY_PITCH_RATE	"Pitch Rate (±NNN.N)"
51	1	Always	,	"Delimiter"

#	Length	Inclusion	Content	Element Description
52	4-6	Always	IRU_RDC_BODY_YAW_RATE	"Yaw Rate (±NNN.N)"
53	2	Always	<CR><LF>	"Delimiter"
54	3 or 6	Always	FUEL_SW_L_CUTOFF	"Left engine fuel control (RUN / CUTOFF)"
55	1	Always	/	"Delimiter"
56	3 or 6	Always	FUEL_SW_R_CUTOFF	"Right engine fuel control (RUN / CUTOFF)"
57	1	Always	,	"Delimiter"
58	3-5	Always	EICAS_ENG_N1_SPD_L	"Left engine N1 (NNN.N)"
59	1	Always	/	"Delimiter"
60	3-5	Always	EICAS_ENG_N1_SPD_R	"Right engine N1 (NNN.N)"
61	1	Always	,	"Delimiter"
62	3-5	Always	ENG_TPR_ACTUAL_L	"Left engine TPR (NNN.N)"
63	1	Always	/	"Delimiter"
64	3-5	Always	ENG_TPR_ACTUAL_R	"Right engine TPR (NNN.N)"
65	1	Always	,	"Delimiter"
66	3-5	Always	EEC_ENG_OIL_P_L	"Left engine oil pressure (NNN.N)"
67	1	Always	/	"Delimiter"
68	3-5	Always	EEC_ENG_OIL_P_R	"Right engine oil pressure (NNN.N) "
69	1	Always	,	"Delimiter"
70	2-4	Always	EEC_ENG_OIL_T_L	"Left engine oil temperature (±NNN)"
71	1	Always	/	"Delimiter"
72	2-4	Always	EEC_ENG_OIL_T_R	"Right engine oil temperature (±NNN)"
73	1	Always	,	"Delimiter"
74	3-4	Always	L_ENG_OIL_QTY	"Left engine oil quantity (NN.N)"
75	1	Always	/	"Delimiter"
76	3-4	Always	R_ENG_OIL_QTY	"Right engine oil quantity (NN.N)"
77	1	Always	,	"Delimiter"
78	3-5	Always	FUEL_ON_BOARD	"Fuel on board / 1000 (NNN.N)"
79	2	Always	<CR><LF>	"Delimiter"
80	3 or 7	Always	FCM_AP_ENGD	"Autopilot (ENG / NOT ENG)"
81	1	Always	,	"Delimiter"
82	1-2	Always	FCM_FLAP_LVR_POSN	"Flap lever position (NN)"
83	1	Always	,	"Delimiter"
84	1-3	Always	FCM_FLAPS_SLATS_NORM	"Flaps/slats position (NNN)"
85	1	Always	,	"Delimiter"
86	1	Always	*	"777X left FWT placeholder"
87	1	Always	/	"Delimiter"
88	1	Always	*	"777X right FWT placeholder"
89	1	Always	,	"Delimiter"

#	Length	Inclusion	Content	Element Description
90	3	Conditionally	OFF	"Gear status - OFF"
91	2	Conditionally	UP	"Gear status - UP"
92	4	Conditionally	DOWN	"Gear status - DOWN"
93	7	Conditionally	TRANSIT	"Gear status - IN TRANSIT"
94	1	Conditionally	*	"Gear status - INVALID"
95	2	Always	<CR><LF>	"Delimiter"
96	3	Conditionally	OFF	"Wing Anti-Ice Off"
97	4	Conditionally	AUTO	"Wing Anti-Ice Auto"
98	2	Conditionally	ON	"Wing Anti-Ice On"
99	1	Conditionally	*	"Wing Anti-Ice Invalid"
100	1	Always	,	"Delimiter"
101	3-6	Always	LRRA_RADIO_HEIGHT	"LRRA height (NNNN.N)"
102	1	Always	,	"Delimiter"
103	2-3	Always	ECS_APU_RUNNING	"APU (ON / OFF)"
104	2	Always	<CR><LF>	"Delimiter"
105	1	Conditionally	M	"Manual Trigger"
106	1	Conditionally	A	"Automatic Trigger"
107	4	Conditionally	,VS-	"Descent Rate Trigger"
108	4	Conditionally	,VS+	"Climb Rate Trigger"
109	3	Conditionally	,EO	"Engines Out Trigger"
110	5	Conditionally	,CAS-	"CAS Low Trigger"
111	4	Conditionally	,VMO	"V _{MO} Trigger"
112	4	Conditionally	,MMO	"M _{MO} Trigger"
113	3	Conditionally	,PA	"Pitch Angle Trigger"
114	3	Conditionally	,PR	"Pitch Rate Trigger"
115	3	Conditionally	,YR	"Yaw Rate Trigger"
116	2	Conditionally	,G	"Vertical Acceleration Trigger"
117	5	Conditionally	,FUEL	"Fuel Trigger"

Note: Headings, tracks, and bearings (i.e., the NAV_MAG_HDG, NAV_MAG_TRACK, NAV_TRUE_HDG, NAV_TRUE_TRACK, and NAV_WIND_TRUE_BRG parameters) are expressed in raw format, where negative values are west of north. For example, a raw heading of -135° corresponds to a compass heading of 225° and a raw track of -015° corresponds to a compass track of 345°. Accordingly, if a raw direction is negative then it may be converted to a compass direction by adding 360°. No such conversion is necessary if a raw direction is positive.

Note: Flap lever position is expressed in degrees and flaps/slats position is expressed in percent. Table 9-7 correlates flap settings to flap lever positions and flaps/slats positions.

Table 9-7 Flap Settings, Flap Lever Positions, and Flap/Slats Positions

Flap Setting	Flap Lever Position	Flaps/Slats Position
UP	0°	0%
1	11°	20%
5	17°	40%
10	24°	50%
15	31°	60%
17	39°	67%
18	43°	73%
20	51°	80%
25	58°	100%
30	65°	120%

Sample Message

```
QU RTNBOCR
.DDLXCXA 221641
SIT
FI BO718/AN N7873BO
DT DDL BFI 221641 M10A
- EMR06 BOE718/22221641KBFIKBFI
N47365W122066,10000,164142
-035,-035,-020,-020,-059,0,289,1653
10000,+0,250,0.452,-5
+1,+0.0,+2.2,+0.0,+0.0
RUN/RUN,74.0/74.0,49.4/49.4,110.9/110.9,+154/+154,17.5/17.5,44.6
ENG,0,0,*/*,UP
AUTO,*,ON
M
```

Note: In this sample message, the fuel on board is “44.6” (44,600), the LRRA (radio altimeter) height is “**” (invalid), and the automatic emergency report trigger is “M” (manual, based on selection of the AUTOMATIC EMERGENCY REPORTS ON button on the SITUATION REPORT display).

9.4.4 Uplink Decoding Definition

An aircraft operator may send the Automatic Emergency Report Configuration Uplink to set alternative thresholds for the Automatic Emergency Report function. CMF stores these values in NVM in order to retain them across power cycles.

To set an alternative threshold, a new threshold value is inserted in the appropriate position in the uplink. Values are comma-delimited and may be omitted (skipped) in the uplink.

The decoding for the Automatic Emergency Report Configuration Uplink is defined as follows:

- ACARS SMI: CMD
- ACARS Label: RA
- Message Identifier: ~1DEF
- Alert Level: None
- Optional Element Delimiter: “,” (comma) character

Table 9-8 describes the Automatic Emergency Report Configuration Uplink decoding definition.

Table 9-8 Automatic Emergency Report Configuration Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	2	Yes	<none>	"Priority code (XX)"
2	7	Yes	<none>	"Ground address (XXXXXXXX)"
3	5	Yes	<none>	"Message Identifier (~1DEF)"
4	2	Yes	<none>	"Message version (XX)"
5	2	Yes	<none>	"Message date (DD)"
6	4	Yes	<none>	"Message time (HHMM)"
7	2	Yes	<none>	"Delimiter (<CR><LF>)"
8	1-5	No	EMERGENCY_DESRATE_THRESHOLD	"Descent rate threshold (NNNNN)"
9	4	No	EMERGENCY_MMO_THRESHOLD	"Above M _{MO} threshold (N.NN)"
10	1-3	No	EMERGENCY_VMO_THRESHOLD	"Above V _{MO} threshold (NNN)"
11	1-3	No	EMERGENCY_LOW_SPEED_THRESHOLD	"CAS low threshold (NNN)"
12	1-2	No	EMERGENCY_PITCH_THRESHOLD	"Pitch angle threshold (NN)"
13	1-2	No	EMERGENCY_PITCH_RATE_THRESHOLD	"Pitch rate threshold (NN)"
14	1-2	No	EMERGENCY_YAW_RATE_THRESHOLD	"Yaw rate threshold (NN)"
15	3	No	EMERGENCY_VERT_ACCEL_THRESHOLD	"Vertical acceleration threshold (N.N)"
16	1-5	No	EMERGENCY_FUEL_THRESHOLD	"Fuel threshold (NNNNN)"
17	1-2	No	EMERGENCY_TPR_THRESHOLD	"TPR threshold (NN)"
18	1-2	No	EMERGENCY_N1_THRESHOLD	"N1 threshold (NN)"
19	2-3	No	EMERGENCY_FREQUENCY	"Reporting interval (NNN)"
20	2-3	No	MAX_EMERGENCY_RPT	"Reporting time limit (NNN)"
21	1-5	No	EMERGENCY_MIN_ALTITUDE	"Reporting min altitude (NNNNN)"
22	1-5	No	EMERGENCY_CLBRATE_THRESHOLD	"Climb rate threshold (NNNNN)"

All threshold parameters are optional.

The following ranges and units apply to the following parameters:

-
- Descent rate threshold: 0-20000 feet per minute
 - Above M_{MO} threshold: 0.00-0.30 Mach
 - Above V_{MO} threshold: 0-100 knots CAS
 - CAS low threshold: 0-200 knots CAS
 - Pitch angle threshold: 0-45 degrees
 - Pitch rate threshold: 0-30 degrees per second
 - Yaw rate threshold: 0-30 degrees per second
 - Vertical acceleration threshold: 1.0-5.0g
 - Fuel threshold: 0-20000 pounds
 - TPR threshold: 0-25
 - N1 threshold: 0-25%
 - Reporting interval: 60-900 seconds
 - Reporting time limit: 30-360 minutes
 - Reporting minimum altitude: 0-30000 feet
 - Climb rate threshold: 0-20000 feet per minute

Note: Elements 8 through 22 are variable-length elements and accordingly should be terminated with a delimiter character. Pre-defined element delimiters are <CR>, <CR><LF>, and <ETX> and the optional element delimiter defined for this uplink decoding is “,” (comma).

Sample Message

```
QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 016A
- QURTNBOCR~1DEF01190445
6000,0.1,40,115,,,15,3.0,5000,10,10,120,30,7500,6000
```

Note: In this sample message the pitch angle and pitch rate thresholds are omitted.

9.5 Automatic Position Report

This section describes the Automatic Position Report function that the AMI implements. This function includes an uplink decoding definition to enable or disable automatic position reports and set the interval between each report.

As defined in the CMF baseline AMI with default values, the Automatic Position Report function automatically generates position reports at a default interval of 15 minutes when the aircraft is airborne and meets the requirements for ICAO “4D/15” normal aircraft tracking.

Note: There is no display definition for this function, although the CMF PARAMETERS display described in Section 12.6.2.1.1 allows automatic position reports to be enabled or disabled and their interval to be changed.

9.5.1 Logic Unit

The AMI implements a logic unit to generate the Automatic Position Report downlink encoding.

```
-- -----  
--  
-- Version D001  
-- M. Matyas 24-OCT-2017  
-- Automatic position reporting logic based on PERIODIC_POS_RPT.C000.  
-- Added ability to enable/disable automatic position reporting and  
-- configure automatic position reporting interval via uplink.  
--  
-- -----  
--  
  
if AIRCRAFT_IN_AIR and POS_RPT_ENABLE then  
  
    if POS_RPT_COUNTER > (POS_RPT_INTERVAL * 60) then  
        send (BCG_POS_RPT);  
        POS_RPT_COUNTER := 0;  
  
    else  
  
        POS_RPT_COUNTER := POS_RPT_COUNTER + 1;  
  
    end_if;  
  
else  
  
    POS_RPT_COUNTER := 0;  
  
end_if;  
  
-----  
--  
-- END
```

9.5.2 Downlink Encoding Definition

The encoding for the Automatic Position Report downlink is defined as follows:

- ACARS SMI: M16
- ACARS Label: 16
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REPORT_AUTO
- Delete at System Reset: No
- Delete on Buffer Full: Yes

Table 9-9 describes the Automatic Position Report downlink encoding definition.

Table 9-9 Automatic Position Report Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	POS	"Message Identifier"
3	2	Always	02	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	13	Always	AIRCRAFT_POSITION	"Position (XDDMMTXDDMMT)"
13	1	Always	,	"Delimiter"
14	1-5	Always	NAV_ALTITUDE	"Altitude (NNNNN)"
15	1	Always	,	"Delimiter"
16	6	Always	CURRENT_TIME	"Current Time (HHMMSS)"
17	1	Always	,	"Delimiter"
18	4	Always	NAV_MAG_HDG	"Magnetic Heading (±NNN)"
19	1	Always	,	"Delimiter"
20	4	Always	NAV_WIND_TRUE_BRG	"True Wind Bearing (±NNN)"
21	1	Always	,	"Delimiter"
22	1-4	Always	NAV_WIND_MAG	"Wind Speed (NNNN)"
23	1	Always	,	"Delimiter"

#	Length	Inclusion	Content	Element Description
24	1-4	Always	NAV_GROUNDSPEED	"Groundspeed (NNNN)"
25	1	Always	,	"Delimiter"
26	4	Always	FMF_DEST_ETA	"Destination ETA (HHMM)"
27	1	Always	,	"Delimiter"
28	2-6	Always	ADRF_ALTRATE_VOTED	"Altitude Rate (±NNNNN)"
29	1	Always	,	"Delimiter"
30	1-3	Always	ADRF_CAS_VOTED	"CAS (NNN)"
31	1	Always	,	"Delimiter"
32	5	Always	ADRF_MACH_VOTED	"Mach (N.NNN)"
33	1	Always	,	"Delimiter"
34	2-4	Always	ADRF_SAT_VOTED	"SAT (±NNN)"
35	1	Always	,	"Delimiter"
36	3-5	Always	FUEL_ON_BOARD	"Fuel on Board / 1000 (NNN.N)"

Note: Headings, tracks, and bearings (i.e., the NAV_MAG_HDG and NAV_WIND_TRUE_BRG parameters) are expressed in raw format, where negative values are west of north. For example, a raw heading of -135° corresponds to a compass heading of 225° and a raw track of -015° corresponds to a compass track of 345°. Accordingly, if a raw direction is negative then it may be converted to a compass direction by adding 360°. No such conversion is necessary if a raw direction is positive.

Sample Message

```
QU RTNBOCR
.DDLXCXA 261904
M16
FI BO718/AN N7873BO
DT DDL BFI 261904 M08A
- POS02 BOE718/26261904KBFIKLGB
N47280W122020,39003,190414,-035,-067,41,289,2157,+2,251,0.454,-5,34.3
```

Note: In this sample message, the fuel on board is "34.3" (34,300).

9.5.3 Uplink Decoding Definition

An aircraft operator may send the Automatic Position Report Configuration Uplink to enable or disable the Automatic Position Report function and set the interval between automatic position reports. CMF stores the values contained in this uplink in NVM in order to retain them across power cycles.

The decoding for the Automatic Position Report Configuration Uplink is defined as follows:

- ACARS SMI: CMD

- ACARS Label: RA
- Message Identifier: ~1POS
- Alert Level: None
- Optional Element Delimiter: “,” (comma)

Table 9-10 describes the Automatic Position Report Configuration Uplink decoding definition.

Table 9-10 Automatic Position Report Configuration Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	2	Yes	<none>	"Priority code (XX)"
2	7	Yes	<none>	"Ground address (XXXXXXX)"
3	5	Yes	<none>	"Message identifier (~1POS)"
4	2	Yes	<none>	"Delimiter (<CR><LF>)"
5	1	Yes	POS_RPT_ENABLE	"Enable (1) or disable (0)"
6	1	No	<none>	"Delimiter (,)"
7	1-2	No	POS_RPT_INTERVAL	"Interval (5-60 [minutes])"

Sample Message

```
QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 017A
- QURTNBOCR~1POS
1,10
```

Note: In the sample message above, automatic position reports are enabled at a 10-minute interval.

```
QU DDLXCXA
.RTNBOCR 221955
CMD
AN N7873BO/MA 018A
- QURTNBOCR~1POS
0
```

Note: In the sample message above, automatic position reports are disabled and the automatic position report interval is omitted.

9.6 OOOI Reports

This section describes the OOOI Reports function that the AMI implements. This function includes logic to detect OOOI state transitions and perform certain actions at those transitions,

including sending specified downlink encodings as OOOI reports. Of note, this logic requires that an FMF flight plan (route) be entered and activated, specifically the destination airport.

Note: Section 11.2 describes the OOOI Data Loop Back Message function, which allows an aircraft operator to request OOOI-related information from the airplane.

Note: There is no display definition for this function, although the OOOI Test Page display described in Section 12.6.2.1.9 shows OOOI-related parameters.

Table 9-11 summarizes the OOOI state transition logic.

Table 9-11 OOOI State Transition Logic

To From	Out	Off	On	In
Out		Doors Closed and In Air	N/A	Doors Open (Return to Gate)
Off	N/A		On Ground	N/A
On	N/A	Doors Closed and In Air (Touch and Go)		Doors Open
In	Doors Closed and Brake Off	N/A	N/A	
Invalid	On Ground, Doors Closed, and Brake Off	In Air and Doors Closed	N/A	On Ground, Doors Closed, and Brake On or On Ground and Doors Open

In Air is true (and On Ground is false) when:

- calibrated airspeed (CAS) is valid and CAS is greater than 125 knots; or
- CAS is invalid, but ground speed is valid and ground speed is greater than 130 knots; or
- both CAS and ground speed are invalid, but the In Air discrete is true.

Otherwise On Ground is true and In Air is false.

Brake On (parking brake set) and Brake Off (parking brake released) are set based on the status of the parking brake handle.

Doors Open is true if any of the following are set to true:

- DOOR_ENTRY_1L_OPEN
- DOOR_ENTRY_2L_OPEN
- DOOR_ENTRY_3L_OPEN

-
- DOOR_ENTRY_4L_OPEN
 - DOOR_ENTRY_1R_OPEN
 - DOOR_ENTRY_2R_OPEN
 - DOOR_ENTRY_3R_OPEN
 - DOOR_ENTRY_4R_OPEN
 - DOOR_AFT_EE_ACCESS_OPEN
 - DOOR_FWD_EE_ACCESS_OPEN
 - DOOR_FWD_OVERHEAD_OPEN
 - DOOR_FWD_ACCESS_OPEN
 - DOOR_BULK_CARGO_OPEN
 - DOOR_FWD_CARGO_OPEN
 - DOOR_AFT_CARGO_OPEN

Doors Closed is true if all of the above parameters are either false or invalid.

9.6.1 State Logic Unit

The AMI implements a logic unit to detect OOOI state transitions.

```

-- -----
--
-- BCG_OOOI_LOGIC  (OOOI STATE MACHINE)
--
-- This logic unit performs OOOI State Transition Logic:
--     OUT, OFF, ON, IN
--
-- -----
-- LOGIC DEPENDENCY:
--     Dependent on other Logic Units:      BCG_PARAMETERS
--     Is required for other Logic Units:  YES
-- -----
--
-- Revision History
--
--     26-FEB-2007 Original version
--
--     29-APR-2007 PEP 29 APR 07
--                 Revised to include Sandell comments from his
--                 13 APR 07 e-mail to P. Poteet, S. Lee and

```

```

--          M. Henninger. The areas were the doors open/closed
--          and determination of air/ground status.
--
-- 13-JUN-2007 PEP 13 JUN 07
--          Revised to include G.Sandell and T.Dastrup revision
--          to air/ground logic.
--
-- 03-OCT-2007
--          Changed airborne determination from 10 seconds to 3 seconds.
--
-- Version C000 - new versioning - no change in logic
--
-- Version D001 M. Matyas 24-OCT-2017
--          Changed logic that captures parking brake set time to use
--          CURRENT_TIME instead of CURRENT_HHMM.
--
-- Version D002 S. Meisch - 2018 OCT
--          1) Added logic dependency in description comment section
--          2) Moved generic parameters to BCG_PARAMETERS logic unit (runs first)
--          to assure proper sequencing of logic DOORS_ALL_CLOSED,
--          AIRCRAFT_IN_AIR, BRAKE_TIME)
--
-- Version E001 M. Matyas 18-FEB-2019
--          Added logic to determine initial delay type.
--
-- Version F001 M. Matyas 29-SEP-2020
--          Updated references from ESB_DELAY_SEL to DLY_SELECT.
--
-----
-- The following logic unit is an OOOI state machine.
-- It determines the current OOOI state and then
-- checks the criteria for transition to other states.
-- -----
-- BEGIN OOOI STATE MACHINE
-- -----
-- If IN, look for OUT event.
-- -----
if OOOI_STATE = IN then
  if DOORS_ALL_CLOSED and NOT PARKINGBRKLEVERSET then
    OOOI_STATE      := OUT;          -- Doors Closed, Brake Rel
    OOOI_STATUS     := "OUT";       -- Go OUT
    RETURN_TO_GATE := FALSE;
    TOUCH_AND_GO    := FALSE;
    set_invalid (BRAKE_TIME);
  end_if;

```

```

-- -----
-- If OUT, look for OFF or IN event.
-- -----

else_if OOOI_STATE = OUT then
    if DOORS_ALL_CLOSED then
        if AIRCRAFT_IN_AIR then
            OOOI_STATE      := OFF;          -- In Air, Doors Closed
            OOOI_STATUS     := "OFF";        -- Go OFF
            set_invalid (BRAKE_TIME);
        end_if;
    else -- (Doors not Closed)
        OOOI_STATE      := IN;              -- Door Open
        OOOI_STATUS     := "IN";            -- Go IN
        RETURN_TO_GATE := TRUE;             -- Return to Gate
    end_if;

-- -----
-- If OFF, look for ON event.
-- -----

else_if OOOI_STATE = OFF then
    if NOT AIRCRAFT_IN_AIR then
        OOOI_STATE      := ON;              -- On Gnd
        OOOI_STATUS     := "ON";            -- GO ON
        RETURN_TO_GATE := FALSE;            -- Not a Return to Gate
        TOUCH_AND_GO    := FALSE;          -- Not a Touch and Go
        set_invalid (BRAKE_TIME);
    end_if;

-- -----
-- If ON, look for OFF or IN event.
-- -----

else_if OOOI_STATE = ON then
    if DOORS_ALL_CLOSED then
        if AIRCRAFT_IN_AIR then
            OOOI_STATE      := OFF;          -- In Air, Doors Closed
            OOOI_STATUS     := "OFF";        -- Go OFF
            TOUCH_AND_GO    := TRUE;         -- Touch and Go
            RETURN_TO_GATE := FALSE;         -- Not a Return to Gate
            set_invalid (BRAKE_TIME);
        end_if;
    else
        OOOI_STATE      := IN;              -- Door Open
        OOOI_STATUS     := "IN";            -- Go IN
        RETURN_TO_GATE := FALSE;            -- Not a Return to Gate
        TOUCH_AND_GO    := FALSE;          -- Not a Touch and Go
    end_if;

```

```

else

-- -----
-- If OOOI state is invalid, then determine initial OOOI state.
-- -----

    if NOT AIRCRAFT_IN_AIR then
        if DOORS_ALL_CLOSED then
            if PARKINGBRKLEVERSET then
                OOOI_STATE      := IN;      -- On Gnd, Doors Closed, Brake Set
                OOOI_STATUS     := "IN";    -- IN State
                DLY_SELECT      := 1;
            else
                OOOI_STATE      := OUT;     -- On Gnd, Doors Closed, Brake Rel
                OOOI_STATUS     := "OUT";   -- OUT State
                DLY_SELECT      := 2;
            end_if;
        else
            OOOI_STATE      := IN;          -- On Gnd, Door(s) Open
            OOOI_STATUS     := "IN";        -- IN State
            DLY_SELECT      := 1;
        end_if;
    else
        if DOORS_ALL_CLOSED then
            OOOI_STATE      := OFF;         -- In Air, Doors Closed
            OOOI_STATUS     := "OFF";       -- OFF State
            DLY_SELECT      := 3;
        -- -----
        -- else
        --     leave OOOI Invalid             -- In Air, Door(s) Open = INVALID
        -- -----
        end_if;
    end_if;

    RETURN_TO_GATE := FALSE;              -- Not a Return to Gate
    TOUCH_AND_GO   := FALSE;              -- Not a Touch and Go
    set_invalid (BRAKE_TIME);

end_if;

-- -----
--
-- END

```

9.6.2 Events Logic Unit

The AMI implements a logic unit to perform certain actions at OOOI state transitions, including sending specified downlink encodings as OOOI reports.

```
-- -----
-- BCG_OOOI_EVENTS      OOOI EVENT PROCESSING
--    26-FEB-2007        Original version
--
-- This logic unit performs necessary processing at the time an OOOI
-- event occurs.
--
-- DEPENDENCY NOTES:
--    1. BCG_PARAMETERS must run before this logic unit
--    2. BCG_OOOI_LOGIC  must run before this logic unit
--
-- -----
--
-- 02-OCT-2007
--    Changed EOF timer to 9 minutes 50 seconds from when IN event is
--    triggered to assure the logic is executed before CMF resets.
--
-- Version C000 - new versioning - no change in logic
--
-- Version D001 M. Matyas  24-OCT-2017
--    Changed logic that captures OOOI times to use CURRENT_TIME instead
--    of CURRENT_HHMM.
--
-- Version D002 S. Meisch  2018 AUG
--    1) Changed downlink encoding names to include "BCG_" prefix.
--    2) Fixed invalid scheduled date. (CPDB defaults to current date, logic
--    snapshots INIT_SKED_DATE at OUT event.)
--
-- Version E001 M. Matyas  11-FEB-2019
--    Removed unused DEFAULT_STA logic.
--    Added logic to set arrival delay report type at On event.
--    Added logic to calculate flight and block times and fuels.
--
-- Version F001 M. Matyas  29-SEP-2020
--    Added logic to reestablish link with FMF destination airport when
--    touch-and-go occurs.
--    Added logic to capture Off station at Off event and On station
--    at On event.
--    Updated references from ESB_DELAY_SEL to DLY_SELECT and
--    from INIT_DEPT_STA to INIT_ORIG_STA.
```

```

--      Changed logic to enable OOOI reports if FMF destination airport is valid.
--      Added logic to capture flight number if invalid at Out event.
--      Corrected block fuel calculation logic to use IN_FOB (not ON_FOB).
--
-- -----

-- Enable OOOI reports only if a destination airport is entered in FMF:

    if valid (FMF_ACTRTEDEST) then
        OOOI_ENABLED := TRUE;          -- Enable OOOI reports
    end_if;

-- -----

-- OUT EVENT
-- -----

if OOOI_STATE changed to (OUT) then
    INIT_SKED_DATE := INIT_SKED_DATE;    -- Snapshot scheduled date
    OUT_TIME := CURRENT_TIME;            -- Capture OUT time
    OUT_FOB := FUEL_ON_BOARD;            -- Capture fuel on board
    set_invalid (IN_TIME);               -- Invalidate IN time
    set_invalid (IN_FOB);               -- Invalidate IN fuel
    set_invalid (EOF_TIMER);             -- Invalidate end of flight timer
    DLY_SELECT := 2;                    -- Select takeoff delay type

    if not valid (ENTERED_FLTNO) then    -- Capture flight number
        ENTERED_FLTNO := FMF_CMF_FLIGHT_NUMBER; -- if not valid
    end_if;

    if not valid (INIT_SKED_DATE) then  -- Capture scheduled date
        INIT_SKED_DATE := CURRENT_DATE; -- if not valid
    end_if;

    if OOOI_ENABLED then
        send (BCG_OOOI_OUT);            -- Send OUT report
    end_if;

end_if;

-- -----

-- OFF EVENT
-- -----

if OOOI_STATE changed to (OFF) then

    if TOUCH_AND_GO then
        set_default (INIT_DEST_STA);    -- Reestablish link to FMF destination
    end_if;

```

```

    set_invalid (ON_TIME);           -- Invalidate ON time
    set_invalid (ON_FOB);           -- Invalidate ON fuel
    INIT_ORIG_STA := INIT_ORIG_STA; -- Break link with FMF origin
    OFF_STA      := INIT_ORIG_STA;  -- Capture Off station
    OFF_TIME     := CURRENT_TIME;   -- Capture OFF time
    OFF_FOB      := FUEL_ON_BOARD;  -- Capture fuel on board
    DLY_SELECT   := 3;              -- Select enroute delay type

    if OOOI_ENABLED then
        send (BCG_OOOI_OFF);        -- Send OFF report
    end_if;

end_if;

-- -----
-- ON EVENT
-- -----
if OOOI_STATE changed to (ON) then
    ON_TIME      := CURRENT_TIME;   -- Capture ON time
    ON_FOB       := FUEL_ON_BOARD;  -- Capture fuel on board
    ENTERED_FLTNO := ENTERED_FLTNO; -- Break link with FMF flight number
    INIT_DEST_STA := INIT_DEST_STA; -- Break link with FMF destination
    ON_STA       := INIT_DEST_STA;  -- Capture On station
    DLY_SELECT   := 4;              -- Select arrival delay type

    if OOOI_ENABLED then
        send (BCG_OOOI_ON);        -- Send ON report
    end_if;

end_if;

-- -----
-- IN EVENT
-- -----
if OOOI_STATE changed to (IN) then

    if VALID(BRAKE_TIME) then       -- Capture IN time...
        IN_TIME := BRAKE_TIME;      -- either when brake was set...
    else
        IN_TIME := CURRENT_TIME;    -- or else use current time.
    end_if;

    IN_FOB      := FUEL_ON_BOARD;   -- Capture fuel on board
    DLY_SELECT   := 1;              -- Select departure delay type

```

```

if RETURN_TO_GATE then
    if OOOI_ENABLED then
        send (BCG_OOOI_RTG);          -- Send return to gate report
    end_if;
else
    -- Normal IN event
    EOF_TIMER := CURRENT_DATE_TIME + (590); -- Set end of flight timer to
                                           -- 9 minutes and 50 seconds

    if OOOI_ENABLED then
        send (BCG_OOOI_IN);          -- Send IN report
    end_if;

end_if;

end_if;

-- -----
-- Calculate flight and block times and fuels
-- -----
FLIGHT_TIME := ON_TIME - OFF_TIME;      -- Calculate flight time
if FLIGHT_TIME < 0 then
    FLIGHT_TIME := FLIGHT_TIME + 86400; -- Keep flight time positive
end_if;

FLIGHT_FUEL := OFF_FOB - ON_FOB;        -- Calculate flight fuel

BLOCK_TIME := IN_TIME - OUT_TIME;      -- Calculate block time
if BLOCK_TIME < 0 then
    BLOCK_TIME := BLOCK_TIME + 86400;   -- Keep block time positive
end_if;

BLOCK_FUEL := OUT_FOB - IN_FOB;        -- Calculate block fuel

-- -----
--
-- END

```

9.6.3 Out Report Downlink Encoding Definition

If the AMI detects a transition to the Out state, then it sends the following downlink encoding:

- ACARS SMI: M11
- ACARS Label: 11
- Allowable Subnetworks: All

- Downlink Priority: Flight Regularity
- Downlink Buffer: REPORT_OOOI
- Delete at System Reset: No
- Delete on Buffer Full: Yes

Table 9-12 describes the Out Report downlink encoding definition.

Table 9-12 Out Report Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	OUT	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	4	Always	OUT_TIME	"Out Time (HHMM)"
13	4	Always	OUT_FOB	"Out Fuel on Board / 100 (NNNN)"
14	6	Always	INIT_BOARDED_FUEL	"Boarded Fuel (NNNNNN)"

Sample Message

```
QU RTNBOCR
.DDLXCXA 221737
M11
FI BO64/AN N7873BO
DT DDL BFI 221737 M23A
- OUT01 BOE64/22221737KBFIPHOG
17370427080000
```

Note: In this sample message, the fuel on board at the Out event is "0427" (42,700) and the boarded fuel is "080000" (80,000).

9.6.4 Off Report Downlink Encoding Definition

If the AMI detects a transition to the Off state, then it sends the following downlink encoding:

- ACARS SMI: M12

- ACARS Label: 12
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REPORT_OOOI
- Delete at System Reset: No
- Delete on Buffer Full: Yes

Table 9-13 describes the Off Report downlink encoding definition.

Table 9-13 Off Report Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	OFF	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	4	Always	OFF_TIME	"Off Time (HHMM)"
13	4	Always	OFF_FOB	"Off Fuel on Board / 100 (NNNN)"

Sample Message

```

QU RTNBOCR
.DDLXCXA 221742
M12
FI BO64/AN N7873BO
DT DDL BFI 221742 M24A
- OFF01      BOE64/22221742KBFIPHOG
17420425

```

Note: In this sample message, the fuel on board at the Off event is "0425" (42,500).

9.6.5 On Report Downlink Encoding Definition

If the AMI detects a transition to the On state, then it sends the following downlink encoding:

- ACARS SMI: M13
- ACARS Label: 13
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REPORT_OOOI
- Delete at System Reset: No
- Delete on Buffer Full: Yes

Table 9-14 describes the On Report downlink encoding definition.

Table 9-14 On Report Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	ONN	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	4	Always	ON_TIME	"On Time (HHMM)"
13	4	Always	ON_FOB	"On Fuel on Board / 100 (NNNN)"

Sample Message

```

QU RTNBOCR
.DDLXCXA 221829
M13
FI BO64/AN N7873BO
DT DDL BFI 221829 M37A
- ONN01      BOE64/22221829KBFIPHOG
18290327

```

Note: In this sample message, the fuel on board at the On event is "0327" (32,700).

9.6.6 In Report Downlink Encoding Definition

If the AMI detects a transition to the In state, then it sends the following downlink encoding:

- ACARS SMI: M14
- ACARS Label: 14
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REPORT_OOOI
- Delete at System Reset: No
- Delete on Buffer Full: Yes

Table 9-15 describes the In Report downlink encoding definition.

Table 9-15 In Report Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	INN	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	4	Always	IN_TIME	"In Time (HHMM)"
13	4	Always	IN_FOB	"In Fuel on Board / 100 (NNNN)"

Note: IN_TIME is set to the most recent time that the parking brake was set prior to a door opening. If the parking brake was not set prior to a door opening, then IN_TIME is set to the time that the door opened. The In event and the In report downlink occur at the time the door opened.

Sample Message

```
QU RTNBOCR
.DDLXCXA 221831
M14
FI BO64/AN N7873BO
DT DDL BFI 221831 M38A
```

- INN01 BOE64/22221831KBFIPHOG
18310325

Note: In this sample message, the fuel on board at the In event is "0325" (32,500).

9.6.7 Return to Gate Report Downlink Encoding Definition

If the AMI detects a transition to the Return to Gate state (i.e., an In state following an Out state), then it sends the following downlink encoding:

- ACARS SMI: M15
- ACARS Label: 15
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: REPORT_OOOI
- Delete at System Reset: No
- Delete on Buffer Full: Yes

Table 9-16 describes the Return to Gate Report downlink encoding definition.

Table 9-16 Return to Gate Report Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	3	Always	RTG	"Message Identifier"
3	2	Always	01	"Message Version"
4	10	Always	ENTERED_FLTNO	"Flight Number (XXXXXXXXXX)"
5	1	Always	/	"Delimiter"
6	2	Always	INIT_SKED_DATE	"Scheduled Date (DD)"
7	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
8	4	Always	CURRENT_DATE_TIME	"Current Time (HHMM)"
9	4	Always	INIT_ORIG_STA	"Origin Station (XXXX)"
10	4	Always	INIT_DEST_STA	"Destination Station (XXXX)"
11	2	Always	<CR><LF>	"Delimiter"
12	4	Always	OUT_TIME	"Out Time (HHMM)"
13	4	Always	IN_TIME	"In Time (HHMM)"
14	4	Always	IN_FOB	"In Fuel on Board / 100 (NNNN)"

Note: IN_TIME is set to the most recent time that the parking brake was set prior to a door opening. If the parking brake was not set prior to a door opening, then IN_TIME is set to the time that the door opened. The In event and the In report downlink occur at the time the door opened.

Sample Message

QU RTNBOCR
.DDLXCXA 262212
M15
FI BO64/AN N7873BO
DT DDL BFI 262212 M06A
- RTG01 BOE64/26262212KBFIKSYR
220922120454

Note: In this sample message, the fuel on board at the In event is “0454” (45,400).

10 Reminders

The Reminders function enables the flight crew to set reminder messages that the AMI posts to NEW MESSAGES when certain conditions are met. For each reminder, the flight crew may enter a specific message to be posted when the condition is met. The posted message is annunciated by visual “•COMM” and aural high-low chime alerts and displays text and parameters indicating which condition has been met along with the specific message entered by the flight crew.

10.1 Display Definition

Figure 10-1 shows the Reminders display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	REMINDERS	XXXXXXXXX
REMINDER EVENT	REMINDER TEXT	
TIME 1:	--:--	-----
TIME 2:	--:--	-----
TIME 3:	--:--	-----
TIME TO T/D:	--+--	-----
TIME TO DEST:	--+--	-----
WAYPOINT:	-----	-----
LATITUDE:	-> --	-----
LONGITUDE:	-> --	-----
FUEL:	-----	-----
ALTITUDE:	-----	-----
FLIGHT LEVEL:	---	-----
WAYPOINT ETA CHANGE	FROM HH:MM BY - MINUTES	-----

RESET RETURN EXIT		
SCRATCHPAD		

Figure 10-1 Reminders Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “REMINDERS”

Main Area

Table 10-1 describes the Reminders display entry boxes in the reminder event column.

Table 10-1 Reminders Display Reminder Event Column Entry Boxes

Name	Mandatory	Length	Format	Range	Default	Parameter
TIME 1	No	4	HHMM			REMINDER_TIME_1
TIME 2	No	4	HHMM			REMINDER_TIME_2
TIME 3	No	4	HHMM			REMINDER_TIME_3
TIME TO T/D (top of descent)	No	4	HHMM			REMINDER_TIME_TOD
TIME TO DEST	No	4	HHMM			REMINDER_TIME_DEST
WAYPOINT	No	1-7	XXXXXXX			REMINDER_POSN
LATITUDE (direction)	No	1	A	N or S		REMINDER_LAT_DIR
LATITUDE (degrees)	No	2	NN	00-90		REMINDER_LAT
LONGITUDE (direction)	No	1	A	E or W		REMINDER_LONG_DIR
LONGITUDE (degrees)	No	3	NNN	000-180		REMINDER_LONG
FUEL	No	1-5	NNN.N	0.0-280.0		REMINDER_FUEL
ALTITUDE	No	1-5	NNNNN	0-49000		REMINDER_ALTITUDE
FLIGHT LEVEL	No	1-3	NNN	0-490		REMINDER_FL
WAYPOINT ETA CHANGE (by minutes)	No	1	N	1-9	3	REMINDER_ETA_CHANGE

Table 10-2 describes the Reminders display entry boxes in the reminder text column.

Table 10-2 Reminders Display Reminder Text Column Entry Boxes

Name	Mandatory	Length	Format	Range	Default	Parameter
TIME 1	No	1-24	XXX...		“CREW CHANGE”	REMINDER_TIME_TEXT_1
TIME 2	No	1-24	XXX...			REMINDER_TIME_TEXT_2
TIME 3	No	1-24	XXX...			REMINDER_TIME_TEXT_3
TIME TO T/D	No	1-24	XXX...			REMINDER_TIME_TEXT_TOD
TIME TO DEST	No	1-24	XXX...			REMINDER_TIME_TEXT_DEST
WAYPOINT	No	1-24	XXX...			REMINDER_POSN_TEXT
LATITUDE	No	1-24	XXX...			REMINDER_LAT_TEXT

Name	Mandatory	Length	Format	Range	Default	Parameter
LONGITUDE	No	1-24	XXX...			REMINDER_LONG_TEXT
FUEL	No	1-24	XXX...			REMINDER_FUEL_TEXT
ALTITUDE	No	1-24	XXX...			REMINDER_ALTITUDE_TEXT
FLIGHT LEVEL	No	1-24	XXX...			REMINDER_FL_TEXT
WAYPOINT ETA CHANGE	No	1-24	XXX...			REMINDER_ETA_CHANGE_TEXT

The Waypoint ETA Change Reminder data (including the current waypoint ETA) is displayed when the WAYPOINT ETA CHANGE NESB is selected.

Command Buttons

Table 10-3 describes the Reminders display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 10-3 Reminders Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-
4	RESET	Reset	Always	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

10.2 Logic Unit

The AMI implements a logic unit to generate reminder messages that the flight crew has set.

```

-- -----
-- This Logic Unit issues sidelinks as requested on the
-- REMINDERS page, using the text specified on that page
-- for each selection.
--
-- G.Sandell 7-28-2009
-- G.Sandell 12-22-2011      Corrected FMF_ETA calculations to add 24
--                           hours to the Baseline ETA if the ETA time
--                           is less than the current time.
--
--                           Removed "FROM" ETA Entry Box from
--                           Reminders Main display and adjusted logic.
--

```

```

--          Corrected Longitude Reminder by checking
--          that if the Reminder Longitude is not 180
--          degrees or zero degrees that the last and
--          current longitude values must both be
--          positive or negative. Added specific
--          check for when the target Longitude is
--          zero due to positive/negative check for
--          non 180 degree Longitudes.
--
--          In Latitude Reminder changed <= and >= to
--          < and > in the LAST_FMF_POS_LAT comparison.
--
--          Added 3rd and 4th UTC Timers.
--
-- G.Sandell 2-7-2012      Made corrections per 777 tests to
--                          longitude check and to move storage of old
--                          values closer to where they are used.
--
-- G.Sandell 7-17-2012    Made corrections to fix errors.
--
-- G.Sandell 4-17-2014    Added logic to reset targets to default
--                          when entry is invalid (to cover resets).
--
-- G.Sandell 5-13-2014    Revised ETA change logic to not alert 10
--                          seconds either side of waypoint sequence
--
-- G.Sandell 11-5-2014    Revised latitude and longitude passing
--                          logic to allow for invalid data
--
-- -----
-- VERSION C003
--
-- G.Sandell 1-7-2015     Revised latitude and longitude to allow
--                          for last position rounding from double to
--                          single-precision
-- -----
--
-- S Meisch 2018 SEP      Changes to conform with Boeing's Configuration
--                          control processes:
--                          - Logic Unit Name Changed to BCG_REMINDERS
--                          - Display Names Posted to New Messages were
--                          change to "BCG_REMINDERS_xxx"
-- -----
--
-- Version E001 M. Matyas 08-FEB-2019

```

-- Replaced time 4 reminder with flight level reminder.

--

-- -----

--

-- Version F001 M. Matyas 29-SEP-2020

-- Updated references to renamed display components.

--

-- -----

--

-- UDPs used specifically for this function:

-- REMINDER_SIDELINK_TIME1_B

-- REMINDER_SIDELINK_TIME2_B

-- REMINDER_SIDELINK_TIME3_B

-- REMINDER_SIDELINK_FL_B

-- REMINDER_SIDELINK_TIME_TOD_B

-- REMINDER_SIDELINK_TIME_DEST_B

-- REMINDER_SIDELINK_POSN_B

-- REMINDER_SIDELINK_LAT_B

-- REMINDER_SIDELINK_LONG_B

-- REMINDER_SIDELINK_FUEL_B

-- REMINDER_SIDELINK_ALT_B

-- REMINDER_SIDELINK_ETA_B

-- REMINDER_TIME_1

-- REMINDER_TIME_2

-- REMINDER_TIME_3

-- REMINDER_FL

-- REMINDER_TIME_1_DISPLAY

-- REMINDER_TIME_2_DISPLAY

-- REMINDER_TIME_3_DISPLAY

-- REMINDER_FL_DISPLAY

-- REMINDER_TIME_TOD

-- REMINDER_TIME_DEST

-- REMINDER_TIME_TOD_DISPLAY

-- REMINDER_TIME_DEST_DISPLAY

-- OLD_REMINDER_TIME_1

-- OLD_REMINDER_TIME_2

-- OLD_REMINDER_TIME_3

-- REMINDER_TIME_TEXT_1

-- REMINDER_TIME_TEXT_2

-- REMINDER_TIME_TEXT_3

-- REMINDER_FL_TEXT

-- REMINDER_TIME_TEXT_TOD

-- REMINDER_TIME_TEXT_DEST

-- REVISED_REMINDER_TIME_1

-- REVISED_REMINDER_TIME_2

-- REVISED_REMINDER_TIME_3

```
-- REVISED_REMINDER_FL
-- REVISED_REMINDER_TIME_DEST
-- REMINDER_POSN
-- REMINDER_POSN_DISPLAY
-- REMINDER_POSN_TEXT
-- REMINDER_FMF_LAST_WPT
-- REMINDER_FMF_ETA
-- REMINDER_LAT
-- REMINDER_LAT_DIR
-- REMINDER_LAT_DISPLAY
-- REMINDER_LAT_DIR_DISPLAY
-- REMINDER_LAT_TEXT
-- REVISED_LAT
-- LAST_NAV_POS_LAT
-- REMINDER_LONG
-- REMINDER_LONG_DIR
-- REMINDER_LONG_DISPLAY
-- REMINDER_LONG_DIR_DISPLAY
-- REMINDER_LONG_TEXT
-- REVISED_LONG
-- LAST_NAV_POS_LONG
-- REMINDER_FUEL
-- REMINDER_FUEL_DISPLAY
-- REMINDER_FUEL_TEXT
-- REMINDER_ALTITUDE
-- REMINDER_ALTITUDE_DISPLAY
-- REMINDER_ALTITUDE_TEXT
-- REMINDER_ENABLE_ETA_CHANGE
-- REMINDER_ETA_CHANGE
-- REMINDER_ETA_CHANGE_DISPLAY
-- REMINDER_ETA_CHANGE_TEXT
-- ETA_CHANGE
-- REMINDER_ETA_BASELINE
-- REMINDER_ETA_BASELINE_DISPLAY
-- FMF_ETA_DISPLAY
-- LAST_ACTIVE_WPT
--
-- ACPs also used:
-- CURRENT_TIME
-- CURRENT_DATE
-- CURRENT_DATE_TIME
-- FMF_ETA_TOP_OF_DESC
-- FMF_TIME_TO_DEST
-- FMF_LAST_WPT
-- FMF_ACTIVE_WPT
-- FMF_ETA
```

```

-- NAV_POS_LAT
-- NAV_POS_LON
-- TOTAL_FILTERED_FUEL_MASS
-- ADRF_PRESSALT_VOTED
--
-- -----
-- Clear targets when entries are invalid
-- -----

If not valid (REMINDER_TIME_1) and valid (OLD_REMINDER_TIME_1) then
    set_default (OLD_REMINDER_TIME_1);
    set_invalid (REVISED_REMINDER_TIME_1);
end_if;

If not valid (REMINDER_TIME_2) and valid (OLD_REMINDER_TIME_2) then
    set_default (OLD_REMINDER_TIME_2);
    set_invalid (REVISED_REMINDER_TIME_2);
end_if;

If not valid (REMINDER_TIME_3) and valid (OLD_REMINDER_TIME_3) then
    set_default (OLD_REMINDER_TIME_3);
    set_invalid (REVISED_REMINDER_TIME_3);
end_if;

If not valid (REMINDER_LAT_DIR) or not valid (REMINDER_LAT) then
    set_default (REVISED_LAT);
end_if;

If not valid (REMINDER_LONG_DIR) or not valid (REMINDER_LONG) then
    set_default (REVISED_LONG);
end_if;

-- -----
-- Look for the individual times (1, 2, 3).
-- -----
-- -----
-- Time 1
-- -----

If valid (REMINDER_TIME_1) then
    if REMINDER_TIME_1 changed or not valid (OLD_REMINDER_TIME_1) then
        if REMINDER_TIME_1 < CURRENT_TIME then
            REVISED_REMINDER_TIME_1 := REMINDER_TIME_1 + (CURRENT_DATE + 86400);
        else
            REVISED_REMINDER_TIME_1 := REMINDER_TIME_1 + CURRENT_DATE;
        end_if;
    end_if;
    OLD_REMINDER_TIME_1 := REMINDER_TIME_1;
    If CURRENT_DATE_TIME reached (REVISED_REMINDER_TIME_1) then
        REMINDER_SIDELINK_TIME1_B := REMINDER_TIME_TEXT_1;
        REMINDER_TIME_1_DISPLAY := REMINDER_TIME_1;
        post_message (BCG_RMDR_TIME1, visual_and_aural);
    end_if;
end_if;

```

```

        set_default (REMINDER_TIME_1);
        set_default (REMINDER_TIME_TEXT_1);
        set_default (OLD_REMINDER_TIME_1);
        set_invalid (REVISED_REMINDER_TIME_1);
    end_if;
end_if;

-- -----
-- Time 2
-- -----

If valid (REMINDER_TIME_2) then
    if REMINDER_TIME_2 changed or not valid (OLD_REMINDER_TIME_2) then
        if REMINDER_TIME_2 < CURRENT_TIME then
            REVISED_REMINDER_TIME_2 := REMINDER_TIME_2 + (CURRENT_DATE + 86400);
        else
            REVISED_REMINDER_TIME_2 := REMINDER_TIME_2 + CURRENT_DATE;
        end_if;
    end_if;
    OLD_REMINDER_TIME_2 := REMINDER_TIME_2;
    If CURRENT_DATE_TIME reached (REVISED_REMINDER_TIME_2) then
        REMINDER_SIDELINK_TIME2_B := REMINDER_TIME_TEXT_2;
        REMINDER_TIME_2_DISPLAY := REMINDER_TIME_2;
        post_message (BCG_RMDR_TIME2, visual_and_aural);
        set_default (REMINDER_TIME_2);
        set_default (REMINDER_TIME_TEXT_2);
        set_default (OLD_REMINDER_TIME_2);
        set_invalid (REVISED_REMINDER_TIME_2);
    end_if;
end_if;

-- -----
-- Time 3
-- -----

If valid (REMINDER_TIME_3) then
    if REMINDER_TIME_3 changed or not valid (OLD_REMINDER_TIME_3) then
        if REMINDER_TIME_3 < CURRENT_TIME then
            REVISED_REMINDER_TIME_3 := REMINDER_TIME_3 +
                (CURRENT_DATE + 86400);
        else
            REVISED_REMINDER_TIME_3 := REMINDER_TIME_3 + CURRENT_DATE;
        end_if;
    end_if;
    OLD_REMINDER_TIME_3 := REMINDER_TIME_3;
    If CURRENT_DATE_TIME reached (REVISED_REMINDER_TIME_3) then
        REMINDER_SIDELINK_TIME3_B := REMINDER_TIME_TEXT_3;
        REMINDER_TIME_3_DISPLAY := REMINDER_TIME_3;

```

```

        post_message (BCG_RMDR_TIME3, visual_and_aural);
        set_default (REMINDER_TIME_3);
        set_default (REMINDER_TIME_TEXT_3);
        set_default (OLD_REMINDER_TIME_3);
        set_invalid (REVISED_REMINDER_TIME_3);
    end_if;
end_if;

-- -----
-- Look for time to top of descent.
-- -----

If valid (REMINDER_TIME_TOD) then
    If FMF_ETA_TOP_OF_DESC <= REMINDER_TIME_TOD then
        REMINDER_SIDELINK_TIME_TOD_B := REMINDER_TIME_TEXT_TOD;
        REMINDER_TIME_TOD_DISPLAY := REMINDER_TIME_TOD;
        post_message (BCG_RMDR_TIME_TOD, visual_and_aural);
        set_default (REMINDER_TIME_TOD);
        set_default (REMINDER_TIME_TEXT_TOD);
    end_if;
end_if;

-- -----
-- Look for time to destination.
-- -----

If valid (REMINDER_TIME_DEST) then
    If FMF_TIME_TO_DEST <= REMINDER_TIME_DEST then
        REMINDER_SIDELINK_TIME_DEST_B := REMINDER_TIME_TEXT_DEST;
        REMINDER_TIME_DEST_DISPLAY := REMINDER_TIME_DEST;
        post_message (BCG_RMDR_TIME_DEST, visual_and_aural);
        set_default (REMINDER_TIME_DEST);
        set_default (REMINDER_TIME_TEXT_DEST);
    end_if;
end_if;

-- -----
-- Remove blank characters from FMF_LAST_WPT
-- -----

if valid (FMF_LAST_WPT) then

    REMINDER_FMF_LAST_WPT := "";
    REMINDER_FMF_LAST_WPT[1] := FMF_LAST_WPT[1];
    if FMF_LAST_WPT[2] != " " then
        REMINDER_FMF_LAST_WPT[2] := FMF_LAST_WPT[2];
        if FMF_LAST_WPT[3] != " " then
            REMINDER_FMF_LAST_WPT[3] := FMF_LAST_WPT[3];

```

```

        if FMF_LAST_WPT[4] != " " then
            REMINDER_FMF_LAST_WPT[4] := FMF_LAST_WPT[4];
            if FMF_LAST_WPT[5] != " " then
                REMINDER_FMF_LAST_WPT[5] := FMF_LAST_WPT[5];
                if FMF_LAST_WPT[6] != " " then
                    REMINDER_FMF_LAST_WPT[6] := FMF_LAST_WPT[6];
                    if FMF_LAST_WPT[7] != " " then
                        REMINDER_FMF_LAST_WPT[7] := FMF_LAST_WPT[7];
                    end_if;
                end_if;
            end_if;
        end_if;
    end_if;

else

    set_invalid (REMINDER_FMF_LAST_WPT);

end_if;

-- -----
-- Look for sequencing the entered waypoint.
-- -----

If valid (REMINDER_POSN) then
    If REMINDER_POSN = REMINDER_FMF_LAST_WPT then
        REMINDER_SIDE_LINK_POSN_B := REMINDER_POSN_TEXT;
        REMINDER_POSN_DISPLAY := REMINDER_POSN;
        post_message (BCG_RMDR_WPT_PASS, visual_and_aural);
        set_default (REMINDER_POSN);
        set_default (REMINDER_POSN_TEXT);
    end_if;
end_if;

-- -----
-- Look for passing the specified latitude.
-- -----

If valid (NAV_POS_LAT) then
    If valid (REMINDER_LAT_DIR) and valid (REMINDER_LAT) and
        ((REMINDER_LAT_DIR = "N") or (REMINDER_LAT_DIR = "S")) then
        If (REMINDER_LAT_DIR = "N") then REVISED_LAT := REMINDER_LAT;
        else REVISED_LAT := -REMINDER_LAT;
        end_if;
        If ((LAST_NAV_POS_LAT <= REVISED_LAT) and
            (NAV_POS_LAT >= REVISED_LAT)) or

```

```

        ((LAST_NAV_POS_LAT >= REVISED_LAT) and
        (NAV_POS_LAT <= REVISED_LAT)) then
    REMINDER_SIDELINK_LAT_B := REMINDER_LAT_TEXT;
    REMINDER_LAT_DIR_DISPLAY := REMINDER_LAT_DIR;
    REMINDER_LAT_DISPLAY := REMINDER_LAT;
    post_message (BCG_RMDR_LAT, visual_and_aural);
    set_default (REMINDER_LAT_DIR);
    set_default (REMINDER_LAT);
    set_default (REMINDER_LAT_TEXT);
end_if;
end_if;

-----

-- Set up for the next cycle
-----

    LAST_NAV_POS_LAT := NAV_POS_LAT;
end_if;

-----

-- Look for passing the specified longitude.
-----

If valid (NAV_POS_LON) then
    If valid (REMINDER_LONG_DIR) and valid (REMINDER_LONG)and
        ((REMINDER_LONG_DIR = "E") or (REMINDER_LONG_DIR = "W")) then
    If (REMINDER_LONG_DIR = "E") then REVISED_LONG := REMINDER_LONG;
    else REVISED_LONG := - REMINDER_LONG;
        If REVISED_LONG = -180 then
            REVISED_LONG := 180;
        end_if;
    end_if;
    If (((LAST_NAV_POS_LONG <= REVISED_LONG) and
        (NAV_POS_LON >= REVISED_LONG)) and
        ((LAST_NAV_POS_LONG > 0 AND NAV_POS_LON > 0) or
        (LAST_NAV_POS_LONG < 0 AND NAV_POS_LON < 0)))

    or

        (((LAST_NAV_POS_LONG >= REVISED_LONG) and
        (NAV_POS_LON <= REVISED_LONG)) and
        ((LAST_NAV_POS_LONG > 0 AND NAV_POS_LON > 0) or
        (LAST_NAV_POS_LONG < 0 AND NAV_POS_LON < 0)))

    or

        (REVISED_LONG = 180 and
        (((LAST_NAV_POS_LONG < -179) and (NAV_POS_LON > 179)) or

```

```

        ((LAST_NAV_POS_LONG > 179) and (NAV_POS_LON < -179)) or
        (NAV_POS_LON = 180) or (LAST_NAV_POS_LONG = 180)))

or

        (REVISED_LONG = 0 and
        (((LAST_NAV_POS_LONG <= 0) and (NAV_POS_LON >= 0)) or
        ((LAST_NAV_POS_LONG >= 0) and (NAV_POS_LON <= 0)))) then

        REMINDER_SIDELINK_LONG_B := REMINDER_LONG_TEXT;
        REMINDER_LONG_DIR_DISPLAY := REMINDER_LONG_DIR;
        REMINDER_LONG_DISPLAY := REMINDER_LONG;
        post_message (BCG_RMDR_LONG, visual_and_aural);
        set_default (REMINDER_LONG_DIR);
        set_default (REMINDER_LONG);
        set_default (REMINDER_LONG_TEXT);
    end_if;
end_if;

-- -----
-- Set up for the next cycle
-- -----

        LAST_NAV_POS_LONG := NAV_POS_LON;
end_if;

-- -----
-- Look for reaching the specified fuel load.
-- -----

If valid (REMINDER_FUEL) then
    If (REMINDER_FUEL >= TOTAL_FILTERED_FUEL_MASS) then
        REMINDER_SIDELINK_FUEL_B := REMINDER_FUEL_TEXT;
        REMINDER_FUEL_DISPLAY := REMINDER_FUEL;
        post_message (BCG_RMDR_FUEL, visual_and_aural);
        set_default (REMINDER_FUEL);
        set_default (REMINDER_FUEL_TEXT);
    end_if;
end_if;

-- -----
-- Look for reaching the specified altitude. Note that this is
-- based on non-baro-corrected pressure altitude.
-- -----

If valid (REMINDER_ALTITUDE) then
    If ADRF_PRESSALT_VOTED >= REMINDER_ALTITUDE - 150 and
        ADRF_PRESSALT_VOTED <= REMINDER_ALTITUDE + 150 then

```

```

    REMINDER_SIDELINK_ALT_B := REMINDER_ALTITUDE_TEXT;
    REMINDER_ALTITUDE_DISPLAY := REMINDER_ALTITUDE;
    post_message (BCG_RMDR_ALT, visual_and_aural);
    set_default (REMINDER_ALTITUDE);
    set_default (REMINDER_ALTITUDE_TEXT);
end_if;
end_if;

-- -----
-- Look for reaching the specified flight level.
-- Note that this is based on non-baro-corrected pressure
-- altitude.
-- -----

If valid (REMINDER_FL) then
    REMINDER_FL_ALT := REMINDER_FL * 100;

    If ADRF_PRESSALT_VOTED >= REMINDER_FL_ALT - 150 and
        ADRF_PRESSALT_VOTED <= REMINDER_FL_ALT + 150 then
        REMINDER_SIDELINK_FL_B := REMINDER_FL_TEXT;
        REMINDER_FL_DISPLAY := REMINDER_FL;
        post_message (BCG_RMDR_FL, visual_and_aural);
        set_default (REMINDER_FL);
        set_default (REMINDER_FL_TEXT);
    end_if;

end_if;

-- -----
-- Set up the ETA value to compare to the new FMS prediction.
-- -----

If REMINDER_ENABLE_ETA_CHANGE then
-- Invalidate baseline when waypoint is sequenced
    If FMF_ACTIVE_WPT != LAST_ACTIVE_WPT then
        set_invalid (REMINDER_ETA_BASELINE);
        TIME_AT_WPT_CHG := CURRENT_DATE_TIME;
    end_if;
-- Set baseline if FMS-predicted ETA was previously invalid
-- and more than 5 seconds since waypoint sequenced
    If valid (FMF_ETA) and not valid (REMINDER_ETA_BASELINE) and
        ((CURRENT_DATE_TIME > (TIME_AT_WPT_CHG + 5)) or
        not valid (TIME_AT_WPT_CHG)) then
        If FMF_ETA < CURRENT_TIME then
            REMINDER_ETA_BASELINE := FMF_ETA + 86400 + CURRENT_DATE;
        else
            REMINDER_ETA_BASELINE := FMF_ETA + CURRENT_DATE;
        end_if;
    end_if;

```

```

        If not valid (TIME_AT_WPT_CHG) then
            TIME_AT_WPT_CHG := CURRENT_DATE_TIME;
        end_if;
    end_if;
end_if;

-- -----
-- Set up for the next cycle
-- -----

LAST_ACTIVE_WPT := FMF_ACTIVE_WPT;

-- -----
-- Look for the specified change in FMS-predicted ETA
-- -----

If REMINDER_ENABLE_ETA_CHANGE and valid (FMF_ETA)
and valid (REMINDER_ETA_BASELINE) then
    If FMF_ETA < (CURRENT_TIME - 5) then
        REMINDER_FMF_ETA := FMF_ETA + 86400 + CURRENT_DATE;
    else
        REMINDER_FMF_ETA := FMF_ETA + CURRENT_DATE;
    end_if;

-- See if near waypoint sequence (before or after)
TIME_SINCE_WPT_CHG := CURRENT_DATE_TIME - TIME_AT_WPT_CHG;
TIME_TO_WPT_CHG := REMINDER_FMF_ETA - CURRENT_DATE_TIME;
If (TIME_SINCE_WPT_CHG > 10) and (TIME_TO_WPT_CHG > 10) then

    ETA_CHANGE := REMINDER_FMF_ETA - REMINDER_ETA_BASELINE;

    If ETA_CHANGE < 0 then
        ETA_CHANGE := - ETA_CHANGE;
    end_if;
    If ETA_CHANGE >= (REMINDER_ETA_CHANGE*60) then
        REMINDER_SIDELINK_ETA_B := REMINDER_ETA_CHANGE_TEXT;
        REMINDER_ETA_CHANGE_DISPLAY := REMINDER_ETA_CHANGE;
        REMINDER_ETA_BASELINE_DISPLAY := REMINDER_ETA_BASELINE;
        FMF_ETA_DISPLAY := FMF_ETA;
        post_message (BCG_RMDR_WPT_ETA, visual_and_aural);
        set_invalid (REMINDER_ETA_BASELINE);
    end_if;
end_if;

-- Clear up if pilot has turned ETA change monitoring off
If not REMINDER_ENABLE_ETA_CHANGE then
    set_default (REMINDER_ETA_CHANGE_TEXT);

```

```

set_default (REMINDER_ETA_BASELINE);
set_default (REMINDER_ETA_CHANGE);
set_default (TIME_AT_WPT_CHG);
end_if;

```

```

--
-- END

```

10.3 Altitude Reminder Display Definition

Figure 10-2 shows the Altitude Reminder display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	REMINDER ALTITUDE	XXXXXXXXXX
ALTITUDE: NNNNN XXXXXXXXXXXXXXXXXXXXXXXXXXXX		
		CANCEL
SCRATCHPAD		

Figure 10-2 Altitude Reminder Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “REMINDER ALTITUDE”

Main Area

The main area contains:

- A parameter displaying the reminder altitude
- The message text entered when setting the reminder

Table 10-4 describes the Altitude Reminder display parameters.

Table 10-4 Altitude Reminder Display Parameters

Name	Length	Format	Range	Parameter
ALTITUDE	1-5	NNNNN	0-49000	REMINDER_ALTITUDE_DISPLAY
Reminder text	1-24	XXX...		REMINDER_SIDELINK_ALT_B

Command Buttons

Table 10-5 describes the Altitude Reminder display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 10-5 Altitude Reminder Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-
4	-	-	-	-
5	-	-	-	-
6	CANCEL	Cancel	Always	-

10.4 Flight Level Reminder Display Definition

Figure 10-3 shows the Flight Level Reminder display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	REMINDER FLIGHT LEVEL	XXXXXXXXXX
FLIGHT LEVEL: NNN XXXXXXXXXXXXXXXXXXXXXXXXXXXX		
SCRATCHPAD		CANCEL

Figure 10-3 Flight Level Reminder Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “REMINDER FLIGHT LEVEL”

Main Area

The main area contains:

- A parameter displaying the reminder flight level
- The message text entered when setting the reminder

Table 10-6 describes the Flight Level Reminder display parameters.

Table 10-6 Flight Level Reminder Display Parameters

Name	Length	Format	Range	Parameter
FLIGHT LEVEL	3	NNN	000-490	REMINDER_FL_DISPLAY
Reminder text	1-24	XXX...		REMINDER_SIDELINK_FL_B

Command Buttons

Table 10-7 describes the flight level reminder display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 10-7 Flight Level Reminder Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-
4	-	-	-	-
5	-	-	-	-
6	CANCEL	Cancel	Always	-

10.5 Fuel Reminder Display Definition

Figure 10-4 shows the Fuel Reminder display definition.

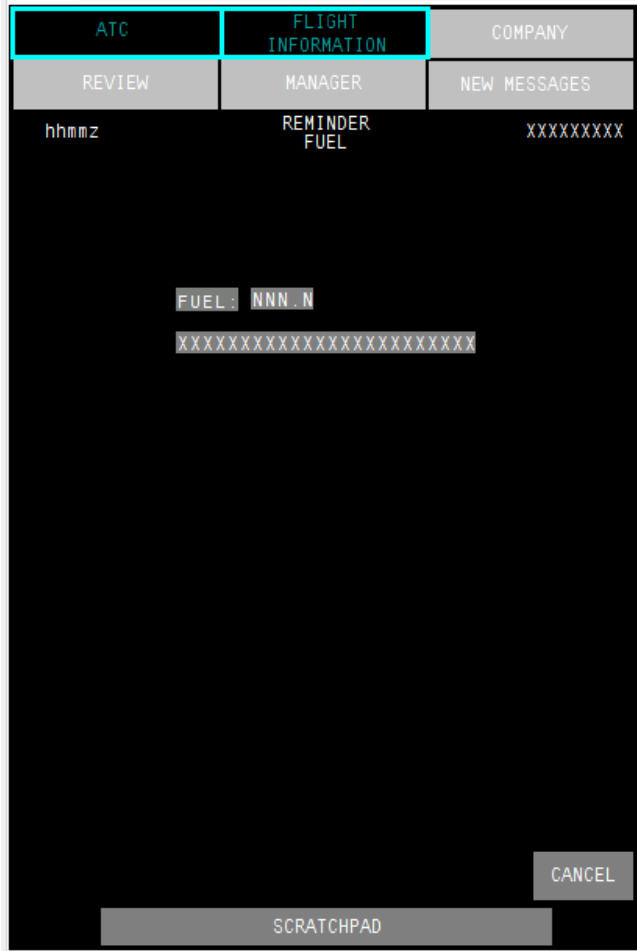


Figure 10-4 Fuel Reminder Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “REMINDER FUEL”

Main Area

The main area contains:

- A parameter displaying the reminder fuel on board
- The message text entered when setting the reminder

Table 10-8 describes the Fuel Reminder display parameters.

Table 10-8 Fuel Reminder Display Parameters

Name	Length	Format	Range	Parameter
FUEL	3-5	NNN.N	0.0-280.0	REMINDER_FUEL_DISPLAY
Reminder text	1-24	XXX...		REMINDER_SIDEINK_FUEL_B

Command Buttons

Table 10-9 describes the Fuel Reminder display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 10-9 Fuel Reminder Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-
4	-	-	-	-
5	-	-	-	-
6	CANCEL	Cancel	Always	-

10.6 Latitude Reminder Display Definition

Figure 10-5 shows the Latitude Reminder display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	REMINDER LATITUDE	XXXXXXXXXX
LATITUDE: NNN XXXXXXXXXXXXXXXXXXXXXXXXXXXX		
		CANCEL
SCRATCHPAD		

Figure 10-5 Latitude Reminder Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “REMINDER LATITUDE”

Main Area

The main area contains:

- A parameter displaying the reminder latitude
- The message text entered when setting the reminder

Table 10-10 describes the Latitude Reminder display parameters.

Table 10-10 Latitude Reminder Display Parameters

Name	Length	Format	Range	Parameter
LATITUDE	1	A	N or S	REMINDER_LAT_DIR_DISPLAY
	2	NN	00-90	REMINDER_LAT_DISPLAY
Reminder text	1-24	XXX...		REMINDER_SIDELINK_LAT_B

Command Buttons

Table 10-11 describes the Latitude Reminder display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 10-11 Latitude Reminder Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-
4	-	-	-	-
5	-	-	-	-
6	CANCEL	Cancel	Always	-

10.7 Longitude Reminder Display Definition

Figure 10-6 shows the Longitude Reminder display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	REMINDER LONGITUDE	XXXXXXXXXX
LONGITUDE: E N N N XXXXXXXXXXXXXXXXXXXXXXXXXXXX		
		CANCEL
SCRATCHPAD		

Figure 10-6 Longitude Reminder Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “REMINDER LONGITUDE”

Main Area

The main area contains:

- A parameter displaying the reminder longitude
- The message text entered when setting the reminder

Table 10-12 describes the Longitude Reminder display parameters.

Table 10-12 Longitude Reminder Display Parameters

Name	Length	Format	Range	Parameter
LONGITUDE	1	A	E or W	REMINDER_LONG_DIR_DISPLAY
	3	NNN	000-180	REMINDER_LONG_DISPLAY
Reminder text	1-24	XXX...		REMINDER_SIDELINK_LONG_B

Command Buttons

Table 10-13 describes the Longitude Reminder display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 10-13 Longitude Reminder Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-
4	-	-	-	-
5	-	-	-	-
6	CANCEL	Cancel	Always	-

10.8 Passing Waypoint Reminder Display Definition

Figure 10-7 shows the Passing Waypoint Reminder display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	REMINDER PASSING WAYPOINT	XXXXXXXXXX
PASSING WAYPOINT: XXXXXX XXXXXXXXXXXXXXXXXXXXXXXXXXXX		
		CANCEL
SCRATCHPAD		

Figure 10-7 Passing Waypoint Reminder Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “REMINDER PASSING WAYPOINT”

Main Area

The main area contains:

- A parameter displaying the reminder waypoint
- The message text entered when setting the reminder

Table 10-14 describes the Passing Waypoint Reminder display parameters.

Table 10-14 Passing Waypoint Reminder Display Parameters

Name	Length	Format	Range	Parameter
PASSING WAYPOINT	1-7	XXXXXXX		REMINDER_POSN_DISPLAY
Reminder text	1-24	XXX...		REMINDER_SIDELINK_POSN_B

Command Buttons

Table 10-15 describes the Passing Waypoint Reminder display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 10-15 Passing Waypoint Reminder Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-
4	-	-	-	-
5	-	-	-	-
6	CANCEL	Cancel	Always	-

10.9 Time Reminder Display Definition

Figure 10-8 shows the Time Reminder display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	REMINDER TIME 1	XXXXXXXXXX
TIME 1: HH:MM XXXXXXXXXXXXXXXXXXXXXXXXXXXX		
		CANCEL
SCRATCHPAD		

Figure 10-8 Time Reminder Display Definition

Note: A total of three reminder times can be set.

Title

The title area contains:

- The current time in HHMMz format
- “REMINDER TIME 1 [or 2 or 3]”

Main Area

The main area contains:

- A parameter displaying the reminder time
- The message text entered when setting the reminder

Table 10-16 describes the Time Reminder display parameters.

Table 10-16 Time Reminder Display Parameters

Name	Length	Format	Range	Parameter
TIME 1 [or 2 or 3]	5	HH:MM		REMINDER_TIME_[1/2/3]_DISPLAY
Reminder text	1-24	XXX...		REMINDER_SIDEINK_TIME[1/2/3]_B

Command Buttons

Table 10-17 describes the Time Reminder display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 10-17 Time Reminder Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-
4	-	-	-	-
5	-	-	-	-
6	CANCEL	Cancel	Always	-

10.10 Time to Destination Reminder Display Definition

Figure 10-9 shows the Time to Destination Reminder display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	REMINDER TIME TO DESTINATION	XXXXXXXXXX
TIME TO DESTINATION: HH+MM XXXXXXXXXXXXXXXXXXXXXXXXXXXX		
		CANCEL
SCRATCHPAD		

Figure 10-9 Time to Destination Reminder Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “REMINDER TIME TO DESTINATION”

Main Area

The main area contains:

- A parameter displaying the reminder time to destination
- The message text entered when setting the reminder

Table 10-18 describes the Time to Destination Reminder display parameters.

Table 10-18 Time to Destination Reminder Display Parameters

Name	Length	Format	Range	Parameter
TIME TO DESTINATION	5	HH+MM		REMINDER_TIME_DEST_DISPLAY
Reminder text	1-24	XXX...		REMINDER_SIDEINK_TIME_DEST_B

Command Buttons

Table 10-19 describes the Time to Destination Reminder display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 10-19 Time to Destination Reminder Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-
4	-	-	-	-
5	-	-	-	-
6	CANCEL	Cancel	Always	-

10.11 Time to Top of Descent Reminder Display Definition

Figure 10-10 shows the Time to Top of Descent Reminder display definition.

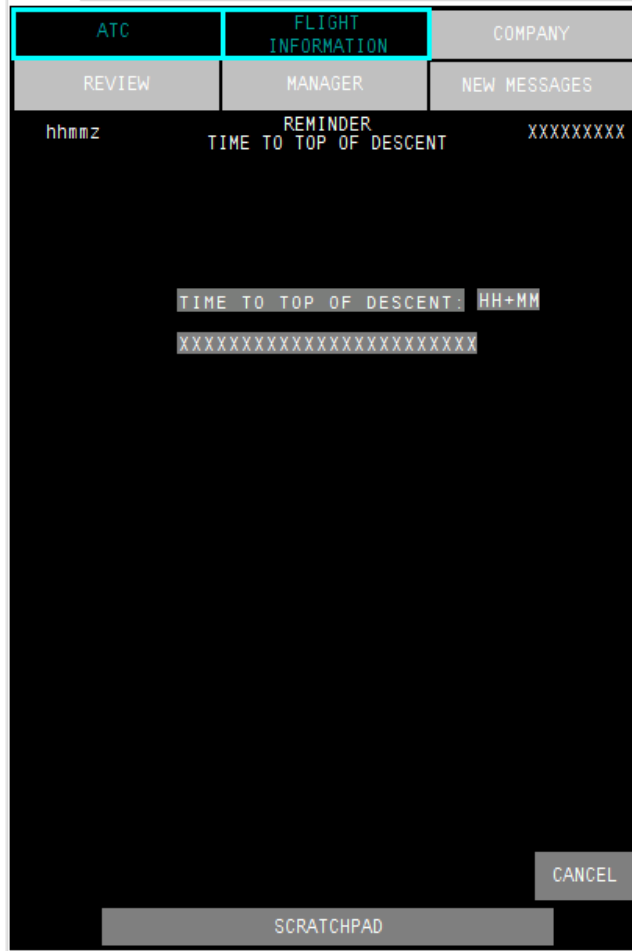


Figure 10-10 Time to Top of Descent Reminder Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “REMINDER TIME TO TOP OF DESCENT”

Main Area

The main area contains:

- A parameter displaying the reminder time to top of descent
- The message text entered when setting the reminder

Table 10-20 describes the Time to Top of Descent Reminder display parameters.

Table 10-20 Time to Top of Descent Reminder Display Parameters

Name	Length	Format	Range	Parameter
TIME TO TOP OF DESCENT	5	HH+MM		REMINDER_TIME_TOD_DISPLAY
Reminder text	1-24	XXX...		REMINDER_SIDEINK_TIME_TOD_B

Command Buttons

Table 10-21 describes the Time to Top of Descent Reminder display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 10-21 Time to Top of Descent Reminder Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-
4	-	-	-	-
5	-	-	-	-
6	CANCEL	Cancel	Always	-

10.12 Waypoint ETA Change Reminder Display Definition

Figure 10-11 shows the Waypoint ETA Change Reminder display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	REMINDER WAYPOINT ETA CHANGE	XXXXXXXXXX
WAYPOINT ETA CHANGE FROM: HH:MM TO: HH:MM XXXXXXXXXXXXXXXXXXXXXXXXXXXX		
		CANCEL
SCRATCHPAD		

Figure 10-11 Waypoint ETA Change Reminder Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “REMINDER WAYPOINT ETA CHANGE”

Main Area

The main area contains:

- “WAYPOINT ETA CHANGE”
- Parameters displaying the original ETA and the reminder ETA
- The message text entered when setting the reminder

Table 10-22 describes the Waypoint ETA Change Reminder display parameters.

Table 10-22 Waypoint ETA Change Reminder Display Parameters

Name	Length	Format	Range	Parameter
FROM	5	HH:MM		REMINDER_ETA_BASELINE_DISPLAY
TO	5	HH:MM		FMF_ETA_DISPLAY
Reminder text	1-24	XXX...		REMINDER_SIDELINK_ETA_B

Command Buttons

Table 10-23 describes the Waypoint ETA Change Reminder display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 10-23 Waypoint ETA Change Reminder Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	-	-	-	-
4	-	-	-	-
5	-	-	-	-
6	CANCEL	Cancel	Always	-

11 Loop Back Messages

The AMI implements several loop back message (also known as command-response message) functions.

11.1 Automatic Emergency Report Status Loop Back Messages

The Automatic Emergency Report Status Uplink enables the aircraft operator to request the AMI to send the Automatic Emergency Report Status Downlink and to set the function to the Armed, On, or Off state. The uplink is not displayed to the flight crew.

11.1.1 Uplink Decoding Definition

The decoding for the Automatic Emergency Report Status Uplink is defined as follows:

- ACARS SMI: CMD
- ACARS Label: RA
- Message Identifier: ~4EMG
- Alert Level: None
- Optional Element Delimiter: None

Table 11-1 describes the Automatic Emergency Report Status Uplink decoding definition.

Table 11-1 Automatic Emergency Report Status Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	9	Yes	RA4_GND_ADDR	"Priority & address (XXXXXXXXXX)"
2	5	Yes	<none>	"Message identifier (~4EMG)"
3	1	No	EMERGENCY_SELECTION	"Automatic emergency report selection (1, 2, or 3)"

If the optional automatic emergency report selection value is included, then the AMI selects the corresponding state for the Automatic Emergency Report function:

- 1: Automatic Emergency Report Function On
- 2: Automatic Emergency Report Function Armed
- 3: Automatic Emergency Report Function Off

When the AMI receives this uplink it sends the Automatic Emergency Report Status Downlink.

Sample Message

```
QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 019A
- QURTNBOCR~4EMG
```

Note: In this sample message the automatic emergency report selection value is omitted.

11.1.2 Downlink Encoding Definition

The encoding for the Automatic Emergency Report Status Downlink is defined as follows:

- ACARS SMI: RDO
- ACARS Label: RB
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_AUTO
- Delete at System Reset: No
- Delete on Buffer Full: Yes

Table 11-2 describes the Automatic Emergency Report Status Downlink encoding definition.

Table 11-2 Automatic Emergency Report Status Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	9	Always	RA4_GND_ADDR	"Priority & address (XXXXXXXXX)"
3	1	Always	/	"Delimiter"
4	5	Always	~4EMG	"Message Identifier"
5	1	Always	<SP>	"Delimiter"
6	18	Conditionally	EMERGENCY MODE OFF	"EMERGENCY_SELECTION = 3"
7	18	Conditionally	EMERGENCY MODE ARM	"EMERGENCY_SELECTION = 2"
8	17	Conditionally	EMERGENCY MODE ON	"EMERGENCY_SELECTION = 1"
9	2	Always	<CR><LF>	"Delimiter"
10	1-5	Always	EMERGENCY_DESRATE_THRESHOLD	"Descent Rate Threshold (NNNNN)"
11	1	Always	,	"Delimiter"
12	4	Always	EMERGENCY_MMO_THRESHOLD	"Above M _{MO} Threshold (N.NN)"
13	1	Always	,	"Delimiter"
14	1-3	Always	EMERGENCY_VMO_THRESHOLD	"Above V _{MO} Threshold (NNN)"

#	Length	Inclusion	Content	Element Description
15	1	Always	,	"Delimiter"
16	1-3	Always	EMERGENCY_LOW_SPEED_THRESHOLD	"CAS Low Threshold (NNN)"
17	1	Always	,	"Delimiter"
18	1-2	Always	EMERGENCY_PITCH_THRESHOLD	"Pitch Angle Threshold (NN)"
19	1	Always	,	"Delimiter"
20	1-2	Always	EMERGENCY_PITCH_RATE_THRESHOLD	"Pitch Rate Threshold (NN)"
21	1	Always	,	"Delimiter"
22	1-2	Always	EMERGENCY_YAW_RATE_THRESHOLD	"Yaw Rate Threshold (NN)"
23	1	Always	,	"Delimiter"
24	3	Always	EMERGENCY_VERT_ACCEL_THRESHOLD	"Vertical Acceleration Threshold (N.N)"
25	1	Always	,	"Delimiter"
26	1-5	Always	EMERGENCY_FUEL_THRESHOLD	"Fuel Threshold (NNNNN)"
27	1	Always	,	"Delimiter"
28	1-2	Always	EMERGENCY_TPR_THRESHOLD	"TPR Threshold (NN)"
29	1	Always	,	"Delimiter"
30	1-2	Always	EMERGENCY_N1_THRESHOLD	"N1 Threshold (NN)"
31	1	Always	,	"Delimiter"
32	2-3	Always	EMERGENCY_FREQUENCY	"Reporting Interval (NNN)"
33	1	Always	,	"Delimiter"
34	2-3	Always	MAX_EMERGENCY_RPT	"Reporting Time Limit (NNN)"
35	1	Always	,	"Delimiter"
36	1-5	Always	EMERGENCY_MIN_ALTITUDE	"Reporting Min Altitude (NNNNN)"
37	1	Always	,	"Delimiter"
38	1-5	Always	EMERGENCY_CLBRATE_THRESHOLD	"Climb Rate Threshold (NNNNN)"

The following ranges and units apply to the following parameters:

- Descent rate threshold: 0-20000 feet per minute
- Above M_{MO} threshold: 0.00-0.30 Mach
- Above V_{MO} threshold: 0-100 knots CAS
- CAS low threshold: 0-200 knots CAS
- Pitch angle threshold: 0-45 degrees
- Pitch rate threshold: 0-30 degrees per second
- Yaw rate threshold: 0-30 degrees per second
- Vertical acceleration threshold: 1.0-5.0g
- Fuel threshold: 0-20000 pounds
- TPR threshold: 0-25

- N1 threshold: 0-25%
- Reporting interval: 60-900 seconds
- Reporting time limit: 30-360 minutes
- Reporting minimum altitude: 0-30000 feet
- Climb rate threshold: 0-20000 feet per minute

Sample Message

```
QU RTNBOCR
.DDLXCXA 221920
RDO
FI BO24/AN N7873BO
DT DDL BFI 221920 M54A
- QURTNBOCR/~4EMG EMERGENCY MODE ARM
6000,0.10,40,115,15,15,15,3.0,5000,10,10,120,30,7500,6000
```

11.2 OOOI Data Loop Back Messages

The OOOI Data Uplink enables the aircraft operator to request the AMI to send the OOOI Data Downlink. The uplink is not displayed to the flight crew.

11.2.1 Uplink Decoding Definition

The decoding for the OOOI Data Uplink is defined as follows:

- ACARS SMI: CMD
- ACARS Label: RA
- Message Identifier: ~2OOI
- Alert Level: None
- Optional Element Delimiter: None

Table 11-3 describes the OOOI Data Uplink decoding definition.

Table 11-3 OOOI Data Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	9	Yes	RA2_GND_ADDR	"Priority & address (XXXXXXXXXX)"
2	5	Yes	<none>	"Message identifier (~2OOI)"

When the AMI receives this uplink it sends the OOOI Data Downlink.

Sample Message

QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 020A
- QURTNBOCR~200I

11.2.2 Downlink Encoding Definition

The encoding for the OOOI Data Downlink is defined as follows:

- ACARS SMI: RDO
- ACARS Label: RB
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_AUTO
- Delete at System Reset: No
- Delete on Buffer Full: Yes

Table 11-4 describes the OOOI data encoding definition.

Table 11-4 OOOI Data Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	9	Always	RA2_GND_ADDR	"Priority & address (XXXXXXXXXX)"
3	1	Always	/	"Delimiter"
4	5	Always	~200I	"Message Identifier"
5	1	Always	<SP>	"Delimiter"
6	1-10	Always	ENTERED_FLTNO	"Current Flight Number (1-10 characters)"
7	1	Always	,	"Delimiter"
8	3-4	Always	INIT_ORIG_STA	"Current Origin Station (3-4 characters)"
9	1	Always	,	"Delimiter"
10	3-4	Always	INIT_DEST_STA	"Current Destination Station (3-4 characters)"
11	1	Always	,	"Delimiter"
12	1-6	Always	INIT_BOARDED_FUEL	"Current Boarded Fuel (NNNNNN)"
13	1	Always	,	"Delimiter"

#	Length	Inclusion	Content	Element Description
14	4	Always	OUT_TIME	"Current Out Time (HHMM)"
15	1	Always	,	"Delimiter"
16	3-5	Always	OUT_FOB	"Current Out Fuel / 1000 (NNN.N)"
17	1	Always	,	"Delimiter"
18	4	Always	OFF_TIME	"Current Off Time (HHMM)"
19	1	Always	,	"Delimiter"
20	3-5	Always	OFF_FOB	"Current Off Fuel / 1000 (NNN.N)"
21	1	Always	,	"Delimiter"
22	4	Always	ON_TIME	"Current On Time (HHMM)"
23	1	Always	,	"Delimiter"
24	3-5	Always	ON_FOB	"Current On Fuel / 1000 (NNN.N)"
25	1	Always	,	"Delimiter"
26	4	Always	IN_TIME	"Current In Time (HHMM)"
27	1	Always	,	"Delimiter"
28	3-5	Always	IN_FOB	"Current In Fuel / 1000 (NNN.N)"
29	2	Always	<CR><LF>	"Delimiter"
30	1-10	Always	PREV_FLIGHT_NUMBER	"Previous Flight Number (1-10 characters)"
31	1	Always	,	"Delimiter"
32	3-4	Always	PREV_ORIG_STA	"Previous Origin Station (3-4 characters)"
33	1	Always	,	"Delimiter"
34	3-4	Always	PREV_DEST_STA	"Previous Destination Station (3-4 characters)"
35	1	Always	,	"Delimiter"
36	1-6	Always	PREV_BOARDED_FUEL	"Previous Boarded Fuel (NNNNNN)"
37	1	Always	,	"Delimiter"
38	4	Always	PREV_OUT_TIME	"Previous Out Time (HHMM)"
39	1	Always	,	"Delimiter"
40	3-5	Always	PREV_OUT_FOB	"Previous Out Fuel / 1000 (NNN.N)"
41	1	Always	,	"Delimiter"
42	4	Always	PREV_OFF_TIME	"Previous Off Time (HHMM)"
43	1	Always	,	"Delimiter"
44	3-5	Always	PREV_OFF_FOB	"Previous Off Fuel / 1000 (NNN.N)"
45	1	Always	,	"Delimiter"
46	4	Always	PREV_ON_TIME	"Previous On Time (HHMM)"
47	1	Always	,	"Delimiter"
48	3-5	Always	PREV_ON_FOB	"Previous On Fuel / 1000 (NNN.N)"
49	1	Always	,	"Delimiter"
50	4	Always	PREV_IN_TIME	"Previous In Time (HHMM)"

#	Length	Inclusion	Content	Element Description
51	1	Always	,	"Delimiter"
52	3-5	Always	PREV_IN_FOB	"Previous In Fuel / 1000 (NNN.N)"

Sample Message

```

QU RTNBOCR
.DDLXCXA 221707
RDO
FI BO24/AN N7873BO
DT DDL BFI 221707 M21A
- QURTNBOCR/~200I
BOE24,KBFI,KBFI,9876,1730,55.9,1734,55.8,1751,52.6,1751,52.9
BOE718,KBFI,KBFI,5432,1635,45.9,1639,45.8,1656,42.6,1656,42.9

```

11.3 Position Report Loop Back Messages

The Position Report Uplink enables the aircraft operator to request the AMI to send the Position Report Downlink. The uplink is not displayed to the flight crew.

11.3.1 Uplink Decoding Definition

The decoding for the Position Report Uplink is defined as follows:

- ACARS SMI: CMD
- ACARS Label: RA
- Message Identifier: ~4POS
- Alert Level: None
- Optional Element Delimiter: None

Table 11-5 describes the Position Report Uplink decoding definition.

Table 11-5 Position Report Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	9	Yes	RA4_GND_ADDR	"Priority & address (XXXXXXXXXX)"
2	5	Yes	<none>	"Message identifier (~4POS)"

When the AMI receives this uplink it sends the Position Report Downlink.

Sample Message

QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 021A
- QURTNBOCR~4POS

11.3.2 Downlink Encoding Definition

The encoding for the Position Report Downlink is defined as follows:

- ACARS SMI: RDO
- ACARS Label: RB
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_AUTO
- Delete at System Reset: No
- Delete on Buffer Full: Yes

Table 11-6 describes the Position Report Downlink encoding definition.

Table 11-6 Position Report Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	9	Always	RA4_GND_ADDR	"Priority & address (XXXXXXXXXX)"
3	1	Always	/	"Delimiter"
4	5	Always	~4POS	"Message Identifier"
5	1	Always	<SP>	"Delimiter"
6	13	Always	AIRCRAFT_POSITION	"Position (XDDMMTXDDMMT)"
7	1	Always	/	"Delimiter"
8	1-5	Always	NAV_ALTITUDE	"Altitude (NNNNN)"
9	1	Always	/	"Delimiter"
10	4	Always	NAV_MAG_HDG	"Magnetic Heading (+/-NNN)"
11	1	Always	/	"Delimiter"
12	1-7	Always	FMF_LAST_WPT	"Last Waypoint (1-7 characters)"
13	1	Always	/	"Delimiter"
14	1-7	Always	FMF_ACTIVE_WPT	"Active Waypoint (1-7 characters)"
15	1	Always	/	"Delimiter"

#	Length	Inclusion	Content	Element Description
16	1-7	Always	FMF_NEXT_WPT	"Next Waypoint (1-7 characters)"
17	1	Always	/	"Delimiter"
18	2	Always	CURRENT_DATE_TIME	"Current Date (DD)"
19	6	Always	CURRENT_DATE_TIME	"Current Time (HHMMSS)"

Note: Headings, tracks, and bearings (i.e., the NAV_MAG_HDG, NAV_MAG_TRACK, NAV_TRUE_HDG, NAV_TRUE_TRACK, NAV_WIND_TRUE_BRG, ND_HEADING_CAPT, and ND_HEADING_FO parameters) are expressed in raw format, where negative values are west of north. For example, a raw heading of -135° corresponds to a compass heading of 225° and a raw track of -015° corresponds to a compass track of 345°. Accordingly, if a raw direction is negative then it may be converted to a compass direction by adding 360°. No such conversion is necessary if a raw direction is positive.

Sample Message

```
QU RTNBOCR
.DDLXCXA 221921
RDO
FI BO24/AN N7873BO
DT DDL BFI 221921 M55A
- QURTNBOCR/~4POS N48083W121405/31004/+170/BLAKO/RADDY/YKM/22192118
```

11.4 Software Configuration Loop Back Messages

The Software Configuration Uplink enables the aircraft operator to request the AMI to send the Software Configuration Downlink. The uplink is not displayed to the flight crew.

11.4.1 Uplink Decoding Definition

The decoding for the Software Configuration Uplink is defined as follows:

- ACARS SMI: CMD
- ACARS Label: RA
- Message Identifier: ~4CFG
- Alert Level: None
- Optional Element Delimiter: None

Table 11-7 describes the Software Configuration Uplink decoding definition.

Table 11-7 Software Configuration Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	9	Yes	RA4_GND_ADDR	"Priority & address (XXXXXXXXXX)"
2	5	Yes	<none>	"Message identifier (~4CFG)"

When the AMI receives this uplink it sends the Software Configuration Downlink.

Sample Message

```
QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 022A
- QURTNBOCR~4CFG
```

11.4.2 Downlink Encoding Definition

The encoding for the Software Configuration Downlink is defined as follows:

- ACARS SMI: RDO
- ACARS Label: RB
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_AUTO
- Delete at System Reset: No
- Delete on Buffer Full: Yes

Table 11-8 describes the Software Configuration Downlink encoding definition.

Table 11-8 Software Configuration Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	9	Always	RA4_GND_ADDR	"Priority & address (XXXXXXXXXX)"
3	1	Always	/	"Delimiter"
4	5	Always	~4CFG	"Message Identifier"
5	1	Always	<SP>	"Delimiter"
6	15	Always	ATS_SOFTWARE_PART_NUMBER	"CMF OPS Part Number"
7	1	Always	/	"Delimiter"
8	10	Always	ATS_DF_FILE_PART_NUMBER	"CMF OPS UADF Part Number"

#	Length	Inclusion	Content	Element Description
9	1	Always	/	"Delimiter"
10	15	Always	AMI_PART_NUMBER	"CMF AMI Part Number"
11	1	Always	/	"Delimiter"
12	10	Always	AMI_DF_FILE_PART_NUMBER	"CMF AMI UADF Part Number"

Sample Message

```

QU RTNBOCR
.DDLXCXA 221920
RDO
FI BO24/AN N7873BO
DT DDL BFI 221920 M53A
- QURTNBOCR/~4CFG HNP5E-AL21-5012/O176061303/BCG-47-8CMF-
A012/A192220610

```

11.5 Subnetwork Status Loop Back Messages

The Subnetwork Status Uplink enables the aircraft operator to request the AMI to send the Subnetwork Status Downlink. The uplink is not displayed to the flight crew.

11.5.1 Uplink Decoding Definition

The decoding for the Subnetwork Status Uplink is defined as follows:

- ACARS SMI: CMD
- ACARS Label: RA
- Message Identifier: ~4SUB
- Alert Level: None
- Optional Element Delimiter: None

Table 11-9 describes the Subnetwork Status Uplink decoding definition.

Table 11-9 Subnetwork Status Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	9	Yes	RA4_GND_ADDR	"Priority & address (XXXXXXXXXX)"
2	5	Yes	<none>	"Message identifier (~4SUB)"

When the AMI receives this uplink it sends the Subnetwork Status Downlink.

Sample Message

QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 023A
- QURTNBOCR~4SUB

11.5.2 Downlink Encoding Definition

The encoding for the Subnetwork Status Downlink is defined as follows:

- ACARS SMI: RDO
- ACARS Label: RB
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_AUTO
- Delete at System Reset: No
- Delete on Buffer Full: Yes

Table 11-10 describes the Subnetwork Status Downlink encoding definition.

Table 11-10 Subnetwork Status Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	9	Always	RA4_GND_ADDR	"Priority & address (XXXXXXXXXX)"
3	1	Always	/	"Delimiter"
4	5	Always	~4SUB	"Message identifier"
5	2	Always	<CR><LF>	"Delimiter"
6	4	Always	VHF:	"VHF flag"
7	4	Conditionally	COMM	"VHF COMM"
8	7	Conditionally	NO COMM	"VHF NO COMM"
9	6	Conditionally	FAILED	"VHF failed"
10	11	Conditionally	NOT INSTALL	"VHF not installed"
11	1	Always	,	"Delimiter"
12	4	Conditionally	LEFT	"VHF-L master"
13	5	Conditionally	RIGHT	"VHF-R master"
14	6	Conditionally	CENTER	"VHF-C master"
15	1	Conditionally	*	"Invalid VHF master"

#	Length	Inclusion	Content	Element Description
16	1	Always	,	"Delimiter"
17	4-5	Always	VHF_VOICE_MODE	"VHF voice/data mode (VOICE or DATA)"
18	1	Always	,	"Delimiter"
19	1-2	Always	CURRENT_REGION	"Current region (NN)"
20	1	Always	,	"Delimiter"
21	7	Always	ACTIVE_DATA_FREQUENCY	"Active VHF frequency (NNN.NNN)"
22	1	Always	,	"Delimiter"
23	1	Always	S	"VDR state prefix"
24	1	Always	VDR_STATE	"VDR state (N)"
25	1	Always	/	"Delimiter"
26	1	Always	P	"VDR substate prefix"
27	1	Always	VDR_SUBSTATE_UDP	"VDR substate (N)"
28	1	Always	,	"Delimiter"
29	5	Always	MODE<SP>	"VDL mode"
30	1	Conditionally	-	"_"
31	1	Conditionally	A	"A"
32	1	Conditionally	2	"2"
33	1	Conditionally	*	"Invalid VDL mode"
34	2	Always	<CR><LF>	"Delimiter"
35	3	Always	HF:	"HF flag"
36	4	Conditionally	COMM	"HF COMM"
37	7	Conditionally	NO COMM	"HF NO COMM"
38	6	Conditionally	FAILED	"HF failed"
39	11	Conditionally	NOT INSTALL	"HF not installed"
40	1	Always	,	"Delimiter"
41	4	Conditionally	LEFT	"HF-L master"
42	5	Conditionally	RIGHT	"HF-R master"
43	1	Conditionally	*	"Invalid HF master"
44	1	Always	,	"Delimiter"
45	4-5	Always	HF_VOICE_MODE	"HF voice/data mode (VOICE or DATA)"
46	1	Always	,	"Delimiter"
47	5-9	Always	HFDR_DATA_CHANNEL_AVAILABLE	"HF datalink availability (AVAIL or NOT AVAIL)"
48	2	Always	<CR><LF>	"Delimiter"
49	7	Always	SATCOM:	"SATCOM flag"
50	4	Conditionally	COMM	"SATCOM COMM"
51	7	Conditionally	NO COMM	"SATCOM NO COMM"
52	6	Conditionally	FAILED	"SATCOM failed"

#	Length	Inclusion	Content	Element Description
53	11	Conditionally	NOT INSTALL	"SATCOM not installed"
54	1	Always	,	"Delimiter"
55	4	Conditionally	LEFT	"SDU-L master"
56	5	Conditionally	RIGHT	"SDU-R master"
57	1	Conditionally	*	"Invalid SDU master"
58	1	Always	,	"Delimiter"
59	5-9	Always	SATCOM_DATA_CHANNEL_AVAILABLE	"SATCOM datalink availability (AVAIL or NOT AVAIL)"
60	2	Always	<CR><LF>	"Delimiter"
61	5	Always	AoIP:	"AoIP flag"
62	5	Conditionally	AVAIL	"AoIP available"
63	9	Conditionally	NOT AVAIL	"AoIP not available"
64	6	Conditionally	FAILED	"AoIP failed"

Sample Message

```

QU RTNBOCR
.DDLXCXA 142229
RDO
FI BO24/AN N7873BO
DT DDL BFIV 142229 M03A
- QURTNBOCR/~4SUB
VHF:COMM,CENTER,DATA,9,136.975,S7/P3,MODE 2
HF:NO COMM,LEFT,DATA,NOT AVAIL
SATCOM:COMM,LEFT,AVAIL
AOIP:FAILED

```

Note: If AoIP is not enabled, then CMF sets its status to FAILED.

11.6 Test Loop Back Messages

The Test Uplink enables the aircraft operator to request the AMI to send the Test Downlink in response in order to verify end-to-end operation of the datalink system. This function provides for a text block containing up to 200 characters that the aircraft operator includes in the uplink and then the AMI includes in the corresponding downlink. The uplink is not displayed to the flight crew.

11.6.1 Uplink Decoding Definition

The decoding for the Test Uplink is defined as follows:

- ACARS SMI: CMD
- ACARS Label: RA
- Message Identifier: ~4LBK
- Alert Level: None
- Optional Element Delimiter: None

Table 11-11 describes the Test Uplink decoding definition.

Table 11-11 Test Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	9	Yes	RA4_GND_ADDR	"Priority & address (XXXXXXXXXX)"
2	5	Yes	<none>	"Message identifier (~4LBK)"
3	1-200	Yes	RA4_TXT_BLK	"Loop back text (1-200 characters)"

When the AMI receives this uplink it sends the Test Downlink.

Sample Message

```
QU DDLXCXA
.RTNBOCR 221954
CMD
AN N7873BO/MA 024A
- QURTNBOCR~4LBKLOOPBACK TEXT
```

11.6.2 Downlink Encoding Definition

The encoding for the Test Downlink is defined as follows:

- ACARS SMI: RDO
- ACARS Label: RB
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_AUTO
- Delete at System Reset: No
- Delete on Buffer Full: Yes

Table 11-12 describes the Test Downlink encoding definition.

Table 11-12 Test Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	9	Always	RA4_GND_ADDR	"Priority & address (XXXXXXXXXX)"
3	1	Always	/	"Delimiter"
4	5	Always	~4LBK	"Message Identifier"
5	1	Always	<SP>	"Delimiter"
6	1-200	Always	RA4_TXT_BLK	"Loop Back Text (1-200 characters)"

Sample Message

QU RTNBOCR
.DDLXCXA 221919
RDO
FI BO24/AN N7873BO
DT DDL BFI 221919 M52A
- QURTNBOCR/~4LBK LOOP BACK TEXT

12 Miscellaneous Functions

This section describes the miscellaneous functions that the AMI implements.

12.1 Airports Processing

The Airports Processing function ensures that the CMF origin and destination airport parameters follow the FMF origin and destination airport parameters as appropriate.

12.1.1 Logic Unit

```
-- -----
--
-- Version F001 M. Matyas 29-SEP-2020
-- Logic from 777 for CMF origin and destination airport parameters to
-- follow equivalent FMF parameters as appropriate.
--
-- -----

-- Origin station:

if (OOOI_STATE = IN and not valid (EOF_TIMER))
or OOOI_STATE = OUT then                                -- Before takeoff

    if not valid (INIT_ORIG_STA) then
        if valid (FMF_ACTRTEORIGIN) then
            INIT_ORIG_STA := FMF_ACTRTEORIGIN;
        end_if;
    else
        if valid (FMF_ACTRTEORIGIN) then
            if (INIT_ORIG_STA != FMF_ACTRTEORIGIN) then
                INIT_ORIG_STA := FMF_ACTRTEORIGIN;
            end_if;
        end_if;
    end_if;

else_if OOOI_STATE = OFF or OOOI_STATE = ON
or (OOOI_STATE = IN and valid (EOF_TIMER)) then          -- After takeoff

    if OOOI_STATE = OFF then
        if not valid (OFF_STA) then
            OFF_STA := FMF_ACTRTEORIGIN;
        end_if;
```

```

end_if;

if not valid (INIT_ORIG_STA) then
    if valid (OFF_STA) then
        INIT_ORIG_STA := OFF_STA;
    end_if;
else
    if valid (OFF_STA) then
        if (INIT_ORIG_STA != OFF_STA) then
            INIT_ORIG_STA := OFF_STA;
        end_if;
    end_if;
end_if;

end_if;

-- -----
--
-- Destination station:

if (OOOI_STATE = IN and not valid (EOF_TIMER))
or OOOI_STATE = OUT or OOOI_STATE = OFF then                -- Before landing

    if not valid (INIT_DEST_STA) then
        if valid (FMF_ACTRTEDEST) then
            INIT_DEST_STA := FMF_ACTRTEDEST;
        end_if;
    else
        if valid (FMF_ACTRTEDEST) then
            if (INIT_DEST_STA != FMF_ACTRTEDEST) then
                INIT_DEST_STA := FMF_ACTRTEDEST;
            end_if;
        end_if;
    end_if;

else_if OOOI_STATE = ON
or (OOOI_STATE = IN and valid (EOF_TIMER)) then                -- After landing

    if OOOI_STATE = ON
and (not FUEL_SW_L_CUTOFF) and (not FUEL_SW_R_CUTOFF) then
        if not valid (ON_STA)
or (valid (ON_STA) and ON_STA != FMF_ACTRTEDEST) then
            ON_STA := FMF_ACTRTEDEST;
        end_if;
    end_if;

```

```

    if not valid (INIT_DEST_STA) then
        if valid (ON_STA) then
            INIT_DEST_STA := ON_STA;
        end_if;
    else
        if valid (ON_STA) then
            if (INIT_DEST_STA != ON_STA) then
                INIT_DEST_STA := ON_STA;
            end_if;
        end_if;
    end_if;
end_if;

```

```

-- -----
--
-- END

```

12.2 End-of-Flight Processing

The End-of-Flight Processing function copies and resets the values of certain user-defined parameters in preparation for the next flight. This function automatically operates 9 minutes and 50 seconds after an In event (before the CMF automatic end of flight reset occurs 10 minutes after the airplane has transitioned from the in-air state to the on-ground state and at least one door has been opened).

12.2.1 Logic Unit

```

-- -----
--
-- VERSION C000 - New versioning - no change in logic
--
-- VERSION D002
--     Added default data for OOOI data
--     Added more comments
--
-- Version E001 M. Matyas 11-FEB-2019
--     Added "set_default (FUEL_ON_BOARD)".
--     Removed unnecessary "set_default (ESB_DELAY_SEL)",
--     "set_default (ARR_LNDG_PILOT)", "set_default (DIV_CODE)", and
--     "set_default (WXR_RQST_TYPE)".
--     Added logic for flight and block times and fuels.
--
-- Version F001 M. Matyas 29-SEP-2020

```

```

--      Added "set_default (EOF_TIMER)", "set_default (OFF_STA)",
--      "set_default (ON_STA)", and "set_default (AHM_INIT_FLAG)".
--      Updated references from INIT_DEPT_STA to INIT_ORIG_STA and
--      from PREV_DEPT_STA to PREV_ORIG_STA.
--
-- -----
--
if CURRENT_DATE_TIME reached (EOF_TIMER)
  and valid (EOF_TIMER) then

  -- Move current flight data to previous flight data:
  PREV_FLIGHT_NUMBER := ENTERED_FLTNO;
  PREV_ORIG_STA      := INIT_ORIG_STA;
  PREV_DEST_STA      := INIT_DEST_STA;
  PREV_BOARDED_FUEL  := INIT_BOARDED_FUEL;
  PREV_OUT_TIME      := OUT_TIME;
  PREV_OUT_FOB       := OUT_FOB;
  PREV_OFF_TIME      := OFF_TIME;
  PREV_OFF_FOB       := OFF_FOB;
  PREV_ON_TIME       := ON_TIME;
  PREV_ON_FOB        := ON_FOB;
  PREV_IN_TIME       := IN_TIME;
  PREV_IN_FOB        := IN_FOB;
  PREV_BLOCK_TIME    := BLOCK_TIME;
  PREV_BLOCK_FUEL    := BLOCK_FUEL;
  PREV_FLIGHT_TIME   := FLIGHT_TIME;
  PREV_FLIGHT_FUEL   := FLIGHT_FUEL;

  -- Initialization parameters:
  set_default (ENTERED_FLTNO);
  set_default (FUEL_ON_BOARD);
  set_default (INIT_SKED_DATE);
  set_default (INIT_ORIG_STA);
  set_default (INIT_DEST_STA);
  set_default (INIT_ALTN_STA_1);
  set_default (INIT_ALTN_STA_2);
  set_default (INIT_FLIGHT_PLAN_TIME);
  set_default (INIT_FOB);
  set_default (INIT_BOARDED_FUEL);
  set_default (INIT_PAY_CAPT);
  set_default (INIT_PAY_FO_1);
  set_default (INIT_PAY_FO_2);
  set_default (INIT_PAY_FO_3);
  set_default (AHM_INIT_FLAG);

  -- OOOI parameters:

```

```

set_default (OOOI_ENABLED);
set_default (OFF_STA);
set_default (ON_STA);

-- OOOI times:
set_default (OUT_TIME);
set_default (OFF_TIME);
set_default (ON_TIME);
set_default (IN_TIME);

-- OOOI fuel on board:
set_default (OUT_FOB);
set_default (OFF_FOB);
set_default (ON_FOB);
set_default (IN_FOB);

-- Flight and block times and fuels:
set_default (FLIGHT_TIME);
set_default (FLIGHT_FUEL);
set_default (BLOCK_TIME);
set_default (BLOCK_FUEL);

set_default (EOF_TIMER);

end_if;

-- -----
-- END
-- -----

```

12.3 Parameters Processing

The Parameters Processing function sets the values of certain parameters, including the following:

- All doors closed/open status
- Air/ground status
- Parking brake set time (for OOOI reports)
- Central Maintenance Computing Function (CMCF) flight phase string
- Aircraft registration
- Fuel on board
- CMF link strength

- FMF destination ETA
- VHF Data Radio (VDR) substate

12.3.1 Logic Unit

```

-----
--
-- Version C000  2014-02-12
--   Added CMCF flight phase integer to text logic used on SENSOR STATUS page.
--
-- Version D001  M. Matyas  24-OCT-2017
--   Commented out logic to calculate CURRENT_HHMM because OOOI logic changed
--   to use CURRENT_TIME instead.
--
-- Version D002  S. Meisch  2018 OCT
--   Moved DOORS_ALL_CLOSED, AIRCRAFT_IN_AIR, BRAKE_TIME parameters
--   from OOOI_LOGIC logic unit to BCG_PARAMETERS logic unit.
--   This logic unit should run before all other logic units.
--   This assures that commonly used parameters are evaluated
--   and updated in the correct sequence.
--
-- Version D003  S. Meisch  2018 DEC
--   Adds patch to suppress increase in FOB after engines have started.
--
-- Version E001  M. Matyas  08-FEB-2019
--   Added FOB invalidity check to logic that suppresses FOB increase
--   after engine start.
--   Added logic to calculate FMF destination ETA UDP based on FMF
--   destination ETE ACP.
--   Added logic to convert VDR substate ACP which reflects A750 binary
--   values to VDR substate UDP which reflects A750 decimal values.
--
-- Version F001  M. Matyas  29-SEP-2020
--   Added logic to set delay type if invalid.
--
-----

-- DOORS_ALL_CLOSED
--   Determine if all DOORS are CLOSED (true)
--   or ANY door is OPEN (false)
--
--   if  ALL_ENTRY_DOORS_CLOSED          = TRUE
--       and DOOR_AFT_EE_ACCESS_OPEN     = FALSE
--       and DOOR_BULK_CARGO_OPEN        = FALSE

```

```

        and DOOR_AFT_CARGO_OPEN          = FALSE
        and DOOR_FWD_ACCESS_OPEN         = FALSE
        and DOOR_FWD_CARGO_OPEN          = FALSE
        and DOOR_FWD_EE_ACCESS_OPEN      = FALSE
        and DOOR_FWD_OVERHEAD_OPEN       = FALSE then
            DOORS_ALL_CLOSED := TRUE;
    else
        DOORS_ALL_CLOSED := FALSE;
    end_if;

-- -----
--
-- On power up, determine air/ground status
-- AIRCRAFT_IN_AIR defaults to invalid at power-up
--
    if not valid (AIRCRAFT_IN_AIR) then
        if (AIRBORNE) then
            AIRCRAFT_IN_AIR := TRUE;
        else
            AIRCRAFT_IN_AIR := FALSE;
        end_if;
    end_if;

-- Use AIRBORNE (weight-on-wheels) if valid and showing on-ground or in-air
-- for 3 seconds

    if for 3 seconds (AIRBORNE) then
        AIRCRAFT_IN_AIR := TRUE;
    end_if;

    if for 3 seconds (not AIRBORNE) then
        AIRCRAFT_IN_AIR := FALSE;
    end_if;

-- Use CAS if AIRBORNE not valid for 3 seconds (and CAS is valid)

    if for 3 seconds (not valid (AIRBORNE)) then
        if valid (ADRF_CAS_VOTED) then
            if ADRF_CAS_VOTED > 125.0 then
                AIRCRAFT_IN_AIR := TRUE;
            else AIRCRAFT_IN_AIR := FALSE;
            end_if;
        end_if;

-- Otherwise use groundspeed if valid

        else if valid (NAV_GROUNDSPED) then

```

```

        if NAV_GROUNDSPEED > 130.0 then
            AIRCRAFT_IN_AIR := TRUE;
        else AIRCRAFT_IN_AIR := FALSE;
            end_if;

    end_if;
end_if;

-- And if none of the three are valid, AIRCRAFT_IN_AIR status remains unchanged
--
-- -----
--
-- BRAKE_TIME
--   Capture parking brake set time:
--
    if PARKINGBRKLEVERSET changed to (TRUE) then
        BRAKE_TIME := CURRENT_TIME;
    end_if;

-- -----
--
-- Set CMCF flight phase string:
--
    if CMCF_FLIGHT_PHASE = 1 then
        CMCF_FLT_PHASE_TEXT := " 1 - POWER ON";

    else_if CMCF_FLIGHT_PHASE = 2 then
        CMCF_FLT_PHASE_TEXT := " 2 - ENGINE START";

    else_if CMCF_FLIGHT_PHASE = 3 then
        CMCF_FLT_PHASE_TEXT := " 3 - TAXI OUT";

    else_if CMCF_FLIGHT_PHASE = 4 then
        CMCF_FLT_PHASE_TEXT := " 4 - TAKEOFF";

    else_if CMCF_FLIGHT_PHASE = 5 then
        CMCF_FLT_PHASE_TEXT := " 5 - INITIAL CLIMB";

    else_if CMCF_FLIGHT_PHASE = 6 then
        CMCF_FLT_PHASE_TEXT := " 6 - CLIMB";

    else_if CMCF_FLIGHT_PHASE = 7 then
        CMCF_FLT_PHASE_TEXT := " 7 - ENROUTE";

    else_if CMCF_FLIGHT_PHASE = 8 then

```

```

    CMCF_FLT_PHASE_TEXT := " 8 - DESCENT";

else_if CMCF_FLIGHT_PHASE = 9    then
    CMCF_FLT_PHASE_TEXT := " 9 - APPROACH";

else_if CMCF_FLIGHT_PHASE = 10   then
    CMCF_FLT_PHASE_TEXT := "10 - GO AROUND";

else_if CMCF_FLIGHT_PHASE = 11   then
    CMCF_FLT_PHASE_TEXT := "11 - FLARE";

else_if CMCF_FLIGHT_PHASE = 12   then
    CMCF_FLT_PHASE_TEXT := "12 - ROLLOUT";

else_if CMCF_FLIGHT_PHASE = 13   then
    CMCF_FLT_PHASE_TEXT := "13 - TAXI IN";

else_if CMCF_FLIGHT_PHASE = 14   then
    CMCF_FLT_PHASE_TEXT := "14 - ENGINE SHUTDOWN";

else_if CMCF_FLIGHT_PHASE = 15   then
    CMCF_FLT_PHASE_TEXT := "15 - MAINTENANCE";

else_if CMCF_FLIGHT_PHASE = 0    then
    CMCF_FLT_PHASE_TEXT := "0";

end_if;

-- -----
--
-- REGISTRATION_NO assigned to UDP and updated real time.
--
REGISTRATION_NO := AIRCRAFT_REGISTRATION_MARK;
--
-- -----
--
-- 1) Convert TOTAL_FILTERED_FUEL_MASS from integer to real
--     integer: TOTAL_FILTERED_FUEL_MASS
--     real:    FUEL_ON_BOARD
--
-- 2) Suppress increase in fuel on board after engines have started
--     real:    FUEL_ON_BOARD_CHECK (used to detect increase in fuel on board)
--
FUEL_ON_BOARD_CHECK := TOTAL_FILTERED_FUEL_MASS;    -- actual fuel

-- Check for increase in fuel on board after engines start

```

```

if not valid (FUEL_ON_BOARD) or (FUEL_SW_L_CUTOFF and FUEL_SW_R_CUTOFF) then
    -- FUEL_ON_BOARD not initialized or both engines off
    FUEL_ON_BOARD := FUEL_ON_BOARD_CHECK;          -- update FUEL_ON_BOARD

else      -- FUEL_ON_BOARD initialized or both engines on
    if FUEL_ON_BOARD > FUEL_ON_BOARD_CHECK then
        FUEL_ON_BOARD := FUEL_ON_BOARD_CHECK;      -- update FUEL_ON_BOARD

        -- else current fuel on board increased after engine start, do not update
        -- FUEL_ON_BOARD
    end_if;

end_if;

-- -----
--
-- CMF_LINK_STRENGTH is a representation of ACARS goodness.

if VHF_LINK_STATUS = 1
    then VHF_STRENGTH := 4;
    else VHF_STRENGTH := 0;
end_if;

if HF_LINK_STATUS = 1
    then HF_STRENGTH := 1;
    else HF_STRENGTH := 0;
end_if;

if SATCOM_LINK_STATUS = 1
    then SATCOM_STRENGTH := 2;
    else SATCOM_STRENGTH := 0;
end_if;

CMF_LINK_STRENGTH := VHF_STRENGTH + HF_STRENGTH + SATCOM_STRENGTH;

-- -----
--
-- Calculate FMF destination ETA UDP based on FMF destination ETE ACP:

if valid (FMF_TIME_TO_DEST) then
    FMF_DEST_ETA := FMF_TIME_TO_DEST + CURRENT_TIME;

else set_invalid (FMF_DEST_ETA);

end_if;

```

```

-----
--
-- Convert VDR substate ACP (which reflects A750 binary values) to VDR substate
-- UDP (which reflects A750 decimal values):

    if VDR_SUBSTATE_ACP = 0 then
        VDR_SUBSTATE_UDP := 1;

    else_if VDR_SUBSTATE_ACP = 1 then
        VDR_SUBSTATE_UDP := 3;

    else_if VDR_SUBSTATE_ACP = 2 then
        VDR_SUBSTATE_UDP := 2;

    else set_invalid (VDR_SUBSTATE_UDP);

    end_if;

-- -----
--
-- Set delay type if invalid:

    if not valid (DLY_SELECT) then

        if OOOI_STATE = IN then
            DLY_SELECT          := 1;          -- Select departure delay type

        else_if OOOI_STATE = OUT then
            DLY_SELECT          := 2;          -- Select takeoff delay type

        else_if OOOI_STATE = OFF then
            DLY_SELECT          := 3;          -- Select enroute delay type

        else_if OOOI_STATE = ON then
            DLY_SELECT          := 4;          -- Select arrival delay type

        end_if;

    end_if;

-- -----
--
-- END

```

12.4 POA Autotune

The “Plain Old” ACARS (POA) Autotune function enables the aircraft operator to send an uplink to command CMF to tune to a specified POA frequency.

12.4.1 Uplink Decoding Definition

The decoding for the POA Autotune Uplink is defined as follows:

- ACARS SMI: M4;
- ACARS Label: 4;
- Message Identifier: None
- Alert Level: None
- Optional Element Delimiter: None

Table 12-1 describes the POA Autotune Uplink decoding definition.

Table 12-1 POA Autotune Uplink Decoding Definition

#	Length	Required	Destination Parameter	Element Description
1	6	Yes	POA_AUTOTUNE_FREQUENCY	"POA autotune frequency (118000-136975)"

Note: Supported frequencies are those in the range from 118.000 MHz to 136.975 MHz with 8.33 kHz frequency spacing.

Sample Message

```
QU DDLXCXA
.RTNBOCR 221954
M4;
AN N7873BO/MA 025A
- 131550
```

12.4.2 Logic Unit

```
-- -----
-- BCG_POA_AUTOTUNE      M. Matyas  24-OCT-2017
--
-- Write POA autotune frequency from POA autotune uplink to autotune
-- frequency ACP:
-- -----
--
```

```

if POA_AUTOTUNE_ENABLE_WRITE = TRUE then
  if valid(POA_AUTOTUNE_FREQUENCY) then
    if ACTIVE_DATA_FREQUENCY != POA_AUTOTUNE_FREQUENCY then
      AUTOTUNE_FREQUENCY := POA_AUTOTUNE_FREQUENCY;
      POA_AUTOTUNE_ENABLE_WRITE := FALSE;
    end_if;
  end_if;
end_if;
--
-- -----
-- END:          BCG_POA_AUTOTUNE
-- -----

```

12.5 Reject Downlinks

The AMI implements three downlinks to reject uplinks for indicated reasons.

12.5.1 Invalid Reject Downlink

CMF sends the Invalid Reject Downlink when it receives an uplink that does not comply with the associated decoding definition.

12.5.1.1 Downlink Encoding Definition

The Invalid Reject Downlink is defined as follows:

- ACARS SMI: A8X
- ACARS Label: 8X
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_AUTO
- Delete at System Reset: No
- Delete on Buffer Full: Yes

Table 12-2 describes the Invalid Reject Downlink encoding definition.

Table 12-2 Invalid Reject Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	2	Always	UPLINK_LABEL	"Uplink Label (XX)"
3	1	Always	<SP>	"Delimiter"

#	Length	Inclusion	Content	Element Description
4	7	Always	INVALID	"Rejection Reason"
5	1	Always	<SP>	"Delimiter"
6	1-50	Always	UPLINK_TEXT	"Uplink text (1-50 characters)"

Sample Message

```

QU RTNBOCR
.DDLXCXA 221839
A8X
FI BO64/AN N7873BO
DT DDL BFI 221839 M41A
- RA INVALID QURTNBOCR~1CAA101190436
CREW ADVISORY...

```

12.5.2 Unable to Deliver Reject Downlink

CMF sends the Unable to Deliver Reject Downlink when it receives an uplink intended for a particular datalink end system but CMF cannot deliver the uplink to the end system.

12.5.2.1 Downlink Encoding Definition

The Unable to Deliver Reject Downlink is defined as follows:

- ACARS SMI: REJ
- ACARS Label: HX
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_AUTO
- Delete at System Reset: No
- Delete on Buffer Full: Yes

Table 12-3 describes the Unable to Deliver Reject Downlink encoding definition.

Table 12-3 Unable to Deliver Reject Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	1	Always	.	"Delimiter"
3	2	Always	EXTERNAL_UPLINK_LABEL	"Uplink Label (XX)"
4	1	Always	<SP>	"Delimiter"

#	Length	Inclusion	Content	Element Description
5	1-50	Always	EXTERNAL_UPLINK_TEXT	"Uplink text (1-50 characters)"

Sample Message

```

QU RTNBOCR
.DDLXCXA 221839
REJ
FI BO64/AN N7873BO
DT DDL BFI 221839 M52A
- .H1 - #T6EMU-R UPLINK TEST

```

12.5.3 Unable to Process Reject Downlink

CMF sends the Unable to Process Reject Downlink when it receives an uplink that it is unable to process (for example, when it receives a printer uplink but the printer is unavailable).

12.5.3.1 Downlink Encoding Definition

The Unable to Process Reject Downlink is defined as follows:

- ACARS SMI: A8X
- ACARS Label: 8X
- Allowable Subnetworks: All
- Downlink Priority: Flight Regularity
- Downlink Buffer: RESPONSE_AUTO
- Delete at System Reset: No
- Delete on Buffer Full: Yes

Table 12-4 describes the Unable to Process Reject Downlink encoding definition.

Table 12-4 Unable to Process Reject Downlink Encoding Definition

#	Length	Inclusion	Content	Element Description
1	10	Always	<seq_no><airline_ID><flight_no>	Standard ACARS downlink header
2	2	Always	UPLINK_LABEL	"Uplink Label (XX)"
3	1	Always	<SP>	"Delimiter"
4	6	Always	UNABLE	"Rejection Reason"
5	1	Always	<SP>	"Delimiter"
6	1-50	Always	UPLINK_TEXT	"Uplink text (1-50 characters)"

Sample Message

```
QU RTNBOCR  
.DDLXCXA 221839  
A8X  
FI BO64/AN N7873BO  
DT DDL BFI 221839 M42A  
- C1 UNABLE PRINTER UPLINK...
```

12.6 Sensor Status

The Sensor Status function enables the flight crew to view certain aircraft sensor information and other relevant data. This data includes aircraft parameters which are available to CMF and are accessible via the A/C PARAMS command button on the SENSOR STATUS display. This command button is password-protected with a password of "GORDON".

Note: The password must be entered into the scratchpad every time the display is accessed.

12.6.1 Display Definition

Figure 12-1 shows the Sensor Status display definition.

ATC

FLIGHT INFORMATION

COMPANY

REVIEW

MANAGER

NEW MESSAGES

hhmmz

SENSOR STATUS

XXXXXXXX

AIRCRAFT REGISTRATION: XXXXXXX

FLIGHT NUMBER: XXXXXXXXXX

ACARS FLIGHT IDENTIFIER: XXXXXX

TIME & DATE: HH:MM:SS DD-MMM-YYYY

POSITION: XDD°MM.T XDDD°MM.T

ALTITUDE: NNNNN FT

DOOR STATE: XXX

ALL DOORS: XXXXXXXXXX

PARKING BRAKE: XXXXXXXX

AIR/GROUND STATE: XXXXXX

FLIGHT PHASE: XXXXXXXXXXXXXXXXXXXX

ENGINES (L/R): XXX / XXX

A/C PARAMS

RETURN

EXIT

SCRATCHPAD

Figure 12-1 Sensor Status Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “SENSOR STATUS”

Main Area

Table 12-5 describes the Sensor Status display parameters.

Table 12-5 Sensor Status Display Parameters

Name	Length	Format	Range	Parameter
AIRCRAFT REGISTRATION	1-7	XXXXXXXX		AIRCRAFT_REGISTRATION_MARK
FLIGHT NUMBER	1-10	XXXXXXXXXXXX		FMF_CMF_FLIGHT_NUMBER

Name	Length	Format	Range	Parameter
ACARS FLIGHT IDENTIFIER	3-6	XXXXXX		ACARS_FLIGHT_IDENTIFIER
TIME & DATE	20	HH:MM:SS DD-MMM-YYYY		CURRENT_DATE_TIME
POSITION	18	XDD°MM.T XDDD°MM.T		AIRCRAFT_POSITION
ALTITUDE	1-5	NNNNN		NAV_ALTITUDE
OOOI STATE	2-3	XXX	OUT, OFF, ON, or IN	OOOI_STATUS
ALL DOORS	6-10	XXXXXXXXXX	CLOSED or NOT CLOSED	DOORS_ALL_CLOSED
PARKING BRAKE	3-8	XXXXXXXX	SET or RELEASED	PARKINGBRKLEVERSET
AIR/GROUND STATE	3-6	XXXXXX	AIR or GROUND	AIRCRAFT_IN_AIR
FLIGHT PHASE	5-20	XXX...	See below	CMCF_FLT_PHASE_TEXT
ENGINES (L/R)	2-3 / 2-3	XXX / XXX	ON or OFF	FUEL_SW_L_CUTOFF / FUEL_SW_R_CUTOFF

Possible flight phase indications are:

- 1 - POWER ON
- 2 - ENGINE START
- 3 - TAXI OUT
- 4 - TAKEOFF
- 5 - INITIAL CLIMB
- 6 - CLIMB
- 7 - ENROUTE
- 8 - DESCENT
- 9 - APPROACH
- 10 - GO AROUND
- 11 - FLARE
- 12 - ROLLOUT
- 13 - TAXI IN
- 14 - ENGINE SHUTDOWN
- 15 - MAINTENANCE

Command Buttons

Table 12-6 describes the Sensor Status display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 12-6 Sensor Status Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	A/C PARAMS	Show AIRCRAFT PARAMETERS display	Always	-
4	-	-	-	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

12.6.2 Aircraft Parameters Menu

The Aircraft Parameters Menu is accessed from the Sensor Status display and provides access to specific aircraft parameter displays.

12.6.2.1 Display Definition

Figure 12-2 shows the Aircraft Parameters Menu display definition.

[illegible]

Figure 12-2 Aircraft Parameters Menu Display Definition

12.6.2.1.1 CMF Parameters Display Definition

Figure 12-3 shows the CMF Parameters page 1 of 5 display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	CMF PARAMETERS	XXXXXXXXX
GENERAL		
CMF POSITION (CCS):	RIGHT	
TIME SINCE POWER ON:	NNNN MIN	
CURRENT DATE:	DD-MMM-YYYY	
CURRENT TIME:	HH:MM:SS	
CURRENT DATE-TIME:	DD-MMM-YYYY HH:MM:SS	
CURRENT DAY:	SUNDAYAY	
POSITION:	XDD°MM.T XDD°MM.T	
AIRCRAFT AIRBORNE:	XXX	
AIRCRAFT TYPE:	XXXX	
LOGIC UNIT MESSAGES:	XXXXXXXXX	
PRINT CHAR PER LINE:	NNN	
ADS-C MIN RPT INT:	NNN SEC	
		RETURN EXIT
SCRATCHPAD		

Figure 12-3: CMF Parameters Page 1 of 5 Display Definition

Figure 12-4 shows the CMF Parameters page 2 of 5 display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	CMF PARAMETERS	XXXXXXXXXX
PART NUMBERS		
CMF OPS:	XXXXXXXXXXXXXXXXXX	
CMF ATS DF:	XXXXXXXXXX	
CMF AMI:	XXXXXXXXXXXXXXXXXX	
CMF AMI DF:	XXXXXXXXXX	
AUTO POSITION REPORTS (IN AIR ONLY)		
☐ ENABLE		
INTERVAL:	-- MIN	
COUNTER:	NNNN SEC	
AHM FLIGHT PHASE REPORTS		
☐ ENABLE		
AOIP LINK ADVISORIES		
☐ ENABLE		
		RETURN EXIT
SCRATCHPAD		

Figure 12-4 CMF Parameters Page 2 of 5 Display Definition

Of note, the CMF Parameters page 2 of 5 display definition shows and enables modification of the following values, each of which are stored in NVM in order to retain the values across CMF power cycles:

- Automatic position reports enable/disable
- Automatic position reports interval (in a range from 5 to 60 minutes)
- AHM flight phase reports enable/disable
- AoIP link advisories enable/disable

Figure 12-5 shows the CMF Parameters page 3 of 5 display definition.

Figure 12-5 CMF Parameters Page 3 of 5 Display Definition

NEW

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	CMF PARAMETERS	XXXXXXXXXX
ATN ATN: XXXXXXXX VER (DCA): NN HEX (41 = AIRLINE) (C1 = NON-AIRLINE) AIRLINE ADM (DCA): XXX (ICAO AIRLINE ID) NON-AIRLINE ADM (DCA): NNNNNN HEX (ICAO ADDRESS) ICAO ADDRESS (DCA): NNNNNN HEX NNNNNNNN OCT		
		RETURN EXIT
SCRATCHPAD		

Figure 12-6 CMF Parameters Page 4 of 5 Display Definition

Figure 12-7 shows the CMF Parameters page 5 of 5 display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	CMF PARAMETERS	XXXXXXXXXX
LINKS		
CMF LINK STATUS:	COMMEDM	
VHF LINK STATUS:	COMMEDMTALLED	
MASTER VDR:	LEFTTR	
VHF VOICE/DATA MODE:	XXXXX	
CURRENT REGION:	NN	
ACTIVE VHF FREQUENCY:	NNN NNN MHZ	
VDR STATE/SUBSTATE:	SN / PN	
VDL MODE:	MODE	
HF LINK STATUS:	COMMEDMTALLED	
MASTER HFDR:	LEFTT	
HF VOICE/DATA MODE:	XXXXX	
HF DATALINK:	XXXXXXXXXXXXXX	
SATCOM LINK STATUS:	COMMEDMTALLED	
MASTER SDU:	LEFTT	
SATCOM DATALINK:	XXXXXXXXXXXXXX	
A0IP LINK STATUS:	FAILEDBLEABLE	
		RETURN EXIT
SCRATCHPAD		

Figure 12-7 CMF Parameters Page 5 of 5 Display Definition

12.6.2.1.2 FMF and Navigation Parameters Display Definition

Figure 12-8 shows the Flight Management Function (FMF) and Navigation Parameters page 1 of 3 display definition.

ATC		FLIGHT INFORMATION		COMPANY	
REVIEW		MANAGER		NEW MESSAGES	
hhmmz		FMF & NAV PARAMETERS		XXXXXXXXXX	
FLIGHT NUMBER:		XXXXXXXXXX			
ORIG:		XXXX			
ORIG RWY:		XXXX			
LAST WPT:		XXXXXXX			
LAST WPT ATA:		HH:MM:SS			
LAST WPT ALT:		NNNNN FT			
ACT WPT:		XXXXXXX			
ACT WPT ETA:		HH:MM:SS			
ACT WPT DIST:		NNNN.NN NM			
NEXT WPT:		XXXXXXX			
1ST ATC COMP RPT WPT:		XXXXXXX			
1ST ATC COMP RPT WPT ETA:		HH:MM:SS			
2ND ATC COMP RPT WPT:		XXXXXXX			
DEST:		XXXX			
DEST RWY:		XXXX			
DEST ETE/ETA:		HH+MM+SS /	HH:MM:SS		
DEST DIST:		NNNNN.N NM			
				RETURN	EXIT
SCRATCHPAD					

Figure 12-8 FMF and Navigation Parameters Page 1 of 3 Display Definition

Figure 12-9 shows the FMF and Navigation Parameters page 2 of 3 display definition.

ATC	FLIGHT INFORMATION	COMPANY																																							
REVIEW	MANAGER	NEW MESSAGES																																							
hhmmz	FMF & NAV PARAMETERS	XXXXXXXXX																																							
<table border="0"> <tr> <td>FLIGHT PHASE:</td> <td>DESCENT:</td> <td></td> </tr> <tr> <td>TRANS ALT:</td> <td>NNNN FT</td> <td></td> </tr> <tr> <td>ENGINE-OUT MAX ALT:</td> <td>NNNN FT</td> <td></td> </tr> <tr> <td>TOC ETE:</td> <td>HH+MM+SS</td> <td></td> </tr> <tr> <td>TOD ETE:</td> <td>HH+MM+SS</td> <td></td> </tr> <tr> <td>ZERO FUEL WEIGHT:</td> <td>NNNNNN XX</td> <td></td> </tr> <tr> <td>GROSS WEIGHT:</td> <td>NNNNNN XX</td> <td></td> </tr> <tr> <td>CG:</td> <td>NN .N % MAC</td> <td></td> </tr> <tr> <td>COST INDEX:</td> <td>NNNN</td> <td></td> </tr> <tr> <td>FUEL REMAINING:</td> <td>NNNNNN XX</td> <td></td> </tr> <tr> <td>FUEL AT DEST:</td> <td>NNNNNN XX</td> <td></td> </tr> <tr> <td>FUEL USED LEFT ENGINE:</td> <td>NNNNNN XX</td> <td></td> </tr> <tr> <td>FUEL USED RIGHT ENGINE:</td> <td>NNNNNN XX</td> <td></td> </tr> </table>			FLIGHT PHASE:	DESCENT:		TRANS ALT:	NNNN FT		ENGINE-OUT MAX ALT:	NNNN FT		TOC ETE:	HH+MM+SS		TOD ETE:	HH+MM+SS		ZERO FUEL WEIGHT:	NNNNNN XX		GROSS WEIGHT:	NNNNNN XX		CG:	NN .N % MAC		COST INDEX:	NNNN		FUEL REMAINING:	NNNNNN XX		FUEL AT DEST:	NNNNNN XX		FUEL USED LEFT ENGINE:	NNNNNN XX		FUEL USED RIGHT ENGINE:	NNNNNN XX	
FLIGHT PHASE:	DESCENT:																																								
TRANS ALT:	NNNN FT																																								
ENGINE-OUT MAX ALT:	NNNN FT																																								
TOC ETE:	HH+MM+SS																																								
TOD ETE:	HH+MM+SS																																								
ZERO FUEL WEIGHT:	NNNNNN XX																																								
GROSS WEIGHT:	NNNNNN XX																																								
CG:	NN .N % MAC																																								
COST INDEX:	NNNN																																								
FUEL REMAINING:	NNNNNN XX																																								
FUEL AT DEST:	NNNNNN XX																																								
FUEL USED LEFT ENGINE:	NNNNNN XX																																								
FUEL USED RIGHT ENGINE:	NNNNNN XX																																								
		RETURN EXIT																																							
SCRATCHPAD																																									

Figure 12-9 FMF and Navigation Parameters Page 2 of 3 Display Definition

Figure 12-10 shows the FMF and Navigation Parameters page 3 of 3 display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	FMF & NAV PARAMETERS	XXXXXXXXX
DATE (MM/DD/YYYY):	NN / NN / NNNN	
TIME (HH:MM:SS):	NN : NN : NN	
ALTITUDE:	NNNN FT	
LATITUDE ("+" = N):	±NN.NN °	
LONGITUDE ("+" = E):	±NNN.NN °	
MAG HDG ("-" = W OF N):	±NNN °	
MAG TRK ("-" = W OF N):	±NNN °	
TRUE HDG ("-" = W OF N):	±NNN °	
TRUE TRK ("-" = W OF N):	±NNN °	
GROUND SPEED:	NNNN KT	
WIND SPEED:	NNNN KT	
WIND TRUE BRG ("-" = W OF N):	±NNN °	
RETURN EXIT		
SCRATCHPAD		

Figure 12-10 FMF and Navigation Parameters Page 3 of 3 Display Definition

Note: Headings, tracks, and bearings (i.e., the NAV_MAG_HDG, NAV_MAG_TRACK, NAV_TRUE_HDG, NAV_TRUE_TRACK, NAV_WIND_TRUE_BRG, ND_HEADING_CAPT, and ND_HEADING_FO parameters) are expressed in raw format, where negative values are west of north. For example, a raw heading of -135° corresponds to a compass heading of 225° and a raw track of -015° corresponds to a compass track of 345°. Accordingly, if a raw direction is negative then it may be converted to a compass direction by adding 360°. No such conversion is necessary if a raw direction is positive.

12.6.2.1.3 Engine and FQMS Parameters Display Definition

Figure 12-11 shows the Engine and FQMS Parameters display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	ENGINE & FQMS PARAMETERS	XXXXXXXXXX
ENGINE TYPE: INVALID		
	LEFT	RIGHT
ENGINE FUEL CONTROL:	XXXXXX	XXXXXX
ENGINE START SWITCH:	START	START
ENGINE N1 SPEED:	NNN.N %	NNN.N %
ENGINE TPR:	NNN.N	NNN.N
ENGINE OIL PRESS:	NNN.N PSI	NNN.N PSI
ENGINE OIL TEMP:	±NNN °C	±NNN °C
ENGINE OIL QTY:	NN.N XX	NN.N XX
TOTAL FILTERED FUEL MASS: NNNNNN XX		
LEFT MAIN FUEL DENSITY:	N.NNN	XXXXXX
CENTER FUEL DENSITY:	N.NNN	XXXXXX
RIGHT MAIN FUEL DENSITY:	N.NNN	XXXXXX
	RETURN	EXIT
SCRATCHPAD		

Figure 12-11 Engine and FQMS Parameters Display Definition

12.6.2.1.4 ADRF and IRU Parameters Display Definition

Figure 12-12 shows the Air Data Reference Function (ADRF) and Inertial Reference Unit (IRU) Parameters display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	ADRF & IRU PARAMETERS	XXXXXXXXX
ADRF PRESS ALT VOTED: NNNNN FT		
ADRF ALT RATE VOTED: ±NNNNN FT/MIN		
ADRF CAS VOTED: NNN KT		
ADRF TAS VOTED: NNN KT		
ADRF MACH VOTED: N.NNN MACH		
ADRF SAT VOTED: ±NNN.N °C		
ADRF TAT VOTED: ±NNN.N °C		
IRU VERT SPEED: ±NNNNN FT/MIN		
IRU VERT ACCEL: ±N.N G		
IRU PITCH ANGLE: ±NNN.N °		
IRU PITCH RATE: ±NNN.N °/SEC		
IRU YAW RATE: ±NNN.N °/SEC		
		RETURN EXIT
SCRATCHPAD		

Figure 12-12 ADRF and IRU Parameters Display Definition

12.6.2.1.5 FCM Parameters Display Definition

Figure 12-13 shows the Flight Control Module (FCM) Parameters display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	FCM PARAMETERS	XXXXXXXXXX
AUTOPILOT:	XXXXXXXXXXXX	
FCM LAND 2:	XXXXXXXXXX	
FCM LAND 3:	XXXXXXXXXX	
FCM NO AUTOLAND:	XXXXXXXXXX	
TOGA SWITCH:	XXXXXXXXXX	
FLAP LEVER POSITION:	NN	
FLAPS/SLATS:	NNN	
		RETURN EXIT
SCRATCHPAD		

Figure 12-13 FCM Parameters Display Definition

12.6.2.1.6 PSS Parameters Display Definition

Figure 12-14 shows the Proximity Sensing System (PSS) Parameters display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	PSS PARAMETERS	XXXXXXXXXX
ENTRY DOORS	LEFT	RIGHT
1:	XXXXXX	XXXXXX
2:	XXXXXX	XXXXXX
3:	XXXXXX	XXXXXX
4:	XXXXXX	XXXXXX
FWD ACCESS DOOR:		XXXXXX
FWD OVERHEAD DOOR:		XXXXXX
FWD EE ACCESS DOOR:		XXXXXX
FWD CARGO DOOR:		XXXXXX
AFT EE ACCESS DOOR:		XXXXXX
AFT CARGO DOOR:		XXXXXX
BULK CARGO DOOR:		XXXXXX
ALL DOORS CLOSED (CMF):		XXX
		RETURN EXIT
SCRATCHPAD		

Figure 12-14 PSS Parameters Display Definition

12.6.2.1.7 Miscellaneous Parameters Display Definition

Figure 12-15 shows the Miscellaneous Parameters display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	MISC PARAMETERS	XXXXXXXXXX
CHCF FLIGHT LEG NUMBER:		NNNN
CHCF FLIGHT PHASE:		NN
(CHF)		XXXXXXXXXXXXXXXXXXXX
DCA CAPT ND HDG ("-" = W OF N):		±NNN °
DCA F/O ND HDG ("-" = W OF N):		±NNN °
DCA UNITS:		XXXXXX
CIS PRINTER STATUS:		NOT READY
CIS PRINT QUEUE:		XXXXXXXXXX
PARKING BRAKE LEVER:		XXXXXXXXXX
WING ANTI-ICE SWITCH:		AOFF
LRRR HEIGHT:		NNNN.N FT
LANDING GEAR:		DIN TRANSIT
		RETURN EXIT
SCRATCHPAD		

Figure 12-15 Miscellaneous Parameters Display Definition

Note: Headings, tracks, and bearings (i.e., the NAV_MAG_HDG, NAV_MAG_TRACK, NAV_TRUE_HDG, NAV_TRUE_TRACK, NAV_WIND_TRUE_BRG, ND_HEADING_CAPT, and ND_HEADING_FO parameters) are expressed in raw format, where negative values are west of north. For example, a raw heading of -135° corresponds to a compass heading of 225° and a raw track of -015° corresponds to a compass track of 345°. Accordingly, if a raw direction is negative then it may be converted to a compass direction by adding 360°. No such conversion is necessary if a raw direction is positive.

12.6.2.1.8 Automatic Emergency Report Parameters Display Definition

Figure 12-16 shows the Automatic Emergency Report Parameters page 1 of 2 display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	AUTOMATIC EMERGENCY REPORT PARAMETERS	
◆ AUTOMATIC EMERGENCY REPORTS ON		
◆ AUTOMATIC EMERGENCY REPORTS ARMED		
◆ AUTOMATIC EMERGENCY REPORTS OFF		
LAST SEL: OFF STATE: XXXXXX LAST STATE: XXXXXX RADIO ALT: XXXXXXXXXXXX MIN ALT: NNNNN FT DETECT ENABLED: XXX INTERVAL: NNNN SEC COUNTER: NNNN SEC MAX RPT TIME: NNN MIN MANUAL: XXX		
		RETURN EXIT
SCRATCHPAD		

Figure 12-16 Automatic Emergency Report Parameters Page 1 of 2 Display Definition

Figure 12-17 shows the Automatic Emergency Report Parameters page 2 of 2 display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES

hmmz AUTOMATIC EMERGENCY REPORT PARAMETERS XXXXXXXX

THRESHOLDS

☒ ALTITUDE RATE

☒ CLIMB RATE: NNNNN FT/MIN

☒ DESCENT RATE: NNNNN FT/MIN

☒ SPEED

☒ CAS LOW: NNN KT

☒ VMO: NNN + NNN KT

☒ HMO: MN.NN + N.NN

☒ ATTITUDE

☒ PITCH ANGLE: NN °

☒ PITCH RATE: NN °/SEC

☒ YAW RATE: NN °/SEC

☒ VERT ACCEL: N.N G

☒ ENGINES

☒ LEFT N1: NN % LEFT TPR: NN

☒ RIGHT N1: NN % RIGHT TPR: NN

☒ FUEL: NNNNN LBS

RETURN EXIT

SCRATCHPAD

Figure 12-17 Automatic Emergency Report Parameters Page 2 of 2 Display Definition

12.6.2.1.9 OOOI Parameters Display Definition

Figure 12-18 shows the OOOI Parameters display definition.

ATC		FLIGHT INFORMATION		COMPANY	
REVIEW		MANAGER		NEW MESSAGES	
hhmmz		000I PARAMETERS		XXXXXXXXXX	
ENABLED:	XXX			CURR	PREV
STATE:	N/XXX			HH:MM	HH:MM
RTG:	XXX	OUT:		NNN.N	NNN.N
TOUCH+GO:	XXX			HH:MM	HH:MM
ALL DOORS:	XXXXXXXX	OFF:		NNN.N	NNN.N
P BRAKE:	XXX/HH:MM:SS			HH:MM	HH:MM
A/G STATE:	XXXXXX	ON:		NNN.N	NNN.N
DLY TYPE:	N			HH:MM	HH:MM
EOF:	HH:MM:SS	IN:		NNN.N	NNN.N
FMF ORIG:	XXXX			HH:MM	HH:MM
FMF DEST:	XXXX	FLT:		NNN.N	NNN.N
CMF ORIG:	XXXX			HH:MM	HH:MM
CMF DEST:	XXXX	BLK:		NNN.N	NNN.N
OFF STA:	XXXX				
ON STA:	XXXX				
FQMS FUEL MASS:	NNNNNN	DIVERT AUTO:		XXX	
CMF FOB CHECK:	NNNNNN	AHM DVRT:		XXX	
CMF FOB:	NNNNNN	AHM INIT:		XXX	
		RETURN		EXIT	
SCRATCHPAD					

Figure 12-18 OOOI Parameters Display Definition

12.6.2.1.10 Reminders Parameters Display Definition

Figure 12-19 shows the Reminders Parameters display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	REMINDERS PARAMETERS	XXXXXXXXXX
PASSING WAYPOINT		
XXXXXXX	REMINDER_FMF_LAST_WPT	
XXXXXXX	REMINDER_POSN	
WAYPOINT ETA CHANGE		
XXX	REMINDER_ENABLE_ETA_CHANGE	
XXXXXXX	FMF_ACTIVE_WPT	
XXXXXXX	LAST_ACTIVE_WPT	
HH:MM:SS	REMINDER_ETA_BASELINE	
HH:MM:SS	TIME_AT_WPT_CHG	
HH:MM:SS	CURRENT_DATE_TIME	
HH:MM:SS	FMF_ETA	
HH:MM:SS	REMINDER_FMF_ETA	
NNNNN	TIME_SINCE_WPT_CHG	
NNNNN	TIME_TO_WPT_CHG	
NNNNN	ETA_CHANGE	
N / NNN	REMINDER_ETA_CHANGE	
		RETURN EXIT
SCRATCHPAD		

Figure 12-19 Reminders Parameters Display Definition

12.7 Times and Fuels

The Times and Fuels function tracks and displays flight identifier, departure and destination airports, fuel boarded, OOOI times and fuels, flight times and fuels, and block times and fuels information for the current and previous flights. This function consists mainly of a display definition, although it relies on the OOOI logic units that Section 9.6.1 and Section 9.6.2 describe and also on the end-of-flight logic unit that Section 12.2.1 describes.

12.7.1 Display Definition

Figure 12-20 shows the Times and Fuels display definition.

ATC	FLIGHT INFORMATION	COMPANY
REVIEW	MANAGER	NEW MESSAGES
hhmmz	TIMES & FUELS	XXXXXXXXXX
	CURRENT FLIGHT	PREVIOUS FLIGHT
	XXXXXXXXXX	XXXXXXXXXX
	XXXX-XXXX	XXXX-XXXX
FUEL BOARDED:	NNNNNN	NNNNNN
OUT TIME:	HH:MM	HH:MM
OUT FUEL:	NNN.N	NNN.N
OFF TIME:	HH:MM	HH:MM
OFF FUEL:	NNN.N	NNN.N
ON TIME:	HH:MM	HH:MM
ON FUEL:	NNN.N	NNN.N
IN TIME:	HH:MM	HH:MM
IN FUEL:	NNN.N	NNN.N
FLIGHT TIME:	HH+MM	HH+MM
FLIGHT FUEL:	NNN.N	NNN.N
BLOCK TIME:	HH+MM	HH+MM
BLOCK FUEL:	NNN.N	NNN.N
	PRINT	RETURN EXIT
	SCRATCHPAD	

Figure 12-20 Times and Fuels Display Definition

Title

The title area contains:

- The current time in HHMMz format
- “TIMES & FUELS”

Main Area

The main area contains two columns of display-only data. The first column contains data for the current flight, while the second column contains corresponding data for the previous flight.

Table 12-7 describes the Times and Fuels display current flight column parameters. The AMI moves the values of these parameters to the parameters in the previous flight column 9 minutes and 50 seconds after an In event (before the CMF automatic end of flight reset occurs 10 minutes after the In event).

Table 12-7 Times and Fuels Display Current Flight Column Parameters

Name	Format	Parameter
none	XXXXXXXXXX	ENTERED_FLTNO
none	XXXX	INIT_ORIG_STA
none	XXXX	INIT_DEST_STA
FUEL BOARDED	NNNNNN	INIT_BOARDED_FUEL
OUT TIME	HH:MM	OUT_TIME
OUT FUEL	NNN.N	OUT_FOB
OFF TIME	HH:MM	OFF_TIME
OFF FUEL	NNN.N	OFF_FOB
ON TIME	HH:MM	ON_TIME
ON FUEL	NNN.N	ON_FOB
IN TIME	HH:MM	IN_TIME
IN FUEL	NNN.N	IN_FOB
FLIGHT TIME	HH+MM	FLIGHT_TIME
FLIGHT FUEL	NNN.N	FLIGHT_FUEL
BLOCK TIME	HH+MM	BLOCK_TIME
BLOCK FUEL	NNN.N	BLOCK_FUEL

Table 12-8 describes the Times and Fuels display PREVIOUS FLIGHT column parameters. CMF stores the values of these parameters in NVM in order to retain them across power cycles.

Table 12-8 Times and Fuels Display Previous Flight Column Parameters

Name	Format	Parameter
none	XXXXXXXXXX	PREV_FLIGHT_NUMBER
none	XXXX	PREV_ORIG_STA
none	XXXX	PREV_DEST_STA
FUEL BOARDED	NNNNNN	PREV_BOARDED_FUEL
OUT TIME	HH:MM	PREV_OUT_TIME
OUT FUEL	NNN.N	PREV_OUT_FOB
OFF TIME	HH:MM	PREV_OFF_TIME
OFF FUEL	NNN.N	PREV_OFF_FOB
ON TIME	HH:MM	PREV_ON_TIME
ON FUEL	NNN.N	PREV_ON_FOB
IN TIME	HH:MM	PREV_IN_TIME
IN FUEL	NNN.N	PREV_IN_FOB
FLIGHT TIME	HH+MM	PREV_FLIGHT_TIME
FLIGHT FUEL	NNN.N	PREV_FLIGHT_FUEL
BLOCK TIME	HH+MM	PREV_BLOCK_TIME
BLOCK FUEL	NNN.N	PREV_BLOCK_FUEL

Command Buttons

Table 12-9 describes the Times and Fuels display command buttons. If neither condition is satisfied, then the button is not displayed.

Table 12-9 Times and Fuels Display Command Buttons

Position	Legend	Function	Displayed	Displayed as Inhibited
1	-	-	-	-
2	-	-	-	-
3	PRINT	Print	Always	-
4	-	-	-	-
5	RETURN	Return	Always	-
6	EXIT	Exit	Always	-

12.7.2 Print Definition

Figure 12-21 shows the Times and Fuels print definition.

TIME: hhmmz	TIMES & FUELS		A/C REG: xxxxxxxx
DATE: dd-MMM-yyyy			FLT NO: xxxxxxxxxx
	CURRENT FLIGHT	PREVIOUS FLIGHT	
	xxxxxxxxxx	xxxxxxxxxx	
	xxxx-xxxx	xxxx-xxxx	
FUEL BOARDED:	nnnnnn	nnnnnn	
OUT TIME:	hh:mm	hh:mm	
OUT FUEL:	nnn.n	nnn.n	
OFF TIME:	hh:mm	hh:mm	
OFF FUEL:	nnn.n	nnn.n	
ON TIME:	hh:mm	hh:mm	
ON FUEL:	nnn.n	nnn.n	
IN TIME:	hh:mm	hh:mm	
IN FUEL:	nnn.n	nnn.n	
FLIGHT TIME:	hh:mm	hh:mm	
FLIGHT FUEL:	nnn.n	nnn.n	
BLOCK TIME:	hh:mm	hh:mm	
BLOCK FUEL:	nnn.n	nnn.n	

Figure 12-21 Times and Fuels Print Definition

13 Supporting Components

The AMI implements several supporting (background) components that do not directly provide any operational functions.

13.1 Customer Parameter Database

The Customer Parameter Database component defines the aircraft parameters (ACPs) and user-defined parameters (UDPs) that are available for other components to reference. It also defines certain attributes of each parameter, including whether CMF should set the parameter to its default value when CMF resets, whether CMF should store the parameter's value in NVM, and the initial value of the parameter. Table 13-1 describes the customer parameter database that the CMF baseline AMI implements.

Table 13-1 Customer Parameter Database

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
ACP	ACARS_FLIGHT_IDENTIFIER	String	<n/a>	<n/a>	<ACARS_FLIGHT_IDENTIFIER>
ACP	ACARS_FLIGHT_NUMBER	String	<n/a>	<n/a>	<ACARS_FLIGHT_NUMBER>
ACP	ACTIVE_DATA_FREQUENCY	Real	<n/a>	<n/a>	<ACTIVE_DATA_FREQUENCY>
ACP	ADRF_ALTRATE_VOTED	Real	<n/a>	<n/a>	<ADRF_ALTRATE_VOTED>
ACP	ADRF_CAS_VOTED	Real	<n/a>	<n/a>	<ADRF_CAS_VOTED>
ACP	ADRF_MACH_VOTED	Real	<n/a>	<n/a>	<ADRF_MACH_VOTED>
ACP	ADRF_PRESSALT_VOTED	Real	<n/a>	<n/a>	<ADRF_PRESSALT_VOTED>
ACP	ADRF_SAT_VOTED	Real	<n/a>	<n/a>	<ADRF_SAT_VOTED>
ACP	ADRF_TAS_VOTED	Real	<n/a>	<n/a>	<ADRF_TAS_VOTED>
ACP	ADRF_TAT_VOTED	Real	<n/a>	<n/a>	<ADRF_TAT_VOTED>
ACP	ADS_MIN_REPORTING_INTERVAL	Real	<n/a>	<n/a>	<ADS_MIN_REPORTING_INTERVAL>
UDP	AGM_ADDR	String	Yes	No	Invalid
UDP	AGM_TEXT_1	String	Yes	No	Invalid
UDP	AGM_TEXT_2	String	Yes	No	Invalid

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
UDP	AGM_TEXT_3	String	Yes	No	Invalid
UDP	AGM_TEXT_4	String	Yes	No	Invalid
UDP	AGM_TEXT_5	String	Yes	No	Invalid
UDP	AGM_TEXT_6	String	Yes	No	Invalid
UDP	AGM_TEXT_7	String	Yes	No	Invalid
UDP	AGM_TEXT_8	String	Yes	No	Invalid
UDP	AHM_DVRT_FLAG	Boolean	Yes	No	FALSE
UDP	AHM_DVRT_PREV_DEST	String	Yes	No	Invalid
UDP	AHM_FLIGHT_PHASE_ENABLE	Boolean	No	Yes	TRUE
UDP	AHM_FLIGHT_PHASE_IDENT	String	Yes	No	Invalid
UDP	AHM_INIT_FLAG	Boolean	No	Yes	FALSE
ACP	AIRBORNE	Boolean	<n/a>	<n/a>	<AIRBORNE>
UDP	AIRCRAFT_IN_AIR	Boolean	Yes	No	Invalid
ACP	AIRCRAFT_POSITION	String	<n/a>	<n/a>	<AIRCRAFT_POSITION>
ACP	AIRCRAFT_REGISTRATION_MARK	String	<n/a>	<n/a>	<AIRCRAFT_REGISTRATION_MARK>
ACP	AIRCRAFT_TYPE	String	<n/a>	<n/a>	<AIRCRAFT_TYPE>
ACP	AIRLINE_IDENTIFIER	String	<n/a>	<n/a>	<AIRLINE_IDENTIFIER>
ACP	ALL_ENTRY_DOORS_CLOSED	Boolean	<n/a>	<n/a>	<ALL_ENTRY_DOORS_CLOSED>
ACP	AMI_DF_FILE_PART_NUMBER	String	<n/a>	<n/a>	<AMI_DF_FILE_PART_NUMBER>
ACP	AMI_PART_NUMBER	String	<n/a>	<n/a>	<AMI_PART_NUMBER>
ACP	ANTI_ICE_SW_POS	Integer	<n/a>	<n/a>	<ANTI_ICE_SW_POS>
UDP	AOIP_LINK_ADV_ENABLE	Boolean	No	Yes	FALSE
UDP	AOIP_LINK_ADV_STATUS	Integer	Yes	No	Invalid
UDP	AOIP_LINK_ADV_TIME	Integer	Yes	No	Invalid
ACP	AOIP_LINK_STATUS	Integer	<n/a>	<n/a>	<AOIP_LINK_STATUS>
UDP	ARR_AUTOLAND	Integer	Yes	No	Invalid
UDP	ARR_AUTOLND_QUAL	Integer	Yes	No	Invalid

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
UDP	ARR_FOB	Real	Yes	No	<IN_FOB>
UDP	ARR_IN_TIME	Integer	Yes	No	<IN_TIME>
UDP	ARR_LAND_CAT_CND	Integer	Yes	No	Invalid
UDP	ARR_LNDG_PILOT	Integer	Yes	No	Invalid
UDP	ARR_ON_TIME	Integer	Yes	No	<ON_TIME>
UDP	ARR_SUPLMT_TXT_1	String	Yes	No	Invalid
UDP	ARR_SUPLMT_TXT_2	String	Yes	No	Invalid
UDP	ARR_SUPLMT_TXT_3	String	Yes	No	Invalid
ACP	ATN_ENABLED	Boolean	<n/a>	<n/a>	<ATN_BASELINE_1>
ACP	ATS_DF_FILE_PART_NUMBER	String	<n/a>	<n/a>	<ATS_DF_FILE_PART_NUMBER>
ACP	ATS_SOFTWARE_PART_NUMBER	String	<n/a>	<n/a>	<SOFTWARE_PART_NUMBER>
ACP	AUTOTUNE_FREQUENCY	Real	<n/a>	<n/a>	<AUTOTUNE_FREQUENCY>
UDP	BLOCK_FUEL	Real	No	Yes	Invalid
UDP	BLOCK_TIME	Integer	No	Yes	Invalid
UDP	BRAKE_TIME	Integer	Yes	No	Invalid
UDP	CAA_CHK_ITEM_NO	String	Yes	No	Invalid
UDP	CAA_CHK_LINE_NO	String	Yes	No	Invalid
UDP	CAA_OTHER	String	Yes	No	Invalid
UDP	CAA_REASON	Integer	Yes	No	Invalid
ACP	CENTER_FUEL_DENSITY	Real	<n/a>	<n/a>	<CENTER_FUEL_DENSITY>
ACP	CMCF_A664_FLIGHT_LEG_NUMBER	Integer	<n/a>	<n/a>	<CMCF_FLIGHT_LEG_NUMBER>
ACP	CMCF_FLIGHT_PHASE	Integer	<n/a>	<n/a>	<CMCF_FLIGHT_PHASE>
UDP	CMCF_FLIGHT_PHASE_CURRENT	Integer	Yes	No	Invalid
UDP	CMCF_FLT_PHASE_TEXT	String	Yes	No	
ACP	CMF_CABINET_POSITION	Integer	<n/a>	<n/a>	<CMF_CABINET_POSITION>
ACP	CMF_LINK_STATUS	Integer	<n/a>	<n/a>	<CMU_LINK_STATUS>
UDP	CMF_LINK_STRENGTH	Integer	Yes	No	0

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
ACP	CURRENT_DATE	Integer	<n/a>	<n/a>	<CURRENT_DATE>
ACP	CURRENT_DATE_TIME	Integer	<n/a>	<n/a>	<CURRENT_DATE_TIME>
ACP	CURRENT_DAY	Integer	<n/a>	<n/a>	<CURRENT_DAY>
ACP	CURRENT_REGION	Integer	<n/a>	<n/a>	<CURRENT_REGION>
ACP	CURRENT_TIME	Integer	<n/a>	<n/a>	<CURRENT_TIME>
UDP	CUR_EMERGENCY_STATE	Boolean	Yes	No	FALSE
ACP	DEFAULT_CHARACTERS_PER_LINE	Integer	<n/a>	<n/a>	<DEFAULT_CHARACTERS_PER_LINE>
UDP	DEPT_TIME	Integer	Yes	No	<OUT_TIME>
UDP	DISP_MSG_TITLE	String	Yes	No	Invalid
UDP	DIV_AUTO_FLAG	Boolean	Yes	No	FALSE
UDP	DIV_DEST	String	Yes	No	<FMF_ACTRTEDEST>
UDP	DIV_DEST_ETA	Integer	Yes	No	<FMF_DEST_ETA>
UDP	DIV_DEST_FUEL	Real	Yes	No	<FMF_FUEL_AT_DEST>
UDP	DIV_FUEL_ON_BOARD	Real	Yes	No	<FUEL_ON_BOARD>
UDP	DIV_PREV_DEST	String	Yes	No	Invalid
UDP	DIV_REASON	Integer	Yes	No	Invalid
UDP	DIV_REASON_OTHER	String	Yes	No	Invalid
UDP	DIV_TXT_1	String	Yes	No	Invalid
UDP	DIV_TXT_2	String	Yes	No	Invalid
UDP	DIV_TXT_3	String	Yes	No	Invalid
UDP	DLY_ARRIVAL_RESN	Integer	Yes	No	Invalid
UDP	DLY_ARRIVAL_TEXT	String	Yes	No	Invalid
UDP	DLY_DEPT_RESN	Integer	Yes	No	Invalid
UDP	DLY_DEPT_TEXT	String	Yes	No	Invalid
UDP	DLY_ENROUTE_RESN	Integer	Yes	No	Invalid
UDP	DLY_ENROUTE_TEXT	String	Yes	No	Invalid
UDP	DLY_FUEL_ON_BD	Real	Yes	No	<FUEL_ON_BOARD>

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
UDP	DLY_SELECT	Integer	No	No	Invalid
UDP	DLY_SUPLMT_TXT_1	String	Yes	No	Invalid
UDP	DLY_SUPLMT_TXT_2	String	Yes	No	Invalid
UDP	DLY_SUPLMT_TXT_3	String	Yes	No	Invalid
UDP	DLY_TAKEOFF_RESN	Integer	Yes	No	Invalid
UDP	DLY_TAKEOFF_TEXT	String	Yes	No	Invalid
UDP	DOORS_ALL_CLOSED	Boolean	Yes	No	Invalid
ACP	DOOR_AFT_CARGO_OPEN	Boolean	<n/a>	<n/a>	<DOOR_AFT_CARGO_OPEN>
ACP	DOOR_AFT_EE_ACCESS_OPEN	Boolean	<n/a>	<n/a>	<DOOR_AFT_EE_ACCESS_OPEN>
ACP	DOOR_BULK_CARGO_OPEN	Boolean	<n/a>	<n/a>	<DOOR_BULK_CARGO_OPEN>
ACP	DOOR_ENTRY_1L_OPEN	Boolean	<n/a>	<n/a>	<DOOR_ENTRY_1L_OPEN>
ACP	DOOR_ENTRY_1R_OPEN	Boolean	<n/a>	<n/a>	<DOOR_ENTRY_1R_OPEN>
ACP	DOOR_ENTRY_2L_OPEN	Boolean	<n/a>	<n/a>	<DOOR_ENTRY_2L_OPEN>
ACP	DOOR_ENTRY_2R_OPEN	Boolean	<n/a>	<n/a>	<DOOR_ENTRY_2R_OPEN>
ACP	DOOR_ENTRY_3L_OPEN	Boolean	<n/a>	<n/a>	<DOOR_ENTRY_3L_OPEN>
ACP	DOOR_ENTRY_3R_OPEN	Boolean	<n/a>	<n/a>	<DOOR_ENTRY_3R_OPEN>
ACP	DOOR_ENTRY_4L_OPEN	Boolean	<n/a>	<n/a>	<DOOR_ENTRY_4L_OPEN>
ACP	DOOR_ENTRY_4R_OPEN	Boolean	<n/a>	<n/a>	<DOOR_ENTRY_4R_OPEN>
ACP	DOOR_FWD_ACCESS_OPEN	Boolean	<n/a>	<n/a>	<DOOR_FWD_ACCESS_OPEN>
ACP	DOOR_FWD_CARGO_OPEN	Boolean	<n/a>	<n/a>	<DOOR_FWD_CARGO_OPEN>
ACP	DOOR_FWD_EE_ACCESS_OPEN	Boolean	<n/a>	<n/a>	<DOOR_FWD_EE_ACCESS_OPEN>
ACP	DOOR_FWD_OVERHEAD_OPEN	Boolean	<n/a>	<n/a>	<DOOR_FWD_OVERHEAD_OPEN>
ACP	ECS_APU_RUNNING	Boolean	<n/a>	<n/a>	<ECS_APU_RUNNING>
ACP	EEC_ENG_OIL_P_L	Real	<n/a>	<n/a>	<EEC_ENG_OIL_P_L>
ACP	EEC_ENG_OIL_P_R	Real	<n/a>	<n/a>	<EEC_ENG_OIL_P_R>
ACP	EEC_ENG_OIL_T_L	Real	<n/a>	<n/a>	<EEC_ENG_OIL_T_L>
ACP	EEC_ENG_OIL_T_R	Real	<n/a>	<n/a>	<EEC_ENG_OIL_T_R>

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
ACP	EICAS_ENG_N1_SPD_L	Real	<n/a>	<n/a>	<EICAS_ENG_N1_SPD_L>
ACP	EICAS_ENG_N1_SPD_R	Real	<n/a>	<n/a>	<EICAS_ENG_N1_SPD_R>
UDP	EMAIL_DL_CC_1_DOMAIN	String	Yes	No	Invalid
UDP	EMAIL_DL_CC_1_LOCAL	String	Yes	No	Invalid
UDP	EMAIL_DL_CC_2_DOMAIN	String	Yes	No	Invalid
UDP	EMAIL_DL_CC_2_LOCAL	String	Yes	No	Invalid
UDP	EMAIL_DL_CC_3_DOMAIN	String	Yes	No	Invalid
UDP	EMAIL_DL_CC_3_LOCAL	String	Yes	No	Invalid
UDP	EMAIL_DL_REPLY_SUBJECT	String	Yes	No	<EMAIL_UL_SUBJECT>
UDP	EMAIL_DL_REPLY_TO_DOMAIN	String	Yes	No	<EMAIL_UL_FROM_DOMAIN>
UDP	EMAIL_DL_REPLY_TO_LOCAL	String	Yes	No	<EMAIL_UL_FROM_LOCAL>
UDP	EMAIL_DL_REPLY_TXT_1	String	Yes	No	Invalid
UDP	EMAIL_DL_REPLY_TXT_2	String	Yes	No	Invalid
UDP	EMAIL_DL_REPLY_TXT_3	String	Yes	No	Invalid
UDP	EMAIL_DL_REPLY_TXT_4	String	Yes	No	Invalid
UDP	EMAIL_DL_REPLY_TXT_5	String	Yes	No	Invalid
UDP	EMAIL_DL_REPLY_TXT_6	String	Yes	No	Invalid
UDP	EMAIL_DL_REPLY_TXT_7	String	Yes	No	Invalid
UDP	EMAIL_DL_REPLY_TXT_8	String	Yes	No	Invalid
UDP	EMAIL_DL_REPLY_TXT_9	String	Yes	No	Invalid
UDP	EMAIL_DL_SUBJECT	String	Yes	No	Invalid
UDP	EMAIL_DL_TO_DOMAIN	String	Yes	No	Invalid
UDP	EMAIL_DL_TO_LOCAL	String	Yes	No	Invalid
UDP	EMAIL_DL_TXT_1	String	Yes	No	Invalid
UDP	EMAIL_DL_TXT_2	String	Yes	No	Invalid
UDP	EMAIL_DL_TXT_3	String	Yes	No	Invalid
UDP	EMAIL_DL_TXT_4	String	Yes	No	Invalid

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
UDP	EMAIL_DL_TXT_5	String	Yes	No	Invalid
UDP	EMAIL_DL_TXT_6	String	Yes	No	Invalid
UDP	EMAIL_DL_TXT_7	String	Yes	No	Invalid
UDP	EMAIL_DL_TXT_8	String	Yes	No	Invalid
UDP	EMAIL_DL_TXT_9	String	Yes	No	Invalid
UDP	EMAIL_UL_FROM_DOMAIN	String	Yes	No	Invalid
UDP	EMAIL_UL_FROM_LOCAL	String	Yes	No	Invalid
UDP	EMAIL_UL_SUBJECT	String	Yes	No	Invalid
UDP	EMERGENCY_CLBRATE_THRESHOLD	Integer	No	Yes	7500
UDP	EMERGENCY_COUNTER	Integer	Yes	No	Invalid
UDP	EMERGENCY_DESRATE_THRESHOLD	Integer	No	Yes	7500
UDP	EMERGENCY_ENABLED	Boolean	Yes	Yes	FALSE
UDP	EMERGENCY_FREQUENCY	Integer	No	Yes	60
UDP	EMERGENCY_FUEL_THRESHOLD	Integer	No	Yes	5500
UDP	EMERGENCY_LOW_SPEED_THRESHOLD	Integer	No	Yes	100
UDP	EMERGENCY_MIN_ALTITUDE	Integer	No	Yes	10000
UDP	EMERGENCY_MMO_THRESHOLD	Real	No	Yes	0.05
UDP	EMERGENCY_N1_THRESHOLD	Integer	No	Yes	5
UDP	EMERGENCY_PITCH_RATE_THRESHOLD	Real	No	Yes	10
UDP	EMERGENCY_PITCH_THRESHOLD	Real	No	Yes	20
UDP	EMERGENCY_SELECTION	Integer	Yes	No	2
UDP	EMERGENCY_STATE_ATTITUDE	Boolean	Yes	No	FALSE
UDP	EMERGENCY_STATE_CLB	Boolean	Yes	No	FALSE
UDP	EMERGENCY_STATE_DES	Boolean	Yes	No	FALSE
UDP	EMERGENCY_STATE_EO	Boolean	Yes	No	FALSE
UDP	EMERGENCY_STATE_FUEL	Boolean	Yes	No	FALSE
UDP	EMERGENCY_STATE_SPD	Boolean	Yes	No	FALSE

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
UDP	EMERGENCY_STATE_VERT_RATE	Boolean	Yes	No	FALSE
UDP	EMERGENCY_TPR_THRESHOLD	Real	No	Yes	5
UDP	EMERGENCY_VERT_ACCEL_THRESHOLD	Real	No	Yes	2.5
UDP	EMERGENCY_VMO_THRESHOLD	Integer	No	Yes	30
UDP	EMERGENCY_YAW_RATE_THRESHOLD	Real	No	Yes	10
UDP	EMER_SITU_TXT_1	String	Yes	No	Invalid
UDP	EMER_SITU_TXT_2	String	Yes	No	Invalid
UDP	EMER_SITU_TXT_3	String	Yes	No	Invalid
UDP	END_EMERGENCY	Boolean	Yes	No	FALSE
UDP	ENGINE_OUT_LEFT	Boolean	Yes	No	FALSE
UDP	ENGINE_OUT_RIGHT	Boolean	Yes	No	FALSE
ACP	ENGINE_TYPE	Integer	<n/a>	<n/a>	<ENGINE_TYPE>
ACP	ENG_L_START_SW	Integer	<n/a>	<n/a>	<ENG_L_START_SW>
ACP	ENG_R_START_SW	Integer	<n/a>	<n/a>	<ENG_R_START_SW>
ACP	ENG_TPR_ACTUAL_L	Real	<n/a>	<n/a>	<ENG_TPR_ACTUAL_L>
ACP	ENG_TPR_ACTUAL_R	Real	<n/a>	<n/a>	<ENG_TPR_ACTUAL_R>
UDP	ENTERED_FLTNO	String	Yes	No	<FMF_CMF_FLIGHT_NUMBER>
UDP	EOF_TIMER	Integer	No	Yes	Invalid
UDP	EST_IN_TIME	Integer	Yes	No	Invalid
UDP	EST_OFF_TIME	Integer	Yes	No	Invalid
UDP	EST_ON_TIME	Integer	Yes	No	<FMF_DEST_ETA>
UDP	EST_OUT_TIME	Integer	Yes	No	Invalid
UDP	ETA_CHANGE	Integer	Yes	No	Invalid
UDP	ETA_ETA	Integer	Yes	No	<FMF_DEST_ETA>
UDP	ETA_FUEL_ON_BD	Real	Yes	No	<FUEL_ON_BOARD>
UDP	ETA_SUPLMT_TXT_1	String	Yes	No	Invalid
UDP	ETA_SUPLMT_TXT_2	String	Yes	No	Invalid

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
UDP	ETA_SUPLMT_TXT_3	String	Yes	No	Invalid
ACP	EXTERNAL_UPLINK_LABEL	String	<n/a>	<n/a>	<EXTERNAL_UPLINK_LABEL>
ACP	EXTERNAL_UPLINK_TEXT	Text Block	<n/a>	<n/a>	<EXTERNAL_UPLINK_TEXT>
ACP	FCM_AP_ENGD	Boolean	<n/a>	<n/a>	<FCM_AP_ENGD>
ACP	FCM_FLAPS_SLATS_NORM	Real	<n/a>	<n/a>	<FCM_FLAPS_SLATS_NORM>
ACP	FCM_FLAP_LVR_POSN	Real	<n/a>	<n/a>	<FCM_FLAP_LVR_POSN>
ACP	FCM_LAND_2_ASA	Boolean	<n/a>	<n/a>	<FCM_LAND_2_ASA>
ACP	FCM_LAND_3_ASA	Boolean	<n/a>	<n/a>	<FCM_LAND_3_ASA>
ACP	FCM_NO_AUTOLAND_ASA	Boolean	<n/a>	<n/a>	<FCM_NO_AUTOLAND_ASA>
ACP	FCM_TOGA_SW_VOTED	Boolean	<n/a>	<n/a>	<FCM_TOGA_SW_VOTED>
UDP	FLIGHT_FUEL	Real	No	Yes	Invalid
UDP	FLIGHT_TIME	Integer	No	Yes	Invalid
ACP	FMF_ACTIVE_WPT	String	<n/a>	<n/a>	<FMF_ACTIVE_WPT>
ACP	FMF_ACTRTEDEST	String	<n/a>	<n/a>	<FMF_ACTRTEDEST>
ACP	FMF_ACTRTEORIGIN	String	<n/a>	<n/a>	<FMF_ACTRTEORIGIN>
ACP	FMF_ALT_AT_LAST_WYPT	Real	<n/a>	<n/a>	<FMF_ALT_AT_LAST_WYPT>
ACP	FMF_ATA_LAST_WYPT	Integer	<n/a>	<n/a>	<FMF_ATA_LAST_WYPT>
ACP	FMF_CG	Real	<n/a>	<n/a>	<FMF_CG>
ACP	FMF_CMF_FLIGHT_NUMBER	String	<n/a>	<n/a>	<ACARS_10_CHAR_FLIGHT_NUMBER>
ACP	FMF_COST_INDEX	Real	<n/a>	<n/a>	<FMF_COST_INDEX>
UDP	FMF_DEST_ETA	Integer	Yes	No	Invalid
ACP	FMF_DEST_RNWDYID	String	<n/a>	<n/a>	<FMF_DEST_RNWDYID>
ACP	FMF_DIST_TO_DEST	Real	<n/a>	<n/a>	<FMF_DIST_TO_DEST>
ACP	FMF_DIST_TO_GO	Real	<n/a>	<n/a>	<FMF_DIST_TO_GO>
ACP	FMF_ENG_OUT_MAX_ALT	Real	<n/a>	<n/a>	<FMF_ENG_OUT_MAX_ALT>
ACP	FMF_ETA	Integer	<n/a>	<n/a>	<ETA_ACTIVE_WAYPOINT>
UDP	FMF_ETA_DISPLAY	Integer	Yes	No	Invalid

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
ACP	FMF_ETA_TOP_OF_DESC	Integer	<n/a>	<n/a>	<ETA_TOP_OF_DESCENT>
ACP	FMF_FIRST_ATC_COMPULSORY_WPT	String	<n/a>	<n/a>	<FMF_FIRST_ATC_COMPULSORY_WPT>
ACP	FMF_FIRST_ATC_COMPULS_WPT_ETA	Integer	<n/a>	<n/a>	<ETA_FIRST_ATC_COMPULS_WPT>
ACP	FMF_FLIGHT_PHASE	Integer	<n/a>	<n/a>	<FMF_FLIGHT_PHASE>
ACP	FMF_FUEL_AT_DEST	Real	<n/a>	<n/a>	<FUEL_AT_DESTINATION>
ACP	FMF_FUEL_REMAINING_WEIGHT	Real	<n/a>	<n/a>	<CUR_FUELREMAIN_WT>
ACP	FMF_FUEL_USED_L	Real	<n/a>	<n/a>	<FUEL_USED_L>
ACP	FMF_FUEL_USED_R	Real	<n/a>	<n/a>	<FUEL_USED_R>
ACP	FMF_GROSS_WEIGHT	Real	<n/a>	<n/a>	<GROSS_WEIGHT>
ACP	FMF_LAST_WPT	String	<n/a>	<n/a>	<FMF_LAST_WPT>
ACP	FMF_NEXT_WPT	String	<n/a>	<n/a>	<FMF_NEXT_WPT>
ACP	FMF_ORIGIN_RWY_ID	String	<n/a>	<n/a>	<FMF_ORG_RWY_ID>
ACP	FMF_SECOND_ATC_COMPULS_WPT	String	<n/a>	<n/a>	<FMF_SECOND_ATC_COMPULS_WPT>
ACP	FMF_TIME_FOR_CLIMB	Integer	<n/a>	<n/a>	<TIME_FOR_CLIMB>
ACP	FMF_TIME_TO_DEST	Integer	<n/a>	<n/a>	<TIME_TO_DESTINATION>
ACP	FMF_TRANSITION_ALT	Real	<n/a>	<n/a>	<FMF_TRANSITION_ALT>
ACP	FMF_ZERO_FUEL_WT	Real	<n/a>	<n/a>	<ZERO_FUEL_WEIGHT>
UDP	FRL_CHK_ITEM_NO	String	Yes	No	Invalid
UDP	FRL_CHK_LINE_NO	String	Yes	No	Invalid
UDP	FRL_OTHER	String	Yes	No	Invalid
UDP	FRL_REASON	Integer	Yes	No	Invalid
UDP	FTREL_CO RTE	String	Yes	No	Invalid
UDP	FTREL_FLT_LVL	Integer	Yes	No	Invalid
UDP	FTREL_REQST_1	String	Yes	No	Invalid
UDP	FTREL_REQST_2	String	Yes	No	Invalid
UDP	FTREL_REQST_3	String	Yes	No	Invalid
UDP	FTREL_ROUTE	Boolean	Yes	No	Invalid

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
UDP	FTREL_RTE_WIND	Boolean	Yes	No	Invalid
UDP	FTREL_WIND	Boolean	Yes	No	Invalid
UDP	FUEL_ON_BOARD	Real	No	No	Invalid
UDP	FUEL_ON_BOARD_CHECK	Real	Yes	No	Invalid
ACP	FUEL_SW_L_CUTOFF	Boolean	<n/a>	<n/a>	<FUEL_SW_L_CUTOFF>
ACP	FUEL_SW_R_CUTOFF	Boolean	<n/a>	<n/a>	<FUEL_SW_R_CUTOFF>
UDP	GATE_REQUEST_1	String	Yes	No	Invalid
UDP	GATE_REQUEST_2	String	Yes	No	Invalid
UDP	GATE_REQUEST_3	String	Yes	No	Invalid
UDP	GATE_RQST_FLTNO	String	Yes	No	<ENTERED_FLTNO>
UDP	GTA_OTHER	String	Yes	No	Invalid
UDP	GTA_REASON	Integer	Yes	No	Invalid
ACP	HFDR_DATA_CHANNEL_AVAILABLE	Boolean	<n/a>	<n/a>	<HFDR_DATA_CHANNEL_AVAILABLE>
ACP	HF_LINK_STATUS	Integer	<n/a>	<n/a>	<HF_LINK_STATUS>
UDP	HF_STRENGTH	Integer	Yes	No	0
ACP	HF_VOICE_MODE	Boolean	<n/a>	<n/a>	<HF_VOICE_MODE>
ACP	ICAO_AIRCRAFT_ADDRESS	Integer	<n/a>	<n/a>	<ICAO_AIRCRAFT_ADDRESS>
UDP	IN	Integer	Yes	No	4
UDP	INIT_ALTN_STA_1	String	No	Yes	Invalid
UDP	INIT_ALTN_STA_2	String	No	Yes	Invalid
UDP	INIT_BOARDED_FUEL	Real	No	Yes	Invalid
UDP	INIT_DEST_STA	String	Yes	No	<FMF_ACTRTEDEST>
UDP	INIT_FLIGHT_PLAN_TIME	Integer	No	Yes	Invalid
UDP	INIT_FOB	Real	Yes	No	<FUEL_ON_BOARD>
UDP	INIT_ORIG_STA	String	Yes	No	<FMF_ACTRTEORIGIN>
UDP	INIT_PAY_CAPT	String	No	Yes	Invalid
UDP	INIT_PAY_FO_1	String	No	Yes	Invalid

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
UDP	INIT_PAY_FO_2	String	No	Yes	Invalid
UDP	INIT_PAY_FO_3	String	No	Yes	Invalid
UDP	INIT_SKED_DATE	Integer	No	Yes	Invalid
UDP	IN_FOB	Real	No	Yes	Invalid
UDP	IN_TIME	Integer	No	Yes	Invalid
ACP	IRU_RDC_BODY_PITCH_RATE	Real	<n/a>	<n/a>	<IRU_RDC_BODY_PITCH_RATE>
ACP	IRU_RDC_BODY_YAW_RATE	Real	<n/a>	<n/a>	<IRU_RDC_BODY_YAW_RATE>
ACP	IRU_RDC_INERTIAL_VERT_SPEED	Real	<n/a>	<n/a>	<IRU_RDC_INERTIAL_VERT_SPEED>
ACP	IRU_RDC_PITCH_ANGLE	Real	<n/a>	<n/a>	<IRU_RDC_PITCH_ANGLE>
ACP	IRU_RDC_VERT_ACCELERATION	Real	<n/a>	<n/a>	<IRU_RDC_VERT_ACCELERATION>
ACP	LANDING_GEAR	Integer	<n/a>	<n/a>	<LANDING_GEAR>
UDP	LAST_ACTIVE_WPT	String	Yes	No	Invalid
UDP	LAST_EMERGENCY_SELECTION	Integer	Yes	No	2
UDP	LAST_EMERGENCY_STATE	Boolean	Yes	No	FALSE
UDP	LAST_NAV_POS_LAT	Real	Yes	No	Invalid
UDP	LAST_NAV_POS_LONG	Real	Yes	No	Invalid
UDP	LDS_CAPTAIN_CODE	String	Yes	No	Invalid
UDP	LDS_CAPTAIN_ID	String	Yes	No	Invalid
UDP	LDS_CONFIRM_REVIEW	Integer	Yes	No	Invalid
UDP	LDS_DATE	String	Yes	No	Invalid
UDP	LDS_EDNO	String	Yes	No	Invalid
UDP	LDS_FLIGHTNO	String	Yes	No	Invalid
UDP	LDS_TYPE	String	Yes	No	Invalid
ACP	LEFT_MAIN_FUEL_DENSITY	Real	<n/a>	<n/a>	<LEFT_MAIN_FUEL_DENSITY>
ACP	LOGIC_UNIT_MESSAGES_ENABLED	Boolean	<n/a>	<n/a>	<LOGIC_UNIT_MESSAGES_ENABLED>
ACP	LRRA_RADIO_HEIGHT	Real	<n/a>	<n/a>	<LRRA_RADIO_HEIGHT>
ACP	L_ENG_OIL_QTY	Real	<n/a>	<n/a>	<L_ENG_OIL_QTY>

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
ACP	MAINT_CONFIG_ADM_AIRLINE	String	<n/a>	<n/a>	<MAINT_CONFIG_ADM_AIRLINE>
ACP	MAINT_CONFIG_ADM_NONAIRLINE	Integer	<n/a>	<n/a>	<MAINT_CONFIG_ADM_NONAIRLINE>
ACP	MAINT_CONFIG_VER_ID	Integer	<n/a>	<n/a>	<MAINT_CONFIG_VER_ID>
UDP	MAINT_TEXT_1	String	Yes	No	Invalid
UDP	MAINT_TEXT_2	String	Yes	No	Invalid
UDP	MAINT_TEXT_3	String	Yes	No	Invalid
UDP	MAINT_TEXT_4	String	Yes	No	Invalid
UDP	MAINT_TEXT_5	String	Yes	No	Invalid
UDP	MAINT_TEXT_6	String	Yes	No	Invalid
UDP	MANUAL_EMERGENCY	Boolean	Yes	No	Invalid
ACP	MASTER_HFDR	Integer	<n/a>	<n/a>	<MASTER_HFDR>
ACP	MASTER_SDU	Integer	<n/a>	<n/a>	<MASTER_SDU>
ACP	MASTER_VDR	Integer	<n/a>	<n/a>	<MASTER_VDR>
UDP	MAX_EMERGENCY_RPT	Integer	No	Yes	60
ACP	METRIC_ENABLE_ON	Boolean	<n/a>	<n/a>	<METRIC_ENABLE_ON>
UDP	MISC_CODE	String	Yes	No	Invalid
UDP	MISC_TEXT_1	String	Yes	No	Invalid
UDP	MISC_TEXT_2	String	Yes	No	Invalid
UDP	MISC_TEXT_3	String	Yes	No	Invalid
UDP	MISC_TEXT_4	String	Yes	No	Invalid
UDP	MISC_TEXT_5	String	Yes	No	Invalid
UDP	MISC_TEXT_6	String	Yes	No	Invalid
UDP	MMO	Real	Yes	No	0.9
ACP	NAV_ALTITUDE	Real	<n/a>	<n/a>	<NAV_ALTITUDE>
ACP	NAV_DATE_DAY	Integer	<n/a>	<n/a>	<NAV_DATE_DAY>
ACP	NAV_DATE_MONTH	Integer	<n/a>	<n/a>	<NAV_DATE_MONTH>
ACP	NAV_DATE_YEAR	Integer	<n/a>	<n/a>	<NAV_DATE_YEAR>

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
ACP	NAV_GROUNDSPEED	Real	<n/a>	<n/a>	<NAV_GROUNDSPEED>
ACP	NAV_MAG_HDG	Real	<n/a>	<n/a>	<NAV_MAG_HDG>
ACP	NAV_MAG_TRACK	Real	<n/a>	<n/a>	<NAV_MAG_TRACK>
ACP	NAV_POS_LAT	Real	<n/a>	<n/a>	<NAV_POS_LAT>
ACP	NAV_POS_LON	Real	<n/a>	<n/a>	<NAV_POS_LON>
ACP	NAV_TRUE_HDG	Real	<n/a>	<n/a>	<NAV_TRUE_HDG>
ACP	NAV_TRUE_TRACK	Real	<n/a>	<n/a>	<NAV_TRUE_TRACK>
ACP	NAV_UTC_HOURS	Integer	<n/a>	<n/a>	<NAV_UTC_HOURS>
ACP	NAV_UTC_MINUTES	Integer	<n/a>	<n/a>	<NAV_UTC_MINUTES>
ACP	NAV_UTC_SECONDS	Integer	<n/a>	<n/a>	<NAV_UTC_SECONDS>
ACP	NAV_WIND_MAG	Real	<n/a>	<n/a>	<NAV_WIND_MAG>
ACP	NAV_WIND_TRUE_BRG	Real	<n/a>	<n/a>	<NAV_WIND_TRUE_BRG>
ACP	ND_HEADING_CAPT	Real	<n/a>	<n/a>	<ND_HEADING_CAPT>
ACP	ND_HEADING_FO	Real	<n/a>	<n/a>	<ND_HEADING_FO>
UDP	NEW_EMERGENCY	Boolean	Yes	No	FALSE
UDP	NOTAM_ENROUTE	String	Yes	No	Invalid
UDP	NOTAM_REQST_1	String	Yes	No	Invalid
UDP	NOTAM_REQST_2	String	Yes	No	Invalid
UDP	NOTAM_REQST_3	String	Yes	No	Invalid
UDP	NOTAM_SEND	Boolean	Yes	No	Invalid
UDP	NOTAM_STA_1	String	Yes	No	Invalid
UDP	NOTAM_STA_2	String	Yes	No	Invalid
UDP	NOTAM_UL_TITLE	String	Yes	No	Invalid
UDP	OFF	Integer	Yes	No	2
UDP	OFF_FOB	Real	No	Yes	Invalid
UDP	OFF_STA	String	No	Yes	Invalid
UDP	OFF_TIME	Integer	No	Yes	Invalid

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
UDP	OLD_REMINDER_TIME_1	Integer	Yes	No	Invalid
UDP	OLD_REMINDER_TIME_2	Integer	Yes	No	Invalid
UDP	OLD_REMINDER_TIME_3	Integer	Yes	No	Invalid
UDP	ON	Integer	Yes	No	3
UDP	ON_FOB	Real	No	Yes	Invalid
UDP	ON_STA	String	No	Yes	Invalid
UDP	ON_TIME	Integer	No	Yes	Invalid
UDP	OOOI_ENABLED	Boolean	No	No	Invalid
UDP	OOOI_STATE	Integer	Yes	No	Invalid
UDP	OOOI_STATUS	String	Yes	No	Invalid
UDP	OUT	Integer	Yes	No	1
UDP	OUT_FOB	Real	No	Yes	Invalid
UDP	OUT_TIME	Integer	No	Yes	Invalid
UDP	OVER_ACCEL	Boolean	Yes	No	FALSE
UDP	OVER_CAS	Boolean	Yes	No	FALSE
UDP	OVER_MACH	Boolean	Yes	No	FALSE
UDP	OVER_PITCH	Boolean	Yes	No	FALSE
UDP	OVER_PITCH_RATE	Boolean	Yes	No	FALSE
UDP	OVER_YAW_RATE	Boolean	Yes	No	FALSE
ACP	PARKINGBRKLEVERSET	Boolean	<n/a>	<n/a>	<PARKING_BRK_LEVER_SET>
UDP	POA_AUTOTUNE_ENABLE_WRITE	Boolean	Yes	No	FALSE
UDP	POA_AUTOTUNE_FREQUENCY	Real	Yes	No	Invalid
UDP	POS_RPT_COUNTER	Integer	Yes	No	0
UDP	POS_RPT_ENABLE	Boolean	No	Yes	TRUE
UDP	POS_RPT_INTERVAL	Integer	No	Yes	15
UDP	PREV_BLOCK_FUEL	Real	No	Yes	Invalid
UDP	PREV_BLOCK_TIME	Integer	No	Yes	Invalid

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
UDP	PREV_BOARDED_FUEL	Real	No	Yes	Invalid
UDP	PREV_DEST_STA	String	No	Yes	Invalid
UDP	PREV_FLIGHT_FUEL	Real	No	Yes	Invalid
UDP	PREV_FLIGHT_NUMBER	String	No	Yes	Invalid
UDP	PREV_FLIGHT_TIME	Integer	No	Yes	Invalid
UDP	PREV_IN_FOB	Real	No	Yes	Invalid
UDP	PREV_IN_TIME	Integer	No	Yes	Invalid
UDP	PREV_OFF_FOB	Real	No	Yes	Invalid
UDP	PREV_OFF_TIME	Integer	No	Yes	Invalid
UDP	PREV_ON_FOB	Real	No	Yes	Invalid
UDP	PREV_ON_TIME	Integer	No	Yes	Invalid
UDP	PREV_ORIG_STA	String	No	Yes	Invalid
UDP	PREV_OUT_FOB	Real	No	Yes	Invalid
UDP	PREV_OUT_TIME	Integer	No	Yes	Invalid
ACP	PRINTER_STATUS	Integer	<n/a>	<n/a>	<PRINTER_STATUS>
ACP	PRINT_QUEUE_FULL	Boolean	<n/a>	<n/a>	<PRINT_QUEUE_FULL>
UDP	PRINT_TXT_BLK	Text Block	Yes	No	Invalid
UDP	PR_ACT_CTR	String	Yes	No	Invalid
UDP	PR_NEXT_CTR	String	Yes	No	Invalid
UDP	PR_ORIGINATOR	String	Yes	No	Invalid
UDP	PR_POSITION	String	Yes	No	Invalid
UDP	PR_TEXT_1	String	Yes	No	Invalid
UDP	PR_TEXT_10	String	Yes	No	Invalid
UDP	PR_TEXT_11	String	Yes	No	Invalid
UDP	PR_TEXT_12	String	Yes	No	Invalid
UDP	PR_TEXT_13	String	Yes	No	Invalid
UDP	PR_TEXT_14	String	Yes	No	Invalid

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
UDP	PR_TEXT_15	String	Yes	No	Invalid
UDP	PR_TEXT_16	String	Yes	No	Invalid
UDP	PR_TEXT_17	String	Yes	No	Invalid
UDP	PR_TEXT_18	String	Yes	No	Invalid
UDP	PR_TEXT_2	String	Yes	No	Invalid
UDP	PR_TEXT_3	String	Yes	No	Invalid
UDP	PR_TEXT_4	String	Yes	No	Invalid
UDP	PR_TEXT_5	String	Yes	No	Invalid
UDP	PR_TEXT_6	String	Yes	No	Invalid
UDP	PR_TEXT_7	String	Yes	No	Invalid
UDP	PR_TEXT_8	String	Yes	No	Invalid
UDP	PR_TEXT_9	String	Yes	No	Invalid
UDP	PR_UTC_DATE	Integer	Yes	No	Invalid
UDP	PR_UTC_TIME	Integer	Yes	No	Invalid
UDP	RA1_TITLE	String	Yes	No	Invalid
UDP	RA2_GND_ADDR	String	Yes	No	Invalid
UDP	RA4_GND_ADDR	String	Yes	No	Invalid
UDP	RA4_TXT_BLK	Text Block	Yes	No	Invalid
UDP	RADIO_ALT_REACHED	Boolean	No	Yes	Invalid
UDP	REGISTRATION_NO	String	Yes	No	Invalid
UDP	REMINDER_ALTITUDE	Integer	Yes	No	Invalid
UDP	REMINDER_ALTITUDE_DISPLAY	Integer	Yes	No	Invalid
UDP	REMINDER_ALTITUDE_TEXT	String	Yes	No	Invalid
UDP	REMINDER_ENABLE_ETA_CHANGE	Boolean	Yes	No	Invalid
UDP	REMINDER_ETA_BASELINE	Integer	Yes	No	Invalid
UDP	REMINDER_ETA_BASELINE_DISPLAY	Integer	Yes	No	Invalid
UDP	REMINDER_ETA_CHANGE	Integer	Yes	No	3

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
UDP	REMINDER_ETA_CHANGE_DISPLAY	Integer	Yes	No	Invalid
UDP	REMINDER_ETA_CHANGE_TEXT	String	Yes	No	Invalid
UDP	REMINDER_FL	Integer	Yes	No	Invalid
UDP	REMINDER_FL_ALT	Integer	Yes	No	Invalid
UDP	REMINDER_FL_DISPLAY	Integer	Yes	No	Invalid
UDP	REMINDER_FL_TEXT	String	Yes	No	Invalid
UDP	REMINDER_FMF_ETA	Integer	Yes	No	Invalid
UDP	REMINDER_FMF_LAST_WPT	String	Yes	No	Invalid
UDP	REMINDER_FUEL	Integer	Yes	No	Invalid
UDP	REMINDER_FUEL_DISPLAY	Integer	Yes	No	Invalid
UDP	REMINDER_FUEL_TEXT	String	Yes	No	Invalid
UDP	REMINDER_LAT	Integer	Yes	No	Invalid
UDP	REMINDER_LAT_DIR	String	Yes	No	Invalid
UDP	REMINDER_LAT_DIR_DISPLAY	String	Yes	No	Invalid
UDP	REMINDER_LAT_DISPLAY	Integer	Yes	No	Invalid
UDP	REMINDER_LAT_TEXT	String	Yes	No	Invalid
UDP	REMINDER_LONG	Integer	Yes	No	Invalid
UDP	REMINDER_LONG_DIR	String	Yes	No	Invalid
UDP	REMINDER_LONG_DIR_DISPLAY	String	Yes	No	Invalid
UDP	REMINDER_LONG_DISPLAY	Integer	Yes	No	Invalid
UDP	REMINDER_LONG_TEXT	String	Yes	No	Invalid
UDP	REMINDER_POSN	String	Yes	No	Invalid
UDP	REMINDER_POSN_DISPLAY	String	Yes	No	Invalid
UDP	REMINDER_POSN_TEXT	String	Yes	No	Invalid
UDP	REMINDER_SIDELINK_ALT_B	String	Yes	No	Invalid
UDP	REMINDER_SIDELINK_ETA_B	String	Yes	No	Invalid
UDP	REMINDER_SIDELINK_FL_B	String	Yes	No	Invalid

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
UDP	REMINDER_SIDELINK_FUEL_B	String	Yes	No	Invalid
UDP	REMINDER_SIDELINK_LAT_B	String	Yes	No	Invalid
UDP	REMINDER_SIDELINK_LONG_B	String	Yes	No	Invalid
UDP	REMINDER_SIDELINK_POSN_B	String	Yes	No	Invalid
UDP	REMINDER_SIDELINK_TIME1_B	String	Yes	No	Invalid
UDP	REMINDER_SIDELINK_TIME2_B	String	Yes	No	Invalid
UDP	REMINDER_SIDELINK_TIME3_B	String	Yes	No	Invalid
UDP	REMINDER_SIDELINK_TIME_DEST_B	String	Yes	No	Invalid
UDP	REMINDER_SIDELINK_TIME_TOD_B	String	Yes	No	Invalid
UDP	REMINDER_TIME_1	Integer	Yes	No	Invalid
UDP	REMINDER_TIME_1_DISPLAY	Integer	Yes	No	Invalid
UDP	REMINDER_TIME_2	Integer	Yes	No	Invalid
UDP	REMINDER_TIME_2_DISPLAY	Integer	Yes	No	Invalid
UDP	REMINDER_TIME_3	Integer	Yes	No	Invalid
UDP	REMINDER_TIME_3_DISPLAY	Integer	Yes	No	Invalid
UDP	REMINDER_TIME_DEST	Integer	Yes	No	Invalid
UDP	REMINDER_TIME_DEST_DISPLAY	Integer	Yes	No	Invalid
UDP	REMINDER_TIME_TEXT_1	String	Yes	No	CREW CHANGE
UDP	REMINDER_TIME_TEXT_2	String	Yes	No	Invalid
UDP	REMINDER_TIME_TEXT_3	String	Yes	No	Invalid
UDP	REMINDER_TIME_TEXT_DEST	String	Yes	No	Invalid
UDP	REMINDER_TIME_TEXT_TOD	String	Yes	No	Invalid
UDP	REMINDER_TIME_TOD	Integer	Yes	No	Invalid
UDP	REMINDER_TIME_TOD_DISPLAY	Integer	Yes	No	Invalid
UDP	RETURN_TO_GATE	Boolean	Yes	No	Invalid
UDP	REVISED_LAT	Integer	Yes	No	Invalid
UDP	REVISED_LONG	Integer	Yes	No	Invalid

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
UDP	REVISED_REMINDER_TIME_1	Integer	Yes	No	Invalid
UDP	REVISED_REMINDER_TIME_2	Integer	Yes	No	Invalid
UDP	REVISED_REMINDER_TIME_3	Integer	Yes	No	Invalid
ACP	RIGHT_MAIN_FUEL_DENSITY	Real	<n/a>	<n/a>	<RIGHT_MAIN_FUEL_DENSITY>
ACP	R_ENG_OIL_QTY	Real	<n/a>	<n/a>	<R_ENG_OIL_QTY>
ACP	SATCOM_DATA_CHANNEL_AVAILABLE	Boolean	<n/a>	<n/a>	<SATCOM_DATA_CHANNEL_AVAILABLE>
ACP	SATCOM_LINK_STATUS	Integer	<n/a>	<n/a>	<SATCOM_LINK_STATUS>
UDP	SATCOM_STRENGTH	Integer	Yes	No	0
UDP	TIME_AT_WPT_CHG	Integer	Yes	No	Invalid
ACP	TIME_SINCE_POWER_ON	Integer	<n/a>	<n/a>	<TIME_SINCE_POWER_ON>
UDP	TIME_SINCE_WPT_CHG	Integer	Yes	No	Invalid
UDP	TIME_TO_WPT_CHG	Integer	Yes	No	Invalid
UDP	TKO_FLAP	Integer	Yes	No	Invalid
UDP	TKO_FUEL	Real	Yes	No	<OFF_FOB>
UDP	TKO_N1	Real	Yes	No	Invalid
UDP	TKO_PILOT	Integer	Yes	No	Invalid
UDP	TKO_SUPLMT_TXT_1	String	Yes	No	Invalid
UDP	TKO_SUPLMT_TXT_2	String	Yes	No	Invalid
UDP	TKO_SUPLMT_TXT_3	String	Yes	No	Invalid
UDP	TKO_TIME	Integer	Yes	No	<OFF_TIME>
UDP	TKO_TPR	Integer	Yes	No	Invalid
ACP	TOTAL_FILTERED_FUEL_MASS	Integer	<n/a>	<n/a>	<TOTAL_FILTERED_FUEL_MASS>
UDP	TOUCH_AND_GO	Boolean	Yes	No	Invalid
UDP	UNDER_CAS	Boolean	Yes	No	FALSE
ACP	UPLINK_LABEL	String	<n/a>	<n/a>	<UPLINK_LABEL>
ACP	UPLINK_TEXT	Text Block	<n/a>	<n/a>	<UPLINK_TEXT>
UDP	UPLINK_TXT_BLK	Text Block	Yes	No	Invalid

Class	Parameter Name	Type	Default Upon Reset	Save In NVM	Default/Aliased Parameter or Initial Value
ACP	VDR_STATE	Integer	<n/a>	<n/a>	<VDR_STATE>
ACP	VDR_SUBSTATE_ACP	Integer	<n/a>	<n/a>	<VDR_SUBSTATE>
UDP	VDR_SUBSTATE_UDP	Integer	Yes	No	Invalid
ACP	VHF_LINK_STATUS	Integer	<n/a>	<n/a>	<VHF_LINK_STATUS>
UDP	VHF_STRENGTH	Integer	Yes	No	0
ACP	VHF_SUBNETWORK_TYPE	Integer	<n/a>	<n/a>	<VHF_SUBNETWORK_TYPE>
UDP	VHF_VOICE_FREQ	Real	Yes	No	Invalid
ACP	VHF_VOICE_MODE	Boolean	<n/a>	<n/a>	<VHF_VOICE_MODE>
UDP	VHF_VOICE_TXT_BLK	Text Block	Yes	No	Invalid
UDP	VMO	Integer	Yes	No	360
UDP	WAB_CHK_ITEM_NO	String	Yes	No	Invalid
UDP	WAB_CHK_LINE_NO	String	Yes	No	Invalid
UDP	WAB_OTHER	String	Yes	No	Invalid
UDP	WAB_REASON	Integer	Yes	No	Invalid
UDP	WTBAL_RQST_TXT_1	String	Yes	No	Invalid
UDP	WTBAL_RQST_TXT_2	String	Yes	No	Invalid
UDP	WTBAL_RQST_TXT_3	String	Yes	No	Invalid
UDP	WXR_STA_1	String	Yes	No	Invalid
UDP	WXR_STA_2	String	Yes	No	Invalid
UDP	WXR_STA_3	String	Yes	No	Invalid
UDP	WXR_STA_4	String	Yes	No	Invalid
UDP	WXR_TYPE_RQST	Integer	Yes	No	Invalid
UDP	WX_TITLE	String	Yes	No	Invalid

13.2 Datalink Options

This section describes the various datalink options selected for the CMF baseline AMI in the Datalink Options component.

13.2.1 VHF Region Tables

The VHF Region Tables control the VHF radio tuning function in CMF and contain data for up to 20 user-defined regions, plus Other (used when the airplane is outside of a user-defined region) and Default (used when CMF does not know the airplane's position) regions. These tables are defined as follows in the CMF baseline AMI:

```
Region Name ..... "GUAM"
  Frequencies
    1   131.550
    2   131.725
  Areas
    Top Left Corner   Bottom Right Corner
    -----
    N20.00 E142.00     N18.00 E149.00
    N18.00 E140.00     N11.00 E150.00
    N11.00 E141.00     N 9.00 E149.00
  Preferences
    1   POA
    2   None
    3   None
    4   None
    5   None
    6   None
    7   None
    8   None
  AOA Service Providers
    1   None
    2   None
    3   None
    4   None

Region Name ..... "KYRGYZSTN"
  Frequencies
    1   131.450
    2   131.725
    3   131.825
  Areas
```

Top Left Corner	Bottom Right Corner
-----	-----
N48.00 E 79.00	N47.00 E 86.00
N47.00 E 79.00	N42.00 E 81.00
N42.00 E 73.00	N36.00 E 81.00

Preferences

- 1 POA
- 2 None
- 3 None
- 4 None
- 5 None
- 6 None
- 7 None
- 8 None

AOA Service Providers

- 1 None
- 2 None
- 3 None
- 4 None

Region Name "RUSSIA"

Frequencies

- 1 131.725
- 2 131.550

Areas

Top Left Corner	Bottom Right Corner
-----	-----
N73.00 E 77.00	N54.00 E166.00
N54.00 E 77.00	N53.00 E115.00
N53.00 E 86.00	N52.00 E115.00
N52.00 E 86.00	N50.50 E 99.00
N54.00 E130.00	N52.00 E166.00
N52.00 E151.00	N49.00 E166.00

Preferences

- 1 POA
- 2 None
- 3 None
- 4 None
- 5 None
- 6 None
- 7 None
- 8 None

AOA Service Providers

- 1 None
- 2 None
- 3 None

4 None

Region Name "ANDAMAN"

Frequencies

1 131.725
2 131.825
3 131.450
4 131.550

Areas

Top Left Corner	Bottom Right Corner
-----	-----
N15.00 E 93.00	N14.00 E 95.00
N14.00 E 93.00	N 9.00 E 96.00
N 9.00 E 93.00	N 8.00 E 95.00

Preferences

1 POA
2 None
3 None
4 None
5 None
6 None
7 None
8 None

AOA Service Providers

1 None
2 None
3 None
4 None

Region Name "ASIA/AUST"

Frequencies

1 131.450
2 131.550
3 131.725

Areas

Top Left Corner	Bottom Right Corner
-----	-----
N54.00 E115.00	N52.00 E130.00
N52.00 E 99.00	N50.50 E151.00
N50.50 E 86.00	N47.00 E151.00
N47.00 E 81.00	N32.00 E151.00
N32.00 E 96.00	N28.00 E151.00
N28.00 E 96.00	N22.50 E135.00
N22.50 E 92.00	N 3.00 E135.00
N 3.00 E 92.00	S 5.00 E153.00
S 5.00 E101.00	S15.00 W145.00

S15.00 E109.00 S50.00 W145.00

Preferences

- 1 AOA
- 2 POA
- 3 None
- 4 None
- 5 None
- 6 None
- 7 None
- 8 None

AOA Service Providers

- 1 SITA 2
- 2 ARINC 1
- 3 AVICOM JAPAN 960
- 4 None

Region Name "MIDL EAST"

Frequencies

- 1 131.725
- 2 131.475

Areas

Top Left Corner	Bottom Right Corner
-----	-----
N37.00 E 42.00	N35.00 E 48.00
N35.00 E 40.00	N34.00 E 48.00
N34.00 E 40.00	N28.00 E 69.00
N28.00 E 37.00	N25.00 E 69.00
N25.00 E 36.00	N21.00 E 69.00
N21.00 E 36.00	N17.00 E 63.00
N17.00 E 43.00	N15.00 E 63.00
N15.00 E 46.00	N13.50 E 63.00

Preferences

- 1 AOA
- 2 POA
- 3 None
- 4 None
- 5 None
- 6 None
- 7 None
- 8 None

AOA Service Providers

- 1 SITA 2
- 2 ARINC 1
- 3 None
- 4 None

Region Name "EUR/AFRIC"

Frequencies

1 131.725
2 131.825

Areas

Top Left Corner	Bottom Right Corner
-----	-----
N83.00 W 11.00	N60.00 E 77.00
N60.00 W 17.00	N53.00 E 77.00
N53.00 W 17.00	N47.00 E 86.00
N47.00 W 17.00	N43.00 E 81.00
N43.00 W 36.00	N32.00 E 81.00
N32.00 W 28.00	N22.50 E 96.00
N22.50 W 28.00	N15.00 E 92.00
N15.00 W 28.00	N 8.00 E 93.00
N 8.00 W 28.00	N 0.00 E 92.00
N 0.00 E 2.00	S38.00 E 62.00

Preferences

1 AOA
2 POA
3 None
4 None
5 None
6 None
7 None
8 None

AOA Service Providers

1 SITA 2
2 ARINC 1
3 None
4 None

Region Name "N.ATLANTC"

Frequencies

1 131.550
2 131.725

Areas

Top Left Corner	Bottom Right Corner
-----	-----
N71.00 W 59.00	N70.00 W 44.00
N70.00 W 59.00	N68.00 W 28.00
N68.00 W 59.00	N60.00 W 11.00
N60.00 W 53.00	N59.00 W 17.00
N59.00 W 53.00	N56.00 W 36.00

Preferences

1 AOA

2 POA
3 None
4 None
5 None
6 None
7 None
8 None
AOA Service Providers
1 ARINC 1
2 SITA 2
3 None
4 None

Region Name "AMERICAS"

Frequencies

1 131.550
2 131.725

Areas

Top Left Corner	Bottom Right Corner
-----	-----
N76.00 E166.00	N60.00 W 59.00
N60.00 E166.00	N56.00 W 53.00
N56.00 E166.00	N47.00 W 46.00
N47.00 W131.00	N41.00 W 46.00
N41.00 W131.00	N30.00 W 55.00
N30.00 W121.00	N11.00 W 55.00
N11.00 W 95.00	N 5.00 W 48.00
N 5.00 W 95.00	S 5.00 W 28.00
S 5.00 W 85.00	S28.00 W 28.00
S28.00 W 85.00	S59.00 W 45.00

Preferences

1 AOA
2 POA
3 None
4 None
5 None
6 None
7 None
8 None

AOA Service Providers
1 ARINC 1
2 SITA 2
3 BRAZIL 961
4 None

Region Name "HAWAII"

Frequencies

1 131.550
2 131.725

Areas

Top Left Corner	Bottom Right Corner
-----	-----
N26.00 W164.00	N24.00 W153.00
N24.00 W164.00	N17.00 W150.00
N17.00 W160.00	N15.00 W151.00

Preferences

1 AOA
2 POA
3 None
4 None
5 None
6 None
7 None
8 None

AOA Service Providers

1 ARINC 1
2 SITA 2
3 None
4 None

Region Name "Other"

Frequencies

No Frequencies

Preferences

1 None
2 None
3 None
4 None
5 None
6 None
7 None
8 None

AOA Service Providers

1 None
2 None
3 None
4 None

Region Name "Default"

Frequencies

1 131.550
2 131.725

3	131.450
4	131.825
5	131.475

Preferences

1	AOA
2	POA
3	None
4	None
5	None
6	None
7	None
8	None

AOA Service Providers

1	AVICOM JAPAN 960
2	SITA 2
3	ARINC 1
4	BRAZIL 961

Figure 13-1 graphically depicts the user-defined VHF region tables that are defined in the CMF baseline AMI.

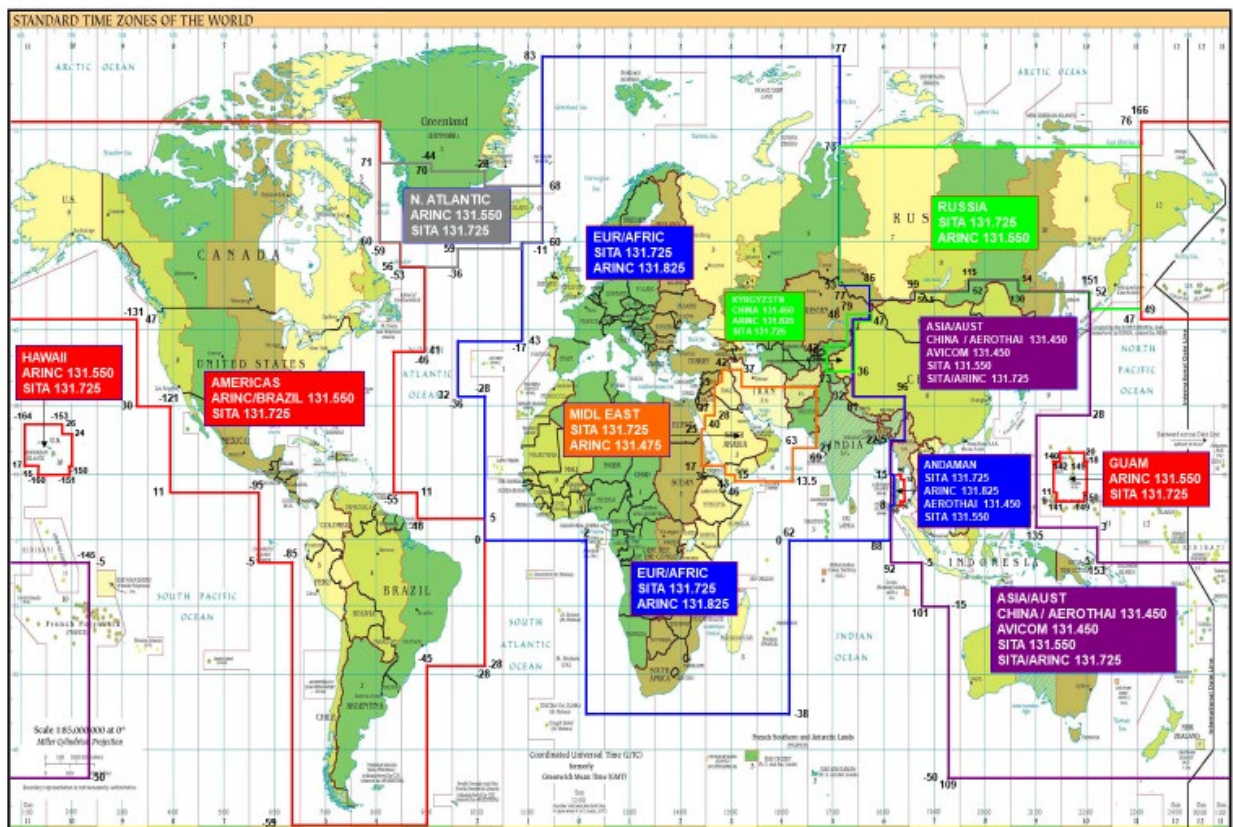


Figure 13-1 User-Defined VHF Region Tables Graphical Depiction

13.2.2 Timers

Table 13-2 describes the Datalink Options component timers and their values in the CMF baseline AMI.

Table 13-2 Datalink Options Timers

Timer	Value
Normal Scan Timer [seconds]	126
Fast Scan Timer [seconds]	15
Mode A Prekey Length [milliseconds]	50
POA Contact Timer (VAT1) [seconds]	126
POA Tracker Timer (VAT3) [seconds]	600
SATCOM Tracker Timer (SAT3) [minutes]	240
AolP Message Lifetime [minutes]	9

13.2.3 Counters

Table 13-3 describes the Datalink Options component counters and their values in the CMF baseline AMI.

Table 13-3 Datalink Options Counters

Counter	Value
Logical Channel Counter Expiration	3
POA Transmission Counter Expiration	3
SATCOM Transmission Counter Expiration	2
HF Transmission Counter Expiration	2
ACARS over Aviation VHF Link Control (AVLC) (AOA) Transmission Counter Expiration	2

Note: The transmission counter values defined in the CMF AMI apply only to non-ATS downlink messages; CMF operational software defines the transmission counter values for ATS downlink messages.

13.2.4 Enables

Table 13-4 describes the Datalink Options component enables and their values in the CMF baseline AMI.

Table 13-4 Datalink Options Enables

Enable	Value
POA Tracker Timer Enabled	Yes
SATCOM Tracker Timer Enabled	Yes

Enable	Value
Media Advisory Enabled	Yes

13.2.4.1 POA Tracker Timer

When enabled, CMF sends the POA tracker message (label Q0 link test downlink) whenever the POA tracker timer (VAT3) expires. This enables the service provider to track the location of the aircraft so that uplinks can be sent by the aircraft operator without prior knowledge of its general location.

In the CMF baseline AML this option is set to enable POA tracker messages.

13.2.4.2 SATCOM Tracker Timer

When enabled, CMF sends the SATCOM tracker message (label Q0 link test downlink) whenever the SATCOM tracker timer (SAT3) expires. This informs the service provider that the CMF still has a SATCOM connection.

In the CMF baseline AML this option is set to enable SATCOM tracker timer messages.

13.2.4.3 Media Advisory Enable

Media advisory messages are downlink messages that CMF automatically transmits when VHF, HF, or SATCOM communications are lost or become established. These messages provide an indication to the recipient on the ground of which subnetworks are available to CMF and are used by some DSPs as a means of routing some types of uplink messages (in particular, ATC uplink messages).

This option enables operators to prevent CMF from sending media advisory messages, except that CMF operational software always sends them when it has a FANS CPDLC connection.

In the CMF baseline AML this option is set to enable the transmission of media advisory messages.

13.2.5 Subnetwork Preference Table

The Subnetwork Preferences Table in the Datalink Options component enables aircraft operators to determine the relative preference of each subnetwork from among VHF, HF data link (HFDL), SATCOM, and AoIP choices for each type of downlink. This affects CMF's choice of subnetwork for delivery of the downlinks, subject to the definition of allowable subnetworks for each downlink type.

Table 13-5 describes the subnetwork preferences defined in the CMF baseline AML.

Table 13-5 Subnetwork Preferences

Message Type/Source	First Preference	Second Preference	Third Preference	Fourth Preference
AOC ("Company")	VHF	SATCOM	HFDL	NONE
ATC	VHF	SATCOM	HFDL	NONE
FIS	VHF	SATCOM	HFDL	NONE
FMF	VHF	SATCOM	HFDL	NONE
Electronic Flight Bag (EFB)	VHF	SATCOM	HFDL	NONE
Central Maintenance Computing Function (CMCF)	VHF	SATCOM	HFDL	NONE
Airplane Condition Monitoring Function (ACMF)	VHF	SATCOM	HFDL	NONE
CDS (Common Display System), including Displays and Crew Alerting (DCA) maintenance displays	VHF	SATCOM	HFDL	NONE
Cabin Systems	VHF	SATCOM	HFDL	NONE

Note: On 787, Cabin Systems are In-Flight Entertainment (IFE), Engine Monitoring Unit (EMU), Cabin Air Conditioning and Temperature Control System (CACTCS), and Controller Server Module (CSM) systems.

13.3 Downlink Buffers

The Downlink Buffers component defines the different buffers (queues) in which CMF stores downlink encodings before sending them. CMF is capable of buffering up to 50 total downlink encodings in up to 10 total downlink buffers. Table 13-6 describes the Downlink Buffers component that the CMF baseline AML implements and Table A-2 lists the buffer to which each downlink encoding is assigned.

Table 13-6 Downlink Buffers

Identifier	Size (Encodings)
MESSAGE	6
REPORT_AUTO	10
REPORT_CREW	8
REPORT_OOOI	8
REQUEST	6
RESPONSE_AUTO	6
RESPONSE_CREW	6

13.4 LRU Messaging Tables

This section describes the Line-Replaceable Unit (LRU, i.e., datalink end system) Messaging Tables component that the CMF baseline AML implements.

13.4.1 Default Uplink Routing Table

The Default Uplink Routing Table shows the datalink end systems to which CMF operational software routes uplink messages containing specified labels and sublabels. Table 13-7 describes the Default Uplink Routing Table that is visible in the CMF baseline AML.

Table 13-7 Default Uplink Routing Table

Uplink Label	Uplink Sublabel	Uplink SMI	Destination System
C1	none	AGM	Printer
H1	CF	CFD	CMCF
H1	DF	DFD	ACMF
H1	EI	ENG	Displays (DCA)
H1	F1	EF1	EFB-L
H1	F2	EF2	EFB-R
H1	M1	FML	FMF (master)
H1	M2	FMR	FMF (master)
H1	M3	FM3	FMF (master)
H1	MD	FMD	FMF (master)
H1	T1	TT1	Cabin System 1 (IFE-1)
H1	T2	TT2	Cabin System 2 (IFE-2)
H1	T3	TT3	Cabin System 3 (IFE-3)
H1	T4	TT4	Cabin System 4 (IFE-4)
H1	T5	TT5	Cabin System 5 (EMU-L)
H1	T6	TT6	Cabin System 6 (EMU-R)
H1	T7	TT7	Cabin System 7 (CACTCS)
H1	T8	TT8	Cabin System 8 (CSM)

13.4.2 Customizable Uplink Routing Table

The Customizable Uplink Routing Table defines the datalink end systems to which CMF routes uplink messages containing specified labels and sublabels in addition to those defined by CMF operational software and visible in the default uplink routing table. The CMF baseline AML does not contain any entries in the Customizable Uplink Routing Table.

13.4.3 Downlink Label Table

The Downlink Label Table defines which labels and sublabels CMF applies to downlink messages from specified datalink end systems. Table 13-8 describes the Downlink Label Table defined in the CMF baseline AML, including the SMI applicable to each entry.

Table 13-8 Downlink Label Table

Downlink Label	Downlink Sublabel	Downlink SMI	Source System
H1	CF	CFD	CMCF
H1	DF	DFD	ACMF
H1	EI	ENG	Displays (DCA)
H1	F1	EF1	EFB-L
H1	F2	EF2	EFB-R
H1	M1	FML	FMF-L
H1	M2	FMR	FMF-R
H1	M3	FM3	FMF-C
H1	T1	TT1	Cabin System 1 (IFE-1)
H1	T2	TT2	Cabin System 2 (IFE-2)
H1	T3	TT3	Cabin System 3 (IFE-3)
H1	T4	TT4	Cabin System 4 (IFE-4)
H1	T5	TT5	Cabin System 5 (EMU-L)
H1	T6	TT6	Cabin System 6 (EMU-R)
H1	T7	TT7	Cabin System 7 (CACTCS)
H1	T8	TT8	Cabin System 8 (CSM)

13.5 Review Logs

The Review Logs component defines up to eight REVIEW menu categories into which CMF places a display associated with a message after the flight crew has displayed (and if necessary, responded to) or sent the message. Table 13-9 describes the REVIEW menu categories that the CMF baseline AMI implements.

Table 13-9 REVIEW Menu Categories

Identifier	Title
REVIEW_RECEIVED	COMPANY UPLINKS...
REVIEW_SENT	COMPANY DOWNLINKS...
REVIEW_WX_NOTAM	WEATHER & NOTAMS...
REVIEW_LDS	LOADSHEETS...

Table 13-10 lists the REVIEW menu category assignments for each display component in the CMF baseline AMI.

Table 13-10 REVIEW Menu Category Display Assignments

Display	REVIEW Menu Category
ADRF and IRU Parameters	None
Aircraft Parameters Menu	None
Altitude Reminder	None
Arrival Report	COMPANY DOWNLINKS...
ATC Datalink Problem Report	COMPANY DOWNLINKS...
Automatic Emergency Report Parameters	None
Automatic Flight Initialization Request	COMPANY DOWNLINKS...
Automatic Flight Initialization Uplink	COMPANY UPLINKS...
CMF Parameters	None
Company Menu	None
Crew Advisory Uplink	COMPANY UPLINKS...
Crew Requests Menu	None
Delay Report	COMPANY DOWNLINKS...
Departure Report	COMPANY DOWNLINKS...
Diversion Report	COMPANY DOWNLINKS...
Display Command Uplink	COMPANY UPLINKS...
Engine and FQMS Parameters	None
ETA Report	COMPANY DOWNLINKS...
E-Mail Downlink	COMPANY DOWNLINKS...
E-Mail Reply	COMPANY DOWNLINKS...
E-Mail Uplink	COMPANY UPLINKS...
FCM Parameters	None
Flight Level Reminder	None
Flight Release Request	COMPANY DOWNLINKS...
Flight Release Uplink	COMPANY UPLINKS...
FMF and Navigation Parameters	None
Free-Text Uplink	COMPANY UPLINKS...

Display	REVIEW Menu Category
Fuel Reminder	None
Gate Assignment Request	COMPANY DOWNLINKS...
Gate Assignment Uplink	COMPANY UPLINKS...
Latitude Reminder	None
Loadsheet Uplink	LOADSHEETS...
Longitude Reminder	None
Maintenance Report	COMPANY DOWNLINKS...
Manual Flight Initialization	None
Message to Ground	COMPANY DOWNLINKS...
Miscellaneous Code Report	COMPANY DOWNLINKS...
Miscellaneous Parameters	None
NOTAMs Request	WEATHER & NOTAMS...
NOTAMs Uplink	WEATHER & NOTAMS...
OOOI Test Page	None
Passing Waypoint Reminder	None
PSS Parameters	None
Reminders	None
Sensor Status	None
Situation Report	COMPANY DOWNLINKS...
Time Reminder	None
Time to Destination Reminder	None
Time to Top of Descent Reminder	None
Times and Fuels	None
VHF Voice Contact Request	COMPANY UPLINKS...
Waypoint ETA Change Reminder	None
Weather Request	WEATHER & NOTAMS...
Weather Uplink	WEATHER & NOTAMS...
Weight and Balance Request	COMPANY DOWNLINKS...
Weight and Balance Uplink	COMPANY UPLINKS...

Appendix A Message Attributes

This appendix describes certain attributes of the messages defined in the baseline AML.

A.1 Uplink Decoding Definition Attributes

Table A-1 lists certain attributes of the baseline AML uplink decoding definitions. Of note:

- “SMI” and “Label” respectively indicate the Standard Message Identifier (SMI) and label defined by ARINC Specification 620 Appendix C for each uplink decoding.
- Pre-defined element delimiters are <CR>, <CR><LF>, and <ETX>. Many uplink decoding definitions also have optional element delimiters.

Table A-1 Uplink Decoding Definition Attributes

Uplink Decoding	SMI	Label	Message Identifier	Alert Level	Optional Element Delimiter	Notes
<i>Flight Initialization</i>						
Automatic Flight Initialization Uplink	CMD	RA	~1INI	Medium	“/” (forward slash)	
<i>Weather and NOTAMs</i>						
Weather Uplink	CMD	RA	~1WXR	Medium	“,” (comma)	Text block limited to 3486 characters
NOTAMs Uplink	CMD	RA	~1NTM	Medium	“,” (comma)	Text block limited to 3486 characters
<i>Messaging</i>						
Display Command Uplink	CMD	RA	~1RA1	Medium	“,” (comma)	Text block limited to 3480 characters
E-Mail Uplink	CMD	RA	~1EML	Medium	“,” (comma)	Text block limited to 3432 characters
Free-Text Uplink	CMD	RA	~1DIS	Medium	“,” (comma)	Text block limited to 3480 characters
Printer Uplink	AGM	C1	None	Medium	“,” (comma)	Text block limited to 3503 characters; Printer Uplink print definition printed upon receipt
VHF Voice Contact Request	GVR	54	None	Medium	“,” (comma)	Text block limited to 200 characters
<i>Request-Response Exchanges</i>						

Uplink Decoding	SMI	Label	Message Identifier	Alert Level	Optional Element Delimiter	Notes
Crew Advisory Uplink	CMD	RA	~1CAA	Medium	“,” (comma)	Text block limited to 3496 characters
Flight Release Uplink	CMD	RA	~1FRL	Medium	“,” (comma)	Text block limited to 3496 characters
Gate Assignment Uplink	CMD	RA	~1GTA	Medium	“,” (comma)	Text block limited to 3496 characters
Loadsheel Uplink	M1L	1L	LDS	Medium	“/” (forward slash)	Text block limited to 3481 characters
Weight and Balance Uplink	CMD	RA	~1WAB	Medium	“,” (comma)	Text block limited to 3496 characters
Automatic Reports						
AHM Flight Phase Report Configuration Uplink	CMD	RA	~1AHM	None	“,” (comma)	
AoIP Link Advisory Configuration Uplink	CMD	RA	~AOI	None	“,” (comma)	
Automatic Emergency Report Configuration Uplink	CMD	RA	~1DEF	None	“,” (comma)	
Automatic Position Report Configuration Uplink	CMD	RA	~1POS	None	“,” (comma)	
Loop Back Messages						
Automatic Emergency Report Status Loop Back Uplink	CMD	RA	~4EMG	None	none	Automatic Emergency Report Status Loop Back Downlink sent upon receipt
OOOI Data Loop Back Uplink	CMD	RA	~2OOI	None	none	OOOI Data Loop Back Downlink sent upon receipt
Position Report Loop Back Uplink	CMD	RA	~4POS	None	none	Position Report Loop Back Downlink sent upon receipt
Software Configuration Loop Back Uplink	CMD	RA	~4CFG	None	none	Software Configuration Loop Back Downlink sent upon receipt
Subnetwork Status Loop Back Uplink	CMD	RA	~4SUB	None	none	Subnetwork Status Loop Back Downlink sent upon receipt
Test Loop Back Uplink	CMD	RA	~4LBK	None	none	Text block limited to 200 characters; Test Loop Back Downlink sent upon receipt
Miscellaneous Functions						
POA Autotune Uplink	M4;	4;	None	None	None	

Note: The Medium alert level provides a “●COMM” visual alert and a high-low chime aural alert. The Low alert level provides no visual alert but provides a high-low chime aural alert. The None alert level provides no visual alert and no aural alert.

A.2 Downlink Encoding Definition Attributes

Table A-2 lists certain attributes of the baseline AMI downlink encoding definitions. Of note:

- “SMI” and “Label” respectively indicate the SMI and label described by ARINC Specification 620 Appendix C for each downlink encoding.
- “Downlink Priority” indicates the downlink priority for each downlink encoding, which CMF uses to prioritize any queued downlink encodings. The highest priority is 1 Emergency and the lowest priority is 9 Low Priority Administrative. If two or more queued downlink encodings have the same downlink priority, then CMF prioritizes the oldest encoding.
- “Downlink Buffer” indicates the downlink buffer for each downlink encoding, which is the buffer in which CMF queues the downlink encoding. Table 13-6 describes the baseline AMI downlink buffers.
- “Delete on Reset” indicates whether CMF deletes each queued downlink encoding from the relevant downlink buffer when a CMF system reset occurs. If this value is “Yes”, then CMF deletes the queued downlink encoding when a CMF system reset occurs. If this value is “No”, then CMF retains the queued downlink encoding when a CMF system reset occurs.
- “Delete on Full” indicates whether CMF deletes each queued downlink encoding from the relevant downlink buffer when a new downlink encoding is queued. If this value is “Yes”, then CMF deletes the oldest queued downlink encoding in order to accept the new downlink encoding. If this value is “No”, then CMF deletes the newest queued downlink encoding in order to retain the oldest queued downlink encoding.

Table A-2 Downlink Encoding Definition Attributes

Downlink Encoding	SMI	Label	Message Identifier and Version	Downlink Priority	Downlink Buffer	Delete on Reset	Delete on Full
<i>Flight Initialization</i>							
Automatic Flight Initialization Request	M20	20	INI01	4 Flight Regularity	REQUEST	Yes	Yes
<i>Weather and NOTAMs</i>							
Weather Request	WXR	5U	none	4 Flight Regularity	REQUEST	Yes	Yes
NOTAMs Request	M38	38	NTM01	4 Flight Regularity	REQUEST	Yes	Yes
<i>Messaging</i>							
E-Mail Downlink	EML	E1	EML02	4 Flight Regularity	MESSAGE	Yes	Yes

Downlink Encoding	SMI	Label	Message Identifier and Version	Downlink Priority	Downlink Buffer	Delete on Reset	Delete on Full
E-Mail Reply	EML	E1	EML02	4 Flight Regularity	MESSAGE	Yes	Yes
Message to Ground	A81	81	none	4 Flight Regularity	MESSAGE	Yes	Yes
Request-Response Exchanges							
Crew Advisory Accept	M40	40	CAA01	4 Flight Regularity	RESPONSE_CREW	Yes	Yes
Crew Advisory Reject	M40	40	CAA01	4 Flight Regularity	RESPONSE_CREW	Yes	Yes
Crew Advisory Reject with Reason	M40	40	CAA01	4 Flight Regularity	RESPONSE_CREW	Yes	Yes
Flight Release Request	M35	35	FRL01	4 Flight Regularity	REQUEST	Yes	Yes
Flight Release Accept	M45	45	FRL01	4 Flight Regularity	RESPONSE_CREW	Yes	Yes
Flight Release Reject	M45	45	FRL01	4 Flight Regularity	RESPONSE_CREW	Yes	Yes
Flight Release Reject with Reason	M45	45	FRL01	4 Flight Regularity	RESPONSE_CREW	Yes	Yes
Gate Assignment Request	M36	36	GTA01	4 Flight Regularity	REQUEST	Yes	Yes
Gate Assignment Accept	M46	46	GTA01	4 Flight Regularity	RESPONSE_CREW	Yes	Yes
Gate Assignment Reject	M46	46	GTA01	4 Flight Regularity	RESPONSE_CREW	Yes	Yes
Gate Assignment Reject with Reason	M46	46	GTA01	4 Flight Regularity	RESPONSE_CREW	Yes	Yes
Loadsheet Response	M1L	1L	none	4 Flight Regularity	RESPONSE_CREW	Yes	Yes
Weight and Balance Request	M31	31	WAB01	4 Flight Regularity	REQUEST	Yes	Yes
Weight and Balance Accept	M41	41	WAB01	4 Flight Regularity	RESPONSE_CREW	Yes	Yes
Weight and Balance Reject	M41	41	WAB01	4 Flight Regularity	RESPONSE_CREW	Yes	Yes
Weight and Balance Reject with Reason	M41	41	WAB01	4 Flight Regularity	RESPONSE_CREW	Yes	Yes
Manual Reports							
Arrival Report	M28	28	LND02	4 Flight Regularity	REPORT_CREW	Yes	Yes
ATC Datalink Problem Report	A82	82	CRA02	4 Flight Regularity	REPORT_CREW	Yes	Yes
Delay Report	M21	21	Identifier DPD, DTO, DNR, or DAR; version 02	4 Flight Regularity	REPORT_CREW	Yes	Yes
Departure Report	M24	24	TKO01	4 Flight Regularity	REPORT_CREW	Yes	Yes
Manual Diversion Report	M25	25	DIV02	4 Flight Regularity	REPORT_CREW	Yes	Yes
ETA Report	M26	26	ETA01	4 Flight Regularity	REPORT_CREW	Yes	Yes
Maintenance Report	M2A	2A	MNT01	4 Flight Regularity	REPORT_CREW	Yes	Yes

Downlink Encoding	SMI	Label	Message Identifier and Version	Downlink Priority	Downlink Buffer	Delete on Reset	Delete on Full
Miscellaneous Code Report	M29	29	MSC01	4 Flight Regularity	REPORT_CREW	Yes	Yes
Situation Report	HJK	00	SIT02	1 Emergency	REPORT_CREW	No	No
Automatic Reports							
AHM Flight Phase Report	M49	49	Identifier variable per Section 9.1.2; version 01	4 Flight Regularity	REPORT_AUTO	No	Yes
AoIP Link Advisory	M19	19	AOI02	4 Flight Regularity	REPORT_AUTO	Yes	Yes
Automatic Diversion Report	M25	25	DIV02	4 Flight Regularity	REPORT_AUTO	Yes	Yes
Automatic Emergency Report	SIT	99	EMR06	1 Emergency	REPORT_AUTO	No	No
Automatic Position Report	M16	16	POS02	4 Flight Regularity	REPORT_AUTO	No	Yes
Out Report	M11	11	OUT01	4 Flight Regularity	REPORT_OOOI	No	Yes
Off Report	M12	12	OFF01	4 Flight Regularity	REPORT_OOOI	No	Yes
On Report	M13	13	ONN01	4 Flight Regularity	REPORT_OOOI	No	Yes
In Report	M14	14	INN01	4 Flight Regularity	REPORT_OOOI	No	Yes
Return to Gate Report	M15	15	RTG01	4 Flight Regularity	REPORT_OOOI	No	Yes
Loop Back Messages							
Automatic Emergency Report Status Loop Back Downlink	RDO	RB	~4EMG	4 Flight Regularity	RESPONSE_AUTO	No	Yes
OOOI Data Loop Back Downlink	RDO	RB	~2OOI	4 Flight Regularity	RESPONSE_AUTO	No	Yes
Position Report Loop Back Downlink	RDO	RB	~4POS	4 Flight Regularity	RESPONSE_AUTO	No	Yes
Software Configuration Loop Back Downlink	RDO	RB	~4CFG	4 Flight Regularity	RESPONSE_AUTO	No	Yes
Subnetwork Status Loop Back Downlink	RDO	RB	~4SUB	4 Flight Regularity	RESPONSE_AUTO	No	Yes
Test Loop Back Downlink	RDO	RB	~4LBK	4 Flight Regularity	RESPONSE_AUTO	No	Yes
Miscellaneous Functions							
Invalid Reject	A8X	8X	none	4 Flight Regularity	RESPONSE_AUTO	No	Yes
Unable to Deliver Reject	REJ	HX	none	4 Flight Regularity	RESPONSE_AUTO	No	Yes
Unable to Process Reject	A8X	8X	none	4 Flight Regularity	RESPONSE_AUTO	No	Yes

Abbreviations and Acronyms

AAC	Aeronautical Administrative Communications
ACARS	Aircraft Communications Addressing and Reporting System
ACMF	Airplane Condition Monitoring Function
ADRF	Air Data Reference Function
ADS-C	Automatic Dependent Surveillance – Contract
AHM	Airplane Health Management
AMI	Airline Modifiable Information
AOA	ACARS over AVLK
AOC	Aeronautical Operational Communications
AoIP	ACARS-over-IP
ATC	Air Traffic Control
ATN	Aeronautical Telecommunications Network
ATS	Air Traffic Services
AVLC	Aviation VHF Link Control
BPv	Block Point version
B1	Baseline 1
CACTCS	Cabin Air Conditioning and Temperature Control System
CAS	calibrated airspeed
CCR	Common Computing Resource
CCS	Common Computing System
CDS	Common Display System
CMCF	Central Maintenance Computing Function

CMF	Communications Management Function
CMU	Communications Management Unit
CPDLC	Controller-Pilot Data Link Communications
CSM	Controller Server Module
DCA	Displays and Crew Alerting
DSP	datalink service provider
D-ATIS	Digital Automated Terminal Information Service
EFB	Electronic Flight Bag
EICAS	Engine Indicating and Crew Alerting System
EMU	Engine Monitoring Unit
ESB	exclusive selector button
ETA	estimated time of arrival
EUROCAE	European Organisation for Civil Aviation Equipment
FANS	Future Air Navigation System
FCM	Flight Control Module
FIS	Flight Information Services
FMF	Flight Management Function
FQMS	Fuel Quantity Monitoring System
FWT	folding wingtip (777X)
GBST	Ground-Based Software Tool
GE	General Electric
GGM	Graphics Generator Module
HF	high frequency
HFDL	HF data link

IATA	International Air Transport Association
ICAO	International Civil Aviation Organization
IFE	In-Flight Entertainment
IP	Internet Protocol
IPDB	Input Parameter Database
IRU	Inertial Reference Unit
LSAP	loadable software application part
MFD	Multi-Function Display
M _{MO}	Mach, maximum operating
MSN	message sequence number
NESB	non-exclusive selector button
NOTAM	Notices to Airmen
NVM	non-volatile memory
OOOI	Out, Off, On, In
OPS	operational software
OSS	Option Selection Software
POA	“Plain Old” ACARS
PSS	Proximity Sensing System
SAT	static air temperature
SATCOM	satellite communications
SMI	Standard Message Identifier
TAS	true airspeed
TAT	total air temperature
TWIP	Terminal Weather Information for Pilots

UADF	User Application Definition File
VDR	VHF Data Radio
VHF	very high frequency
V _{MO}	Speed, maximum operating

Revision Record

Release/Revision	NEW	
Limitations	None additional.	
Description of Change	Based on D613Z019-01 Rev H to describe 787 CMF baseline AMI version F001 (part number BCG45-8CMF-A013). Notable AMI changes are those described in Section 1.3. Added Section 1.1.1 to describe execution order of logic units, Section 2.1 to describe start display, and Section 2.8 to describe special encodings. Added note that if AoIP is not enabled then the AMI does not generate AoIP link advisories in Section 2.8.2. Added descriptions of the conditions under which the AMI enables and disables the automatic emergency report function and begins sending automatic emergency reports in Section 9.4. Corrected description of automatic emergency report selections in Section 11.1.1. Added SMIs to Table 13-8 and 13-9. Added alert levels to Table A-1. Added message versions to Table A-2. Corrected text block lengths in uplink decoding definitions throughout. Enhanced element descriptions in uplink decoding definitions and downlink encoding definitions throughout. Made editorial clarifications throughout.	
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